MATHEMATICAL METHODS

Gilbert Strang - Linear Algebra and Its Applications, 4th Ed.

Syllabus-only edition

Only the portions listed in the supplied course outline are retained.

Included syllabus portions

The sequence below follows the course-outline categories. Original textbook pages are preserved; material outside the listed sections is omitted.

Course block	Included textbook material
Foundation of LA	Chapter 1, Sections 1.1-1.2
Systems of linear equations	Chapter 1, Sections 1.1-1.3; Chapter 2, Sections 2.1-2.4 and 2.6
Inner Product Spaces	Chapter 3, Sections 3.1-3.4
Spectral Theory	Chapter 5, Sections 5.1-5.2
Quadratic Forms	Chapter 6, Sections 6.1-6.2
Solutions	Solutions to Selected Exercises retained

FOUNDATION OF LINEAR ALGEBRA

Strang Chapter 1, Sections 1.1-1.2

Chapter 1

Matrices and Gaussian Elimination

1.1 Introduction

This book begins with the central problem of linear algebra: *solving linear equations*. The most important ease, and the simplest, is when the number of unknowns equals the number of equations. We have n **equations in** n **unknowns**, starting with $n = 2$:

$$\begin{array}{lrcrcl} \textbf{Two equations} & 1x & + & 2y & = & 3 \\ \textbf{Two unknowns} & 4x & + & 5y & = & 6. \end{array} \tag{1}$$

The unknowns are x and y. I want to describe two ways, *elimination* and *determinants*, to solve these equations. Certainly x and y are determined by the numbers 1, 2, 3, 4, 5, 6. The question is how to use those six numbers to solve the system.

1. **Elimination** Subtract 4 times the first equation from the second equation. This eliminates x from the second equation. and it leaves one equation for y:

$$(\textbf{equation 2}) - 4(\textbf{equation 1}) \qquad -3y = -6. \tag{2}$$

   Immediately we know $y = 2$. Then x comes from the first equation $1x + 2y = 3$:

$$\textbf{Back-substitution} \qquad 1x + 2(2) = 3 \quad \text{gives} \quad x = -1. \tag{3}$$

   Proceeding carefully, we cheek that x and y also solve the second equation. This should work and it does: 4 times ($x = -1$) plus 5 times ($y = 2$) equals 6.

2. **Determinants** The solution $y = 2$ depends completely on those six numbers in the equations. There most be a formula for y (and also x) It is a "ratio of determinants" and I hope you will allow me to write it down directly:

$$y = \frac{\begin{vmatrix} 1 & 3 \\ 4 & 6 \end{vmatrix}}{\begin{vmatrix} 1 & 2 \\ 4 & 5 \end{vmatrix}} = \frac{1 \cdot 6 - 3 \cdot 4}{1 \cdot 5 - 2 \cdot 4} = \frac{-6}{-3} = 2. \tag{4}$$

That could seem a little mysterious, unless you already know about 2 by 2 determinants. They gave the same answer $y = 2$, coming from the same ratio of -6 to -3. If we stay with determinants (which we don't plan to do), there will be a similar formula to compute the other unknown, x:

$$x = \frac{\begin{vmatrix} 3 & 2 \\ 6 & 5 \end{vmatrix}}{\begin{vmatrix} 1 & 2 \\ 4 & 5 \end{vmatrix}} = \frac{3 \cdot 5 - 2 \cdot 6}{1 \cdot 5 - 2 \cdot 4} = \frac{3}{-3} = -1. \tag{5}$$

Let me compare those two approaches, looking ahead to real problems when n is much larger ($n = 1000$ is a very moderate size in scientific computing). The truth is that direct use of the determinant formula for 1000 equations would be a total disaster. It would use the million numbers on the left sides correctly, but not efficiently. We will find that formula (Cramer's Rule) in Chapter 4, but we want a good method to solve 1000 equations in Chapter 1.

That good method is *Gaussian Elimination*. This is the algorithm that is constantly used to solve large systems of equations. From the examples in a textbook ($n = 3$ is close to the upper limit on the patience of the author and reader) too might not see much difference. Equations (2) and (4) used essentially the same steps to find $y = 2$. Certainly x came faster by the back-substitution in equation (3) than the ratio in (5). For larger n there is absolutely no question. Elimination wins (and this is even the best way to compute determinants).

The idea of elimination is deceptively simple—you will master it after a few examples. It will become the basis for half of this book, simplifying a matrix so that we can understand it. Together with the mechanics of the algorithm, we want to explain four deeper aspects in this chapter. They are:

1. Linear equations lead to ***geometry of planes***. It is not easy to visualize a nine-dimensional plane in ten-dimensional space. It is harder to see ten of those planes, intersecting at the solution to ten equations—but somehow this is almost possible. Our example has two lines in Figure 1.1, meeting at the point $(x, y) = (-1, 2)$. Linear algebra moves that picture into ten dimensions, where the intuition has to imagine the geometry (and gets it right)

2. We move to ***matrix notation***, writing the n unknowns as a vector x and the n equations as $Ax = b$. We multiply A by "elimination matrices" to reach an upper triangular matrix U. Those steps factor A into L times U, where L is lower triangular. I will write down A and its factors for our example, and explain them at the right time:

$$\textbf{Factorization} \qquad A = \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & -3 \end{bmatrix} = L \textbf{ times } U. \tag{6}$$

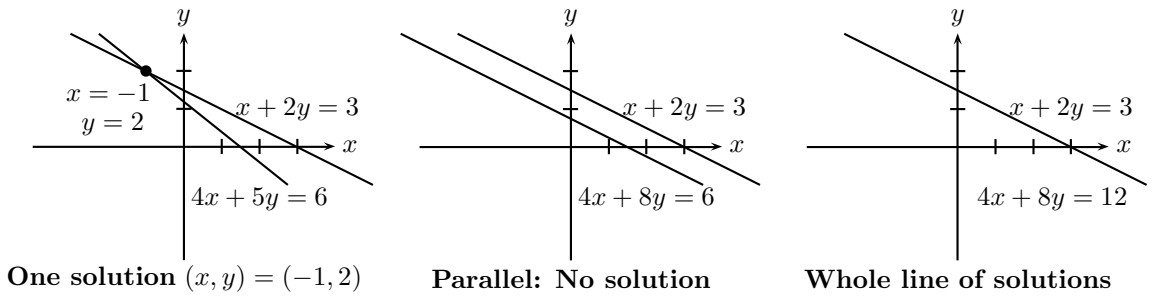


Figure 1.1: The example has one solution. Singular cases have none or too many.

First we have to introduce matrices and vectors and the rules for multiplication. Every matrix has a ***transpose*** A^{T}. This matrix has an ***inverse*** A^{-1}.

3. In most cases elimination goes forward without difficulties. The matrix has an inverse and the system $Ax = b$ has one solution. In exceptional cases the method will *break down*—either the equations were written in the wrong order, which is easily fixed by exchanging them, or the equations don't have a unique solution.

   That ***singular case*** will appear if 8 replaces 5 in our example:

$$\begin{array}{lrcrcr} \textbf{Singular case} & 1x & + & 2y & = & 3 \\ \textbf{Two parallel lines} & 4x & + & 8y & = & 6. \end{array} \tag{7}$$

   Elimination still innocently subtracts 4 times the first equation from the second. But look at the result!

$$(\textbf{equation 2}) - 4(\textbf{equation 1}) \qquad 0 = -6.$$

   This singular case has ***no solution***. Other singular cases have ***infinitely many solutions***. (Change 6 to 12 in the example, and elimination will lead to $0 = 0$. Now y can have *any value*,) When elimination breaks down, we want to find every possible solution.

4. We need a rough count of the ***number of elimination steps*** required to solve a system of size n. The computing cost often determines the accuracy in the model. A hundred equations require a third of a million steps (multiplications and subtractions). The computer can do those quickly, but not many trillions. And already after a million steps, roundoff error could be significant. (Some problems are sensitive; others are not.) Without trying for full detail, we want to see large systems that arise in practice, and how they are actually solved.

The final result of this chapter will be an elimination algorithm that is about as efficient as possible. It is essentially the algorithm that is in constant use in a tremendous variety of applications. And at the same time, understanding it in terms of *matrices*—the coefficient matrix A, the matrices E for elimination and P for row exchanges, and the

final factors L and U—is an essential foundation for the theory. I hope you will enjoy this book and this course.

1.2 The Geometry of Linear Equations

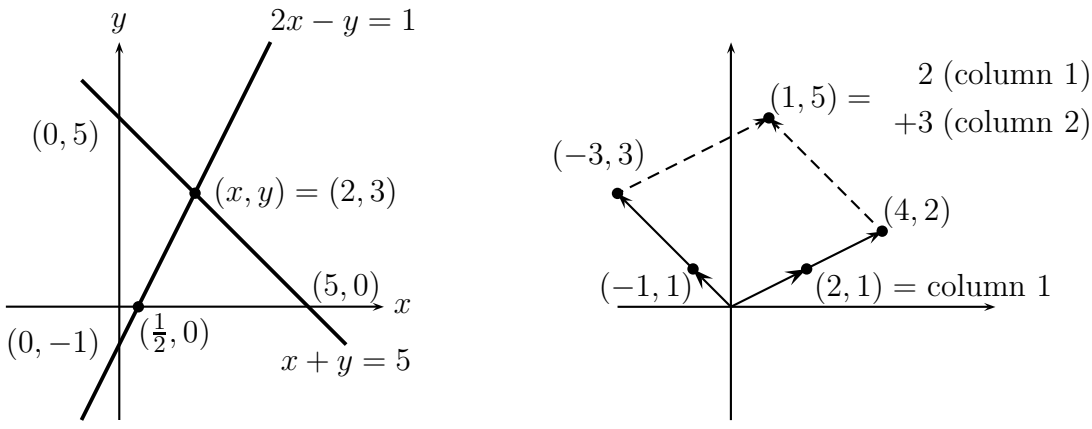
The way to understand this subject is by example. We begin with two extremely humble equations, recognizing that you could solve them without a course in linear algebra. Nevertheless I hope you will give Gauss a chance:

$$\begin{aligned} 2x &- y = 1 \\ x &+ y = 5. \end{aligned}$$

We can look at that system *by rows* or *by columns*. We want to see them both.

The first approach concentrates on the separate equations (the ***rows***). That is the most familiar, and in two dimensions we can do it quickly. The equation $2x - y = 1$ is represented by a *straight line* in the x-y plane. The line goes through the points $x = 1$, $y = 1$ and $x = \frac{1}{2}$, $y = 0$ (and also through $(2,3)$ and all intermediate points). The second equation $x + y = 5$ produces a second line (Figure 1.2a). Its slope is $dy/dx = -1$ and it crosses the first line at the solution.

The point of intersection lies on both lines. It is the only solution to both equations. That point $x = 2$ and $y = 3$ will soon be found by "elimination."



(a) Lines meet at $x = 2$, $y = 3$ (b) Columns combine with 2 and 3

Figure 1.2: Row picture (two lines) and column picture (combine columns).

The second approach looks at the ***columns*** of the linear system. The two separate equations are really ***one vector equation***:

$$\textbf{Column form} \qquad x\begin{bmatrix} 2 \\ 1 \end{bmatrix} + y\begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}.$$

The problem is ***to find the combination of the column vectors on the left side that produces the vector on the right side***. Those vectors $(2,1)$ and $(-1,1)$ are represented by the bold lines in Figure 1.2b. The unknowns are the numbers x and y that multiply the column vectors. The whole idea can be seen in that figure, where 2 times column 1 is added to 3 times column 2. Geometrically this produces a famous parallelogram. Algebraically it produces the correct vector $(1,5)$, on the right side of our equations. The column picture confirms that $x = 2$ and $y = 3$.

More time could be spent on that example, but I would rather move forward to $n = 3$. Three equations are still manageable, and they have much more variety:

$$\textbf{Three planes} \qquad \begin{array}{rcrcrcr} 2u & + & v & + & w & = & 5 \\ 4u & - & 6v & & & = & -2 \\ -2u & + & 7v & + & 2w & = & 9. \end{array} \tag{1}$$

Again we can study the rows or the columns, and we start with the rows. Each equation describes a ***plane*** in three dimensions. The first plane is $2u+v+w=5$, and it is sketched in Figure 1.3. It contains the points $(\frac{5}{2},0,0)$ and $(0,5,0)$ and $(0,0,5)$. It is determined by any three of its points—provided they do not lie on a line.

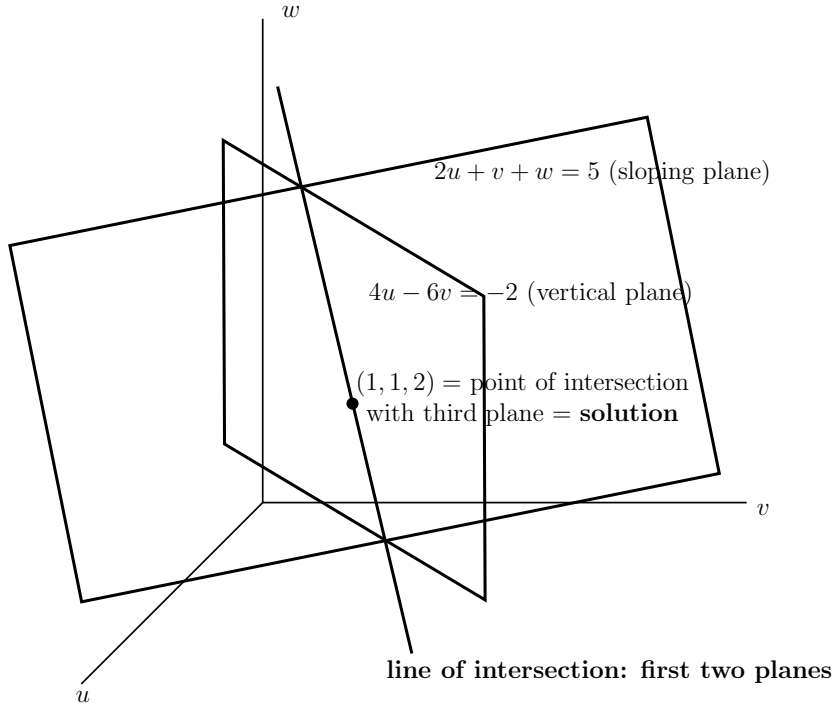


Figure 1.3: The row picture: three intersecting planes from three linear equations.

Changing 5 *to* 10*, the plane* $2u+v+w=10$ *would be parallel to this one*. It contains $(5,0,0)$ and $(0,10,0)$ and $(0,0,10)$, twice as far from the origin—which is the center point $u = 0$, $v = 0$, $w = 0$. Changing the right side moves the plane parallel to itself, and the plane $2u+v+w=0$ goes through the origin.

The second plane is $4u-6v=-2$. It is drawn vertically, because w can take any value. The coefficient of w is zero, but this remains a plane in 3-space. (The equation $4u=3$, or even the extreme case $u=0$, would still describe a plane.) The figure shows the intersection of the second plane with the first. That intersection is a line. *In three dimensions a line requires two equations*; in n dimensions it will require $n-1$.

Finally the third plane intersects this line in a point. The plane (not drawn) represents the third equation $-2u+7v+2w=9$, and it crosses the line at $u=1$, $v=1$, $w=2$. That triple intersection point $(1,1,2)$ solves the linear system.

How does this row picture extend into n dimensions? The n equations will contain n unknowns. The first equation still determines a "plane." It is no longer a two-dimensional plane in 3-space; somehow it has "dimension" $n-1$. It must be flat and extremely thin within n-dimensional space, although it would look solid to us.

If time is the fourth dimension, then the plane $t=0$ cuts through four-dimensional space and produces the three-dimensional universe we live in (or rather, the universe as it was at $t=0$). Another plane is $z=0$, which is also three-dimensional; it is the ordinary x-y plane taken over all time. Those three-dimensional planes will intersect! They share the ordinary x-y plane at $t=0$. We are down to two dimensions, and the next plane leaves a line. Finally a fourth plane leaves a single point. It is the intersection point of 4 planes in 4 dimensions, and it solves the 4 underlying equations.

I will be in trouble if that example from relativity goes any further. The point is that linear algebra can operate with any number of equations. The first equation produces an $(n-1)$-dimensional plane in n dimensions, The second plane intersects it (we hope) in a smaller set of "dimension $n-2$." Assuming all goes well, every new plane (every new equation) reduces the dimension by one. At the end, when all n planes are accounted for, the intersection has dimension zero. It is a *point*, it lies on all the planes, and its coordinates satisfy all n equations. It is the solution!

Column Vectors and Linear Combinations

We turn to the columns. This time the vector equation (the same equation as (1)) is

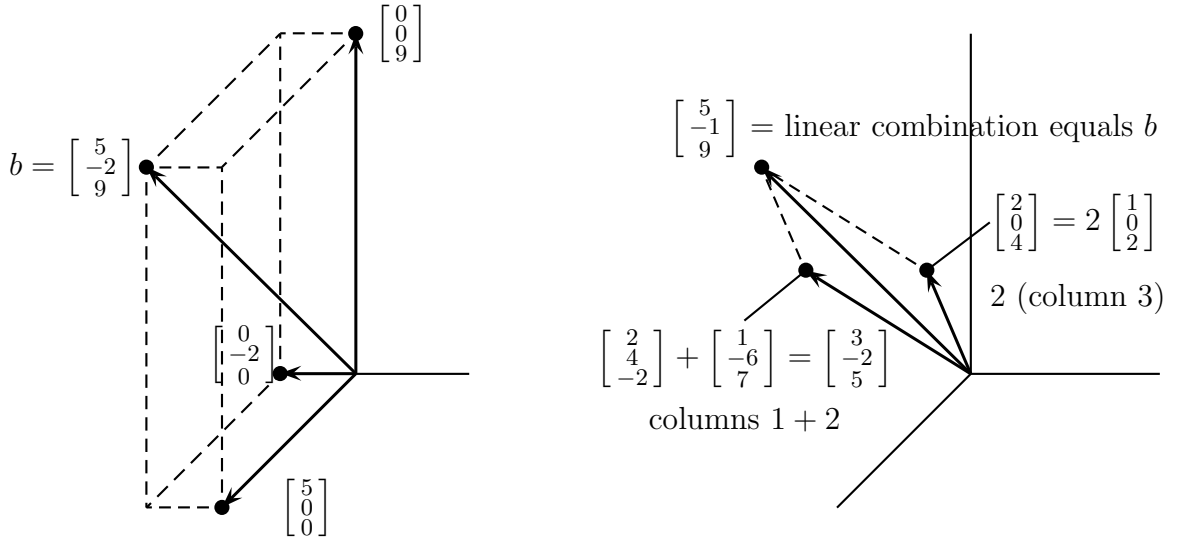
$$\textbf{Column form} \qquad u\begin{bmatrix}2\\4\\-2\end{bmatrix}+v\begin{bmatrix}1\\-6\\7\end{bmatrix}+w\begin{bmatrix}1\\0\\2\end{bmatrix}=\begin{bmatrix}5\\-2\\9\end{bmatrix}=b. \tag{2}$$

Those are *three-dimensional column vectors*. *The vector b is identified with the point whose coordinates are* 5, -2, 9. Every point in three-dimensional space is matched to a vector, and vice versa. That was the idea of Descartes, who turned geometry into algebra by working with the coordinates of the point. We can write the vector in a column, or we can list its components as $b=(5,-2,9)$, or we can represent it geometrically by an arrow from the origin. You can choose *the arrow*, or *the point*, or *the three numbers*. In six dimensions it is probably easiest to choose the six numbers.

We use parentheses and commas when the components are listed horizontally, and square brackets (with no commas) when a column vector is printed vertically. What really matters is ***addition of vectors*** and ***multiplication by a scalar*** (a number). In Figure 1.4a you see a vector addition, component by component:

$$\textbf{Vector addition} \qquad \begin{bmatrix}5\\0\\0\end{bmatrix}+\begin{bmatrix}0\\-2\\0\end{bmatrix}+\begin{bmatrix}0\\0\\9\end{bmatrix}=\begin{bmatrix}5\\-2\\9\end{bmatrix}.$$

In the right-hand figure there is a multiplication by 2 (and if it had been -2 the vector



(a) Add vectors along axes (b) Add columns $1+2+(3+3)$

Figure 1.4: The column picture: linear combination of columns equals b.

would have gone in the reverse direction):

$$\textbf{Multiplication by scalars} \qquad 2\begin{bmatrix}1\\0\\2\end{bmatrix}=\begin{bmatrix}2\\0\\4\end{bmatrix}, \quad -2\begin{bmatrix}1\\0\\2\end{bmatrix}=\begin{bmatrix}-2\\0\\-4\end{bmatrix}.$$

Also in the right-hand figure is one of the central ideas of linear algebra. It uses *both* of the basic operations; vectors are *multiplied by numbers and then added*. The result is called a ***linear combination***, and this combination solves our equation:

$$\textbf{Linear combination} \qquad \mathbf{1}\begin{bmatrix}2\\4\\-2\end{bmatrix}+\mathbf{1}\begin{bmatrix}1\\-6\\7\end{bmatrix}+\mathbf{2}\begin{bmatrix}1\\0\\2\end{bmatrix}=\begin{bmatrix}5\\-2\\9\end{bmatrix}.$$

Equation (2) asked for multipliers u, v, w that produce the right side b. Those numbers are $u=\mathbf{1}$, $v=\mathbf{1}$, $w=\mathbf{2}$. They give the correct combination of the columns. They also gave the point $(1,1,2)$ in the row picture (where the three planes intersect).

Our true goal is to look beyond two or three dimensions into n dimensions. With n equations in n unknowns, there are n planes in the row picture. There are n vectors in the column picture, plus a vector b on the right side. The equations ask for a ***linear combination of the n columns that equals b***. For certain equations that will be impossible. Paradoxically, the way to understand the good case is to study the bad one. Therefore we look at the geometry exactly when it breaks down, in the **singular case**.

Row picture: Intersection of planes	***Column picture***: Combination of columns

The Singular Case

Suppose we are again in three dimensions, and the three planes in the row picture *do not intersect*. What can go wrong? One possibility is that two planes may be parallel. The equations $2u+v+w=5$ and $4u+2v+2w=11$ are inconsistent—and parallel planes give no solution (Figure 1.5a shows an end view). In two dimensions, parallel lines are the only possibility for breakdown. But three planes in three dimensions can be in trouble without being parallel.

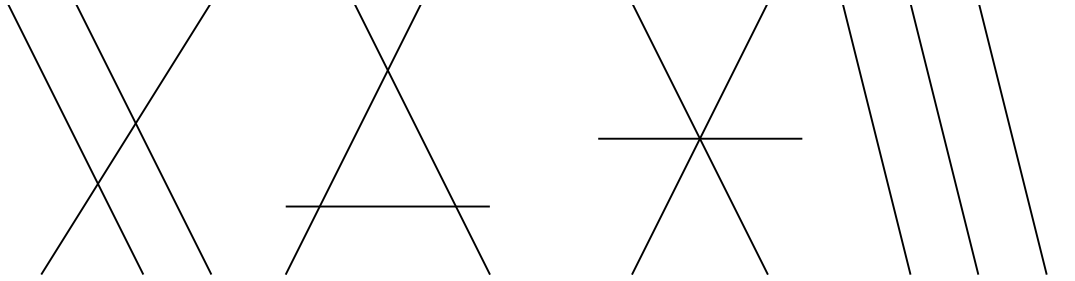


Figure 1.5: Singular cases: no solution for (a), (b), or (d), an infinity of solutions for (c).

The most common difficulty is shown in Figure 1.5b. From the end view the planes form a triangle. Every pair of planes intersects in a line, and those lines are parallel. The third plane is not parallel to the other planes, but it is parallel to their line of intersection. This corresponds to a singular system with $b=(2,5,6)$:

No solution, as in Figure 1.5b

$$\begin{array}{rcrcrcl} u & + & v & + & w & = & 2 \\ 2u & & & + & 3w & = & 5 \\ 3u & + & v & + & 4w & = & 6. \end{array} \tag{3}$$

The first two left sides add up to the third. On the right side that fails: $2+5\neq 6$. Equation 1 plus equation 2 minus equation 3 is the impossible statement $0=1$. Thus the equations are ***inconsistent***, as Gaussian elimination will systematically discover.

Another singular system, close to this one, has an **infinity of solutions**. When the 6 in the last equation becomes 7, the three equations combine to give $0 = 0$. Now the third equation is the sum of the first two. In that case the three planes have a whole *line in common* (Figure 1.5c). Changing the right sides will move the planes in Figure 1.5b parallel to themselves, and for $b = (2,5,7)$ the figure is suddenly different. The lowest plane moved up to meet the others, and there is a line of solutions. Problem 1.5c is still singular, but now it suffers from ***too many solutions*** instead of too few.

The extreme case is three parallel planes. For most right sides there is no solution (Figure 1.5d). For special right sides (like $b = (0,0,0)$!) there is a whole plane of solutions—because the three parallel planes move over to become the same.

What happens to the ***column picture*** when the system is singular? it has to go wrong; the question is how, There are still three columns on the left side of the equations, and we try to combine them to produce b. Stay with equation (3):

Singular case: Column picture
Three columns in the same plane
Solvable only for b in that plane

$$u\begin{bmatrix}1\\2\\3\end{bmatrix} + v\begin{bmatrix}1\\0\\1\end{bmatrix} + w\begin{bmatrix}1\\3\\4\end{bmatrix} = b. \qquad (4)$$

For $b = (2,5,7)$ this was possible; for $b = (2,5,6)$ it was not. The reason is that ***those three columns lie in a plane***. Then every combination is also in the plane (which goes through the origin). If the vector b is not in that plane, no solution is possible (Figure 1.6). That is by far the most likely event; a singular system generally has no solution. But there is a chance that b *does* lie in the plane of the columns. In that case there are too many solutions; the three columns can be combined in *infinitely many ways* to produce b. That column picture in Figure 1.6b corresponds to the row picture in Figure 1.5c.

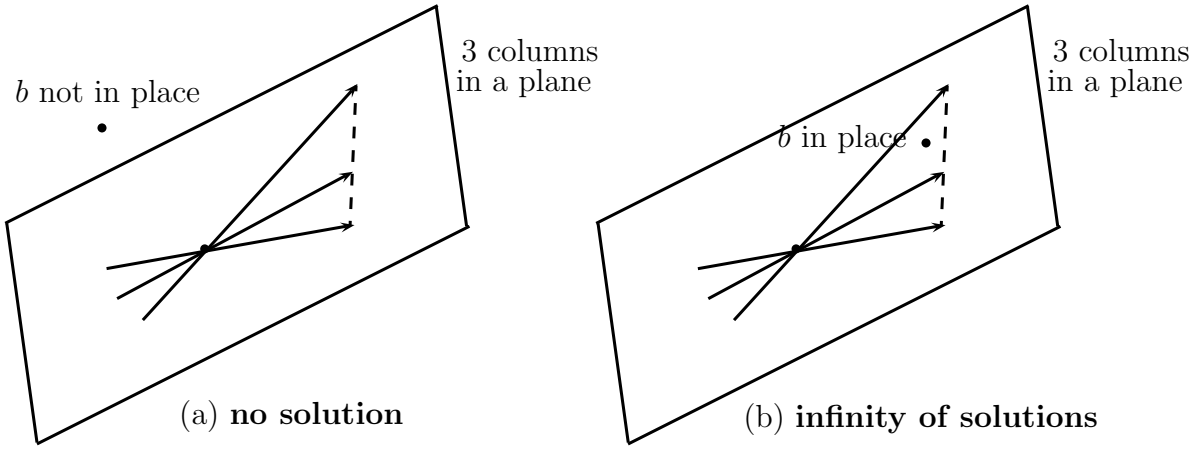


Figure 1.6: Singular cases: b outside or inside the plane with all three columns.

How do we know that the three columns lie in the same plane? One answer is to find a combination of the columns that adds to zero. After some calculation, it is $u = 3$, $v = 1$, $w = -2$. Three times column 1 equals column 2 plus twice column 3. Column 1 is in

the plane of columns 2 and 3. Only two columns are independent.

The vector $b = (2,5,7)$ is in that plane of the columns—it is column 1 plus column 3—so (1, 0, 1) is a solution. *We can add an multiple of the combination* $(3,-1,-2)$ *that gives* $b = 0$. So there is a whole line of solutions—as we know from the row picture.

The truth is that we *knew* the columns would combine to give zero, because the rows did. That is a fact of mathematics, not of computation—and it remains true in dimension n. ***If the*** n ***planes have no point in common, or infinitely many points, then the*** n ***columns lie in the same plane***.

If the row picture breaks down, so does the column picture. That brings out the difference between Chapter 1 and Chapter 2. This chapter studies the most important problem—the *nonsingular* case—where there is one solution and it has to be found. Chapter 2 studies the general case, where there may be many solutions or none. In both cases we cannot continue without a decent notation (*matrix notation*) and a decent algorithm (*elimination*). After the exercises, we start with elimination.

Problem Set 1.2

1. For the equations $x + y = 4$, $2x - 2y = 4$, draw the row picture (two intersecting lines) and the column picture (combination of two columns equal to the column vector $(4,4)$ on the right side).

2. Solve to find a combination of the columns that equals b:

Triangular system

$$\begin{aligned} u - v - w &= b_1 \\ v + w &= b_2 \\ w &= b_3. \end{aligned}$$

3. (Recommended) Describe the intersection of the three planes $u+v+w+z=6$ and $u+w+z=4$ and $u+w=2$ (all in four-dimensional space). Is it a line or a point or an empty set? What is the intersection if the fourth plane $u=-1$ is included? Find a fourth equation that leaves us with no solution.

4. Sketch these three lines and decide if the equations are solvable:

3 by 2 system

$$\begin{aligned} x + 2y &= 2 \\ x - y &= 2 \\ y &= 1. \end{aligned}$$

What happens if all right-hand sides are zero? Is there any nonzero choice of right-hand sides that allows the three lines to intersect at the same point?

5. Find two points on the line of intersection of the three planes $t = 0$ and $z = 0$ and $x+y+z+t = 1$ in four-dimensional space.

6. When $b = (2,5,7)$, find a solution (u,v,w) to equation (4) different from the solution $(1,0,1)$ mentioned in the text.

7. Give two more right-hand sides in addition to $b = (2,5,7)$ for which equation (4) can be solved. Give two more right-hand sides in addition to $b = (2,5,6)$ for which it cannot be solved.

8. Explain why the system

$$\begin{array}{rcrcrcl} u & + & v & + & w & = & 2 \\ u & + & 2v & + & 3w & = & 1 \\ & & v & + & 2w & = & 0 \end{array}$$

is singular by finding a combination of the three equations that adds up to $0 = 1$. What value should replace the last zero on the right side to allow the equations to have solutions—and what is one of the solutions?

9. The column picture for the previous exercise (singular system) is

$$u\begin{bmatrix}1\\1\\0\end{bmatrix} + v\begin{bmatrix}1\\2\\1\end{bmatrix} + w\begin{bmatrix}1\\3\\2\end{bmatrix} = b.$$

Show that the three columns on the left lie in the same plane by expressing the third column as a combination of the first two. What are all the solutions (u,v,w) if b is the zero vector $(0,0,0)$?

10. (Recommended) Under what condition on y_1, y_2, y_3 do the points $(0,y_1)$, $(1,y_2)$, $(2,y_3)$ lie on a straight line?

11. These equations are certain to have the solution $x = y = 0$. For which values of a is there a whole line of solutions?

$$\begin{array}{rcrcl} ax & + & 2y & = & 0 \\ 2x & + & ay & = & 0 \end{array}$$

12. Starting with $x + 4y = 7$, find the equation for the parallel line through $x = 0$, $y = 0$. Find the equation of another line that meets the first at $x = 3$, $y = 1$.

Problems 13–15 are a review of the row and column pictures.

13. Draw the two pictures in two planes for the equations $x - 2y = 0$, $x + y = 6$.

14. For two linear equations in three unknowns x, y, z, the row picture will show (2 or 3) (lines or planes) in (two or three)-dimensional space. The column picture is in (two or three)-dimensional space. The solutions normally lie on a _____.

15. For four linear equations in two unknowns x and y, the row picture shows four _____. The column picture is in _____-dimensional space. The equations have no solution unless the vector on the right-hand side is a combination of _____.

16. Find a point with $z = 2$ on the intersection line of the planes $x + y + 3z = 6$ and $x - y + z = 4$. Find the point with $z = 0$ and a third point halfway between.

17. The first of these equations plus the second equals the third:

$$\begin{array}{rcrcrcl} x & + & y & + & z & = & 2 \\ x & + & 2y & + & z & = & 3 \\ 2x & + & 3y & + & 2z & = & 5. \end{array}$$

The first two planes meet along a line. The third plane contains that line, because if x, y, z satisfy the first two equations then they also _____. The equations have infinitely many solutions (the whole line **L**). Find three solutions.

18. Move the third plane in Problem 17 to a parallel plane $2x + 3y + 2z = 9$. Now the three equations have no solution—*why not*? The first two planes meet along the line **L**, but the third plane doesn't _____ that line.

19. In Problem 17 the columns are $(1,1,2)$ and $(1,2,3)$ and $(1,1,2)$. This is a "singular case" because the third column is _____. Find two combinations of the columns that give $b = (2,3,5)$. This is only possible for $b = (4,6,c)$ if $c =$ _____.

20. Normally 4 "planes" in four-dimensional space meet at a _____. Normally 4 column vectors in four-dimensional space can combine to produce b. What combination of $(1,0,0,0)$, $(1,1,0,0)$, $(1,1,1,0)$, $(1,1,1,1)$ produces $b = (3,3,3,2)$? What 4 equations for x, y, z, t are you solving?

21. When equation 1 is added to equation 2, which of these are changed: the planes in the row picture, the column picture, the coefficient matrix, the solution?

22. If (a,b) is a multiple of (c,d) with $abcd \neq 0$, *show that* (a,c) *is a multiple of* (b,d). This is surprisingly important: call it a challenge question. You could use numbers first to see how a, b, c, and d are related. The question will lead to:

If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ has dependent rows then it has dependent columns.

23. In these equations, the third column (multiplying w) is the *same* as the right side b. The column form of the equations *immediately* gives what solution for (u,v,w)?

$$\begin{array}{rcrcrcl} 6u & + & 7v & + & 8w & = & 8 \\ 4u & + & 5v & + & 9w & = & 9 \\ 2u & - & 2v & + & 7w & = & 7. \end{array}$$

SYSTEMS OF LINEAR EQUATIONS - CHAPTER 1

Strang Chapter 1, Section 1.3 (Sections 1.1-1.2 appear immediately before this block)

1.3 An Example of Gaussian Elimination

The way to understand elimination is by example. We begin in three dimensions:

$$\textbf{Original system} \qquad \begin{array}{rcrcrcr} 2u & + & v & + & w & = & 5 \\ 4u & - & 6v & & & = & -2 \\ -2u & + & 7v & + & 2w & = & 9. \end{array} \tag{1}$$

The problem is to find the unknown values of u, v, and w, and we shall apply Gaussian elimination. (Gauss is recognized as the greatest of all mathematicians, but certainly not because of this invention, which probably took him ten minutes. Ironically, it is the most frequently used of all the ideas that bear his name.) The method starts by ***subtracting multiples of the first equation from the other equations***. The goal is to eliminate u from the last two equations. This requires that we

(a) subtract 2 times the first equation from the second

(b) subtract -1 times the first equation from the third.

$$\textbf{Equivalent system} \qquad \begin{array}{rcrcrcr} \mathbf{2}u & + & v & + & w & = & 5 \\ & - & \mathbf{8}v & - & 2w & = & -12 \\ & & 8v & + & 3w & = & 14. \end{array} \tag{2}$$

The coefficient 2 is the ***first pivot***. Elimination is constantly dividing the pivot into the numbers underneath it, to find out the right multipliers.

The pivot for the **second stage of elimination** is -8. We now ignore the first equation. A multiple of the second equation will be subtracted from the remaining equations (in this case there is only the third one) so as to eliminate v. We add the second equation to the third or, in other words, we

(c) subtract -1 times the second equation from the third.

The elimination process is now complete, at least in the "forward" direction:

$$\textbf{Triangular system} \qquad \begin{array}{rcrcrcr} \mathbf{2}u & + & v & + & w & = & 5 \\ & - & \mathbf{8}v & - & 2w & = & -12 \\ & & & & \mathbf{1}w & = & 2. \end{array} \tag{3}$$

This system is solved backward, bottom to top. The last equation gives $w = 2$. Substituting into the second equation, we find $v = 1$. Then the first equation gives $u = 1$. This process is called ***back-substitution***.

To repeat: Forward elimination produced the pivots 2, -8, 1. It subtracted multiples of each row from the rows beneath, It reached the "triangular" system (3), which is solved in reverse order: Substitute each newly computed value into the equations that are waiting.

Remark. One good way to write down the forward elimination steps is to include the right-hand side as an extra column. There is no need to copy u and v and w and $=$ at every step, so we are left with the bare minimum:

$$\begin{bmatrix} 2 & 1 & 1 & 5 \\ 4 & -6 & 0 & -2 \\ -2 & 7 & 2 & 9 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 8 & 3 & 14 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

At the end is the triangular system, ready for back-substitution. You may prefer this arrangement, which guarantees that operations on the left-hand side of the equations are also done on the right-hand side—because *both sides are there together*.

In a larger problem, forward elimination takes most of the effort. We use multiples of the first equation to produce zeros below the first pivot. Then the second column is cleared out below the second pivot. The forward step is finished when the system is triangular; equation n contains only the last unknown multiplied by the last pivot. Back-substitution yields the complete solution in the opposite order—beginning with the last unknown, then solving for the next to last, and eventually for the first.

By definition, ***pivots cannot be zero***. We need to divide by them.

The Breakdown of Elimination

Under what circumstances could the process break down? Something must go wrong in the singular case, and something might go wrong in the nonsingular case. This may seem a little premature—after all, we have barely got the algorithm working. But the possibility of breakdown sheds light on the method itself.

The answer is: With a full set of n pivots, there is only one solution. The system is non singular, and it is solved by forward elimination and back-substitution. But ***if a zero appears*** in a pivot position, elimination has to stop—either temporarily or permanently. The system might or might not be singular.

If the first coefficient is zero, in the upper left corner, the elimination of u from the other equations will be impossible. The same is true at every intermediate stage. Notice that a zero can appear in a pivot position, even if the original coefficient in that place was not zero. Roughly speaking, ***we do not know whether a zero will appear until we try***, by actually going through the elimination process.

In many cases this problem can be cured, and elimination can proceed. Such a system still counts as nonsingular; it is only the algorithm that needs repair. In other cases a breakdown is unavoidable. Those incurable systems are singular, they have no solution or else infinitely many, and a full set of pivots cannot be found.

Example 1. Nonsingular (cured by exchanging equations 2 and 3)

$$\begin{array}{rcrcrcl} u & + & v & + & w & = & __ \\ 2u & + & 2v & + & 5w & = & __ \\ 4u & + & 6v & + & 8w & = & __ \end{array} \rightarrow \begin{array}{rcrcrcl} u & + & v & + & w & = & __ \\ & & & & 3w & = & __ \\ & & 2v & + & 4w & = & __ \end{array} \rightarrow \begin{array}{rcrcrcl} u & + & v & + & w & = & __ \\ & & 2v & + & 4w & = & __ \\ & & & & 3w & = & __ \end{array}$$

The system is now triangular, and back-substitution will solve it.

Example 2. Singular (incurable)

$$\begin{array}{rcrcrcl} u & + & v & + & w & = & ____ \\ 2u & + & 2v & + & 5w & = & ____ \\ 4u & + & 4v & + & 8w & = & ____ \end{array} \longrightarrow \begin{array}{rcrcrcl} u & + & v & + & w & = & ____ \\ & & & & 3w & = & ____ \\ & & & & 4w & = & ____ \end{array}$$

There is no exchange of equations that can avoid zero in the second pivot position. The equations themselves may be solvable or unsolvable. If the last two equations are $3w = 6$ and $4w = 7$, there is no solution. If those two equations happen to be consistent—as in $3w = 6$ and $4w = 8$—then this singular case has an infinity of solutions. We know that $w = 2$, but the first equation cannot decide both u and v.

Section 1.5 will discuss row exchanges when the system is not singular. Then the exchanges produce a full set of pivots. Chapter 2 admits the singular case, and limps forward with elimination. The $3w$ can still eliminate the $4w$, and we will call 3 the second pivot. (There won't be a third pivot.) For the present we trust all n pivot entries to be nonzero, without changing the order of the equations. That is the best case, with which we continue.

The Cost of Elimination

Our other question is very practical. *How many separate arithmetical operations does elimination require, for n equations in n unknowns?* If n is large, a computer is going to take our place in carrying out the elimination. Since all the steps are known, we should be able to predict the number of operations.

For the moment, ignore the right-hand sides of the equations, and count only the operations on the left. These operations are of two kinds. We divide by the pivot to find out what multiple (say ℓ) of the pivot equation is to be subtracted. When we do this subtraction, we continually meet a "multiply-subtract" combination; the terms in the pivot equation are multiplied by ℓ, and then subtracted from another equation.

Suppose we call each division, and each multiplication-subtraction, one operation. In column 1, *it takes n operations for every zero we achieve*—one to find the multiple ℓ, and the other to find the new entries along the row. There are $n-1$ rows underneath the first one, so the first stage of elimination needs $n(n-1) = n^2 - n$ operations. (Another approach to $n^2 - n$ is this: *All n^2 entries need to be changed, except the n in the first row.*) Later stages are faster because the equations are shorter.

When the elimination is down to k equations, only k^2-k operations are needed to clear out the column below the pivot—by the same reasoning that applied to the first stage, when k equaled n. Altogether, the total number of operations is the sum of k^2-k over all values of k from 1 to n:

$$\textbf{Left side} \qquad (1^2+\cdots+n^2)-(1+\cdots+n) = \frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2}$$
$$= \frac{n^3-n}{3}.$$

Those are standard formulas for the sums of the first n numbers and the first n squares. Substituting $n=1$ and $n=2$ and $n=100$ into the formula $\frac{1}{3}(n^3-n)$, forward elimination can take no steps or two steps or about a third of a million steps:

If n is at all large, ***a good estimate for the number of operations is*** $\frac{1}{3}n^3$**.**

If the size is doubled, and few of the coefficients are zero, the cost is multiplied by 8.

Back-substitution is considerably faster. The last unknown is found in only one operation (a division by the last pivot). The second to last unknown requires two operations, and so on. Then the total for back-substitution is $1+2+\cdots+n$.

Forward elimination also acts on the right-hand side (subtracting the same multiples as on the left to maintain correct equations). This starts with $n-1$ subtractions of the first equation. Altogether ***the right-hand side is responsible for*** n^2 ***operations***—much less than the $n^3/3$ on the left. The total for forward and back is

$$\textbf{Right side} \qquad [(n-1)+(n-2)+\cdots+1]+[1+2+\cdots+n] = n^2.$$

Thirty years ago, almost every mathematician would have guessed that a general system of order n could not be solved with much fewer than $n^3/3$ multiplications. (There were even theorems to demonstrate it, but they did not allow for all possible methods.) Astonishingly, that guess has been proved wrong. *There now exists a method that requires only* $Cn^{\log_2 7}$ *multiplications!* It depends on a simple fact: Two combinations of two vectors in two-dimensional space would seem to take 8 multiplications, but they can be done in 7. That lowered the exponent from $\log_2 8$, which is 3, to $\log_2 7 \approx 2.8$. This discovery produced tremendous activity to find the smallest possible power of n. The exponent finally fell (at IBM) below 2.376. Fortunately for elimination, the constant C is so large and the coding is so awkward that the new method is largely (or entirely) of theoretical interest. The newest problem is the cost with *many processors in parallel*.

Problem Set 1.3

Problems 1–9 are about elimination on 2 by 2 systems.

1. What multiple ℓ of equation 1 should be subtracted from equation 2?

$$\begin{aligned} 2x &+ 3y = 1 \\ 10x &+ 9y = 11. \end{aligned}$$

After this elimination step, write down the upper triangular system and circle the two pivots. The numbers 1 and 11 have no influence on those pivots.

2. Solve the triangular system of Problem 1 by back-substitution, y before x. Verify that x times $(2,10)$ plus y times $(3,9)$ equals $(1,11)$. If the right-hand side changes to $(4,44)$, what is the new solution?

3. What multiple of equation 2 should be *subtracted* from equation 3?

$$\begin{aligned} 2x &- 4y = 6 \\ -x &+ 5y = 0. \end{aligned}$$

After this elimination step, solve the triangular system. If the right-hand side changes to $(-6,0)$, what is the new solution?

4. What multiple ℓ of equation 1 should be subtracted from equation 2?

$$\begin{aligned} ax &+ by = f \\ cx &+ dy = g. \end{aligned}$$

The first pivot is a (assumed nonzero). Elimination produces what formula for the second pivot? What is y? The second pivot is missing when $ad = bc$.

5. Choose a right-hand side which gives no solution and another right-hand side which gives infinitely many solutions. What are two of those solutions?

$$\begin{aligned} 3x &+ 2y = 10 \\ 6x &+ 4y = ____. \end{aligned}$$

6. Choose a coefficient b that makes this system singular. Then choose a right-hand side g that makes it solvable. Find two solutions in that singular case.

$$\begin{aligned} 2x &+ by = 16 \\ 4x &+ 8y = g. \end{aligned}$$

7. For which numbers a does elimination break down (a) permanently, and (b) temporarily?

$$\begin{aligned} ax &+ 3y = -3 \\ 4x &+ 6y = 6. \end{aligned}$$

Solve for x and y after fixing the second breakdown by a row exchange.

8. For which three numbers k does elimination break down? Which is fixed by a row exchange? In each case, is the number of solutions 0 or 1 or ∞?

$$\begin{aligned} kx + 3y &= 6 \\ 3x + ky &= -6. \end{aligned}$$

9. What test on b_1 and b_2 decides whether these two equations allow a solution? How many solutions will they have? Draw the column picture.

$$\begin{aligned} 3x - 2y &= b_1 \\ 6x - 4y &= b_2. \end{aligned}$$

Problems 10–19 study elimination on 3 by 3 systems (and possible failure).

10. Reduce this system to upper triangular form by two row operations:

$$\begin{aligned} 2x + 3y + z &= 8 \\ 4x + 7y + 5z &= 20 \\ -2y + 2z &= 0. \end{aligned}$$

Circle the pivots. Solve by back-substitution for z, y, x.

11. Apply elimination (circle the pivots) and back-substitution to solve

$$\begin{aligned} 2x - 3y &= 3 \\ 4x - 5y + z &= 7 \\ 2x - y - 3z &= 5. \end{aligned}$$

List the three row operations: Subtract ____ times row ____ from row ____.

12. Which number d forces a row exchange, and what is the triangular system (not singular) for that d? Which d makes this system singular (no third pivot)?

$$\begin{aligned} 2x + 5y + z &= 0 \\ 4x + dy + z &= 2 \\ y - z &= 3. \end{aligned}$$

13. Which number b leads later to a row exchange? Which b leads to a missing pivot? In that singular case find a nonzero solution x, y, z.

$$\begin{aligned} x + by &= 0 \\ x - 2y - z &= 0 \\ y + z &= 0. \end{aligned}$$

14. (a) Construct a 3 by 3 system that needs two row exchanges to reach a triangular form and a solution.

(b) Construct a 3 by 3 system that needs a row exchange to keep going, but breaks down later.

15. If rows 1 and 2 are the same, how far can you get with elimination (allowing row exchange)? If columns 1 and 2 are the same, which pivot is missing?

$$\begin{aligned} 2x-y+z&=0 \qquad & 2x+2y+z&=0\\ 2x-y+z&=0 \qquad & 4x+4y+z&=0\\ 4x+y+z&=2 \qquad & 6x+6y+z&=2. \end{aligned}$$

16. Construct a 3 by 3 example that has 9 different coefficients on the left-hand side, but rows 2 and 3 become zero in elimination. How many solutions to your system with $b=(1,10,100)$ and how many with $b=(0,0,0)$?

17. Which number q makes this system singular and which right-hand side t gives it infinitely many solutions? Find the solution that has $z=1$.

$$\begin{aligned} x+4y-2z&=1\\ x+7y-6z&=6\\ 3y+qz&=t. \end{aligned}$$

18. (Recommended) It is impossible for a system of linear equations to have exactly two solutions. *Explain why*.

(a) If (x,y,z) and (X,Y,Z) are two solutions, what is another one?

(b) If 25 planes meet at two points, where else do they meet?

19. Three planes can fail to have an intersection point, when no two planes are parallel. The system is singular if row 3 of A is a ____ of the first two rows. Find a third equation that can't be solved if $x+y+z=0$ and $x-2y-z=1$.

Problems 20–22 move up to 4 by 4 and n by n.

20. Find the pivots and the solution for these four equations:

$$\begin{aligned} 2x+y&=0\\ x+2y+z&=0\\ y+2z+t&=0\\ z+2t&=5. \end{aligned}$$

21. If you extend Problem 20 following the 1, 2, 1 pattern or the -1, 2, -1 pattern, what is the fifth pivot? What is the nth pivot?

22. Apply elimination and back-substitution to solve

$$\begin{aligned} 2u+3v&=0\\ 4u+5v+w&=3\\ 2u-v-3w&=5. \end{aligned}$$

What are the pivots? List the three operations in which a multiple of one row is subtracted from another.

23. For the system

$$\begin{aligned} u + v + w &= 2 \\ u + 3v + 3w &= 0 \\ u + 3v + 5w &= 2, \end{aligned}$$

what is the triangular system after forward elimination, and what is the solution?

24. Solve the system and find the pivots when

$$\begin{aligned} 2u - v &= 0 \\ -u + 2v - w &= 0 \\ - v + 2w - z &= 0 \\ - w + 2z &= 5. \end{aligned}$$

You may carry the right-hand side as a fifth column (and omit writing u, v, w, z until the solution at the end).

25. Apply elimination to the system

$$\begin{aligned} u + v + w &= -2 \\ 3u + 3v - w &= 6 \\ u - v + w &= -1. \end{aligned}$$

When a zero arises in the pivot position, exchange that equation for the one below it and proceed. What coefficient of v in the third equation, in place of the present -1, would make it impossible to proceed—and force elimination to break down?

26. Solve by elimination the system of two equations

$$\begin{aligned} x - y &= 0 \\ 3x + 6y &= 18. \end{aligned}$$

Draw a graph representing each equation as a straight line in the x-y plane; the lines intersect at the solution. Also, add one more line—the graph of the new second equation which arises after elimination.

27. Find three values of a for which elimination breaks down, temporarily or permanently, in

$$\begin{aligned} au + u &= 1 \\ 4u + av &= 2. \end{aligned}$$

Breakdown at the first step can be fixed by exchanging rows—but not breakdown at the last step.

28. True or false:

(a) If the third equation starts with a zero coefficient (it begins with $0u$) then no multiple of equation 1 will be subtracted from equation 3.

SYSTEMS OF LINEAR EQUATIONS - CHAPTER 2

Strang Chapter 2, Sections 2.1-2.4

Chapter 2

Vector Spaces

2.1 Vector Spaces and Subspaces

Elimination can simplify, one entry at a time, the linear system $Ax = b$. Fortunately it also simplifies the theory. The basic questions of *existence* and *uniqueness*—Is there one solution, or no solution, or an infinity of solutions?—are much easier to answer after elimination, We need to devote one more section to those questions, to find every solution for an m by n system. Then that circle of ideas will be complete.

But elimination produces only one kind of understanding of $Ax = b$. Our chief object is to achieve a different and deeper understanding. This chapter may be more difficult than the first one. It goes to the heart of linear algebra.

For the concept of a ***vector space***, we start immediately with the most important spaces. They are denoted by $\mathbf{R}^1, \mathbf{R}^2, \mathbf{R}^3, \ldots$; the space $\mathbf{R}^n$ consists of *all column vectors with n components*. (We write $\mathbf{R}$ because the components are real numbers.) $\mathbf{R}^2$ is represented by the usual x-y plane; the two components of the vector become the x and y coordinates of the corresponding point. The three components of a vector in $\mathbf{R}^3$ give a point in three-dimensional space. The one-dimensional space $\mathbf{R}^1$ is a line.

The valuable thing for linear algebra is that the extension to n dimensions is so straightforward. For a vector in $\mathbf{R}^7$ we just need the seven components, even if the geometry is hard to visualize. Within all vector spaces, two operations are possible:

> ***We can add any two vectors, and we can multiply all vectors by scalars.***
> **In other words, we can take linear combinations.**

Addition obeys the commutative law $x + y = y + x$; there is a "zero vector" satisfying $0 + x = x$; and there is a vector "$-x$" satisfying $-x + x = 0$. Eight properties (including those three) are fundamental; the full list is given in Problem 5 at the end of this section. ***A real vector space is a set of*** *vectors* ***together with rules for vector addition and multiplication by real numbers***. Addition and multiplication must produce vectors in the space, and they must satisfy the eight conditions.

Normally our vectors belong to one of the spaces $\mathbf{R}^n$; they are ordinary column vectors. If $x = (1,0,0,3)$, then $2x$ (and also $x+x$) has components 2, 0, 0, 6. The formal definition allows other things to be "vectors"-provided that addition and scalar multiplication are all right. We give three examples:

1. *The infinite-dimensional space* $\mathbf{R}^\infty$. Its vectors have infinitely many components, as in $x = (1,2,1,2,\ldots)$. The laws for $x+y$ and cx stay unchanged.

2. *The space of 3 by 2 matrices*. In this case the "vectors" are matrices! We can add two matrices, and $A+B = B+A$, and there is a zero matrix, and so on. This space is almost the same as $\mathbf{R}^6$. (The six components are arranged in a rectangle instead of a column.) Any choice of m and n would give, as a similar example, the vector space of all m by n matrices.

3. *The space of functions* $f(x)$. Here we admit all functions f that are defined on a fixed interval, say $0 \le x \le 1$. The space includes $f(x) = x^2$, $g(x) = \sin x$, their sum $(f+g)(x) = x^2 + \sin x$, and all multiples like $3x^2$ and $-\sin x$. The vectors are functions, and the dimension is somehow a larger infinity than for $\mathbf{R}^\infty$.

Other examples are given in the exercises, but the vector spaces we need most are somewhere else—***they are inside the standard spaces*** $\mathbf{R}^n$. We want to describe them and explain why they are important. Geometrically, think of the usual three-dimensional $\mathbf{R}^3$ and choose any plane through the origin. ***That plane is a vector space in its own right***. If we multiply a vector in the plane by 3, or -3, or any other scalar, we get a vector in the same plane. If we add two vectors in the plane, their sum stays in the plane. This plane through $(0,0,0)$ illustrates one of the most fundamental ideas in linear algebra; it is a ***subspace*** of the original space $\mathbf{R}^3$.

Definition. A ***subspace*** of a vector space is a nonempty subset that satisfies the requirements for a vector space: ***Linear combinations stay in the subspace***.

(i) If we add any vectors x and y in the subspace, $x+y$ is *in the subspace*.

(ii) If we multiply any vector x in the subspace by any scalar c, cx is *in the subspace*.

Notice our emphasis on the word ***space***. A sub*space* is a subset that is "closed" under addition and scalar multiplication. Those operations follow the rules of the host space, keeping us ***inside the subspace***. The eight required properties are satisfied in the larger space and will automatically be satisfied in every subspace. Notice in particular that ***the zero vector will belong to every subspace***. That comes from rule (ii): Choose the scalar to be $c = 0$.

The smallest subspace $\mathbf{Z}$ contains only one vector, the zero vector. It is a "zero-dimensional space," containing only the point at the origin. Rules (i) and (ii) are satisfied,

since the sum $0+0$ is in this one-point space, and so are all multiples $c0$. *This is the smallest possible vector space*: the empty set is not allowed. At the other extreme. the largest subspace is the whole of the original space. If the original space is $\mathbf{R}^3$, then the possible subspaces are easy to describe: $\mathbf{R}^3$ itself, any plane through the origin, any line through the origin, or the origin (the zero vector) alone.

The distinction between a subset and a subspace is made clear by examples. In each case, can you add vectors and multiply by scalars, without leaving the space?

Example 1. Consider all vectors in $\mathbf{R}^2$ whose components are positive or zero. This subset is the first quadrant of the x-y plane; the coordinates satisfy $x \geq 0$ and $y \geq 0$. It is *not a subspace*, even though it contains zero and addition does leave us within the subset. Rule (ii) is violated, since if the scalar is -1 and the vector is $[1 \;\; 1]$, the multiple $cx = [-1 \;\; -1]$ is in the third quadrant instead of the first.

If we include the third quadrant along with the first, scalar multiplication is all right. Every multiple cx will stay in this subset. However, rule (i) is now violated, since adding $[1 \;\; 2] + [-2 \;\; -1]$ gives $[-1 \;\; 1]$, which is not in either quadrant. The smallest subspace containing the first quadrant is the whole space $\mathbf{R}^2$.

Example 2. Start from the vector space of 3 by 3 matrices. One possible subspace is the set of *lower triangular matrices*. Another is the set of *symmetric matrices*. $A+B$ and cA are lower triangular if A and B are lower triangular, and they are symmetric if A and B are symmetric. Of course, the zero matrix is in both subspaces.

The Column Space of A

We now come to the key examples, the **column space** and the **nullspace** of a matrix A. ***The column space contains all linear combinations of the columns of*** A. It is a subspace of $\mathbf{R}^m$. We illustrate by a system of $m = 3$ equations in $n = 2$ unknowns:

$$\textbf{Combination of columns equals } b \qquad \begin{bmatrix} 1 & 0 \\ 5 & 4 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}. \tag{1}$$

With $m > n$ we have more equations than unknowns—and *usually there will be no solution*. The system will be solvable only for a very "thin" subset of all possible b's. One way of describing this thin subset is so simple that it is easy to overlook.

2A The system $Ax = b$ is solvable if and only if the vector b can be expressed as a combination of the columns of A. Then b is in the column space.

This description involves nothing more than a restatement of $Ax = b$, *by columns*:

$$\textbf{Combination of columns} \qquad u \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix} + v \begin{bmatrix} 0 \\ 4 \\ 4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}. \tag{2}$$

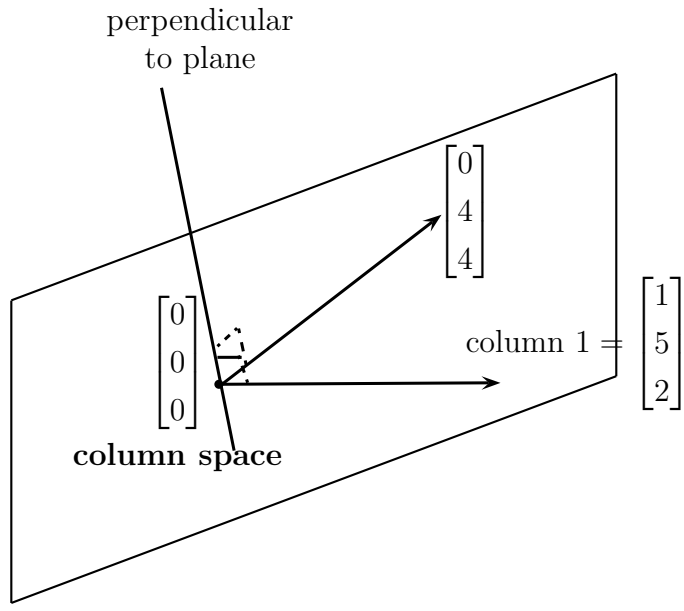


Figure 2.1: The column space $\boldsymbol{C}(A)$, a plane in three-dimensional space.

These are the same three equations in two unknowns. Now the problem is: Find numbers u and v that multiply the first and second columns to produce b. The system is solvable exactly when such coefficients exist, and the vector (u, v) is the solution x.

We are saying that the attainable right-hand sides b are *all combinations of the columns of A*. One possible right-hand side is the first column itself; the weights are $u = 1$ and $v = 0$. Another possibility is the second column: $u = 0$ and $v = 1$. A third is the right-hand side $b = 0$. With $u = 0$ and $v = 0$, the vector $b = 0$ will always be attainable.

We can describe *all combinations* of the two columns geometrically: *$Ax = b$ can be solved if and only if b lies in the **plane** that is spanned by the two column vectors* (Figure 2.1). This is the thin set of attainable b. If b lies off the plane, then it is not a combination of the two columns. In that case $Ax = b$ has no solution.

What is important is that this plane is not just a subset of $\mathbf{R}^3$ it is a subspace. It is the ***column space*** of A, consisting of ***all combinations of the columns***. It is denoted by $\boldsymbol{C}(A)$. Requirements (i) and (ii) for a subspace of $\mathbf{R}^m$ are easy to check:

(i) Suppose b and b' lie in the column space, so that $Ax = b$ for some x and $Ax' = b'$ for some x'. Then $A(x + x') = b + b'$, so that $b + b'$ is also a combination of the columns. The column space of all attainable vectors b is closed under addition.

(ii) If b is in the column space $\boldsymbol{C}(A)$, so is any multiple cb. If some combination of columns produces b (say $Ax = b$), then multiplying that combination by c will produce cb. In other words, $A(cx) = cb$.

For another matrix A, the dimensions in Figure 2.1 may be very different. The smallest possible column space (one vector only) comes from the zero matrix $A = 0$. The

only combination of the columns is $b = 0$. At the other extreme, suppose A is the 5 by 5 identity matrix. Then $C(I)$ is the whole of $\mathbf{R}^5$; the five columns of I can combine to produce any five-dimensional vector b. This is not at all special to the identity matrix. *Any 5 by 5 matrix that is nonsingular will have the whole of* $\mathbf{R}^5$ *as its column space.* For such a matrix we can solve $Ax = b$ by Gaussian elimination; there are five pivots. Therefore every b is in $C(A)$ for a nonsingular matrix.

You can see how Chapter 1 is contained in this chapter. There we studied n by n matrices whose column space is $\mathbf{R}^n$. Now we allow singular matrices, and rectangular matrices of any shape. Then $C(A)$ can be somewhere between the zero space and the whole space $\mathbf{R}^m$. Together with its perpendicular space, it gives one of our two approaches to understanding $Ax = b$.

The Nullspace of A

The second approach to $Ax = b$ is "dual" to the first. We are concerned not only with attainable right-hand sides b, but also with the solutions x that attain them. The right-hand side $b = 0$ always allows the solution $x = 0$, but there may be infinitely many other solutions. (There always are, if there are more unknowns than equations, $n > m$.) ***The solutions to*** $Ax = 0$ ***form a vector space—the nullspace of*** A.

> The ***nullspace*** of a matrix consists of all vectors x such that $Ax = 0$. It is denoted by $N(A)$. It is a subspace of $\mathbf{R}^n$, just as the column space was a subspace of $\mathbf{R}^m$.

Requirement (i) holds: If $Ax = 0$ and $Ax' = 0$, then $A(x+x') = 0$. Requirement (ii) also holds: If $Ax = 0$ then $A(cx) = 0$. Both requirements fail if the right-hand side is not zero! Only the solutions to a *homogeneous* equation ($b = 0$) form a subspace. The nullspace is easy to find for the example given above; it is as small as possible:

$$\begin{bmatrix} 1 & 0 \\ 5 & 4 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The first equation gives $u = 0$, and the second equation then forces $v = 0$. The nullspace contains only the vector $(0,0)$. This matrix has "independent columns"—a key idea that comes soon.

The situation is changed when a third column is a combination of the first two:

Larger nullspace $$B = \begin{bmatrix} 1 & 0 & 1 \\ 5 & 4 & 9 \\ 2 & 4 & 6 \end{bmatrix}.$$

B has the same column space as A. The new column lies in the plane of Figure 2.1; it is the sum of the two column vectors we started with. But the nullspace of B contains the

vector $(1,1,-1)$ and automatically contains any multiple $(c,c,-c)$:

Nullspace is a line
$$\begin{bmatrix} 1 & 0 & 1 \\ 5 & 4 & 9 \\ 2 & 4 & 6 \end{bmatrix} \begin{bmatrix} c & c & -c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The nullspace of B is the line of all points $x = c$, $y = c$, $z = -c$. (The line goes through the origin, as any subspace must.) We want to be able, for any system $Ax = b$, to find $\boldsymbol{C}(A)$ and $\boldsymbol{N}(A)$: all attainable right-hand sides b and all solutions to $Ax = 0$.

The vectors b are in the column space and the vectors x are in the nullspace. We shall compute the dimensions of those subspaces and a convenient set of vectors to generate them. We hope to end up by understanding all *four* of the subspaces that are intimately related to each other and to A—the column space of A, the nullspace of A, and their two perpendicular spaces.

Problem Set 2.1

1. Construct a subset of the x-y plane $\mathbf{R}^2$ that is

(a) closed under vector addition and subtraction, but not scalar multiplication.

(b) closed under scalar multiplication but not under vector addition.

Hint: Starting with u and v, add and subtract for (a). Try cu and cv for (b).

2. Which of the following subsets of $\mathbf{R}^3$ are actually subspaces?

(a) The plane of vectors (b_1, b_2, b_3) with first component $b_1 = 0$.

(b) The plane of vectors b with $b_1 = 1$.

(c) The vectors b with $b_2 b_3 = 0$ (this is the union of two subspaces, the plane $b_2 = 0$ and the plane $b_3 = 0$).

(d) All combinations of two given vectors $(1,1,0)$ and $(2,0,1)$.

(e) The plane of vectors (b_1, b_2, b_3) that satisfy $b_3 - b_2 + 3b_1 = 0$.

3. Describe the column space and the nullspace of the matrices

$$A = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 0 & 3 \\ 1 & 2 & 3 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

4. What is the smallest subspace of 3 by 3 matrices that contains all symmetric matrices *and* all lower triangular matrices? What is the largest subspace that is contained in both of those subspaces?

5. Addition and scalar multiplication are required to satisfy these eight rules:

1. $x+y=y+x$.
2. $x+(y+z)=(x+y)+z$.
3. There is a unique "zero vector" such that $x+0=x$ for all x.
4. For each x there is a unique vector $-x$ such that $x+(-x)=0$.
5. $1x=x$.
6. $(c_1c_2)x=c_1(c_2x)$.
7. $c(x+y)=cx+cy$.
8. $(c_1+c_2)x=c_1x+c_2x$.

(a) Suppose addition in $\mathbf{R}^2$ adds an extra 1 to each component, so that $(3,1)+(5,0)$ equals $(9,2)$ instead of $(8,1)$. With scalar multiplication unchanged, which rules are broken?

(b) Show that the set of all positive real numbers, with $x+y$ and cx redefined to equal the usual xy and x^c, is a vector space. What is the "zero vector"?

(c) Suppose $(x_1,x_2)+(y_1,y_2)$ is defined to be (x_1+y_2,x_2+y_1). With the usual $cx=(cx_1,cx_2)$, which of the eight conditions are not satisfied?

6. Let $\mathbf{P}$ be the plane in 3-space with equation $x+2y+z=6$. What is the equation of the plane $\mathbf{P}_0$ through the origin parallel to $\mathbf{P}$? Are $\mathbf{P}$ and $\mathbf{P}_0$ subspaces of $\mathbf{R}^3$?

7. Which of the following are subspaces of $\mathbf{R}^\infty$?

(a) All sequences like $(1,0,1,0,\ldots)$ that include infinitely many zeros.

(b) All sequences $(x1,x2,\ldots)$ with $x_j=0$ from some point onward.

(c) All decreasing sequences: $x_{j+1}\le x_j$ for each j.

(d) All convergent sequences: the x_j have a limit as $j\to\infty$.

(e) All arithmetic progressions: $x_{j+1}-x_j$ is the same for all j.

(f) All geometric progressions $(x_1,kx_1,k^2x_1,\ldots)$ allowing all k and x_1.

8. Which of the following descriptions are correct? The solutions x of

$$Ax=\begin{bmatrix}1&1&1\\1&0&2\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}$$

form

(a) a plane.

(b) a line.

(c) a point.

(d) a subspace.

(e) the nullspace of A.

(f) the column space of A.

9. Show that the set of nonsingular 2 by 2 matrices is not a vector space. Show also that the set of *singular* 2 by 2 matrices is not a vector space.

10. The matrix $A = \begin{bmatrix} 2 & -2 \\ 2 & -2 \end{bmatrix}$ is a "vector" in the space **M** of all 2 by 2 matrices. Write the zero vector in this space, the vector $\frac{1}{2}A$, and the vector $-A$. What matrices are in the smallest subspace containing A?

11. (a) Describe a subspace of **M** that contains $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ but not $B = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$.

(b) If a subspace of **M** contains A and B, must it contain I?

(c) Describe a subspace of **M** that contains no nonzero diagonal matrices.

12. The functions $f(x) = x^2$ and $g(x) = 5x$ are "vectors" in the vector space **F** of all real functions. The combination $3f(x) - 4g(x)$ is the function $h(x) =$ ____. Which rule is broken if multiplying $f(x)$ by c gives the function $f(cx)$?

13. If the sum of the "vectors" $f(x)$ and $g(x)$ in **F** is defined to be $f(g(x))$, then the "zero vector" is $g(x) = x$. Keep the usual scalar multiplication $cf(x)$, and find two rules that are broken.

14. Describe the smallest subspace of the 2 by 2 matrix space **M** that contains

(a) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$.

(b) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

(c) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$.

(d) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$.

15. Let **P** be the plane in $\mathbf{R}^3$ with equation $x + y - 2z = 4$. The origin $(0,0,0)$ is not in **P**! Find two vectors in **P** and check that their sum is not in **P**.

16. $\mathbf{P}_0$ is the plane through $(0,0,0)$ parallel to the plane **P** in Problem 15. What is the equation for $\mathbf{P}_0$? Find two vectors in $\mathbf{P}_0$ and check that their sum is in $\mathbf{P}_0$.

17. The four types of subspaces of $\mathbf{R}^3$ are planes, lines, R^3 itself, or **Z** containing only $(0,0,0)$.

(a) Describe the three types of subspaces of $\mathbf{R}^2$.

(b) Describe the five types of subspaces of $\mathbf{R}^4$.

18. (a) The intersection of two planes through $(0,0,0)$ is probably a ____ but it could be a ____. It can't be the zero vector **Z**!

(b) The intersection of a plane through $(0,0,0)$ with a line through $(0,0,0)$ is probably a ____ but it could be a ____.

(c) If **S** and **T** are subspaces of $\mathbf{R}^5$, their intersection $\mathbf{S} \cap \mathbf{T}$ (vectors in both subspaces) is a subspace of $\mathbf{R}^5$. *Check the requirements on $x+y$ and cx.*

19. Suppose **P** is a plane through $(0,0,0)$ and **L** is a line through $(0,0,0)$. The smallest vector space containing both **P** and **L** is either ____ or ____.

20. True or false for $\mathbf{M}$ = all 3 by 3 matrices (check addition using an example)?

(a) The skew-symmetric matrices in $\mathbf{M}$ (with $A^{\mathrm{T}} = -A$) form a subspace.

(b) The unsymmetric matrices in $\mathbf{M}$ (with $A^{\mathrm{T}} \neq A$) form a subspace.

(c) The matrices that have $(1,1,1)$ in their nullspace form a subspace.

Problems 21–30 are about column spaces $C(A)$ and the equation $Ax = b$.

21. Describe the column spaces (lines or planes) of these particular matrices:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 0 & 0 \end{bmatrix}.$$

22. For which right-hand sides (find a condition on b_1, b_2, b_3) are these systems solvable?

(a) $\begin{bmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ -1 & -4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$. (b) $\begin{bmatrix} 1 & 4 \\ 2 & 9 \\ -1 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$.

23. Adding row 1 of A to row 2 produces B. Adding column 1 to column 2 produces C. A combination of the columns of ____ is also a combination of the columns of A. Which two matrices have the same column ____?

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}.$$

24. For which vectors (b_1, b_2, b_3) do these systems have a solution?

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

25. (Recommended) If we add an extra column b to a matrix A, then the column space gets larger unless ____. Give an example in which the column space gets larger and an example in which it doesn't. Why is $Ax = b$ solvable exactly when the column space *doesn't* get larger by including b?

26. The columns of AB are combinations of the columns of A. This means: *The column space of AB is contained in* (possibly equal to) *the column space of A.* Give an example where the column spaces of A and AB are not equal.

27. If A is any 8 by 8 invertible matrix, then its column space is ____. Why?

28. True or false (with a counterexample if false)?

(a) The vectors b that are not in the column space $C(A)$ form a subspace.

(b) If $C(A)$ contains only the zero vector, then A is the zero matrix.

(c) The column space of $2A$ equals the column space of A.

(d) The column space of $A-I$ equals the column space of A.

29. Construct a 3 by 3 matrix whose column space contains $(1,1,0)$ and $(1,0,1)$ but not $(1,1,1)$. Construct a 3 by 3 matrix whose column space is only a line.

30. If the 9 by 12 system $Ax=b$ is solvable for every b, then $C(A)=$ ____.

31. Why isn't $\mathbf{R}^2$ a subspace of $\mathbf{R}^3$?

2.2 Solving $Ax=0$ and $Ax=b$

Chapter 1 concentrated on square invertible matrices. There was one solution to $Ax=b$ and it was $x=-A^{-1}b$. That solution was found by elimination (not by computing A^{-1}). A rectangular matrix brings new possibilities—U may not have a full set of pivots. This section goes onward from U to a reduced form R—**the simplest matrix that elimination can give**. R reveals all solutions immediately.

For an invertible matrix, the nullspace contains only $x=0$ (multiply $Ax=0$ by A^{-1}). The column space is the whole space ($Ax=b$ has a solution for every b). The new questions appear when the nullspace contains *more than the zero vector* and/or the column space contains *less than all vectors*:

1. Any vector x_n in the nullspace can be added to a particular solution x_p. The solutions to all linear equations have this form, $x=x_p+x_n$:

   Complete solution $\quad Ax_p=b$ **and** $\quad Ax_n=0$ $\quad$ **produce** $\quad A(x_p+x_n)=b$**.**

2. When the column space doesn't contain every b in $\mathbf{R}^m$, we need the conditions on b that make $Ax=b$ solvable.

A 3 by 4 example will be a good size. We will write down all solutions to $Ax=0$. We will find the conditions for b to lie in the column space (so that $Ax=b$ is solvable). The 1 by 1 system $0x=b$, one equation and one unknown, shows two possibilities:

$0x=b$ has *no solution* unless $b=0$. The column space of the 1 by 1 zero matrix contains only $b=0$.

$0x=0$ has *infinitely many solutions*. The nullspace contains *all* x. A particular solution is $x_p=0$, and the complete solution is $x=x_p+x_n=0+(\text{any } x)$.

Simple, I admit. If you move up to 2 by 2, it's more interesting. The matrix $\begin{bmatrix} \mathbf{1} & \mathbf{1} \\ \mathbf{2} & \mathbf{2} \end{bmatrix}$ is not invertible: $y+z=b_1$ and $2y+2z=b_2$ usually have no solution.

There is ***no solution*** unless $b_2 = 2b_1$. The column space of A contains only those b's, the multiples of $(1,2)$.

When $b_2 = 2b_1$ there are ***infinitely many solutions***. A particular solution to $y+z=2$ and $2y+2z=4$ is $x_p = (1,1)$. The nullspace of A in Figure 2.2 contains $(-1,1)$ and all its multiples $x_n = (-c,c)$:

$$\textbf{Complete solution} \qquad \begin{array}{rcrcl} y & + & z & = & 2 \\ 2y & + & 2z & = & 4 \end{array} \quad \text{is solved by} \quad x_p + x_n = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1-c \\ 1+c \end{bmatrix}.$$

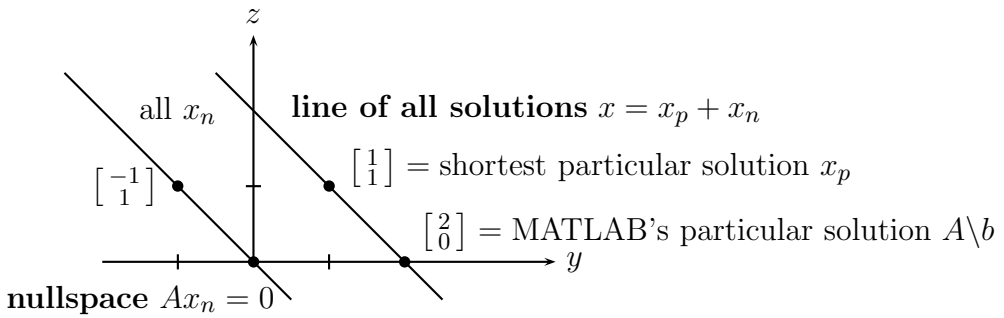


Figure 2.2: The parallel lines of solutions to $Ax_n = 0$ and $\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$.

Echelon Form U and Row Reduced Form R

We start by simplifying this 3 by 4 matrix, first to U and then further to R:

$$\textbf{Basic example} \qquad A = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \\ -1 & -3 & 3 & 4 \end{bmatrix}.$$

The pivot $a_{11} = 1$ is nonzero. The usual elementary operations will produce zeros in the first column below this pivot. The bad news appears in column 2:

$$\textbf{No pivot in column 2} \qquad A \to \begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & \mathbf{0} & 3 & 3 \\ 0 & \mathbf{0} & 6 & 6 \end{bmatrix}.$$

The candidate for the second pivot has become zero: *unacceptable*. We look below that zero for a nonzero entry—intending to carry out a row exchange. In this case the *entry below it is also zero*. If A were square, this would signal that the matrix was singular. With a rectangular matrix, we must expect trouble anyway, and there is no reason to stop.

All we can do is to *go on to the next column*, where the pivot entry is 3. Subtracting twice the second row from the third, we arrive at U:

$$\textbf{Echelon matrix } U \qquad U = \begin{bmatrix} \mathbf{1} & 3 & 3 & 2 \\ 0 & 0 & \mathbf{3} & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Strictly speaking, we proceed to the fourth column. A zero is in the third pivot position, and nothing can be done. U is upper triangular, but its pivots are not on the main diagonal. The nonzero entries of U have a "staircase pattern," or ***echelon form***. For the 5 by 8 case in Figure 2.3, the starred entries may or may not be zero.

$$U = \begin{bmatrix} \bullet & * & * & * & * & * & * & * \\ 0 & \bullet & * & * & * & * & * & * \\ 0 & 0 & 0 & \bullet & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \bullet \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \qquad R = \begin{bmatrix} \mathbf{1} & \mathbf{0} & * & \mathbf{0} & * & * & * & \mathbf{0} \\ 0 & \mathbf{1} & * & \mathbf{0} & * & * & * & \mathbf{0} \\ 0 & 0 & 0 & \mathbf{1} & * & * & * & \mathbf{0} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \mathbf{1} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Figure 2.3: The entries of a 5 by 8 echelon matrix U and its reduced form R.

We can always reach this echelon form U, with zeros below the pivots:

1. The pivots are the first nonzero entries in their rows.
2. Below each pivot is a column of zeros, obtained by elimination.
3. Each pivot lies to the right of the pivot in the row above. This produces the staircase pattern, and zero rows come last.

Since we started with A and ended with U, the reader is certain to ask: Do we have $A = LU$ as before? There is no reason why not, since the elimination steps have not changed. Each step still subtracts a multiple of one row from a row beneath it. The inverse of each step adds back the multiple that was subtracted. These inverses come in the right order to put the multipliers directly into L:

$$\textbf{Lower triangular} \qquad L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix} \quad \text{and} \quad A = LU.$$

Note that L is square. It has the same number of rows as A and U.

The only operation not required by our example, but needed in general, is row exchange by a permutation matrix P. Since we keep going to the next column when no pivots are available, there is no need to assume that A is nonsingular. Here is $PA = LU$ for all matrices:

2B For any m by n matrix A there is a permutation P, a lower triangular L with unit diagonal, and an m by n echelon matrix U, such that $PA = LU$.

Now comes R. We can go further than U, to make the matrix even simpler. Divide the second row by its pivot 3, so that ***all pivots are*** 1. Then use the pivot row to produce ***zero above the pivot***. This time we subtract a row from a *higher row*. The final result (the best form we can get) is the ***reduced row echelon form*** R:

$$\begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & 0 & \mathbf{1} & \mathbf{1} \\ 0 & 0 & 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} \mathbf{1} & \mathbf{3} & \mathbf{0} & \mathbf{-1} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{1} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} = R.$$

This matrix R is the final result of elimination on A. MATLAB would use the command R = rref(A). Of course rref(R) would give R again!

What is the row reduced form of a square invertible matrix? In that case R is the *identity matrix*. There is a full set of pivots, all equal to 1, with zeros above and below. So rref(A) = I, when A is invertible.

For a 5 by 8 matrix with four pivots, Figure 2.3 shows the reduced form R. **It still contains an identity matrix, in the four pivot rows and four pivot columns**. From R we will quickly find the nullspace of A. $Rx = 0$ has the same solutions as $Ux = 0$ and $Ax = 0$.

Pivot Variables and Free Variables

Our goal is to read off all the solutions to $Rx = 0$. The pivots are crucial:

Nullspace of R (pivot columns in boldface)

$$Rx = \begin{bmatrix} \mathbf{1} & 3 & \mathbf{0} & -1 \\ \mathbf{0} & 0 & \mathbf{1} & 1 \\ \mathbf{0} & 0 & \mathbf{0} & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{u} \\ v \\ \boldsymbol{w} \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The unknowns u, v, w, y go into two groups. One group contains the ***pivot variables***, those that correspond to ***columns with pivots***. The first and third columns contain the pivots, so $\boldsymbol{u}$ and $\boldsymbol{w}$ are the pivot variables. The other group is made up of the ***free variables***, corresponding to ***columns without pivots***. These are the second and fourth columns, so v and y are free variables.

To find the most general solution to $Rx = 0$ (or, equivalently, to $Ax = 0$) we may assign arbitrary values to the free variables. Suppose we call these values simply v and y. The pivot variables are completely determined in terms of v and y:

$$Rx = 0 \qquad \begin{aligned} \boldsymbol{u} + 3v - y &= 0 \quad \text{yields} \quad \boldsymbol{u} = -3v + y \\ \boldsymbol{w} + y &= 0 \quad \text{yields} \quad \boldsymbol{w} = \quad -y \end{aligned} \tag{1}$$

There is a "double infinity" of solutions, with v and y free and independent. The complete solution is a combination of two **special solutions**:

Nullspace contains all combinations of special solutions

$$x = \begin{bmatrix} -3v+y \\ v \\ -y \\ y \end{bmatrix} = v\begin{bmatrix} \mathbf{-3} \\ \mathbf{1} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} + y\begin{bmatrix} \mathbf{1} \\ \mathbf{0} \\ \mathbf{-1} \\ \mathbf{1} \end{bmatrix}. \tag{2}$$

Please look again at this complete solution to $Rx = 0$ and $Ax = 0$. The special solution $(-3,1,0,0)$ has free variables $v = 1$, $y = 0$. The other special solution $(1,0,-1,1)$ has $v = 0$ and $y = 1$. *All solutions are linear combinations of these two*. The best way to find all solutions to $Ax = 0$ is from the special solutions:

1. After reaching $Rx = 0$, identify the pivot variables and free variables.
2. Give one free variable the value 1, set the other free variables to 0, and solve $Rx = 0$ for the pivot variables. This x is a special solution.
3. Every free variable produces its own "special solution" by step 2. The combinations of special solutions form the nullspace—all solutions to $Ax = 0$.

Within the four-dimensional space of all possible vectors x, the solutions to $Ax = 0$ form a ***two-dimensional subspace***—the nullspace of A, In the example, $\boldsymbol{N}(A)$ is generated by the special vectors $(-3,1,0,0)$ and $(1,0,-1,1)$. The combinations of these two vectors produce the whole nullspace.

Here is a little trick. The special solutions are especially easy from R. The numbers 3 and 0 and -1 and 1 lie in the "nonpivot columns" of R. **Reverse their signs to find the pivot variables** (not free) **in the special solutions**. I will put the two special solutions from equation (2) into a nullspace matrix N, so you see this neat pattern:

Nullspace matrix (columns are special solutions)

$$N = \begin{bmatrix} \mathbf{-3} & \mathbf{1} \\ 1 & 0 \\ \mathbf{0} & \mathbf{-1} \\ 0 & 1 \end{bmatrix} \begin{matrix} \text{not free} \\ \text{free} \\ \text{not free} \\ \text{free} \end{matrix}$$

The free variables have values 1 and 0. When the free columns moved to the right-hand side of equation (2), their coefficients 3 and 0 and -1 and 1 switched sign. That determined the pivot variables in the special solutions (the columns of N).

This is the place to recognize one extremely important theorem. Suppose a matrix has more columns than rows, $n > m$. Since m rows can hold at most m pivots, ***there must be at least*** $n - m$ ***free variables***. There will be even more free variables if some rows of R reduce to zero; but no matter what, at least one variable must be free. This free variable can be assigned any value, leading to the following conclusion:

2C If $Ax = 0$ has more unknowns than equations ($n > m$), it has at least one special solution: There are more solutions than the trivial $x = 0$.

There must be infinitely many solutions, since any multiple cx will also satisfy $A(cx) = 0$. The nullspace contains the line through x. And if there are additional free variables, the nullspace becomes more than just a line in n-dimensional space. *The nullspace has the same "dimension" as the number of free variables and special solutions.*

This central idea—the ***dimension*** of a subspace—is made precise in the next section. We count the free variables for the nullspace. We count the pivot variables for the column space!

Solving $Ax = b$, $Ux = c$, and $Rx = d$

The case $b \neq 0$ is quite different from $b = 0$. The row operations on A must act also on the right-hand side (on b). We begin with letters (b_1, b_2, b_3) to find the solvability condition—for b to lie in the column space. Then we choose $b = (1, 5, 5)$ and find all solutions x.

For the original example $Ax = b = (b_1, b_2, b_3)$, apply to both sides the operations that led from A to U. The result is an upper triangular system $Ux = c$:

$$Ux = c \qquad \begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \\ y \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 - 2b_1 \\ b_3 - 2b_2 + 5b_1 \end{bmatrix}. \tag{3}$$

The vector c on the right-hand side, which appeared after the forward elimination steps, is just $L^{-1}b$ as in the previous chapter. Start now with $Ux = c$.

It is not clear that these equations have a solution. The third equation is very much in doubt, because its left-hand side is zero. ***The equations are inconsistent unless*** $b_3 - 2b_2 + 5b_1 = 0$. Even though there are more unknowns than equations, there may be no solution. We know another way of answering the same question: $Ax = b$ can be solved if and only if b lies in the column space of A. This subspace comes from the four columns of A (not of U!):

$$\textbf{Columns of } A \textbf{ "span" the column space} \qquad \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \quad \begin{bmatrix} 3 \\ 6 \\ -3 \end{bmatrix}, \quad \begin{bmatrix} 3 \\ 9 \\ 3 \end{bmatrix}, \quad \begin{bmatrix} 2 \\ 7 \\ 4 \end{bmatrix}.$$

Even though there are four vectors, their combinations only fill out a plane in three-dimensional space. Column 2 is three times column 1. The fourth column equals the third minus the first. *These dependent columns, the second and fourth, are exactly the ones without pivots.*

The column space $C(A)$ can be described in two different ways. On the one hand, it is *the plane generated by columns 1 and 3*. The other columns lie in that plane, and contribute nothing new. Equivalently, it is the plane of all vectors b that satisfy $b_3 - 2b_2 + 5b_1 = 0$; this is the constraint if the system is to be solvable. ***Every column***

satisfies this constraint, so it is forced on b***!*** Geometrically, we shall see that the vector $(5,-2,1)$ is perpendicular to each column.

If b belongs to the column space, the solutions of $Ax=b$ are easy to find. The last equation in $Ux=c$ is $0=0$. To the free variables v and y, we may assign any values, as before. The pivot variables u and w are still determined by back-substitution. For a specific example with $b_3-2b_2+5b_1=0$, choose $b=(1,5,5)$:

$$Ax=b \qquad \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \\ -1 & -3 & 3 & 4 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 5 \end{bmatrix}.$$

Forward elimination produces U on the left and c on the right:

$$Ux=c \qquad \begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}.$$

The last equation is $0=0$, as expected. Back-substitution gives

$$\begin{aligned} 3w+3y=3 \qquad &\text{or} \qquad w=1-y \\ u+3v+3w+2y=1 \qquad &\text{or} \qquad u=-2-3v+y. \end{aligned}$$

Again there is a double infinity of solutions: v and y are free, u and w are not:

Complete solution
$x=x_p+x_n$

$$x=\begin{bmatrix} u \\ v \\ w \\ y \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ 1 \\ 0 \end{bmatrix} + v \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}. \tag{4}$$

This has all solutions to $Ax=0$, plus the new $x_p=(-2,0,1,0)$. That x_p is ***a particular solution*** to $Ax=b$. The last two terms with v and y yield more solutions (because they satisfy $Ax=0$). ***Every solution to*** $Ax=b$ ***is the sum of one particular solution and a solution to*** $Ax=0$:

$$x_{\textbf{complete}} = x_{\textbf{particular}} + x_{\textbf{nullspace}}$$

The particular solution in equation (4) comes from solving the equation *with all free variables set to zero*. That is the only new part, since the nullspace is already computed. When you multiply the highlighted equation by A, you get $Ax_{\text{complete}}=b+0$.

Geometrically, the solutions again fill a two-dimensional surface—but it is not a subspace. It does not contain $x=0$. It is *parallel* to the nullspace we had before, shifted by the particular solution x_p as in Figure 2.2. Equation (4) is a good way to write the answer:

1. Reduce $Ax=b$ to $Ux=c$.

2. With free variables $= 0$, find a particular solution to $Ax_p = b$ and $Ux_p = c$.

3. Find the special solutions to $Ax = 0$ (or $Ux = 0$ or $Rx = 0$). Each free variable, in turn, is 1. Then $x = x_p +$ (any combination x_n of special solutions).

When the equation was $Ax = 0$, the particular solution was the zero vector! It fits the pattern, but $x_{\text{particular}} = 0$ was not written in equation (2). Now x_p is added to the nullspace solutions, as in equation (4).

Question: How does the reduced form R make this solution even clearer? You will see it in our example. Subtract equation 2 from equation 1, and then divide equation 2 by its pivot. On the left-hand side, this produces R, as before. On the right-hand side, these operations change $c = (1,3,0)$ to a new vector $d = (-2,1,0)$:

$$\textbf{Reduced equation} \quad Rx = d \qquad \begin{bmatrix} 1 & 3 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \\ y \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}. \tag{5}$$

Our particular solution x_p, (one choice out of many) has free variables $v = y = 0$. Columns 2 and 4 can be ignored. Then we immediately have $u = -2$ and $w = 1$, exactly as in equation (4). **The entries of d go directly into x_p.** This is because the identity matrix is sitting in the pivot columns of R!

Let me summarize this section, before working a new example. Elimination reveals the pivot variables and free variables. ***If there are r pivots, there are r pivot variables and $n - r$ free variables***. That important number r will be given a name—it is the ***rank of the matrix***.

2D Suppose elimination reduces $Ax = b$ to $Ux = c$ and $Rx = d$, with r pivot rows and r pivot columns. **The rank of those matrices is r.** The last $m - r$ rows of U and R are zero, so there is a solution only if the last $m - r$ entries of c and d are also zero.

The complete solution is $x = x_p + x_n$. One particular solution x_p has all free variables zero. Its pivot variables are the first r entries of d, so $Rx_p = d$.

The nullspace solutions x_n are combinations of $n - r$ special solutions, with one free variable equal to 1. The pivot variables in that special solution can be found in the corresponding column of R (with sign reversed).

You see how the rank r is crucial. It counts the pivot rows in the "row space" and the pivot columns in the column space. There are $n - r$ special solutions in the nullspace. There are $m - r$ solvability conditions on b or c or d.

Another Worked Example

The full picture uses elimination and pivot columns to find the column space, nullspace, and rank. The 3 by 4 matrix A has rank 2:

$$Ax = b \quad \text{is} \quad \begin{array}{rcrcrcrcl} \mathbf{1}x_1 & + & \mathbf{2}x_2 & + & \mathbf{3}x_3 & + & \mathbf{5}x_4 & = & b_1 \\ \mathbf{2}x_1 & + & \mathbf{4}x_2 & + & \mathbf{8}x_3 & + & \mathbf{12}x_4 & = & b_2 \\ \mathbf{3}x_1 & + & \mathbf{6}x_2 & + & \mathbf{7}x_3 & + & \mathbf{13}x_4 & = & b_3 \end{array} \tag{6}$$

1. Reduce $[A \;\; b]$ to $[U \;\; c]$, to reach a triangular system $Ux = c$.
2. Find the condition on b_1, b_2, b_3 to have a solution.
3. Describe the column space of A: Which plane in $\mathbf{R}^3$?
4. Describe the nullspace of A: Which special solutions in $\mathbf{R}^4$?
5. Find a particular solution to $Ax = (0, 6, -6)$ and the complete $x_p + x_n$.
6. Reduce $[U \;\; c]$ to $[R \;\; d]$: Special solutions from R and x_p from d.

Solution. (Notice how the right-hand side is included as an extra column!)

1. The multipliers in elimination are 2 and 3 and -1, taking $[A \;\; b]$ to $[U \;\; c]$.

$$\left[\begin{array}{ccccc} 1 & 2 & 3 & 5 & \boldsymbol{b_1} \\ 2 & 4 & 8 & 12 & \boldsymbol{b_2} \\ 3 & 6 & 7 & 13 & \boldsymbol{b_3} \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 2 & 3 & 5 & \boldsymbol{b_1} \\ 0 & 0 & 2 & 2 & \boldsymbol{b_2 - 2b_1} \\ 0 & 0 & -2 & -2 & \boldsymbol{b_3 - 3b_1} \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 2 & 3 & 5 & \boldsymbol{b_1} \\ 0 & 0 & 2 & 2 & \boldsymbol{b_2 - 2b_1} \\ 0 & 0 & 0 & 0 & \boldsymbol{b_3 + b_2 - 5b_1} \end{array}\right].$$

2. The last equation shows the solvability condition $b_3 + b_2 - 5b_1 = 0$. Then $0 = 0$.
3. The column space of A is the plane containing all combinations of the pivot columns $(1, 2, 3)$ and $(3, 8, 7)$.

   Second description: The column space contains all vectors with $b_3 + b_2 - 5b_1 = 0$. That makes $Ax = b$ solvable, so b is in the column space. *All columns of A pass this test $b_3 + b_2 - 5b_1 = 0$. This is the equation for the plane (in the first description of the column space).*
4. The special solutions in N have free variables $x_2 = 1$, $x_4 = 0$ and $x_2 = 0$, $x_4 = 1$:

   Nullspace matrix
   Special solutions to $Ax = 0$
   Back-substitution in $Ux = 0$
   Just switch signs in $Rx = 0$

$$N = \begin{bmatrix} -2 & -2 \\ 1 & 0 \\ 0 & -1 \\ 0 & 1 \end{bmatrix}.$$

5. Choose $b = (0,6,-6)$, which has $b_3 + b_2 - 5b_1 = 0$. Elimination takes $Ax = b$ to $Ux = c = (0,6,0)$. Back-substitute with free variables $= 0$:

$$\textbf{Particular solution to } Ax_p = (0,6,-6) \qquad x_p = \begin{bmatrix} -9 \\ 0 \\ 3 \\ 0 \end{bmatrix} \begin{matrix} \text{free} \\ \\ \\ \text{free} \end{matrix}$$

The complete solution to $Ax = (0,6,-6)$ is (this x_p) + (all x_n).

6. In the reduced R, the third column changes from $(3,2,0)$ to $(0,1,0)$. The right-hand side $c = (0,6,0)$ becomes $d = (-9,3,0)$. Then -9 and 3 go into x_p:

$$\left[\begin{array}{cc} U & c \end{array}\right] = \left[\begin{array}{cccc|c} 1 & 2 & 3 & 5 & 0 \\ 0 & 0 & 2 & 2 & 6 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \longrightarrow \left[\begin{array}{cc} R & d \end{array}\right] = \left[\begin{array}{cccc|c} \mathbf{1} & 2 & 0 & 2 & \mathbf{-9} \\ 0 & 0 & \mathbf{1} & 1 & \mathbf{3} \\ 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

That final matrix $[R \;\; d]$ **is** $\mathsf{rref}([A \;\; b]) = \mathsf{rref}([U \;\; c])$. The numbers 2 and 0 and 2 and 1 in the free columns of R have opposite sign in the special solutions (the nullspace matrix N). Everything is revealed by $Rx = d$.

Problem Set 2.2

1. Construct a system with more unknowns than equations, but no solution. Change the right-hand side to zero and find all solutions x_n.

2. Reduce A and B to echelon form, to find their ranks. Which variables are free?

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}.$$

Find the special solutions to $Ax = 0$ and $Bx = 0$. Find all solutions.

3. Find the echelon form U, the free variables, and the special solutions:

$$A = \begin{bmatrix} 0 & 1 & 0 & 3 \\ 0 & 2 & 0 & 6 \end{bmatrix}, \qquad b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}.$$

$Ax = b$ is consistent (has a solution) when b satisfies $b_2 =$ ____. Find the complete solution in the same form as equation (4).

4. Carry out the same steps as in the previous problem to find the complete solution of $Mx = b$:

$$M = \begin{bmatrix} 0 & 0 \\ 1 & 2 \\ 0 & 0 \\ 3 & 6 \end{bmatrix}, \qquad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}.$$

5. Write the complete solutions $x = x_p + x_n$ to these systems, as in equation (4):

$$\begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 5 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix} \qquad \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}.$$

6. Describe the set of attainable right-hand sides b (in the column space) for

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix},$$

by finding the constraints on b that turn the third equation into $0 = 0$ (after elimination). What is the rank, and a particular solution?

7. Find the value of c that makes it possible to solve $Ax = b$, and solve it:

$$\begin{aligned} u + v + 2w &= 2 \\ 2u + 3v - w &= 5 \\ 3u + 4v + w &= c. \end{aligned}$$

8. Under what conditions on b_1 and b_2 (if any) does $Ax = b$ have a solution?

$$A = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 0 & 7 \end{bmatrix}, \qquad b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}.$$

Find two vectors in the nullspace of A, and the complete solution to $Ax = b$.

9. (a) Find the special solutions to $Ux = 0$. Reduce U to R and repeat:

$$Ux = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

(b) If the right-hand side is changed from $(0,0,0)$ to $(a,b,0)$, what are all solutions?

10. Find a 2 by 3 system $Ax = b$ whose complete solution is

$$x = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + w \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}.$$

Find a 3 by 3 system with these solutions exactly when $b_1 + b_2 = b_3$.

11. Write a 2 by 2 system $Ax = b$ with many solutions x_n but no solution x_p. (Therefore the system has no solution.) Which b's allow an x_p?

12. Which of these rules give a correct definition of the *rank* of A?

(a) The number of nonzero rows in R.

(b) The number of columns minus the total number of rows.

(c) The number of columns minus the number of free columns.

(d) The number of 1s in R.

13. Find the reduced row echelon forms R and the rank of these matrices:

(a) The 3 by 4 matrix of all 1s.

(b) The 4 by 4 matrix with $a_{ij} = (-1)^{ij}$.

(c) The 3 by 4 matrix with $a_{ij} = (-1)^{j}$.

14. Find R for each of these (block) matrices, and the special solutions:

$$A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 3 \\ 2 & 4 & 6 \end{bmatrix} \qquad B = \begin{bmatrix} A & A \end{bmatrix} \qquad C = \begin{bmatrix} A & A \\ A & 0 \end{bmatrix}.$$

15. If the r pivot variables come first, the reduced R must look like

$$R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix} \qquad \begin{array}{l} I \text{ is } r \text{ by } r \\ F \text{ is } r \text{ by } n-r \end{array}$$

What is the nullspace matrix N containing the special solutions?

16. Suppose all r pivot variables come *last*. Describe the four blocks in the m by n reduced echelon form (the block B should be r by r):

$$R = \begin{bmatrix} A & B \\ C & D \end{bmatrix}.$$

What is the nullspace matrix N of special solutions? What is its shape?

17. (Silly problem) Describe all 2 by 3 matrices A_1 and A_2 with row echelon forms R_1 and R_2, such that $R_1 + R_2$ is the row echelon form of $A_1 + A_2$. Is it true that $R_1 = A_1$ and $R_2 = A_2$ in this case?

18. If A has r pivot columns, then A^{T} has r pivot columns. Give a 3 by 3 example for which the column numbers are different for A and A^{T}.

19. What are the special solutions to $Rx = 0$ and $R^{\mathrm{T}}y = 0$ for these R?

$$R = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 4 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad R = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

20. *If A has rank r, then it has an r by r submatrix S that is invertible.* Find that submatrix S from the pivot rows and pivot columns of each A:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 4 \end{bmatrix} \qquad A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix} \qquad A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

21. Explain why the pivot rows and pivot columns of A (not R) always give an r by r invertible submatrix of A.

22. Find the ranks of AB and AM (rank 1 matrix times rank 1 matrix):

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 1.5 & 6 \end{bmatrix} \quad \text{and} \quad M = \begin{bmatrix} 1 & b \\ c & bc \end{bmatrix}.$$

23. Multiplying the rank 1 matrices $A = uv^{\mathrm{T}}$ and $B = wz^{\mathrm{T}}$ gives uz^{T} times the number ____. AB has rank 1 unless ____ $= 0$.

24. Every column of AB is a combination of the columns of A. Then the dimensions of the column spaces give $\mathrm{rank}(AB) \leq \mathrm{rank}(A)$.

Problem: Prove also that $\mathrm{rank}(AB) \leq \mathrm{rank}(B)$.

25. (Important) Suppose A and B are n by n matrices, and $AB = I$. Prove from $\mathrm{rank}(AB) \leq \mathrm{rank}(A)$ that the rank of A is n. So A is invertible and B must be its two-sided inverse. Therefore $BA = I$ (*which is not so obvious!*).

26. If A is 2 by 3 and C is 3 by 2, show from its rank that $CA \neq I$. Give an example in which $AC = I$. For $m < n$, a right inverse is not a left inverse.

27. Suppose A and B have the *same* reduced-row echelon form R. Explain how to change A to B by elementary row operations. So B equals an ____ matrix times A.

28. Every m by n matrix of rank r reduces to (m by r) times (r by n):

$$A = (\text{pivot columns of } A)(\text{first } r \text{ rows of } R) = (\mathbf{COL})(\mathbf{ROW}).$$

Write the 3 by 4 matrix A at the start of this section as the product of the 3 by 2 matrix from the pivot columns and the 2 by 4 matrix from R:

$$A = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \\ -1 & -3 & 3 & 4 \end{bmatrix}$$

29. Suppose A is an m by n matrix of rank r. Its reduced echelon form is R. Describe exactly the *reduced row echelon form of* R^{T} (not A^{T}).

30. (Recommended) Execute the six steps following equation (6) to find the column space and nullspace of A and the solution to $Ax = b$:

$$A = \begin{bmatrix} 2 & 4 & 6 & 4 \\ 2 & 5 & 7 & 6 \\ 2 & 3 & 5 & 2 \end{bmatrix} \qquad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix}.$$

31. For every c, find R and the special solutions to $Ax = 0$:

$$A = \begin{bmatrix} 1 & 1 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 1 & c & 2 & 2 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1-c & 2 \\ 0 & 2-c \end{bmatrix}.$$

32. What is the nullspace matrix N (of special solutions) for A, B, C?

$$A = \begin{bmatrix} I & I \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} I & I \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} I & I & I \end{bmatrix}.$$

Problems 33–36 are about the solution of $Ax = b$. Follow the steps in the text to x_p and x_n. Reduce the augmented matrix $[A\ \ b]$.

33. Find the complete solutions of

$$\begin{aligned} x + 3y + 3z &= 1 \\ 2x + 6y + 9z &= 5 \\ -x - 3y + 3z &= 5 \end{aligned} \qquad \text{and} \qquad \begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ 0 & 0 & 2 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}.$$

34. Under what condition on b_1, b_2, b_3 is the following system solvable? Include b as a fourth column in $[A\ \ b]$. Find all solutions when that condition holds:

$$\begin{aligned} x + 2y - 2z &= b_1 \\ 2x + 5y - 4z &= b_2 \\ 4x + 9y - 8z &= b_3. \end{aligned}$$

35. What conditions on b_1, b_2, b_3, b_4 make each system solvable? *Solve for* x:

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 2 & 5 \\ 3 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} \qquad \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 2 & 5 & 7 \\ 3 & 9 & 12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}.$$

36. Which vectors (b_1, b_2, b_3) are in the column space of A? Which combinations of the rows of A give zero?

$$\text{(a)} \quad A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 6 & 3 \\ 0 & 2 & 5 \end{bmatrix} \qquad \text{(b)} \quad A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 2 & 4 & 8 \end{bmatrix}.$$

37. Why can't a 1 by 3 system have $x_p = (2,4,0)$ and $x_n =$ any multiple of $(1,1,1)$?

38. (a) If $Ax = b$ has two solutions x_1 and x_2, find two solutions to $Ax = 0$.

(b) Then find another solution to $Ax = b$.

39. Explain why all these statements are false:

(a) The complete solution is any linear combination of x_p and x_n.

(b) A system $Ax = b$ has at most one particular solution.

(c) The solution x_p with all free variables zero is the shortest solution (minimum length $\|x\|$). (Find a 2 by 2 counterexample.)

(d) If A is invertible there is no solution x_n in the nullspace.

40. Suppose column 5 of U has no pivot. Then x_5 is a ____ variable. The zero vector (is) (is not) the only solution to $Ax = 0$. If $Ax = b$ has a solution, then it has ____ solutions.

41. If you know x_p (free variables $= 0$) and all special solutions for $Ax = b$, find x_p and all special solutions for these systems:

$$Ax = 2b \qquad \begin{bmatrix} A & A \end{bmatrix} \begin{bmatrix} x \\ X \end{bmatrix} = b \qquad \begin{bmatrix} A \\ A \end{bmatrix} \begin{bmatrix} x \end{bmatrix} = \begin{bmatrix} b \\ b \end{bmatrix}.$$

42. If $Ax = b$ has infinitely many solutions, why is it impossible for $Ax = B$ (new right-hand side) to have only one solution? Could $Ax = B$ have no solution?

43. Choose the number q so that (if possible) the ranks are (a) 1, (b) 2, (c) 3:

$$A = \begin{bmatrix} 6 & 4 & 2 \\ -3 & -2 & -1 \\ 9 & 6 & q \end{bmatrix} \qquad \text{and} \qquad B = \begin{bmatrix} 3 & 1 & 3 \\ q & 2 & q \end{bmatrix}.$$

44. Give examples of matrices A for which the number of solutions to $Ax = b$ is

(a) 0 or 1, depending on b.

(b) ∞, regardless of b.

(c) 0 or ∞, depending on b.

(d) 1, regardless of b.

45. Write all known relations between r and m and n if $Ax = b$ has

(a) no solution for some b.

(b) infinitely many solutions for every b.

(c) exactly one solution for some b, no solution for other b.

(d) exactly one solution for every b.

46. Apply Gauss-Jordan elimination (right-hand side becomes extra column) to $Ux = 0$ and $Ux = c$. Reach $Rx = 0$ and $Rx = d$:

$$\begin{bmatrix} U & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 & \mathbf{0} \\ 0 & 0 & 4 & \mathbf{0} \end{bmatrix} \qquad \text{and} \qquad \begin{bmatrix} U & c \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 & \mathbf{5} \\ 0 & 0 & 4 & \mathbf{8} \end{bmatrix}.$$

Solve $Rx = 0$ to find x_n (its free variable is $x_2 = 1$). Solve $Rx = d$ to find x_p (its free variable is $x_2 = 0$).

47. Apply elimination with the extra column to reach $Rx = 0$ and $Rx = d$:

$$\begin{bmatrix} U & 0 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 6 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad \text{and} \qquad \begin{bmatrix} U & c \end{bmatrix} = \begin{bmatrix} 3 & 0 & 6 & \mathbf{9} \\ 0 & 0 & 2 & \mathbf{4} \\ 0 & 0 & 0 & \mathbf{5} \end{bmatrix}.$$

Solve $Rx = 0$ (free variable $= 1$). What are the solutions to $Rx = d$?

48. Reduce to $Ux = c$ (Gaussian elimination) and then $Rx = d$:

$$Ax = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 1 & 3 & 2 & 0 \\ 2 & 0 & 4 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 10 \end{bmatrix} = b.$$

Find a particular solution x_p and all nullspace solutions x_n.

49. Find A and B with the given property or explain why you can't.

(a) The only solution to $Ax = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ is $x = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

(b) The only solution to $Bx = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ is $x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

50. The complete solution to $Ax = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ is $x = \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Find A.

51. The nullspace of a 3 by 4 matrix A is the line through $(2,3,1,0)$.

(a) What is the *rank* of A and the complete solution to $Ax = 0$?

(b) What is the exact row reduced echelon form R of A?

52. Reduce these matrices A and B to their ordinary echelon forms U:

$$\text{(a)} \quad A = \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 1 & 2 & 3 & 6 & 9 \\ 0 & 0 & 1 & 2 & 3 \end{bmatrix} \qquad \text{(b)} \quad B = \begin{bmatrix} 2 & 4 & 2 \\ 0 & 4 & 4 \\ 0 & 8 & 8 \end{bmatrix}.$$

Find a special solution for each free variable and describe every solution to $Ax = 0$ and $Bx = 0$. Reduce the echelon forms U to R, and draw a box around the identity matrix in the pivot rows and pivot columns.

53. True or False? (Give reason if true, or counterexample to show it is false.)

(a) A square matrix has no free variables.

(b) An invertible matrix has no free variables.

(c) An m by n matrix has no more than n pivot variables.

(d) An m by n matrix has no more than m pivot variables.

54. Is there a 3 by 3 matrix with no zero entries for which $U = R = I$?

55. Put as many 1s as possible in a 4 by 7 echelon matrix U and in a *reduced* form R whose pivot columns are 2, 4, 5.

56. Suppose column 4 of a 3 by 5 matrix is all 0s. Then x_4 is certainly a ____ variable. The special solution for this variable is the vector $x =$ ____.

57. Suppose the first and last columns of a 3 by 5 matrix are the same (nonzero). Then ____ is a free variable. Find the special solution for this variable.

58. The equation $x - 3y - z = 0$ determines a plane in $\mathbf{R}^3$. What is the matrix A in this equation? Which are the free variables? The special solutions are $(3,1,0)$ and ____. The parallel plane $x - 3y - z = 12$ contains the particular point $(12,0,0)$. All points on this plane have the following form (fill in the first components):

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

59. Suppose column 1 + column 3 + column 5 = 0 in a 4 by 5 matrix with four pivots. Which column is sure to have no pivot (and which variable is free)? What is the special solution? What is the nullspace?

Problems 60–66 ask for matrices (if possible) with specific properties.

60. Construct a matrix whose nullspace consists of all combinations of $(2,2,1,0)$ and $(3,1,0,1)$.

61. Construct a matrix whose nullspace consists of all multiples of $(4,3,2,1)$.

62. Construct a matrix whose column space contains $(1,1,5)$ and $(0,3.1)$ and whose nullspace contains $(1,1,2)$.

63. Construct a matrix whose column space contains $(1,1,0)$ and $(0,1,1)$ and whose nullspace contains $(1,0,1)$ and $(0,0,1)$.

64. Construct a matrix whose column space contains $(1,1,1)$ and whose nullspace is the line of multiples of $(1,1,1,1)$.

65. Construct a 2 by 2 matrix whose nullspace equals its column space.

66. Why does no 3 by 3 matrix have a nullspace that equals its column space?

67. The reduced form R of a 3 by 3 matrix with randomly chosen entries is almost sure to be ____. What R is virtually certain if the random A is 4 by 3?

68. Show by example that these three statements are generally false:

(a) A and A^{T} have the same nullspace.

(b) A and A^{T} have the same free variables.

(c) If R is the reduced form $\mathsf{rref}(A)$ then R^{T} is $\mathsf{rref}(A^{\mathrm{T}})$.

69. If the special solutions to $Rx = 0$ are in the columns of these N, go backward to find the nonzero rows of the reduced matrices R:

$$N = \begin{bmatrix} 2 & 3 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad N = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad \text{and} \quad N = \begin{bmatrix} \\ \\ \end{bmatrix} \quad \text{(empty 3 by 1).}$$

70. Explain why A and $-A$ always have the same reduced echelon form R.

2.3 Linear Independence, Basis, and Dimension

By themselves, the numbers m and n give an incomplete picture of the true size of a linear system. The matrix in our example had three rows and four columns, but the third row was only a combination of the first two. After elimination it became a zero row, It had no effect on the homogeneous problem $Ax = 0$. The four columns also failed to be independent, and the column space degenerated into a two-dimensional plane.

The important number that is beginning to emerge (the true size) is the ***rank*** r. The rank was introduced as the *number of pivots* in the elimination process. Equivalently, the final matrix U has r nonzero rows. This definition could be given to a computer. But it would be wrong to leave it there because the rank has a simple and intuitive meaning: *The rank counts the number of genuinely independent rows in the matrix A*. We want definitions that are mathematical rather than computational.

The goal of this section is to explain and use four ideas:

1. Linear independence or dependence.
2. Spanning a subspace.
3. Basis for a subspace (a set of vectors).
4. Dimension of a subspace (a number).

The first step is to define ***linear independence***. Given a set of vectors $v_1, \dots, v_k$, we look at their combinations $c_1v_1 + c_2v_2 + \cdots + c_kv_k$. The trivial combination, with all weights $c_i = 0$, obviously produces the zero vector: $0v_1 + \cdots + 0v_k = 0$. The question is whether this is the *only way* to produce zero. If so, the vectors are independent.

If any other combination of the vectors gives zero, they are *dependent*.

2E Suppose $c_1v_1 + \cdots + c_kv_k = 0$ only happens when $c_1 = \cdots = c_k = 0$. Then the vectors $v_1, \dots, v_k$ are ***linearly independent***. If any c's are nonzero, the v's are ***linearly dependent***. One vector is a combination of the others.

Linear dependence is easy to visualize in three-dimensional space, when all vectors go out from the origin. Two vectors are dependent if they lie on the same line. *Three vectors are dependent if they lie in the same plane*. A random choice of three vectors, without any special accident, should produce linear independence (not in a plane). Four vectors are always linearly dependent in $\mathbf{R}^3$.

Example 1. If $v_1 =$ zero vector, then the set is linearly dependent. We may choose $c_1 = 3$ and all other $c_i = 0$; this is a nontrivial combination that produces zero.

Example 2. The columns of the matrix

$$A = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 5 \\ -1 & -3 & 3 & 0 \end{bmatrix}$$

are linearly dependent, since the second column is three times the first. The combination of columns with weights -3, 1, 0, 0 gives a column of zeros.

The rows are also linearly dependent; row 3 is two times row 2 minus five times row 1. (This is the same as the combination of b_1, b_2, b_3, that had to vanish on the right-hand side in order for $Ax = b$ to be consistent. Unless $b_3 - 2b_2 + 5b_1 = 0$, the third equation would not become $0 = 0$.)

Example 3. The columns of this triangular matrix are linearly *independent*:

No zeros on the diagonal $\qquad A = \begin{bmatrix} 3 & 4 & 2 \\ 0 & 1 & 5 \\ 0 & 0 & 2 \end{bmatrix}.$

Look for a combination of the columns that makes zero:

$$\text{Solve } Ac = 0 \qquad c_1 \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 2 \\ 5 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

We have to show that c_1, c_2, c_3 ***are all forced to be zero***. The last equation gives $c_3 = 0$. Then the next equation gives $c_2 = 0$, and substituting into the first equation forces $c_1 = 0$. The only combination to produce the zero vector is the trivial combination. ***The nullspace of*** A ***contains only the zero vector*** $c_1 = c_2 = c_3 = 0$.

> ***The columns of*** A ***are independent exactly when*** $N(A) = \{\textbf{zero vector}\}$.

A similar reasoning applies to the rows of A, which are also independent. Suppose

$$c_1(3,4,2) + c_2(0,1,5) + c_3(0,0,2) = (0,0,0).$$

From the first components we find $3c_1 = 0$ or $c_1 = 0$. Then the second components give $c_2 = 0$, and finally $c_3 = 0$.

The nonzero rows of any echelon matrix U must be independent. Furthermore, if we pick out *the columns that contain the pivots*, they also are linearly independent. In our earlier example, with

Two independent rows
Two independent columns

$$U = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

the pivot columns 1 and 3 are independent. No set of three columns is independent, and certainly not all four. It is true that columns 1 and 4 are also independent, but if that last 1 were changed to 0 they would be dependent. *It is the columns with pivots that are guaranteed to be independent*. The general rule is this:

2F The r nonzero rows of an echelon matrix U and a reduced matrix R are linearly independent. So are the r columns that contain pivots.

Example 4. The columns of the n by n identity matrix are independent:

$$I = \begin{bmatrix} 1 & 0 & \cdot & 0 \\ 0 & 1 & \cdot & 0 \\ \cdot & \cdot & \cdot & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

These columns $e_1, \ldots, e_n$ represent unit vectors in the coordinate directions; in $\mathbf{R}^4$,

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \qquad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \qquad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \qquad e_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

Most sets of four vectors in $\mathbf{R}^4$ are independent. Those e's might be the safest.

To check any set of vectors $v_1, \ldots, v_n$ for independence, put them in the columns of A. Then solve the system $Ac = 0$; the vectors are dependent if there is a solution other than $c = 0$. With no free variables (*rank n*), there is no nullspace except $c = 0$; the vectors are independent. If the rank is less than n, at least one free variable can be nonzero and the columns are dependent.

One case has special importance. Let the n vectors have m components, so that A is an m by n matrix. Suppose now that $n > m$. There are too many columns to be independents There cannot be n pivots, since there are not enough rows to hold them. The rank will be less than n. Every system $Ac = 0$ with more unknowns than equations has solutions $c \neq 0$.

2G A set of n vectors in $\mathbf{R}^m$ must be linearly dependent if $n > m$.

The reader will recognize this as a disguised form of 2C: Every m by n system $Ax = 0$ has nonzero solutions if $n > m$.

Example 5. These three columns in $\mathbf{R}^2$ cannot be independent:

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 3 & 2 \end{bmatrix}.$$

To find the combination of the columns producing zero we solve $Ac = 0$:

$$A \to U = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

If we give the value 1 to the free variable c_3, then back-substitution in $Uc = 0$ gives $c_2 = -1$, $c_1 = 1$. With these three weights, the first column minus the second plus the third equals zero: Dependence.

Spanning a Subspace

Now we define what it means for a set of vectors to *span a space*. The column space of A is *spanned* by the columns. **Their combinations produce the whole space**:

2H If a vector space $\mathbf{V}$ consists of all linear combinations of $w_1, \ldots, w_\ell$, then these vectors ***span*** the space. Every vector v in $\mathbf{V}$ is some combination of the w's:

Every v comes from w's $\qquad v = c_1 w_1 + \cdots + c_\ell w_\ell$ for some coefficients c_i.

It is permitted that a different combination of w's could give the same vector v. The c's need not be unique, because the spanning set might be excessively large—it could include the zero vector, or even all vectors.

Example 6. The vectors $w_1 = (1,0,0)$, $w_2 = (0,1,0)$, and $w_3 = (-2,0,0)$ span a plane (the x-y plane) in $\mathbf{R}^3$. The first two vectors also span this plane, whereas w_1 and w_3 span only a line.

Example 7. The column space of A is exactly ***the space that is spanned by its columns***. The row space is spanned by the rows. The definition is made to order. Multiplying A by any x gives a combination of the columns; it is a vector Ax in the column space.

The coordinate vectors $e_1, \ldots, e_n$ coming from the identity matrix span $\mathbf{R}^n$. Every vector $b = (b_1, \ldots, b_n)$ is a combination of those columns. In this example the weights are the components b_i themselves: $b = b_1 e_1 + \cdots + b_n e_n$. But the columns of other matrices also span $\mathbf{R}^n$!

Basis for a Vector Space

To decide if b is a combination of the columns, we try to solve $Ax = b$. To decide if the columns are independent, we solve $Ax = 0$. ***Spanning involves the column space, and independence involves the nullspace***. The coordinate vectors $e_1, \ldots, e_n$ span $\mathbf{R}^n$ and they are linearly independent. Roughly speaking, ***no vectors in that set are wasted***. This leads to the crucial idea of a ***basis***.

2I A ***basis*** for $\mathbf{V}$ is a sequence of vectors having two properties at once:

1. The vectors are linearly independent (not too many vectors).
2. They span the space $\mathbf{V}$ (not too few vectors).

This combination of properties is absolutely fundamental to linear algebra. It means that every vector in the space is a combination of the basis vectors, because they span. It also means that the combination is unique: If $v = a_1 v_1 + \cdots + a_k v_k$ and also $v = b_1 v_1 + \cdots + b_k v_k$, then subtraction gives $0 = \sum (a_i - b_i) v_i$. Now independence plays its part; every coefficient $a_i - b_i$ must be zero. Therefore $a_i = b_i$. ***There is one and only one way to write v as a combination of the basis vectors***.

We had better say at once that the coordinate vectors $e_1, \ldots, e_n$ are not the only basis for $\mathbf{R}^n$. Some things in linear algebra are unique, but not this. A vector space has ***infinitely many different bases***. Whenever a square matrix is invertible, its columns are independent—and they are a basis for $\mathbf{R}^n$. The two columns of this nonsingular matrix are a basis for $\mathbf{R}^2$:

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$$

Every two-dimensional vector is a combination of those (independent!) columns.

Example 8. The x-y plane in Figure 2.4 is just $\mathbf{R}^2$. The vector v_1 by itself is linearly independent, but it fails to span $\mathbf{R}^2$. The three vectors v_1, v_2, v_3 certainly span $\mathbf{R}^2$, but are not independent. *Any two* of these vectors, say v_1 and v_2, have both properties—they

span, and they are independent. So they form a basis. Notice again that *a vector space does not have a unique basis*.

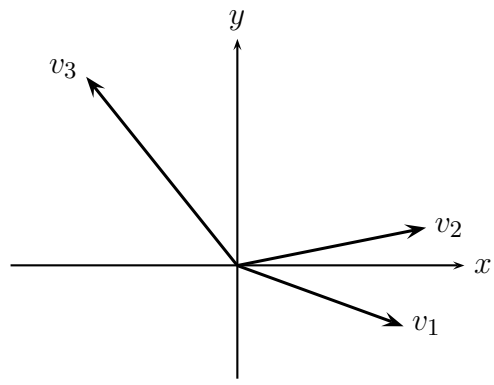


Figure 2.4: A spanning set v_1, v_2, v_3. Bases v_1, v_2 and v_1, v_3 and v_2, v_3.

Example 9. These four columns span the column space of U, but they are not independent:

Echelon matrix $$U = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There are many possibilities for a basis, but we propose a specific choice: ***The columns that contain pivots*** (in this case the first and third, which correspond to the basic variables) ***are a basis for the column space***. These columns are independent, and it is easy to see that they span the space. In fact, the column space of U is just the x-y plane within $\mathbf{R}^3$. $C(U)$ is *not the same* as the column space $C(A)$ before elimination—but the *number* of independent columns didn't change.

To summarize: *The columns of any matrix span its column space*. If they are independent, they are a basis for the column space—whether the matrix is square or rectangular. If we are asking the columns to be a basis for the whole space $\mathbf{R}^n$, then the matrix must be square and invertible.

Dimension of a Vector Space

A space has infinitely many different bases, but there is something common to all of these choices. The ***number of basis vectors*** is a property of the space itself:

> **2J** Any two bases for a vector space $\mathbf{V}$ contain the same number of vectors. This number, which is shared by all bases and expresses the number of "degrees of freedom" of the space, is the ***dimension*** of $\mathbf{V}$.

We have to prove this fact: All possible bases contain the same number of vectors. The x-y plane in Figure 2.4 has two vectors in every basis; its dimension is 2. In three

dimensions we need three vectors, along the x-y-z axes or in three other (linearly independent!) directions. ***The dimension of the space*** $\mathbf{R}^n$ ***is*** n. The column space of U in Example 9 had dimension 2; it was a "two-dimensional subspace of $\mathbf{R}^3$." The zero matrix is rather exceptional, because its column space contains only the zero vector. By convention, the empty set is a basis for that space, and its dimension is zero.

Here is our first big theorem in linear algebra:

2K If $v_1, \ldots, v_m$ and $w_1, \ldots, w_n$ are both bases for the same vector space, then $m = n$. The number of vectors is the same.

Proof. Suppose there are more w's than v's ($n > m$). We will arrive at a contradiction. Since the v's form a basis, they must span the space. *Every* w_j *can be written as a combination of the* v*'s*: If $w_1 = a_{11}v_1 + \cdots + a_{m1}v_m$, this is the first column of a matrix multiplication VA:

$$W = \begin{bmatrix} w_1 & w_2 & \cdots & w_n \end{bmatrix} = \begin{bmatrix} v_1 & \cdots & v_m \end{bmatrix} \begin{bmatrix} a_{11} & & \\ \vdots & & \\ a_{m1} & & \end{bmatrix} = VA.$$

We don't know each a_{ij}, but we know the shape of A (it is m by n). The second vector w_2 is also a combination of the v's. The coefficients in that combination fill the second column of A. The key is that A has a row for every v and a column for every w. A is a short, wide matrix, since $n > m$. ***There is a nonzero solution to*** $Ax = 0$. Then $VAx = 0$ which is $Wx = 0$. *A combination of the* w*'s gives zero!* The w's could not be a basis—so we cannot have $n > m$.

If $m > n$ we exchange the v's and w's and repeat the same steps. The only way to avoid a contradiction is to have $m = n$. This completes the proof that $m = n$. To repeat: The ***dimension of a space*** is the number of vectors in every basis. □

This proof was used earlier to show that every set of $m+1$ vectors in $\mathbf{R}^m$ must be dependent. The v's and w's need not be column vectors—the proof was all about the matrix A of coefficients. In fact we can see this general result: *In a subspace of dimension k, no set of more than k vectors can be independent, and no set of more than k vectors can. span the space*.

There are other "dual" theorems, of which we mention only one. We can start with a set of vectors that is too small or too big, and end up with a basis:

2L Any linearly independent set in $\mathbf{V}$ can be extended to a basis, by adding more vectors if necessary.

Any spanning set in $\mathbf{V}$ can be reduced to a basis, by discarding vectors if necessary.

The point is that a basis is a ***maximal independent set***. It cannot be made larger without losing independence. A basis is also a ***minimal spanning set***. It cannot be made smaller and still span the space.

You must notice that the word "dimensional" is used in two different ways. We speak about a four-dimensional ***vector***, meaning a vector in $\mathbf{R}^4$. Now we have defined a four-dimensional ***subspace***; an example is the set of vectors in $\mathbf{R}^6$ whose first and last components are zero. The members of this four-dimensional subspace are six-dimensional vectors like $(0,5,1,3,4,0)$.

One final note about the language of linear algebra. We never use the terms "basis of a matrix" or "rank of a space" or "dimension of a basis." These phrases have no meaning. It is *the dimension of the column space that equals the rank of the matrix*, as we prove in the coming section.

Problem Set 2.3

Problems 1–10 are about linear independence and linear dependence.

1. Show that v_1, v_2, v_3 are independent but v_1, v_2, v_3, v_4 are dependent:

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \qquad v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \qquad v_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \qquad v_4 = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}.$$

Solve $c_1v_1 + \cdots + c_4v_4 = 0$ or $Ac = 0$. The v's go in the columns of A.

2. Find the largest possible number of independent vectors among

$$v_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix} \quad v_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix} \quad v_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix} \quad v_4 = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix} \quad v_5 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} \quad v_6 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}.$$

This number is the ____ of the space spanned by the v's.

3. Prove that if $a = 0$, $d = 0$, or $f = 0$ (3 cases), the columns of U are dependent:

$$U = \begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix}.$$

4. If a, d, f in Problem 3 are all nonzero, show that the only solution to $Ux = 0$ is $x = 0$. Then U has independent columns.

5. Decide the dependence or independence of

(a) the vectors $(1,3,2)$, $(2,1,3)$, and $(3.2,1)$.

(b) the vectors $(1,-3,2)$, $(2,1,-3)$, and $(-3,2,1)$.

6. Choose three independent columns of U. Then make two other choices. Do the same for A. You have found bases for which spaces?

$$U = \begin{bmatrix} 2 & 3 & 4 & 1 \\ 0 & 6 & 7 & 0 \\ 0 & 0 & 0 & 9 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 2 & 3 & 4 & 1 \\ 0 & 6 & 7 & 0 \\ 0 & 0 & 0 & 9 \\ 4 & 6 & 8 & 2 \end{bmatrix}.$$

7. If w_1, w_2, w_3 are independent vectors, show that the differences $v_1 = w_2 - w_3$, $v_2 = w_1 - w_3$, and $v_3 = w_1 - w_2$ are *dependent*. Find a combination of the v's that gives zero.

8. If w_1, w_2, w_3 are independent vectors, show that the sums $v_1 = w_2 + w_3$, $v_2 = w_1 + w_3$, and $v_3 = w_1 + w_2$ are *independent*. (Write $c_1v_1 + c_2v_2 + c_3v_3 = 0$ in terms of the w's. Find and solve equations for the c's.)

9. Suppose v_1, v_2, v_3, v_4 are vectors in $\mathbf{R}^3$.

(a) These four vectors are dependent because _____.

(b) The two vectors v_1 and v_2 will be dependent if _____.

(c) The vectors v_1 and $(0,0,0)$ are dependent because _____.

10. Find two independent vectors on the plane $x + 2y - 3z - t = 0$ in $\mathbf{R}^4$. Then find three independent vectors. Why not four? This plane is the nullspace of what matrix?

Problems 11–18 are about the space *spanned* by a set of vectors. Take all linear combinations of the vectors

11. Describe the subspace of $\mathbf{R}^3$ (is it a line or a plane or $\mathbf{R}^3$?) spanned by

(a) the two vectors $(1,1,-1)$ and $(-1,-1,1)$.

(b) the three vectors $(0,1,1)$ and $(1,1,0)$ and $(0,0,0)$.

(c) the columns of a 3 by 5 echelon matrix with 2 pivots.

(d) all vectors with positive components.

12. The vector b is in the subspace spanned by the columns of A when there is a solution to _____. The vector c is in the row space of A when there is a solution to _____. *True or false*: If the zero vector is in the row space, the rows are dependent.

13. Find the dimensions of (a) the column space of A, (b) the column space of U, (c) the row space of A, (d) the row space of U. Which two of the spaces are the same?

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 3 & 1 \\ 3 & 1 & -1 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

14. Choose $x = (x_1, x_2, x_3, x_4)$ in $\mathbf{R}^4$. It has 24 rearrangements like (x_2, x_1, x_3, x_4) and (x_4, x_3, x_1, x_2). Those 24 vectors, including x itself, span a subspace $\mathbf{S}$. Find specific vectors x so that the dimension of $\mathbf{S}$ is: (a) 0, (b) 1, (c) 3, (d) 4.

15. $v+w$ and $v-w$ are combinations of v and w. Write v and w as combinations of $v+w$ and $v-w$. The two pairs of vectors ____ the same space. When are they a basis for the same space?

16. Decide whether or not the following vectors are linearly independent, by solving $c_1v_1 + c_2v_2 + c_3v_3 + c_4v_4 = 0$:

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \qquad v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \qquad v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \qquad v_4 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}.$$

Decide also if they span $\mathbf{R}^4$, by trying to solve $c_1v_1 + \cdots + c_4v_4 = (0,0,0,1)$.

17. Suppose the vectors to be tested for independence are placed into the rows instead of the columns of A, How does the elimination process from A to U decide for or against independence?

18. To decide whether b is in the sub space spanned by $w_1, \ldots, w_n$, let the vectors w be the columns of A and try to solve $Ax = b$. What is the result for

(a) $w_1 = (1,1,0)$, $w_2 = (2,2,1)$, $w_3 = (0,0,2)$, $b = (3,4,5)$?

(b) $w_1 = (1,2,0)$, $w_2 = (2,5,0)$, $w_3 = (0,0,2)$, $w_4 = (0,0,0)$, and any b?

Problems 19–37 are about the requirements for a basis.

19. If $v_1, \ldots, v_n$ are linearly independent, the space they span has dimension ____. These vectors are a ____ for that space. If the vectors are the columns of an m by n matrix, then m is ____ than n.

20. Find a basis for each of these subspaces of $\mathbf{R}^4$:

(a) All vectors whose components are equal.

(b) All vectors whose components add to zero.

(c) All vectors that are perpendicular to $(1,1,0,0)$ and $(1,0,1,1)$.

(d) The column space (in $\mathbf{R}^2$) and nullspace (in $\mathbf{R}^5$) of $U = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$.

21. Find three different bases for the column space of U above. Then find two different bases for the row space of U.

22. Suppose $v_1, v_2, \ldots, v_6$ are six vectors in $\mathbf{R}^4$.

(a) Those vectors (do)(do not)(might not) span $\mathbf{R}^4$.

(b) Those vectors (are)(are not)(might be) linearly independent.

(c) Any four of those vectors (are)(are not)(might be) a basis for $\mathbf{R}^4$.

(d) If those vectors are the columns of A, then $Ax = b$ (has) (does not have) (might not have) a solution.

23. The columns of A are n vectors from $\mathbf{R}^m$. If they are linearly independent, what is the rank of A? If they span $\mathbf{R}^m$, what is the rank? If they are a basis for $\mathbf{R}^m$, what then?

24. Find a basis for the plane $x - 2y + 3z = 0$ in $\mathbf{R}^3$. Then find a basis for the intersection of that plane with the xy-plane. Then find a basis for all vectors perpendicular to the plane.

25. Suppose the columns of a 5 by 5 matrix A are a basis for $\mathbf{R}^5$.

(a) The equation $Ax = 0$ has only the solution $x = 0$ because ____.

(b) If b is in $\mathbf{R}^5$ then $Ax = b$ is solvable because ____.

Conclusion: A is invertible. Its rank is 5.

26. Suppose $\mathbf{S}$ is a five-dimensional subspace of $\mathbf{R}^6$. True or false?

(a) Every basis for $\mathbf{S}$ can be extended to a basis for $\mathbf{R}^6$ by adding one more vector.

(b) Every basis for $\mathbf{R}^6$ can be reduced to a basis for $\mathbf{S}$ by removing one vector.

27. U comes from A by subtracting row 1 from row 3:

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Find bases for the two column spaces. Find bases for the two row spaces. Find bases for the two nullspace.

28. True or false (give a good reason)?

(a) If the columns of a matrix are dependent, so are the rows.

(b) The column space of a 2 by 2 matrix is the same as its row space.

(c) The column space of a 2 by 2 matrix has the same dimension as its row space.

(d) The columns of a matrix are a basis for the column space.

29. For which numbers c and d do these matrices have rank 2?

$$A = \begin{bmatrix} 1 & 2 & 5 & 0 & 5 \\ 0 & 0 & c & 2 & 2 \\ 0 & 0 & 0 & d & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} c & d \\ d & c \end{bmatrix}.$$

30. By locating the pivots, find a basis for the column space of

$$U = \begin{bmatrix} 0 & 5 & 4 & 3 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Express each column that is not in the basis as a combination of the basic columns, Find also a matrix A with this echelon form U, but a different column space.

31. Find a counterexample to the following statement: If v_1, v_2, v_3, v_4 is a basis for the vector space $\mathbf{R}^4$, and if $\mathbf{W}$ is a subspace, then some subset of the v's is a basis for $\mathbf{W}$.

32. Find the dimensions of these vector spaces:

(a) The space of all vectors in $\mathbf{R}^4$ whose components add to zero.

(b) The nullspace of the 4 by 4 identity matrix.

(c) The space of all 4 by 4 matrices.

33. Suppose $\mathbf{V}$ is known to have dimension k. Prove that

(a) any k independent vectors in $\mathbf{V}$ form a basis;

(b) any k vectors that span $\mathbf{V}$ form a basis.

In other words, if the number of vectors is known to be correct, either of the two properties of a basis implies the other.

34. Prove that if $\mathbf{V}$ and $\mathbf{W}$ are three-dimensional subspaces of $\mathbf{R}^5$, then $\mathbf{V}$ and $\mathbf{W}$ must have a nonzero vector in common. *Hint*: Start with bases for the two subspaces, making six vectors in all.

35. *True or false?*

(a) If the columns of A are linearly independent, then $Ax = b$ has exactly one solution for every b.

(b) A 5 by 7 matrix never has linearly independent columns,

36. If A is a 64 by 17 matrix of rank 11, how many independent vectors satisfy $Ax = 0$? How many independent vectors satisfy $A^{\mathrm{T}}y = 0$?

37. Find a basis for each of these subspaces of 3 by 3 matrices:

(a) All diagonal matrices.

(b) All symmetric matrices ($A^{\mathrm{T}} = A$).

(c) All skew-symmetric matrices ($A^{\mathrm{T}} = -A$).

Problems 38–42 are about spaces in which the "vectors" are functions.

38. (a) Find all functions that satisfy $\frac{dy}{dx} = 0$.

(b) Choose a particular function that satisfies $\frac{dy}{dx} = 3$.

(c) Find all functions that satisfy $\frac{dy}{dx} = 3$.

39. The cosine space $\mathbf{F}_3$ contains all combinations $y(x) = A\cos x + B\cos 2x + C\cos 3x$. Find a basis for the subspace that has $y(0) = 0$.

40. Find a basis for the space of functions that satisfy

(a) $\dfrac{dy}{dx} - 2y = 0$.

(b) $\dfrac{dy}{dx} - \dfrac{y}{x} = 0$.

41. Suppose $y_1(x)$, $y_2(x)$, $y_3(x)$ are three different functions of x. The vector space they span could have dimension 1, 2, or 3. Give an example of y_1, y_2, y_3 to show each possibility.

42. Find a basis for the space of polynomials $p(x)$ of degree ≤ 3. Find a basis for the subspace with $p(1) = 0$.

43. Write the 3 by 3 identity matrix as a combination of the other five permutation matrices! Then show that those five matrices are linearly independent. (Assume a combi nation gives zero, and check entries to prove each term is zero.) The five permutations are a basis for the subspace of 3 by 3 matrices with row and column sums all equal.

44. *Review*: Which of the following are bases for $\mathbf{R}^3$?

(a) $(1,2,0)$ and $(0,1,-1)$.

(b) $(1,1,-1)$, $(2,3,4)$, $(4,1,-1)$, $(0,1,-1)$.

(c) $(1,2,2)$, $(-1,2,1)$, $(0,8,0)$.

(d) $(1,2,2)$, $(-1,2,1)$, $(0,8,6)$.

45. *Review*: Suppose A is 5 by 4 with rank 4. Show that $Ax = b$ has no solution when the 5 by 5 matrix $[A\ \ b]$ is invertible. Show that $Ax = b$ is solvable when $[A\ \ b]$ is singular.

2.4 The Four Fundamental Subspaces

The previous section dealt with definitions rather than constructions. We know what a basis is, but not how to find one. Now, starting from an explicit description of a subspace, we would like to compute an explicit basis.

Subspaces can be described in two ways. First, we may be given a set of vectors that span the space. (*Example*: The columns span the column space.) Second, we may be

told which conditions the vectors in the space must satisfy. (*Example*: The nullspace consists of all vectors that satisfy $Ax = 0$.)

The first description may include useless vectors (dependent columns). The second description may include repeated conditions (dependent rows). We can't write a basis by inspection, and a systematic procedure is necessary.

The reader can guess what that procedure will be. When elimination on A produces an echelon matrix U or a reduced R, we will find a basis for each of the subspaces associated with A. Then we have to look at the extreme case of **full rank**:

When the rank is as large as possible, $r = n$ or $r = m$ or $r = m = n$, the matrix has a **left-inverse** B *or a* **right-inverse** C *or a* **two-sided** A^{-1}.

To organize the whole discussion, we take each of the four subspaces in turn. Two of them are familiar and two are new.

1. The ***column space*** of A is denoted by $\boldsymbol{C}(A)$. Its dimension is the rank r.
2. The ***nullspace*** of A is denoted by $\boldsymbol{N}(A)$. Its dimension is $n - r$.
3. The ***row space*** of A is the *column space* of A^{T}. It is $\boldsymbol{C}(A^{\mathrm{T}})$, and it is spanned by the rows of A. Its dimension is also r.
4. The ***left nullspace*** of A is the *nullspace* of A^{T}. It contains all vectors y such that $A^{\mathrm{T}}y = 0$, and it is written $\boldsymbol{N}(A^{\mathrm{T}})$. Its dimension is ____.

The point about the last two subspaces is that *they come from* A^{T}. If A is an m by n matrix, you can see which "host" spaces contain the four subspaces by looking at the number of components:

The nullspace $\boldsymbol{N}(A)$ and row space $\boldsymbol{C}(A^{\mathrm{T}})$ are subspaces of $\mathbf{R}^n$.
The left nullspace $\boldsymbol{N}(A^{\mathrm{T}})$ and column space $\boldsymbol{C}(A)$ are subspaces of $\mathbf{R}^m$.

The rows have n components and the columns have m. For a simple matrix like

$$A = U = R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

the column space is the line through $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$. The row space is the line through $[1 \;\; 0 \;\; 0]^{\mathrm{T}}$. It is in $\mathbf{R}^3$. The nullspace is a plane in $\mathbf{R}^3$ and the left nullspace is a line in $\mathbf{R}^2$:

$$\boldsymbol{N}(A) \quad \text{contains} \quad \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \qquad \boldsymbol{N}(A^{\mathrm{T}}) \quad \text{contains} \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Note that all vectors are column vectors. Even the rows are transposed, and the row space of A is the *column* space of A^{T}, Our problem will be to connect the four spaces for U (after elimination) to the four spaces for A:

Basic example

$$U = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{came from} \quad A = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \\ -1 & -3 & 3 & 4 \end{bmatrix}.$$

For novelty, we take the four subspaces in a more interesting order.

3. The row space of A For an echelon matrix like U, the row space is clear. It contains all combinations of the rows, as every row space does—but here the third row contributes nothing. The first two rows are a basis for the row space. A similar rule applies to every echelon matrix U or R, with r pivots and r nonzero rows: ***The nonzero rows are a basis, and the row space has dimension*** r. That makes it easy to deal with the original matrix A.

> **2M** The row space of A has the same dimension r as the row space of U, and it has the same bases, because ***the row spaces of*** A ***and*** U ***(and*** R***) are the same***.

The reason is that each elementary operation leaves the row space unchanged. The rows in U are combinations of the original rows in A. Therefore the row space of U contains nothing new. At the same time, because every step can be reversed, nothing is lost; the rows of A can be recovered from U. It is true that A and U have different rows, but the *combinations* of the rows are identical: *same space*!

Note that we did not start with the m rows of A, which span the row space, and discard $m-r$ of them to end up with a basis. According to 2L, we could have done so. But it might be hard to decide which rows to keep and which to discard, so it was easier just to take the nonzero rows of U.

2. The nullspace of A Elimination simplifies a system of linear equations without changing the solutions. The system $Ax = 0$ is reduced to $Ux = 0$, and this process is reversible. ***The nullspace of*** A ***is the same as the nullspace of*** U ***and*** R. Only r of the equations $Ax = 0$ are independent. Choosing the $n-r$ "special solutions" to $Ax = 0$ provides a definite basis for the nullspace:

> **2N** The nullspace $\boldsymbol{N}(A)$ has dimension $n-r$. The "special solutions" are a basis—each free variable is given the value 1, while the other free variables are 0. Then $Ax = 0$ or $Ux = 0$ or $Rx = 0$ gives the pivot variables by back-substitution.

This is exactly the way we have been solving $Ux = 0$. The basic example above has pivots in columns 1 and 3. Therefore its free variables are the second and fourth v and y.

The basis for the nullspace is

$$\textbf{Special solutions} \quad \begin{array}{rcl} v &=& 1 \\ y &=& 0 \end{array} \quad x_1 = \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix}; \qquad \begin{array}{rcl} v &=& 0 \\ y &=& 1 \end{array} \quad x_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}.$$

Any combination $c_1x_1 + c_2x_2$ has c_1 as its v component, and c_2 as its y component. The only way to have $c_1x_1 + c_2x_2 = 0$ is to have $c_1 = c_2 = 0$, so these vectors are independent. They also span the nullspace; the complete solution is $vx_1 + yx_2$. Thus the $n - r = 4 - 2$ vectors are a basis.

The nullspace is also called the *kernel* of A, and its dimension $n - r$ is the *nullity*.

1. The column space of A The column space is sometimes called the **range**. This is consistent with the usual idea of the range, as the set of all possible values $f(x)$; x is in the domain and $f(x)$ is in the range. In our case the function is $f(x) = Ax$. Its domain consists of all x in $\mathbf{R}^n$; its range is all possible vectors Ax, which is the column space. (In an earlier edition of this book we called it $R(A)$.)

Our problem is to find bases for the column spaces of U and A. ***Those spaces are different*** (just look at the matrices!) but their dimensions are the same.

The first and third columns of U are a basis for its column space. They are the ***columns with pivots***. Every other column is a combination of those two. Furthermore, the same is true of the original A—even though its columns are different. ***The pivot columns of A are a basis for its column space***. The second column is three times the first, just as in U. The fourth column equals (column 3) − (column 1). The same nullspace is telling us those dependencies.

The reason is this: $Ax = 0$ *exactly when* $Ux = 0$. The two systems are equivalent and have the same solutions. The fourth column of U was also (column 3) − (column 1). Every linear dependence $Ax = 0$ among the columns of A is matched by a dependence $Ux = 0$ among the columns of U, with exactly the same coefficients. *If a set of columns of A is independent, then so are the corresponding columns of U, and vice versa.*

To find a basis for the column space $C(A)$, we use what is already done for U. The r columns containing pivots are a basis for the column space of U. We will pick those same r columns in A:

> **2O** The dimension of the column space $C(A)$ equals the rank r, which also equals the dimension of the row space: ***The number of independent columns equals the number of independent rows***. A basis for $C(A)$ is formed by the r columns of A that correspond, in U, to the columns containing pivots.

The row space and the column space have the same dimension r! This is one of the most important theorems in linear algebra. It is often abbreviated as "***row rank*** = ***column rank***." It expresses a result that, for a random 10 by 12 matrix, is not at all

obvious. It also says something about square matrices: *If the rows of a square matrix are linearly independent, then so are the columns* (and vice versa). Again, that does not seem self-evident (at least, not to the author).

To see once more that both the row and column spaces of U have dimension r, consider a typical situation with rank $r = 3$. The echelon matrix U certainly has three independent rows:

$$U = \begin{bmatrix} d_1 & * & * & * & * & * \\ 0 & 0 & 0 & d_2 & * & * \\ 0 & 0 & 0 & 0 & 0 & d_3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

We claim that U also has three independent columns, and no more, The columns have only three nonzero components. If we can show that the pivot columns—the first, fourth, and sixth—are linearly independent, they must be a basis (for the column space of U, not A!). Suppose a combination of these pivot columns produced zero:

$$c_1 \begin{bmatrix} d_1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} * \\ d_2 \\ 0 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} * \\ * \\ d_3 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Working upward in the usual way, c_3 must be zero because the pivot $d_3 \neq 0$, then c_2 must be zero because $d_2 \neq 0$, and finally $c_1 = 0$. This establishes independence and completes the proof. Since $Ax = 0$ if and only if $Ux = 0$, the first, fourth, and sixth columns of A—whatever the original matrix A was, which we do not even know in this example—are a basis for $\boldsymbol{C}(A)$.

The row space and column space both became clear after elimination on A. Now comes the fourth fundamental subspace, which has been keeping quietly out of sight. Since the first three spaces were $\boldsymbol{C}(A)$, $\boldsymbol{N}(A)$, and $\boldsymbol{C}(A^{\mathrm{T}})$, the fourth space must be $\boldsymbol{N}(A^{\mathrm{T}})$, It is the nullspace of the transpose, or the ***left nullspace*** of A. $A^{\mathrm{T}}y = 0$ means $y^{\mathrm{T}}A = 0$, and the vector appears on the left-hand side of A.

4. The left nullspace of A (= the nullspace of A^{T}) If A is an m by n matrix, then A^{T} is n by m. Its nullspace is a subspace of $\mathbf{R}^m$; the vector y has m components. Written as $y^{\mathrm{T}}A = 0$, those components multiply the *rows* of A to produce the zero row:

$$y^{\mathrm{T}}A = \begin{bmatrix} y_1 & \cdots & y_m \end{bmatrix} \begin{bmatrix} & A & \end{bmatrix} = \begin{bmatrix} 0 & \cdots & 0 \end{bmatrix}.$$

The dimension of this nullspace $\boldsymbol{N}(A^{\mathrm{T}})$ is easy to find, For *any* matrix, ***the number of pivot variables plus the number of free variables must match the total number of columns***. For A, that was $r + (n - r) = n$. In other words, rank plus nullity equals n:

$$\text{dimension of } \boldsymbol{C}(A) + \text{dimension of } \boldsymbol{N}(A) = \text{number of columns}.$$

This law applies equally to A^{T}, which has m columns. A^{T} is just as good a matrix as A. But the dimension of its column space is also r, so

$$r + \text{dimension } \left(\boldsymbol{N}(A^{\mathrm{T}})\right) = m. \tag{1}$$

2P The left nullspace $\boldsymbol{N}(A^{\mathrm{T}})$ has dimension $m-r$.

The $m-r$ solutions to $y^{\mathrm{T}}A = 0$ are hiding somewhere in elimination. The rows of A combine to produce the $m-r$ *zero rows* of U. Start from $PA = LU$, or $L^{-1}PA = U$. The last $m-r$ rows of the invertible matrix $L^{-1}P$ must be a basis of y's in the left nullspace—because they multiply A to give the zero rows in U.

In our 3 by 4 example, the zero row was row 3 − 2(row 2) + 5(row 1). Therefore the components of y are 5, −2, 1. This is the same combination as in $b_3 - 2b_2 + 5b_1$ on the right-hand side, leading to $0 = 0$ as the final equation. That vector y is a basis for the left nullspace, which has dimension $m - r = 3 - 2 = 1$. It is the last row of $L^{-1}P$, and produces the zero row in U—and we can often see it without computing L^{-1}. When desperate, it is always possible just to solve $A^{\mathrm{T}}y = 0$.

I realize that so far in this book we have given no reason to care about $\boldsymbol{N}(A^{\mathrm{T}})$. It is correct but not convincing if I write in italics that *the left nullspace is also important*. The next section does better by finding a physical meaning for y from Kirchhoff's Current Law.

Now we know the dimensions of the four spaces. We can summarize them in a table, and it even seems fair to advertise them as the

Fundamental Theorem of Linear Algebra, Part I

1. $\boldsymbol{C}(A) =$ column space of A; dimension r.
2. $\boldsymbol{N}(A) =$ nullspace of A; dimension $n-r$.
3. $\boldsymbol{C}(A^{\mathrm{T}}) =$ row space of A; dimension r.
4. $\boldsymbol{N}(A^{\mathrm{T}}) =$ left nullspace of A; dimension $m-r$.

Example 1. $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$ has $m = n = 2$, and rank $r = 1$.

1. The ***column space*** contains all multiples of $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$. The second column is in the same direction and contributes nothing new.
2. The ***nullspace*** contains all multiples of $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$. This vector satisfies $Ax = 0$.
3. The ***row space*** contains all multiples of $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$. I write it as a column vector, since strictly speaking it is in the column space of A^{T}.
4. The ***left nullspace*** contains all multiples of $y = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$. The rows of A with coefficients −3 and 1 add to zero, so $A^{\mathrm{T}}y = 0$.

In this example *all four subspaces are lines*. That is an accident, coming from $r = 1$ and $n - r = 1$ and $m - r = 1$. Figure 2.5 shows that two pairs of lines are perpendicular. That is no accident!

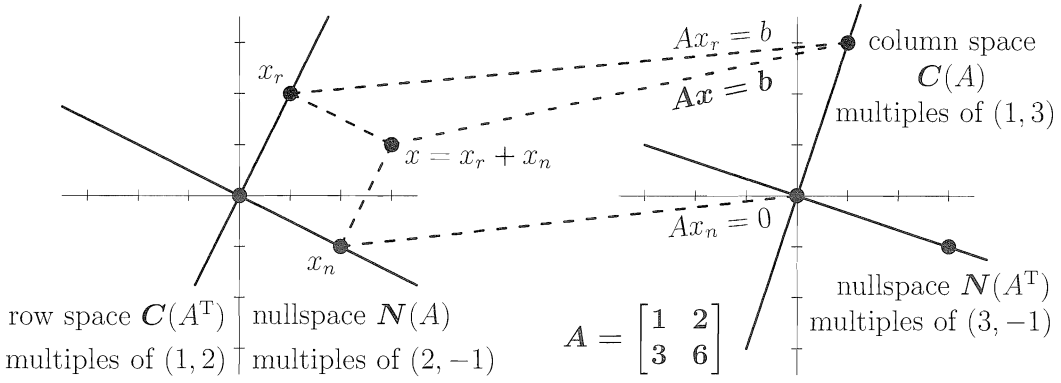


Figure 2.5: The four fundamental subspaces (lines) for the singular matrix A.

If you change the last entry of A from 6 to 7, all the dimensions are different. The column space and row space have dimension $r = 2$. The nullspace and left nullspace contain only the vectors $x = 0$ and $y = 0$. *The matrix is invertible*.

Existence of Inverses

We know that if A has a left-inverse ($BA = I$) and a right-inverse ($AC = I$), then the two inverses are equal: $B = B(AC)(BA)C = C$. Now, from the rank of a matrix, it is easy to decide which matrices actually have these inverses. Roughly speaking, ***an inverse exists only when the rank is as large as possible***.

The rank always satisfies $r \leq m$ and also $r \leq n$. An m by n matrix cannot have more than m independent rows or n independent columns. There is not space for more than m pivots, or more than n. We want to prove that when $r = m$ there is a right-inverse, and $Ax = b$ always has a solution. When $r = n$ there is a left-inverse, and the solution (*if it exists*) is unique.

Only a square matrix can have both $r = m$ and $r = n$, and therefore only a square matrix can achieve both existence and uniqueness. Only a square matrix has a two-sided inverse.

2Q EXISTENCE: Full row rank $r = m$. $Ax = b$ has ***at least*** one solution x for every b if and only if the columns span $\mathbf{R}^m$. Then A has a ***right-inverse*** C such that $AC = I_m$ (m by m). This is possible only if $m \leq n$.

UNIQUENESS: Full column rank $r = n$. $Ax = b$ has ***at most*** one solution x for every b if and only if the columns are linearly independent. Then A has an n by m **left-inverse** B such that $BA = I_n$. This is possible only if $m \geq n$.

In the existence case, one possible solution is $x = Cb$, since then $Ax = ACb = b$. But there will be other solutions if there are other right-inverses. The number of solutions when the columns span $\mathbf{R}^m$ is 1 or ∞.

In the uniqueness case, if there is a solution to $Ax = b$, it has to be $x = BAx = Bb$. But there may be no solution. The number of solutions is 0 or 1.

There are simple formulas for the best left and right inverses, if they exist:

$$\textbf{One-sided inverses} \qquad B = (A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}} \quad \text{and} \quad C = A^{\mathrm{T}}(AA^{\mathrm{T}})^{-1}.$$

Certainly $BA = I$ and $AC = I$. What is not so certain is that $A^{\mathrm{T}}A$ and AA^{T} are actually invertible. We show in Chapter 3 that $A^{\mathrm{T}}A$ does have an inverse if the rank is n, and AA^{T} has an inverse when the rank is m. Thus the formulas make sense exactly when the rank is as large as possible, and the one-sided inverses are found.

Example 2. Consider a simple 2 by 3 matrix of rank 2:

$$A = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 5 & 0 \end{bmatrix}.$$

Since $r = m = 2$, the theorem guarantees a right-inverse C:

$$AC = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 5 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{4} & 0 \\ 0 & \frac{1}{5} \\ c_{31} & c_{32} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

There are many right-inverses because the last row of C is completely arbitrary. This is a case of existence but not uniqueness. The matrix A has no left-inverse because the last column of BA is certain to be zero. The specific right-inverse $C = A^{\mathrm{T}}(AA^{\mathrm{T}})^{-1}$ chooses c_{31} and c_{32} to be zero:

$$\textbf{Best right-inverse} \qquad A^{\mathrm{T}}(AA^{\mathrm{T}})^{-1} = \begin{bmatrix} 4 & 0 \\ 0 & 5 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{16} & 0 \\ 0 & \frac{1}{25} \end{bmatrix} = \begin{bmatrix} \frac{1}{4} & 0 \\ 0 & \frac{1}{5} \\ 0 & 0 \end{bmatrix} = C.$$

This is the *pseudoinverse*—a way of choosing the best C in Section 6.3. The transpose of A yields an example with infinitely many *left*-inverses:

$$BA^{\mathrm{T}} = \begin{bmatrix} \frac{1}{4} & 0 & b_{13} \\ 0 & \frac{1}{5} & b_{23} \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 5 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Now it is the last column of B that is completely arbitrary. The best left-inverse (also the pseudoinverse) has $b_{13} = b_{23} = 0$. This is a "uniqueness case," when the rank is $r = n$. There are no free variables, since $n - r = 0$. If there is a solution it will be the only one. You can see when this example has one solution or no solution:

$$\begin{bmatrix} 4 & 0 \\ 0 & 5 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \qquad \text{is solvable exactly when} \quad b_3 = 0.$$

A rectangular matrix cannot have both existence and uniqueness. If m is different from n, we cannot have $r = m$ and $r = n$.

A square matrix is the opposite. If $m = n$, we cannot have one property *without* the other. A square matrix has a left-inverse if and only if it has a right-inverse. There is only one inverse, namely $B = C = A^{-1}$. *Existence implies uniqueness and uniqueness implies existence, when the matrix is square*. The condition for invertibility is **full rank**: $r = m = n$. Each of these conditions is a necessary and sufficient test:

1. The columns span $\mathbf{R}^n$, so $Ax = b$ has at least one solution for every b.
2. The columns are independent, so $Ax = 0$ has only the solution $x = 0$.

This list can be made much longer, especially if we look ahead to later chapters. Every condition is equivalent to every other, and ensures that A is invertible.

3. The rows of A span $\mathbf{R}^n$.
4. The rows are linearly independent.
5. Elimination can be completed: $PA = LDU$, with all n pivots.
6. The determinant of A is not zero.
7. Zero is not an eigenvalue of A.
8. $A^{\mathrm{T}}A$ is positive definite.

Here is a typical application to polynomials $P(t)$ of degree $n-1$. The only such polynomial that vanishes at $t_1, \ldots, t_n$ is $P(t) \equiv 0$. No other polynomial of degree $n-1$ can have n roots. This is uniqueness, and it implies existence: Given any values $b_1, \ldots, b_n$, there *exists* a polynomial of degree $n-1$ interpolating these values: $P(t_i) = b_i$. The point is that we are dealing with a square matrix; the number n of coefficients in $P(t) = x_1 + x_2 t + \cdots + x_n t^{n-1}$ matches the number of equations:

Interpolation
$P(t_i) = b_i$

$$\begin{bmatrix} 1 & t_1 & t_1^2 & \cdots & t_1^{n-1} \\ 1 & t_2 & t_2^2 & \cdots & t_2^{n-1} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & t_n & t_n^2 & \cdots & t_n^{n-1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}.$$

That *Vandermonde matrix* is n by n and full rank. $Ax = b$ always has a solution—a polynomial can be passed through any b_i at distinct points t_i. Later we shall actually find the determinant of A; it is not zero.

Matrices of Rank 1

Finally comes the easiest case, when the rank is as *small* as possible (except for the zero matrix with rank 0), One basic theme of mathematics is, given something complicated,

to show how it can be broken into simple pieces. For linear algebra, the simple pieces are matrices of ***rank* 1**:

$$\textbf{Rank 1} \qquad A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 2 & 2 \\ 8 & 4 & 4 \\ -2 & -1 & -1 \end{bmatrix} \quad \text{has} \quad r = 1.$$

Every row is a multiple of the first row, so the row space is one-dimensional. In fact, we can write the whole matrix *as the product of a column vector and a row vector*:

$$A = (\textbf{column})(\textbf{row}) \qquad \begin{bmatrix} 2 & 1 & 1 \\ 4 & 2 & 2 \\ 8 & 4 & 4 \\ -2 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 4 \\ -1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \end{bmatrix}.$$

The product of a 4 by 1 matrix and a 1 by 3 matrix is a 4 by 3 matrix. *This product has rank* 1. At the same time, the columns are all multiples of the same column vector; the column space shares the dimension $r = 1$ and reduces to a line.

> ***Every matrix of rank* 1 *has the simple form* $A = uv^{\mathrm{T}} =$ *column times row.***

The rows are all multiples of the same vector v^{T}, and the columns are all multiples of u. The row space and column space are lines—the easiest case.

Problem Set 2.4

1. True or false: If $m = n$, then the row space of A equals the column space. If $m < n$, then the nullspace has a larger dimension than ____.

2. Find the dimension and construct a basis for the four subspaces associated with each of the matrices

$$A = \begin{bmatrix} 0 & 1 & 4 & 0 \\ 0 & 2 & 8 & 0 \end{bmatrix} \qquad \text{and} \qquad U = \begin{bmatrix} 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

3. Find the dimension and a basis for the four fundamental subspaces for

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix} \qquad \text{and} \qquad U = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

4. Describe the four subspaces in three-dimensional space associated with

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

5. If the product AB is the zero matrix, $AB = 0$, show that the column space of B is contained in the nullspace of A. (Also the row space of A is in the left nullspace of B, since each row of A multiplies B to give a zero row.)

6. Suppose A is an m by n matrix of rank r. Under what conditions on those numbers does

(a) A have a two-sided inverse: $AA^{-1} = A^{-1}A = I$?

(b) $Ax = b$ have *infinitely many solutions* for *every* b?

7. Why is there no matrix whose row space and nullspace both contain $(1,1,1)$?

8. Suppose the only solution to $Ax = 0$ (m equations in n unknowns) is $x = 0$. What is the rank and why? The columns of A are linearly ____.

9. Find a 1 by 3 matrix whose nullspace consists of all vectors in $\mathbf{R}^3$ such that $x_1 + 2x_2 + 4x_3 = 0$. Find a 3 by 3 matrix with that same nullspace.

10. If $Ax = b$ always has at least one solution, show that the only solution to $A^{\mathrm{T}}y = 0$ is $y = 0$. *Hint*: What is the rank?

11. If $Ax = 0$ has a nonzero solution, show that $A^{\mathrm{T}}y = f$ fails to be solvable for some right-hand sides f. Construct an example of A and f.

12. Find the rank of A and write the matrix as $A = uv^{\mathrm{T}}$:

$$A = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 6 \end{bmatrix} \qquad \text{and} \qquad A = \begin{bmatrix} 2 & -2 \\ 6 & -6 \end{bmatrix}.$$

13. If a, b, c are given with $a \neq 0$, choose d so that

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = uv^{\mathrm{T}}$$

has rank 1. What are the pivots?

14. Find a left-inverse and/or a right-inverse (when they exist) for

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \quad \text{and} \quad M = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad T = \begin{bmatrix} a & b \\ 0 & a \end{bmatrix}.$$

15. If the columns of A are linearly independent (A is m by n), then the rank is ____, the nullspace is ____, the row space is ____, and there exists a ____-inverse.

16. (*A paradox*) Suppose A has a right-inverse B. Then $AB = I$ leads to $A^{\mathrm{T}}AB = A^{\mathrm{T}}$ or $B(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$. But that satisfies $BA = I$; it is a *left*-inverse. Which step is not justified?

17. Find a matrix A that has $\mathbf{V}$ as its row space, and a matrix B that has $\mathbf{V}$ as its nullspace, if $\mathbf{V}$ is the subspace spanned by

$$\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \qquad \begin{bmatrix} 1 \\ 5 \\ 0 \end{bmatrix}.$$

18. Find a basis for each of the four subspaces of

$$A = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 4 & 6 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

19. If A has the same four fundamental subspaces as B, does $A = cB$?

20. (a) If a 7 by 9 matrix has rank 5, what are the dimensions of the four subspaces? What is the sum of all four dimensions?

(b) If a 3 by 4 matrix has rank 3, what are its column space and left nullspace?

21. Construct a matrix with the required property, or explain why you can't.

(a) Column space contains $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, row space contains $\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \end{bmatrix}$.

(b) Column space has basis $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, nullspace has basis $\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$.

(c) Dimension of nullspace $= 1+$ dimension of left nullspace.

(d) Left nullspace contains $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$, row space contains $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$.

(e) Row space $=$ column space, nullspace $\neq$ left nullspace.

22. Without elimination, find dimensions and bases for the four subspaces for

$$A = \begin{bmatrix} 0 & 3 & 3 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 1 \\ 4 & 4 \\ 5 & 5 \end{bmatrix}.$$

23. Suppose the 3 by 3 matrix A is invertible. Write bases for the four subspaces for A, and also for the 3 by 6 matrix $B = [A \;\; A]$.

24. What are the dimensions of the four subspaces for A, B, and C, if I is the 3 by 3 identity matrix and 0 is the 3 by 2 zero matrix?

$$A = \begin{bmatrix} I & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} I & I \\ 0^{\mathrm{T}} & 0^{\mathrm{T}} \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 0 \end{bmatrix}.$$

25. Which subspaces are the same for these matrices of different sizes?

$$\text{(a)} \quad \begin{bmatrix} A \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} A \\ A \end{bmatrix}. \qquad \text{(b)} \quad \begin{bmatrix} A \\ A \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} A & A \\ A & A \end{bmatrix}.$$

Prove that all three matrices have the same rank r.

26. If the entries of a 3 by 3 matrix are chosen randomly between 0 and 1, what are the most likely dimensions of the four subspaces? What if the matrix is 3 by 5?

27. (Important) A is an m by n matrix of rank r. Suppose there are right-hand sides b for which $Ax = b$ has *no solution*.

(a) What inequalities ($<$ or $\leq$) must be true between m, n, and r?

(b) How do you know that $A^{\mathrm{T}}y = 0$ has a nonzero solution?

28. Construct a matrix with $(1,0,1)$ and $(1,2,0)$ as a basis for its row space and its column space. Why can't this be a basis for the row space and nullspace?

29. Without computing A, find bases for the four fundamental subspaces:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 6 & 1 & 0 \\ 9 & 8 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

30. If you exchange the first two rows of a matrix A, which of the four subspaces stay the same? If $y = (1,2,3,4)$ is in the left nullspace of A, write down a vector in the left nullspace of the new matrix.

31. Explain why $v = (1,0,-1)$ cannot be a row of A and also be in the nullspace.

32. Describe the four subspaces of $\mathbf{R}^3$ associated with

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad I + A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

33. (Left nullspace) Add the extra column b and reduce A to echelon form:

$$\begin{bmatrix} A & b \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 & b_1 \\ 4 & 5 & 6 & b_2 \\ 7 & 8 & 9 & b_3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 & b_1 \\ 0 & -3 & -6 & b_2 - 4b_1 \\ 0 & 0 & 0 & b_3 - 2b_2 + b_1 \end{bmatrix}.$$

A combination of the rows of A has produced the zero row. What combination is it? (Look at $b_3 - 2b_2 + b_1$ on the right-hand side.) Which vectors are in the nullspace of A^{T} and which are in the nullspace of A?

34. Following the method of Problem 33, reduce A to echelon form and look at zero rows. The b column tells which combinations you have taken of the rows:

$$\text{(a)} \quad \begin{bmatrix} 1 & 2 & b_1 \\ 3 & 4 & b_2 \\ 4 & 6 & b_3 \end{bmatrix}. \qquad \text{(b)} \quad \begin{bmatrix} 1 & 2 & b_1 \\ 2 & 3 & b_2 \\ 2 & 4 & b_3 \\ 2 & 5 & b_4 \end{bmatrix}.$$

From the b column after elimination, read off $m-r$ basis vectors in the left nullspace of A (combinations of rows that give zero).

35. Suppose A is the sum of two matrices of rank one: $A = uv^{\mathrm{T}} + wz^{\mathrm{T}}$.

(a) Which vectors span the column space of A?

(b) Which vectors span the row space of A?

(c) The rank is less than 2 if ____ or if ____.

(d) Compute A and its rank if $u = z = (1,0,0)$ and $v = w = (0,0,1)$.

36. Without multiplying matrices, find bases for the row and column spaces of A:

$$A = \begin{bmatrix} 1 & 2 \\ 4 & 5 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} 3 & 0 & 3 \\ 1 & 1 & 2 \end{bmatrix}.$$

How do you know from these shapes that A is not invertible?

37. True or false (with a reason or a counterexample)?

(a) A and A^{T} have the same number of pivots.

(b) A and A^{T} have the same left nullspace.

(c) If the row space equals the column space then $A^{\mathrm{T}} = A$.

(d) If $A^{\mathrm{T}} = -A$ then the row space of A equals the column space.

38. If $AB = 0$, the columns of B are in the nullspace of A. If those vectors are in $\mathbf{R}^n$, prove that $\operatorname{rank}(A) + \operatorname{rank}(B) \le n$.

39. Can tic-tac-toe be completed (5 ones and 4 zeros in A) so that $\operatorname{rank}(A) = 2$ but neither side passed up a winning move?

40. Construct any 2 by 3 matrix of rank 1. Copy Figure 2.5 and put one vector in each subspace (two in the nullspace). Which vectors are orthogonal?

41. Redraw Figure 2.5 for a 3 by 2 matrix of rank $r = 2$. Which subspace is $\boldsymbol{Z}$ (zero vector only)? The nullspace part of any vector x in $\mathbf{R}^2$ is $x_n =$ ____.

SYSTEMS OF LINEAR EQUATIONS - LINEAR TRANSFORMATIONS

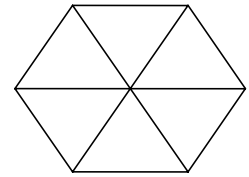
Strang Chapter 2, Section 2.6

16. If there is an edge between every pair of nodes (a complete graph), how many edges are there? The graph has n nodes, and edges from a node to itself are not allowed.

17. For both graphs drawn below, verify *Euler's formula*:

$$\textbf{(\# of nodes)} - \textbf{(\# of edges)} + \textbf{(\# of loops)} = \textbf{1}.$$





18. Multiply matrices to find $A^{\mathrm{T}}A$, and guess how its entries come from the graph:

(a) The diagonal of $A^{\mathrm{T}}A$ tells how many ____ into each node.

(b) The off-diagonals -1 or 0 tell which pairs of nodes are ____.

19. Why does the nullspace of $A^{\mathrm{T}}A$ contain $(1,1,1,1)$? What is its rank?

20. Why does a complete graph with $n = 6$ nodes have $m = 15$ edges? A spanning tree connecting all six nodes has ____ edges. There are $n^{n-2} = 6^4$ spanning trees!

21. The *adjacency matrix* of a graph has $M_{ij} = 1$ if nodes i and j are connected by an edge (otherwise $M_{ij} = 0$). For the graph in Problem 6 with 6 nodes and 4 edges, write down M and also M^2. Why does $(M^2)_{ij}$ count the number of 2-*step paths* from node i to node j?

2.6 Linear Transformations

We know how a matrix moves subspaces around when we multiply by A. The nullspace goes into the zero vector. All vectors go into the column space, since Ax is always a combination of the columns. You will soon see something beautiful—that A takes its row space into its column space, and on those spaces of dimension r it is 100 percent invertible. That is the real action of A. It is partly hidden by nullspaces and left nullspaces, which lie at right angles and go their own way (toward zero).

What matters now is what happens *inside* the space—which means inside n-dimensional space, if A is n by n. That demands a closer look.

Suppose x is an n-dimensional vector. When A multiplies x, it ***transforms*** that vector into a new vector Ax. This happens at every point x of the n-dimensional space $\mathbf{R}^n$. The whole space is transformed, or "mapped into itself," by the matrix A. Figure 2.9 illustrates four transformations that come from matrices:

$A = \begin{bmatrix} c & 0 \\ 0 & c \end{bmatrix}$

1. A multiple of the identity matrix, $A = cI$, ***stretches*** every vector by the same factor c. The whole space expands or contracts (or somehow goes through the origin and out the opposite side, when c is negative).

$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

2. A ***rotation*** matrix turns the whole space around the origin. This example turns all vectors through 90°, transforming every point (x,y) to $(-y,x)$.

$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

3. A ***reflection*** matrix transforms every vector into its image on the opposite side of a mirror. In this example the mirror is the 45° line $y = x$, and a point like $(2,2)$ is unchanged. A point like $(2,-2)$ is reversed to $(-2,2)$. On a combination like $v = (2,2) + (2,-2) = (4,0)$, the matrix leaves one part and reverses the other part. The output is $Av = (2,2) + (-2,2) = (0,4)$
That reflection matrix is also a permutation matrix! It is algebraically so simple, sending (x,y) to (y,x), that the geometric picture was concealed.

$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

4. A ***projection*** matrix takes the whole space onto a lower-dimensional subspace (not invertible). The example transforms each vector (x,y) in the plane to the nearest point $(x,0)$ on the horizontal axis. That axis is the column space of A. The y-axis that projects to $(0,0)$ is the nullspace.

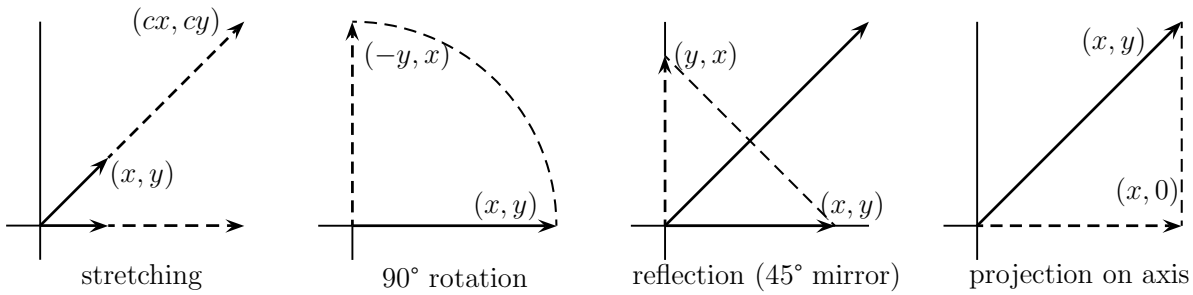


Figure 2.9: Transformations of the plane by four matrices.

Those examples could be lifted into three dimensions. There are matrices to stretch the earth or spin it or reflect it across the plane of the equator (forth pole transforming to south pole). There is a matrix that projects everything onto that plane (both poles to the center). It is also important to recognize that matrices cannot do everything, and some transformations $T(x)$ are *not possible* with Ax:

(i) It is impossible to move the origin, since $A0 = 0$ for every matrix.

(ii) If the vector x goes to x', then $2x$ must go to $2x'$. in general cx must go to cx', since $A(cx) = c(Ax)$.

(iii) If the vectors x and y go to x' and y', then their sum $x+y$ must go to $x'+y'$—since $A(x+y) = Ax+Ay$.

Matrix multiplication imposes those rules on the transformation. The second rule contains the first (take $c = 0$ to get $A0 = 0$). We saw rule (iii) in action when $(4,0)$ was

reflected across the 45° line. It was split into $(2,2)+(2,-2)$ and the two parts were reflected separately. The same could be done for projections: split, project separately, and add the projections. These rules apply to ***any transformation that comes from a matrix***.

Their importance has earned them a name: Transformations that obey rules (i)–(iii) are called ***linear transformations***. The rules can be combined into one requirement:

2T For all numbers c and d and all vectors x and y, matrix multiplication satisfies the rule of linearity:

$$A(cx+dy) = c(Ax)+d(Ay). \tag{1}$$

Every transformation $T(x)$ that meets this requirement is a ***linear transformation***.

Any matrix leads immediately to a linear transformation. The more interesting question is in the opposite direction: *Does every linear transformation lead to a matrix*? The object of this section is to find the answer, yes. This is the foundation of an approach to linear algebra—starting with property (1) and developing its consequences—that is much more abstract than the main approach in this book. We preferred to begin directly with matrices, and now we see how they represent linear transformations.

A transformation need not go from $\mathbf{R}^n$ to the same space $\mathbf{R}^n$. It is absolutely permitted to transform vectors in $\mathbf{R}^n$ to vectors in a different space $\mathbf{R}^m$. That is exactly what is done by an m by n matrix! The original vector x has n components, and the transformed vector Ax has m components. The rule of linearity is equally satisfied by rectangular matrices, so they also produce linear transformations.

Having gone that far, there is no reason to stop. The operations in the linearity condition (1) are addition and scalar multiplication, but x and y need not be column vectors in $\mathbf{R}^n$. Those are not the only spaces. By definition, *any vector space allows the combinations* $cx+dy$—the "vectors" are x and y, but they may actually be polynomials or matrices or functions $x(t)$ and $y(t)$. As long as the transformation satisfies equation (1), it is linear.

We take as examples the spaces $\mathbf{P}_n$, in which the vectors are polynomials $p(t)$ of degree n. They look like $p = a_0 + a_1 t + \cdots + a_n t^n$, and the dimension of the vector space is $n+1$ (because with the constant term, there are $n+1$ coefficients).

Example 1. The operation of ***differentiation***, $A = d/dt$, is linear:

$$Ap(t) = \frac{d}{dt}(a_0 + a_1 t + \cdots + a_n t^n) = a_1 + \cdots + n a_n t^{n-1}. \tag{2}$$

The nullspace of this A is the one-dimensional space of constants: $da_0/dt = 0$. The column space is the n-dimensional space $\mathbf{P}_{n-1}$; the right-hand side of equation (2) is always in that space. The sum of nullity $(= 1)$ and rank $(= n)$ is the dimension of the original space $\mathbf{P}_n$.

Example 2. ***Integration*** from 0 to t is also linear (*it takes* $\boldsymbol{P}_n$ *to* $\boldsymbol{P}_{n+1}$):

$$Ap(t) = \int_0^t (a_0 + \cdots + a_n t^n)dt = a_0 t + \cdots + \frac{a_n}{n+1} t^{n+1}. \tag{3}$$

This time there is no nullspace (except for the zero vector, as always!) but integration does not produce all polynomials in $\mathbf{P}_{n+1}$. The right side of equation (3) has no constant term. Probably the constant polynomials will be the left nullspace.

Example 3. ***Multiplication*** by a fixed polynomial like $2+3t$ is linear:

$$Ap(t) = (2+3t)(a_0 + \cdots + a_n t^n) = 2a_0 + \cdots + 3a_n t^{n+1}.$$

Again this transforms $\mathbf{P}_n$ to $\mathbf{P}_{n+1}$, with no nullspace except $p = 0$.

In these examples (and in almost all examples), linearity is not difficult to verify. It hardly even seems interesting. If it is there, it is practically impossible to miss. Nevertheless, it is the most important property a transformation can have[1]. Of course most transformations are not linear—for example, to square the polynomial ($Ap = p^2$), or to add 1 ($Ap = p+1$), or to keep the positive coefficients ($A(t-t^2) = t$). It will be linear transformations, and *only those*, that lead us back to matrices.

Transformations Represented by Matrices

Linearity has a crucial consequence: ***If we know*** Ax ***for each vector in a basis, then we know*** Ax ***for each vector in the entire space***. Suppose the basis consists of the n vectors $x_1, \ldots, x_n$. Every other vector x is a combination of those particular vectors (they span the space). Then linearity determines Ax:

Linearity If $x = c_1 x_1 + \cdots + c_n x_n$ then $Ax = c_1(Ax_1) + \cdots + c_n(Ax_n)$. (4)

The transformation $T(x) = Ax$ has no freedom left, after it has decided what to do with the basis vectors. The rest is determined by linearity. The requirement (1) for two vectors x and y leads to condition (4) for n vectors $x_1, \ldots, x_n$. The transformation does have a free hand with the vectors in the basis (they are independent). When those are settled, the transformation of every vector is settled.

Example 4. What linear transformation takes x_1 and x_2 to Ax_1 and Ax_2?

$$x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{goes to} \quad Ax_1 = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}; \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \text{goes to} \quad Ax_2 = \begin{bmatrix} 4 \\ 6 \\ 8 \end{bmatrix}.$$

It must be multiplication $T(x) = Ax$ by the matrix

$$A = \begin{bmatrix} 2 & 4 \\ 3 & 6 \\ 4 & 8 \end{bmatrix}.$$

[1] Invertibility is perhaps in second place as an important property.

Starting with a different basis $(1,1)$ and $(2,-1)$, this same A is also the only linear transformation with

$$A\begin{bmatrix}1\\1\end{bmatrix}=\begin{bmatrix}6\\9\\12\end{bmatrix} \qquad \text{and} \qquad A\begin{bmatrix}2\\-1\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}.$$

Next we find matrices that represent differentiation and integration. ***First we must decide on a basis***. For the polynomials of degree 3 there is a natural choice for the four basis vectors:

$$\textbf{Basis for } \mathbf{P}_3 \qquad p_1=1, \quad p_2=t, \quad p_3=t^2, \quad p_4=t^3.$$

That basis is not unique (it never is), but some choice is necessary and this is the most convenient. The derivatives of those four basis vectors are 0, 1, $2t$, $3t^2$:

$$\textbf{Action of } d/dt \qquad Ap_1=0, \quad Ap_2=p_1, \quad Ap_3=2p_2, \quad Ap_4=3p_3. \tag{5}$$

"d/dt" is acting exactly like a matrix, but which matrix? Suppose we were in the usual four-dimensional space with the usual basis—the coordinate vectors $p_1=(1,0,0,0)$, $p_2=(0,1,0,0)$, $p_3=(0,0,1,0)$, $p_4=(0,0,0,1)$. The matrix is decided by equation (5):

$$\textbf{Differentiation matrix} \qquad A_{\text{diff}}=\begin{bmatrix}0&1&0&0\\0&0&2&0\\0&0&0&3\\0&0&0&0\end{bmatrix}.$$

Ap_1 is its first column, which is zero. Ap_2 is the second column, which is p_1. Ap_3 is $2p_2$ and Ap_4 is $3p_3$. The nullspace contains p_1 (the derivative of a constant is zero). The column space contains p_1, p_2, p_3 (the derivative of a cubic is a quadratic). The derivative of a combination like $p=2+t-t^2-t^3$ is decided by linearity, and there is nothing new about that—it is the way we all differentiate. It would be crazy to memorize the derivative of every polynomial.

The matrix can differentiate that $p(t)$, because matrices build in linearity!

$$\frac{dp}{dt}=Ap \longrightarrow \begin{bmatrix}0&1&0&0\\0&0&2&0\\0&0&0&3\\0&0&0&0\end{bmatrix}\begin{bmatrix}2\\1\\-1\\-1\end{bmatrix}=\begin{bmatrix}1\\-2\\-3\\0\end{bmatrix} \longrightarrow 1-2t-3t^2.$$

In short, *the matrix carries all the essential information.* If the basis is known, and the matrix is known, then the transformation of every vector is known.

The coding of the information is simple. To transform a space to itself, one basis is enough. A transformation from one space to another requires a basis for each.

2U Suppose the vectors $x_1,\dots,x_n$ are a basis for the space $\mathbf{V}$, and vectors $y_1,\dots,y_m$ are a basis for $\mathbf{W}$. Each linear transformation T from $\mathbf{V}$ to $\mathbf{W}$ is represented by a matrix A. The jth column is found by applying T to the jth basis vector x_j, and writing $T(x_j)$ as a combination of the y's:

$$\textbf{Column } j \textbf{ of } A \qquad T(x_j) = Ax_j = a_{1j}y_1 + a_{2j}y_2 + \cdots + a_{mj}y_m. \tag{6}$$

For the differentiation matrix, column 1 came from the first basis vector $p_1 = 1$. Its derivative is zero, so column 1 is zero. The last column came from $(d/dt)t^3 = 3t^2$. Since $3t^2 = 0p_1 + 0p_2 + 3p_3 + 0p_4$, the last column contained 0, 0, 3. 0. The rule (6) constructs the matrix, a column at a time.

We do the same for integration. That goes from cubics to quartics, transforming $\mathbf{V} = \mathbf{P}_3$ into $\mathbf{W} = \mathbf{P}_4$, so we need a basis for $\mathbf{W}$. The natural choice is $y_1 = 1$, $y_2 = t$, $y_3 = t^2$, $y_4 = t^3$, $y_5 = t^4$, spanning the polynomials of degree 4. The matrix A will be m by n, or 5 by 4. It comes from applying integration to each basis vector of $\mathbf{V}$:

$$\int_0^t 1\,dt = t \quad \text{or} \quad Ax_1 = y_2, \quad \dots, \quad \int_0^t t^3\,dt = \frac{1}{4}t^4 \quad \text{or} \quad Ax_4 = \frac{1}{4}y_5.$$

$$\textbf{Integration matrix} \qquad A_{\text{int}} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{3} & 0 \\ 0 & 0 & 0 & \frac{1}{4} \end{bmatrix}.$$

Differentiation and integration are *inverse operations*. Or at least integration *followed* by differentiation brings back the original function. To make that happen for matrices, we need the differentiation matrix from quartics down to cubics, which is 4 by 5:

$$A_{\text{diff}} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix} \qquad \text{and} \qquad A_{\text{diff}}A_{\text{int}} = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}.$$

Differentiation is a ***left-inverse*** of integration. Rectangular matrices cannot have two-sided inverses! In the opposite order, $A_{\text{int}}A_{\text{diff}} = I$ cannot be true. The 5 by 5 product has zeros in column 1. The derivative of a constant is zero. In the other columns $A_{\text{int}}A_{\text{diff}}$ is the identity and the integral of the derivative of t^n is t^n.

Rotations Q, Projections P, and Reflections H

This section began with 90° rotations, projections onto the x-axis, and reflections through the 45° line. Their matrices were especially simple:

$$Q = \underset{\text{(rotation)}}{\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}} \qquad P = \underset{\text{(projection)}}{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}} \qquad H = \underset{\text{(reflection)}}{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}}.$$

The underlying linear transformations of the x-y plane are also simple. But rotations through other angles, projections onto other lines, and reflections in other mirrors are almost as easy to visualize, They are still linear transformations, provided that the origin is fixed: $A0 = 0$. They *must* be represented by matrices. Using the natural basis $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, we want to discover those matrices.

1. **Rotation** Figure 2.10 shows rotation through an angle θ. It also shows the effect on the two basis vectors. The first one goes to $(\cos\theta, \sin\theta)$, whose length is still 1; it lies on the "θ-line." The second basis vector $(0,1)$ rotates into $(-\sin\theta, \cos\theta)$. By rule (6), those numbers go into the columns of the matrix (we use c and s for $\cos\theta$ and $\sin\theta$). This family of rotations Q_θ is a perfect chance to test the correspondence between transformations and matrices:

   *Does the **inverse** of Q_θ equal $Q_{-\theta}$* (rotation backward through θ)? *Yes.*

   $$Q_\theta Q_{-\theta} = \begin{bmatrix} c & -s \\ s & c \end{bmatrix} \begin{bmatrix} c & s \\ -s & c \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

   *Does the **square** of Q_θ equal $Q_{2\theta}$* (rotation through a double angle)? *Yes.*

   $$Q_\theta^2 = \begin{bmatrix} c & -s \\ s & c \end{bmatrix} \begin{bmatrix} c & -s \\ s & c \end{bmatrix} = \begin{bmatrix} c^2 - s^2 & -2cs \\ 2cs & c^2 - s^2 \end{bmatrix} = \begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix}.$$

   *Does the **product** of Q_θ and Q_φ equal $Q_{\theta+\varphi}$* (rotation through θ then φ)? *Yes.*

   $$Q_\theta Q_\varphi = \begin{bmatrix} \cos\theta\cos\varphi - \sin\theta\sin\varphi & \cdots \\ \sin\theta\cos\varphi + \cos\theta\sin\varphi & \cdots \end{bmatrix} = \begin{bmatrix} \cos(\theta+\varphi) & \cdots \\ \sin(\theta+\varphi) & \cdots \end{bmatrix}.$$

   The last case contains the first two. The inverse appears when φ is $-\theta$, and the square appears when φ is $+\theta$. All three questions were decided by trigonometric identities (and they give a new way to remember those identities). It was no accident that all the answers were yes. *Matrix multiplication is defined exactly so that **the product of the matrices corresponds to the product of the transformations**.*

   2V Suppose A and B are linear transformations from $\mathbf{V}$ to $\mathbf{W}$ and from $\mathbf{U}$ to $\mathbf{V}$. Their product AB starts with a vector u in $\mathbf{U}$, goes to Bu in $\mathbf{V}$, and

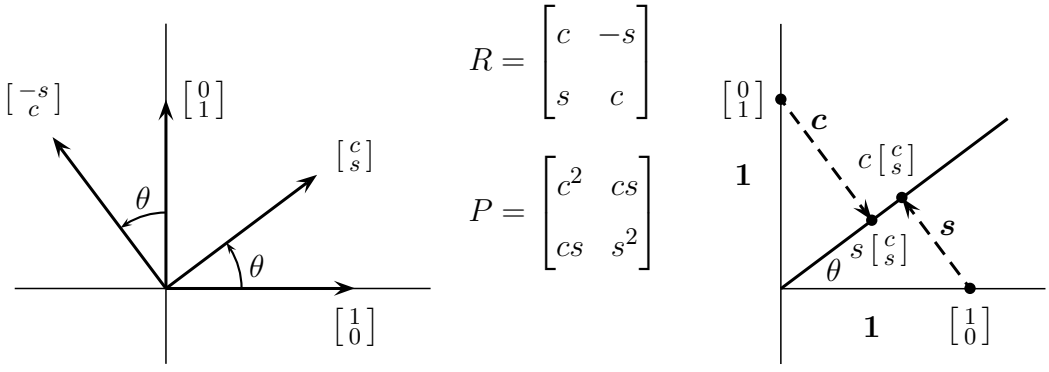


Figure 2.10: Rotation through θ (left). Projection onto the θ-line (right).

finishes with ABu in $\mathbf{W}$. This "composition" AB is again a linear transformation (from $\mathbf{U}$ to $\mathbf{W}$). Its matrix is the product of the individual matrices representing A and B.

For $A_{\text{diff}}A_{\text{int}}$, the composite transformation was the identity (and $A_{\text{int}}A_{\text{diff}}$ annihilated all constants). For rotations, the order of multiplication does not matter. Then $\mathbf{U} = \mathbf{V} = \mathbf{W}$ is the x-y plane, and $Q_\theta Q_\varphi$ is the same as $Q_\varphi Q_\theta$. For a rotation and a reflection, the order makes a difference.

Technical note: To construct the matrices, we need bases for $\mathbf{V}$ and $\mathbf{W}$, and then for $\mathbf{U}$ and $\mathbf{V}$. By keeping the same basis for $\mathbf{V}$, the product matrix goes correctly from the basis in $\mathbf{U}$ to the basis in $\mathbf{W}$. If we distinguish the transformation A from its matrix (call that $[A]$), then the product rule $2V$ becomes extremely concise: $[AB] = [A][B]$. The rule for multiplying matrices in Chapter 1 was totally determined by this requirement—it must match the product of linear transformations.

2. Projection Figure 2.10 also shows the projection of $(1,0)$ onto the θ-line. The length of the projection is $c = \cos\theta$. Notice that the *point* of projection is not (c,s), as I mistakenly thought; that vector has length 1 (it is the rotation), so we must multiply by c. Similarly the projection of $(0,1)$ has length s, and falls at $s(c,s) = (cs,s^2)$, That gives the second column of the projection matrix P:

$$\textbf{Projection onto } \theta\textbf{-line} \qquad P = \begin{bmatrix} c^2 & cs \\ cs & s^2 \end{bmatrix}.$$

This matrix has no inverse, because the transformation has no inverse. Points on the perpendicular line are projected onto the origin; that line is the nullspace of P. Points on the θ-line are projected to themselves! Projecting twice is the same as projecting once, and $P^2 = P$:

$$P^2 = \begin{bmatrix} c^2 & cs \\ cs & s^2 \end{bmatrix}^2 = \begin{bmatrix} c^2(c^2+s^2) & cs(c^2+s^2) \\ cs(c^2+s^2) & s^2(c^2+s^2) \end{bmatrix} = P.$$

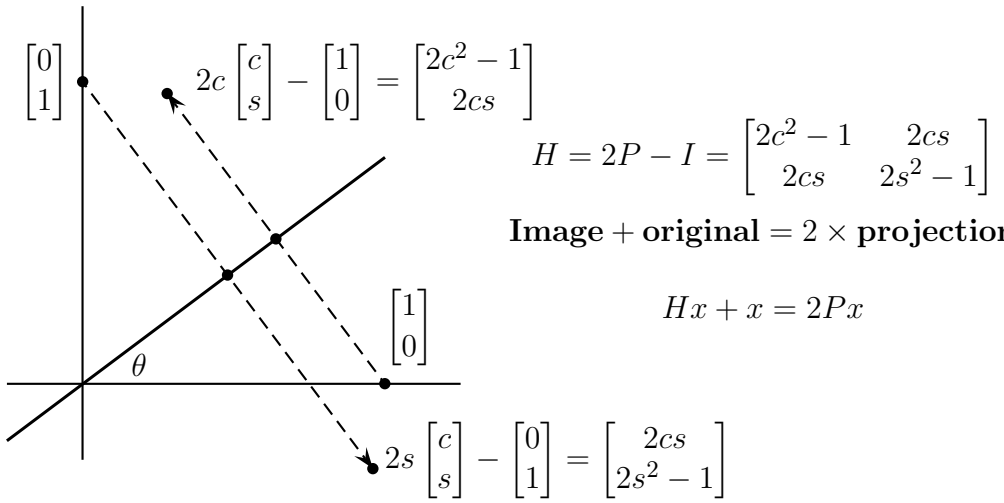


Figure 2.11: Reflection through the θ-line: the geometry and the matrix.

Of course $c^2+s^2=\cos^2\theta+\sin^2\theta=1$. ***A projection matrix equals its own square***.

3. Reflection Figure 2.11 shows the reflection of $(1,0)$ in the θ-line. The length of the reflection equals the length of the original, as it did after rotation—but here the θ-line stays where it is. The perpendicular line reverses direction; all points go straight through the mirror, Linearity decides the rest.

$$\textbf{Reflection matrix} \qquad H=\begin{bmatrix}2c^2-1 & 2cs\\ 2cs & 2s^2-1\end{bmatrix}.$$

This matrix H has the remarkable property $H^2=I$. ***Two reflections bring back the original***. A reflection is its own inverse, $H=H^{-1}$, which is clear from the geometry but less clear from the matrix. One approach is through the relationship of reflections to projections: $H=2P-I$. This means that $Hx+x=2Px$—the image plus the original equals twice the projection. It also confirms that $H^2=I$:

$$H^2=(2P-I)^2=4P^2-4P+I=I, \qquad \text{since} \quad P^2=P.$$

Other transformations Ax can increase the length of x; stretching and shearing are in the exercises. Each example has a matrix to represent it—which is the main point of this section. But there is also the question of choosing a basis, and we emphasize that *the matrix depends on the choice of basis*. Suppose the first basis vector is ***on the θ-line*** and the second basis vector is ***perpendicular***:

(i) The projection matrix is back to $P=\left[\begin{smallmatrix}1&0\\0&0\end{smallmatrix}\right]$. This matrix is constructed as always: its first column comes from the first basis vector (projected to itself). The second column comes from the basis vector that is projected to zero.

(ii) For reflections, that same basis gives $H = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. The second basis vector is reflected onto its negative, to produce this second column. The matrix H is still $2P - I$ when the same basis is used for H and P.

(iii) For rotations, the matrix is not changed. Those lines are still rotated through θ, and $Q = \begin{bmatrix} c & -s \\ s & c \end{bmatrix}$ as before.

The whole question of choosing the best basis is absolutely central, and we come back to it in Chapter 5. The goal is to make the matrix diagonal, as achieved for P and H. To make Q diagonal requires complex vectors, since all real vectors are rotated.

We mention here the effect on the matrix of a change of basis, while the linear transformation stays the same. ***The matrix*** A (or Q or P or H) ***is altered to*** $S^{-1}AS$. Thus a single transformation is represented by different matrices (via different bases, accounted for by S). The theory of eigenvectors will lead to this formula $S^{-1}AS$, and to the best basis.

Problem Set 2.6

1. What matrix has the effect of rotating every vector through 90° and then projecting the result onto the x-axis? What matrix represents projection onto the x-axis followed by projection onto the y-axis?

2. Does the product of 5 reflections and 8 rotations of the x-y plane produce a rotation or a reflection?

3. The matrix $A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ produces a ***stretching*** in the x-direction. Draw the circle $x^2 + y^2 = 1$ and sketch around it the points $(2x, y)$ that result from multiplication by A. What shape is that curve?

4. Every straight line remains straight after a linear transformation. If z is halfway between x and y, show that Az is halfway between Ax and Ay.

5. The matrix $A = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$ yields a ***shearing*** transformation, which leaves the y-axis unchanged. Sketch its effect on the x-axis, by indicating what happens to $(1,0)$ and $(2,0)$ and $(-1,0)$—and how the whole axis is transformed.

6. What 3 by 3 matrices represent the transformations that

(a) project every vector onto the x-y plane?

(b) reflect every vector through the x-y plane?

(c) rotate the x-y plane through 90°, leaving the z-axis alone?

(d) rotate the x-y plane, then x-z, then y-z, through 90°?

(e) carry out the same three rotations, but each one through 180°?

7. On the space $\mathbf{P}_3$ of cubic polynomials, what matrix represents d^2/dt^2? Construct the 4 by 4 matrix from the standard basis $1, t, t^2, t^3$. Find its nullspace and column space. What do they mean in terms of polynomials?

8. From the cubics $\mathbf{P}_3$ to the fourth-degree polynomials $\mathbf{P}_4$, what matrix represents multiplication by $2+3t$? The columns of the 5 by 4 matrix A come from applying the transformation to $1, t, t^2, t^3$.

9. The solutions to the linear differential equation $d^2u/dt^2 = u$ form a vector space (since combinations of solutions are still solutions). Find two independent solutions, to give a basis for that solution space.

10. With initial values $u = x$ and $du/dt = y$ at $t = 0$, what combination of basis vectors in Problem 9 solves $u'' = u$? This transformation from initial values to solution is linear. What is its 2 by 2 matrix (using $x = 1$, $y = 0$ and $x = 0$, $y = 1$ as basis for $\mathbf{V}$, and your basis for $\mathbf{W}$)?

11. Verify directly from $c^2 + s^2 = 1$ that reflection matrices satisfy $H^2 = 1$.

12. Suppose A is a linear transformation from the x-y plane to itself. Why does $A^{-1}(x+y) = A^{-1}x + A^{-1}y$? If A is represented by the matrix M, explain why A^{-1} is represented by M^{-1}.

13. The product $(AB)C$ of linear transformations starts with a vector x and produces $u = Cx$. Then rule 2V applies AB to u and reaches $(AB)Cx$.

(a) Is this result the same as separately applying C then B then A?

(b) Is the result the same as applying BC followed by A? Parentheses are unnecessary and the associative law $(AB)C = A(BC)$ holds for linear transformations. This is the best proof of the same law for matrices.

14. Prove that T^2 is a linear transformation if T is linear (from $\mathbf{R}^3$ to $\mathbf{R}^3$).

15. The space of all 2 by 2 matrices has the four basis "vectors"

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

For the linear transformation of *transposing*, find its matrix A with respect to this basis. Why is $A^2 = I$?

16. Find the 4 by 4 cyclic permutation matrix: (x_1, x_2, x_3, x_4) is transformed to $Ax = (x_2, x_3, x_4, x_1)$. What is the effect of A^2? Show that $A^3 = A^{-1}$.

17. Find the 4 by 3 matrix A that represents a *right shift*: (x_1, x_2, x_3) is transformed to $(0, x_1, x_2, x_3)$. Find also the *left shift* matrix B from $\mathbf{R}^4$ back to $\mathbf{R}^3$, transforming (x_1, x_2, x_3, x_4) to (x_2, x_3, x_4). What are the products AB and BA?

18. In the vector space P_3 of all $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$, let **S** be the subset of polynomials with $\int_0^1 p(x)dx = 0$. Verify that **S** is a subspace and find a basis.

19. A *nonlinear* transformation is invertible if $T(x) = b$ has exactly one solution for every b. The example $T(x) = x^2$ is not invertible because $x^2 = b$ has two solutions for positive b and no solution for negative b. Which of the following transformations (from the real numbers $\mathbf{R}^1$ to the real numbers $\mathbf{R}^1$) are invertible? None are linear, not even (c).

(a) $T(x) = x^3$. (b) $T(x) = e^x$.
(c) $T(x) = x + 11$. (d) $T(x) = \cos x$.

20. What is the axis and the rotation angle for the transformation that takes (x_1, x_2, x_3) into (x_2, x_3, x_1)?

21. A linear transformation must leave the zero vector fixed: $T(0) = 0$. Prove this from $T(v+w) = T(v) + T(w)$ by choosing $w =$ ____. Prove it also from the requirement $T(cv) = cT(v)$ by choosing $c =$ ____

22. Which of these transformations is not linear? The input is $v = (v_1, v_2)$.

(a) $T(v) = (v_2, v_1)$. (b) $T(v) = (v_1, v_1)$.
(c) $T(v) = (0, v_1)$. (d) $T(v) = (0, 1)$.

23. If S and T are linear with $S(v) = T(v) = v$, then $S(T(v)) = v$ or v^2?

24. Suppose $T(v) = v$, except that $T(0, v_2) = (0, 0)$. Show that this transformation satisfies $T(cv) = cT(v)$ but not $T(v+w) = T(v) + T(w)$.

25. Which of these transformations satisfy $T(v+w) = T(v) + T(w)$, and which satisfy $T(cv) = cT(v)$?

(a) $T(v) = v/\|v\|$. (b) $T(v) = v_1 + v_2 + v_3$.
(c) $T(v) = (v_1, 2v_2, 3v_3)$. (d) $T(v) =$ largest component of v.

26. For these transformations of $\mathbf{V} = \mathbf{R}^2$ to $\mathbf{W} = \mathbf{R}^2$, find $T(T(v))$.

(a) $T(v) = -v$.

(b) $T(v) = v + (1, 1)$.

(c) $T(v) = 90°$ rotation $= (-v_2, v_1)$.

(d) $T(v) =$ projection $= \left(\dfrac{v_1 + v_2}{2}, \dfrac{v_1 + v_2}{2} \right)$.

27. The "cyclic" transformation T is defined by $T(v_1, v_2, v_3) = (v_2, v_3, v_1)$. What is $T(T(T(v)))$? What is $T^{100}(v)$?

28. Find the *range* and *kernel* (those are new words for the column space and nullspace) of T.

(a) $T(v_1,v_2)=(v_2,v_1)$. (b) $T(v_1,v_2,v_3)=(v_1,v_2)$.
(c) $T(v_1,v_2)=(0,0)$. (d) $T(v_1,v_2)=(v_1,v_1)$.

29. A linear transformation from $\mathbf{V}$ to $\mathbf{W}$ has an *inverse* from $\mathbf{W}$ to $\mathbf{V}$ when the range is all of $\mathbf{W}$ and the kernel contains only $v=0$. Why are these transformations not invertible?

(a) $T(v_1,v_2)=(v_2,v_2)$ $\quad \mathbf{W}=\mathbf{R}^2$.
(b) $T(v_1,v_2)=(v_1,v_2,v_1+v_2)$ $\quad \mathbf{W}=\mathbf{R}^3$.
(c) $T(v_1,v_2)=v_1$ $\quad \mathbf{W}=\mathbf{R}^1$.

30. Suppose a linear T transforms $(1,1)$ to $(2,2)$ and $(2,0)$ to $(0,0)$. Find $T(v)$ when
(a) $v=(2,2)$. (b) $v=(3,1)$. (c) $v=(-1,1)$. (d) $v=(a,b)$.

Problems 31–35 may be harder. The input space V contains all 2 by 2 matrices M.

31. M is any 2 by 2 matrix and $A=\left[\begin{smallmatrix}1&2\\3&4\end{smallmatrix}\right]$. The linear transformation T is defined by $T(M)=AM$. What rules of matrix multiplication show that T is linear?

32. Suppose $A=\left[\begin{smallmatrix}1&2\\3&6\end{smallmatrix}\right]$. Show that the identity matrix I is not in the range of T. Find a nonzero matrix M such that $T(M)=AM$ is zero.

33. Suppose T transposes every matrix M. Try to find a matrix A that gives $AM=M^{\mathrm{T}}$ for every M. Show that no matrix A will do it. *To professors:* Is this a linear transformation that doesn't come from a matrix?

34. The transformation T that transposes every matrix is definitely linear. Which of these extra properties are true?

(a) $T^2=$ identity transformation.
(b) The kernel of T is the zero matrix.
(c) Every matrix is in the range of T.
(d) $T(M)=-M$ is impossible.

35. Suppose $T(M)=\left[\begin{smallmatrix}1&0\\0&0\end{smallmatrix}\right]\left[M\right]\left[\begin{smallmatrix}0&0\\0&1\end{smallmatrix}\right]$. Find a matrix with $T(M)\neq 0$. Describe all matrices with $T(M)=0$ (the kernel of T) and all output matrices $T(M)$ (the range of T).

Problems 36–40 are about changing the basis

36. (a) What matrix transforms $(1,0)$ into $(2,5)$ and transforms $(0,1)$ to $(1,3)$?
(b) What matrix transforms $(2,5)$ to $(1,0)$ and $(1,3)$ to $(0,1)$?

(c) Why does no matrix transform $(2,6)$ to $(1,0)$ and $(1,3)$ to $(0,1)$?

37. (a) What matrix M transforms $(1,0)$ and $(0,1)$ to (r,t) and (s,u)?

(b) What matrix N transforms (a,c) and (b,d) to $(1,0)$ and $(0,1)$?

(c) What condition on a, b, c, d will make part (b) impossible?

38. (a) How do M and N in Problem 37 yield the matrix that transforms (a,c) to (r,t) and (b,d) to (s,u)?

(b) What matrix transforms $(2,5)$ to $(1,1)$ and $(1,3)$ to $(0,2)$?

39. If you keep the same basis vectors but put them in a different order, the change-of-basis matrix M is a ____ matrix. If you keep the basis vectors in order but change their lengths, M is a ____ matrix.

40. The matrix that transforms $(1,0)$ and $(0,1)$ to $(1,4)$ and $(1,5)$ is $M =$ ____. The combination $a(1,4)+b(1,5)$ that equals $(1,0)$ has $(a,b) = (\quad,\quad)$. How are those new coordinates of $(1,0)$ related to M or M^{-1}?

41. What are the three equations for A, B, C if the parabola $Y = A+Bx+Cx^2$ equals 4 at $x = a$, 5 at $x = b$, and 6 at $x = c$? Find the determinant of the 3 by 3 matrix. For which numbers a, b, c will it be impossible to find this parabola Y?

42. Suppose v_1, v_2, v_3 are eigenvectors for T. This means $T(v_i) = \lambda_i v_i$ for $i = 1,2,3$. What is the matrix for T when the input and output bases are the v's?

43. Every invertible linear transformation can have I as its matrix. For the output basis just choose $w_i = T(v_i)$. Why must T be invertible?

44. Suppose T is reflection across the x-axis and S is reflection across the y-axis. The domain $\mathbf{V}$ is the x-y plane. If $v = (x,y)$ what is $S(T(v))$? Find a simpler description of the product ST.

45. Suppose T is reflection across the 45° line, and S is reflection across the y-axis, If $v = (2,1)$ then $T(v) = (1,2)$. Find $S(T(v))$ and $T(S(v))$. This shows that generally $ST \neq TS$.

46. Show that the product ST of two reflections is a rotation. Multiply these reflection matrices to find the rotation angle:

$$\begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix} \qquad \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{bmatrix}.$$

47. The 4 by 4 *Hadamard matrix* is entirely $+1$ and -1:

$$H = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}.$$

INNER PRODUCT SPACES

Strang Chapter 3, Sections 3.1-3.4

Chapter 3

Orthogonality

3.1 Orthogonal Vectors and Subspaces

A basis is a set of independent vectors that span a space. Geometrically, it is a set of coordinate axes. A vector space is defined without those axes, but every time I think of the x-y plane or three-dimensional space or $\mathbf{R}^n$, the axes are there. They are usually perpendicular! *The coordinate axes that the imagination constructs are practically always orthogonal*. In choosing a basis, we tend to choose an orthogonal basis.

The idea of an orthogonal basis is one of the foundations of linear algebra. We need a basis to convert geometric constructions into algebraic calculations, and we need an orthogonal basis to make those calculations simple. A further specialization makes the basis just about optimal: The vectors should have *length* 1. For an ***orthonormal basis*** (orthogonal unit vectors), we will find

1. the length $\|x\|$ of a vector;

2. the test $x^{\mathrm{T}}y = 0$ for perpendicular vectors; and

3. how to create perpendicular vectors from linearly independent vectors.

More than just vectors, *subspaces* can also be perpendicular. We will discover, so beautifully and simply that it will be a delight to see, that ***the fundamental subspaces meet at right angles***. Those four subspaces are perpendicular in pairs, two in $\mathbf{R}^m$ and two in $\mathbf{R}^n$. That will complete the fundamental theorem of linear algebra.

The first step is to find the ***length of a vector***. It is denoted by $\|x\|$, and in two dimensions it comes from the hypotenuse of a right triangle (Figure 3.1a). The square of the length was given a long time ago by Pythagoras: $\|x\|^2 = x_1^2 + x_2^2$.

In three-dimensional space, $x = (x_1, x_2, x_3)$ is the diagonal of a box (Figure 3.1b). Its length comes from *two* applications of the Pythagorean formula. The two-dimensional case takes care of $(x_1, x_2, 0) = (1, 2, 0)$ across the base. This forms a right angle with the vertical side $(0, 0, x_3) = (0, 0, 3)$. The hypotenuse of the bold triangle (Pythagoras again)

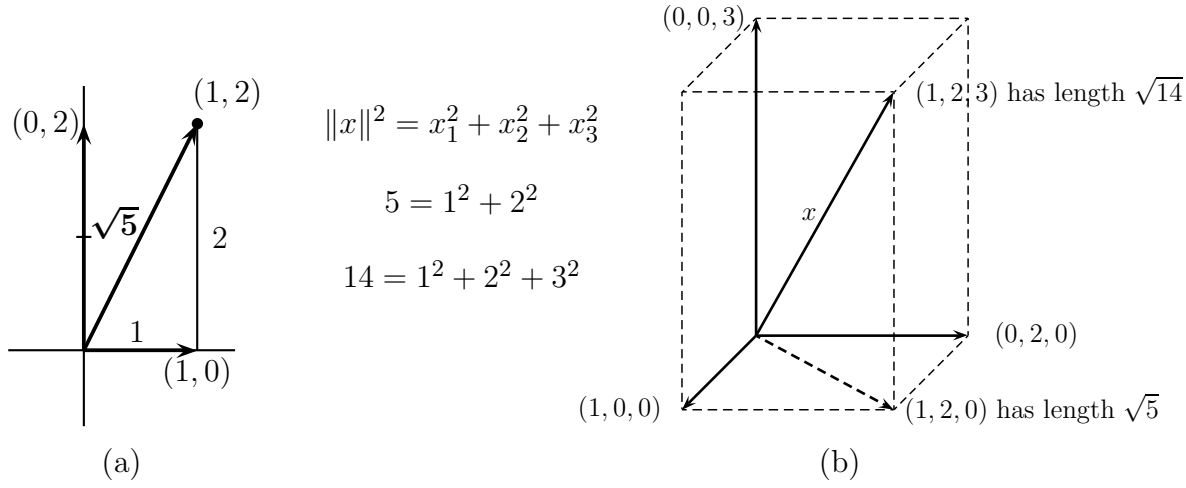


Figure 3.1: The length of vectors (x_1, x_2) and (x_1, x_2, x_3).

is the length $\|x\|$ we want:

$$\textbf{Length in 3D} \qquad \|x\|^2 = 1^2 + 2^2 + 3^2 \quad \text{and} \quad \|x\| = \sqrt{x_1^2 + x_2^2 + x_3^2}.$$

The extension to $x = (x_1, \ldots, x_n)$ in n dimensions is immediate. By Pythagoras $n-1$ times, ***the length*** $\|x\|$ ***in*** $\mathbf{R}^n$ ***is the positive square root of*** $x^{\mathrm{T}}x$:

$$\textbf{Length squared} \qquad \|x\|^2 = x_1^2 + x_2^2 + \cdots + x_n^2 = x^{\mathrm{T}}x. \tag{1}$$

The sum of squares matches $x^{\mathrm{T}}x$—and the length of $x = (1, 2, -3)$ is $\sqrt{14}$:

$$x^{\mathrm{T}}x = \begin{bmatrix} 1 & 2 & -3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} = 1^2 + 2^2 + (-3)^2 = 14.$$

Orthogonal Vectors

How can we decide whether two vectors x and y are perpendicular? What is the test for orthogonality in Figure 3.2? In the plane spanned by x and y, those vectors are orthogonal provided they form a *right triangle*. We go back to $a^2 + b^2 = c^2$:

$$\textbf{Sides of a right triangle} \qquad \|x\|^2 + \|y\|^2 = \|x - y\|^2. \tag{2}$$

Applying the length formula (1), this test for orthogonality in $\mathbf{R}^n$ becomes

$$\left(x_1^2 + \cdots + x_n^2\right) + \left(y_1^2 + \cdots + y_n^2\right) = (x_1 - y_1)^2 + \cdots + (x_n - y_n)^2.$$

The right-hand side has an extra $-2x_iy_i$ from each $(x_i - y_i)^2$:

$$\text{right-hand side} = \left(x_1^2 + \cdots + x_n^2\right) - 2(x_1y_1 + \cdots + x_ny_n) + \left(y_1^2 + \cdots + y_n^2\right).$$

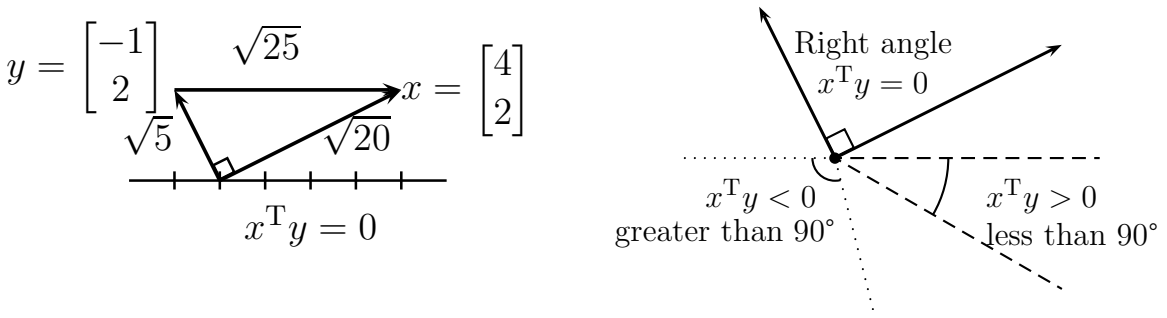


Figure 3.2: A right triangle with $5 + 20 = 25$. Dotted angle 100°, dashed angle 30°.

We have a right triangle when that sum of cross-product terms $x_i y_i$ is zero:

$$\textbf{Orthogonal vectors} \qquad x^{\mathrm{T}}y = x_1y_1 + \cdots + x_ny_n = 0. \tag{3}$$

This sum is $x^{\mathrm{T}}y = \sum x_iy_i = y^{\mathrm{T}}x$, the row vector x^{T} times the column vector y:

$$\textbf{Inner product} \qquad x^{\mathrm{T}}y = \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = x_1y_1 + \cdots + x_ny_n. \tag{4}$$

This number is sometimes called the scalar product or dot product, and denoted by (x,y) or $x \cdot y$. We will use the name ***inner product*** and keep the notation $x^{\mathrm{T}}y$.

3A The inner product $x^{\mathrm{T}}y$ is zero if and only if x and y are orthogonal vectors. If $x^{\mathrm{T}}y > 0$, their angle is less than 90°. If $x^{\mathrm{T}}y < 0$, their angle is greater than 90°.

The length squared is the inner product of x with itself: $x^{\mathrm{T}}x = x_1^2 + \cdots + x_n^2 = \|x\|^2$. The only vector with length zero—the only vector orthogonal to itself—is the zero vector. This vector $x = 0$ is orthogonal to every vector in $\mathbf{R}^n$.

Example 1. $(2,2,-1)$ is orthogonal to $(-1,2,2)$. Both have length $\sqrt{4+4+1} = 3$.

Useful fact: **If nonzero vectors $v_1, \ldots, v_k$ are mutually orthogonal** (every vector is perpendicular to every other), **then those vectors are linearly independent**.

Proof. Suppose $c_1v_1 + \cdots + c_kv_k = 0$. To show that c_1 must be zero, take the inner product of both sides with v_1. Orthogonality of the v's leaves only one term:

$$v_1^{\mathrm{T}}(c_1v_1 + \cdots + c_kv_k) = c_1v_1^{\mathrm{T}}v_1 = 0. \tag{5}$$

The vectors are nonzero, so $v_1^{\mathrm{T}}v_1 \neq 0$ and therefore $c_1 = 0$. The same is true of every c_i. The only combination of the v's producing zero has all $c_i = 0$: *independence*! □

The coordinate vectors $e_1, \ldots, e_n$ in $\mathbf{R}^n$ are the most important orthogonal vectors. Those are the columns of the identity matrix. They form the simplest basis for $\mathbf{R}^n$, and

they are *unit vectors*—each has length $\|e_i\| = 1$. They point along the coordinate axes. If these axes are rotated, the result is a new **orthonormal basis**: a new system of *mutually orthogonal unit vectors*. In $\mathbf{R}^2$ we have $\cos^2\theta + \sin^2\theta = 1$:

Orthonormal vectors in $\mathbf{R}^2$ $\quad v_1 = (\cos\theta, \sin\theta)$ and $v_2 = (-\sin\theta, \cos\theta)$.

Orthogonal Subspaces

We come to the orthogonality of two subspaces. **Every vector** *in one subspace must be orthogonal to* **every vector** *in the other subspace*. Subspaces of $\mathbf{R}^3$ can have dimension 0, 1, 2, or 3. The subspaces are represented by lines or planes through the origin—and in the extreme cases, by the origin alone or the whole space. The subspace $\{0\}$ is orthogonal to all subspaces. A line can be orthogonal to another line, or it can be orthogonal to a plane, but *a plane cannot be orthogonal to a plane*.

I have to admit that the front wall and side wall of a room look like perpendicular planes in $\mathbf{R}^3$. But by our definition, that is not so! There are lines v and w in the front and side walls that do not meet at a right angle. The line along the corner is in *both* walls, and it is certainly not orthogonal to itself.

3B Two subspaces $\mathbf{V}$ and $\mathbf{W}$ of the same space $\mathbf{R}^n$ are *orthogonal* if every vector v in $\mathbf{V}$ is orthogonal to every vector w in $\mathbf{W}$: $v^{\mathrm{T}}w = 0$ for all v and w.

Example 2. Suppose $\mathbf{V}$ is the plane spanned by $v_1 = (1,0,0,0)$ and $v_2 = (1,1,0,0)$. If $\mathbf{W}$ is the line spanned by $w = (0,0,4,5)$, then w is orthogonal to both v's. The line $\mathbf{W}$ will be orthogonal to the whole plane $\mathbf{V}$.

In this case, with subspaces of dimension 2 and 1 in $\mathbf{R}^4$, there is room for a third subspace. The line $\mathbf{L}$ through $z = (0,0,5,-4)$ is perpendicular to $\mathbf{V}$ and $\mathbf{W}$. Then the dimensions add to $2+1+1=4$. What space is perpendicular to all of $\mathbf{V}$, $\mathbf{W}$, and $\mathbf{L}$?

The important orthogonal subspaces don't come by accident, and they come two at a time. In fact orthogonal subspaces are unavoidable: ***They are the fundamental subspaces!*** The first pair is the *nullspace* and *row space*. Those are subspaces of $\mathbf{R}^n$—the rows have n components and so does the vector x in $Ax = 0$. We have to show, using $Ax = 0$, that ***the rows of A are orthogonal to the nullspace vector x.***

3C Fundamental theorem of orthogonality The row space is orthogonal to the nullspace (in $\mathbf{R}^n$). The column space is orthogonal to the left nullspace (in $\mathbf{R}^m$).

First Proof. Suppose x is a vector in the nullspace. Then $Ax = 0$, and this system of m

equations can be written out as rows of A multiplying x:

$$\textbf{Every row is orthogonal to } x \qquad Ax = \begin{bmatrix} \cdots & \textbf{row 1} & \cdots \\ \cdots & \text{row } 2 & \cdots \\ \vdots & \vdots & \vdots \\ \cdots & \text{row } m & \cdots \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ 0 \\ \vdots \\ 0 \end{bmatrix}. \tag{6}$$

The main point is already in the first equation: *row* 1 *is orthogonal to x*. Their inner product is zero; that is equation 1. Every right-hand side is zero, so x is orthogonal to every row. Therefore x is orthogonal to every *combination* of the rows. Each x in the nullspace is orthogonal to each vector in the row space, so $\boldsymbol{N}(A) \perp \boldsymbol{C}(A^{\mathrm{T}})$.

The other pair of orthogonal subspaces comes from $A^{\mathrm{T}}y = 0$, or $y^{\mathrm{T}}A = 0$:

$$y^{\mathrm{T}}A = \begin{bmatrix} y_1 & \cdots & y_m \end{bmatrix} \begin{bmatrix} \textbf{c} & & \text{c} \\ \textbf{o} & & \text{o} \\ \textbf{l} & & \text{l} \\ \textbf{u} & \cdots & \text{u} \\ \textbf{m} & & \text{m} \\ \textbf{n} & & \text{n} \\ & & \\ \textbf{1} & & n \end{bmatrix} = \begin{bmatrix} \mathbf{0} & \cdots & 0 \end{bmatrix}. \tag{7}$$

The vector y is orthogonal to every column. The equation says so, from the zeros on the right-hand side. Therefore y is orthogonal to every combination of the columns. It is orthogonal to the column space, and it is a typical vector in the left nullspace: $\boldsymbol{N}(A^{\mathrm{T}}) \perp \boldsymbol{C}(A)$. This is the same as the first half of the theorem, with A replaced by A^{T}. □

Second Proof. The contrast with this "coordinate-free proof" should be useful to the reader. It shows a more "abstract" method of reasoning. I wish I knew which proof is clearer, and more permanently understood.

If x is in the nullspace then $Ax = 0$. If v is in the row space, it is a combination of the rows: $v = A^{\mathrm{T}}z$ for some vector z. Now, in one line:

$$\textbf{Nullspace} \perp \textbf{Row space} \qquad v^{\mathrm{T}}x = (A^{\mathrm{T}}z)^{\mathrm{T}}x = z^{\mathrm{T}}Ax = z^{\mathrm{T}}0 = 0. \tag{8}$$

□

Example 3. Suppose A has rank 1, so its row space and column space are lines:

$$\textbf{Rank-1 matrix} \qquad A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \\ 3 & 9 \end{bmatrix}.$$

The rows are multiples of $(1,3)$. The nullspace contains $x = (-3,1)$, which is orthogonal to all the rows. The nullspace and row space are perpendicular lines in $\mathbf{R}^2$:

$$\begin{bmatrix} 1 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = 0 \quad \text{and} \quad \begin{bmatrix} 2 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = 0 \quad \text{and} \quad \begin{bmatrix} 3 & 9 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = 0.$$

In contrast, the other two subspaces are in $\mathbf{R}^3$. The column space is the line through $(1,2,3)$. The left nullspace must be the *perpendicular plane* $y_1+2y_2+3y_3=0$. That equation is exactly the content of $y^{\mathrm{T}}A=0$.

The first two subspaces (the two lines) had dimensions $1+1=2$ in the space $\mathbf{R}^2$. The second pair (line and plane) had dimensions $1+2=3$ in the space $\mathbf{R}^3$. In general, *the row space and nullspace have dimensions that add to* $r+(n-r)=n$. The other pair adds to $r+(m-r)=m$. Something more than orthogonality is occurring, and I have to ask your patience about that one further point: **the dimensions**.

It is certainly true that the null space is perpendicular to the row space—but it is not the whole truth. $\boldsymbol{N}(A)$ *contains every vector orthogonal to the row space*. The nullspace was formed from *all* solutions to $Ax=0$.

Definition. Given a subspace $\mathbf{V}$ of $\mathbf{R}^n$, the space of *all* vectors orthogonal to $\mathbf{V}$ is called the **orthogonal complement** of $\mathbf{V}$. It is denoted by $\mathbf{V}^{\perp}=$ "$\mathbf{V}$ perp."

Using this terminology, the nullspace is the orthogonal complement of the row space: $\boldsymbol{N}(A)=(\boldsymbol{C}(A^{\mathrm{T}}))^{\perp}$. At the same time, the row space contains all vectors that are orthogonal to the nullspace. A vector z can't be orthogonal to the nullspace but outside the row space. Adding z as an extra row of A would enlarge the row space, but we know that there is a fixed formula $r+(n-r)=n$:

Dimension formula $\quad$ dim(row space) + dim(nullspace) = number of columns.

Every vector orthogonal to the nullspace is in the row space: $\boldsymbol{C}(A^{\mathrm{T}})=(\boldsymbol{N}(A))^{\perp}$.

The same reasoning applied to A^{T} produces the dual result: *The left nullspace* $\boldsymbol{N}(A^{\mathrm{T}})$ *and the column space* $\boldsymbol{C}(A)$ *are orthogonal complements*. Their dimensions add up to $(m-r)+r=m$, This completes the second half of the fundamental theorem of linear algebra. The first half gave the dimensions of the four subspaces. including the fact that row rank = column rank. Now we know that those subspaces are perpendicular. More than that, the subspaces are orthogonal complements.

3D Fundamental Theorem of Linear Algebra, Part II

The nullspace is the *orthogonal complement* of the row space in $\mathbf{R}^n$.

The left nullspace is the *orthogonal complement* of the column space in $\mathbf{R}^m$.

To repeat, the row space contains everything orthogonal to the nullspace. The column space contains everything orthogonal to the left nullspace. That is just a sentence, hidden in the middle of the book, but *it decides exactly which equations can be solved*! Looked at directly, $Ax=b$ requires b to be in the column space. Looked at indirectly. $Ax=b$ ***requires*** b ***to be perpendicular to the left nullspace***.

3E $\quad Ax=b$ is solvable if and only if $y^{\mathrm{T}}b=0$ whenever $y^{\mathrm{T}}A=0$.

The direct approach was "b must be a combination of the columns." The indirect approach is "b must be orthogonal to every vector that is orthogonal to the columns." That doesn't sound like an improvement (to put it mildly). But if only one or two vectors are orthogonal to the columns. it is much easier to check those one or two conditions $y^{\mathrm{T}}b = 0$. A good example is Kirchhoff's Voltage Law in Section 2.5. Testing for zero around loops is much easier than recognizing combinations of the columns.

When the left-hand sides of $Ax = b$ add to zero, the right-hand sides must, too:

$$\begin{array}{l} x_1 - x_2 = b_1 \\ x_2 - x_3 = b_2 \\ x_3 - x_1 = b_3 \end{array} \quad \text{is solvable if and only if } b_1 + b_2 + b_3 = 0. \quad \text{Here} \quad A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix}.$$

This test $b_1 + b_2 + b_3 = 0$ makes b orthogonal to $y = (1,1,1)$ in the left nullspace. By the Fundamental Theorem, b is a combination of the columns!

The Matrix and the Subspaces

We emphasize that $\mathbf{V}$ and $\mathbf{W}$ can be orthogonal without being complements. Their dimensions can be too small. The line $\mathbf{V}$ spanned by $(0,1,0)$ is orthogonal to the line $\mathbf{W}$ spanned by $(0,0,1)$, but $\mathbf{V}$ is not $\mathbf{W}^{\perp}$. The orthogonal complement of $\mathbf{W}$ is a two-dimensional plane, and the line is only part of $\mathbf{W}^{\perp}$. When the dimensions are right, orthogonal subspaces *are* necessarily orthogonal complements:

$$\text{If} \quad \mathbf{W} = \mathbf{V}^{\perp} \quad \text{then} \quad \mathbf{V} = \mathbf{W}^{\perp} \quad \text{and} \quad \dim\mathbf{V} + \dim\mathbf{W} = n.$$

In other words $\mathbf{V}^{\perp\perp} = \mathbf{V}$. The dimensions of $\mathbf{V}$ and $\mathbf{W}$ are right, and the whole space $\mathbf{R}^n$ is being decomposed into two perpendicular parts (Figure 3.3).

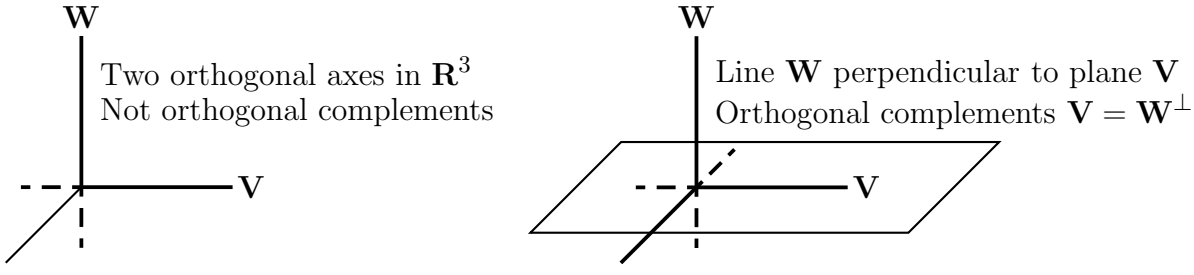


Figure 3.3: Orthogonal complements in $\mathbf{R}^3$: a plane and a line (not two lines).

Splitting $\mathbf{R}^n$ into orthogonal parts will split every vector into $x = v + w$. The vector v is the projection onto the subspace $\mathbf{V}$. The orthogonal component w is the projection of x onto $\mathbf{W}$. The next sections show how to find those projections of x. They lead to what is probably the most important figure in the book (Figure 3.4).

Figure 3.4 summarizes the fundamental theorem of linear algebra. It illustrates the true effect of a matrix—what is happening inside the multiplication Ax. The nullspace

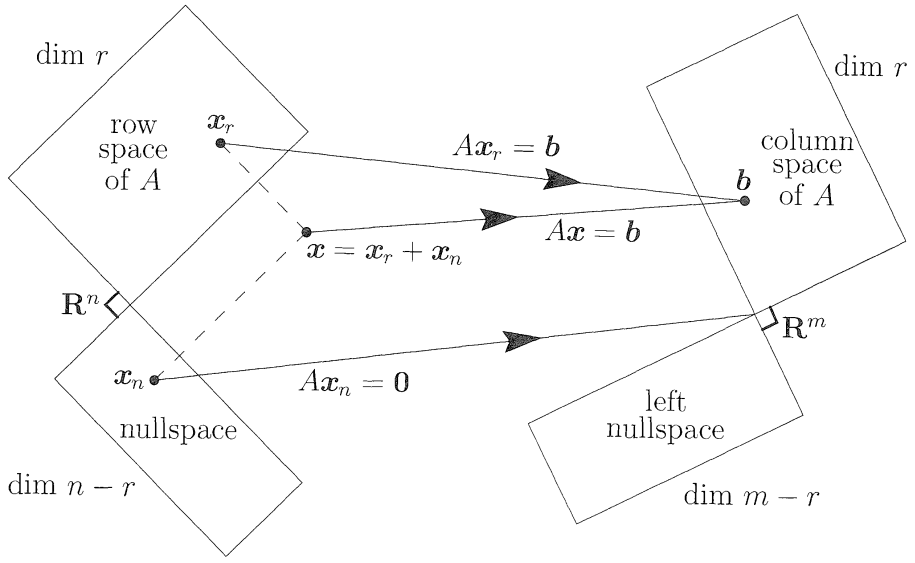


Figure 3.4: The true action $Ax = A(x_{\text{row}} + x_{\text{null}})$ of any m by n matrix.

is carried to the zero vector. Every Ax is in the column space. Nothing is carried to the left nullspace. *The real action is between the row space and column space*, and you see it by looking at a typical vector x. It has a "row space component" and a "nullspace component," with $x = x_r + x_n$. When multiplied by A, this is $Ax = Ax_r + Ax_n$:

The nullspace component goes to zero: $Ax_n = 0$.

The row space component goes to the column space: $Ax_r = Ax$.

Of course everything goes to the column space—the matrix cannot do anything else. I tried to make the row and column spaces the same size, with equal dimension r.

3F From the row space to the column space, A is actually invertible. Every vector b in the column space comes from exactly one vector x_r in the row space.

Proof. Every b in the column space is a combination Ax of the columns. In fact, b is Ax_r, with x_r in the row space, since the nullspace component gives $Ax_n = 0$, If another vector x_r' in the row space gives $Ax_r' = b$, then $A(x_r - x_r') = b - b = 0$. This puts $x_r - x_r'$ in the nullspace and the row space, which makes it orthogonal to itself. Therefore it is zero, and $x_r - x_r'$. Exactly one vector in the row space is carried to b. □

Every matrix transforms its row space onto its column space.

On those r-dimensional spaces A is invertible. On its nullspace A is zero. When A is diagonal, you see the invertible submatrix holding the r nonzeros.

A^{T} goes in the opposite direction, from $\mathbf{R}^m$ to $\mathbf{R}^n$ and from $\boldsymbol{C}(A)$ back to $\boldsymbol{C}(A^{\mathrm{T}})$. Of course the transpose is not the inverse! A^{T} moves the spaces correctly, but not the

individual vectors. That honor belongs to A^{-1} if it exists—and it only exists if $r = m = n$. We cannot ask A^{-1} to bring back a whole nullspace out of the zero vector.

When A^{-1} fails to exist, the best substitute is the ***pseudoinverse*** A^+. This inverts A where that is possible: $A^+Ax = x$ for x in the row space. On the left nullspace, nothing can be done: $A^+y = 0$. Thus A^+ inverts A where it is invertible, and has the same rank r. One formula for A^+ depends on the ***singular value decomposition***—for which we first need to know about eigenvalues.

Problem Set 3.1

1. Find the lengths and the inner product of $x = (1,4,0,2)$ and $y = (2,-2,1,3)$.

2. Give an example in $\mathbf{R}^2$ of linearly independent vectors that are not orthogonal. Also, give an example of orthogonal vectors that are not independent.

3. Two lines in the plane are perpendicular when the product of their slopes is -1. Apply this to the vectors $x = (x_1, x_2)$ and $y = (y_1, y_2)$, whose slopes are x_2/x_1 and y_2/y_1, to derive again the orthogonality condition $x^{\mathrm{T}}y = 0$.

4. How do we know that the ith row of an invertible matrix B is orthogonal to the jth column of B^{-1}, if $i \neq j$?

5. Which pairs are orthogonal among the vectors v_1, v_2, v_3, v_4?

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ -2 \\ 1 \end{bmatrix}, \qquad v_2 = \begin{bmatrix} 4 \\ 0 \\ 4 \\ 0 \end{bmatrix}, \qquad v_3 = \begin{bmatrix} 1 \\ -1 \\ -1 \\ -1 \end{bmatrix}, \qquad v_4 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$

6. Find all vectors in $\mathbf{R}^3$ that are orthogonal to $(1,1,1)$ and $(1,-1,0)$. Produce an orthonormal basis from these vectors (mutually orthogonal unit vectors).

7. Find a vector x orthogonal to the row space of A, and a vector y orthogonal to the column space, and a vector z orthogonal to the nullspace:

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 6 & 4 \end{bmatrix}.$$

8. If $\mathbf{V}$ and $\mathbf{W}$ are orthogonal subspaces, show that the only vector they have in common is the zero vector: $\mathbf{V} \cap \mathbf{W} = \{0\}$.

9. Find the orthogonal complement of the plane spanned by the vectors $(1,1,2)$ and $(1,2,3)$, by taking these to be the rows of A and solving $Ax = 0$. Remember that the complement is a whole line.

10. Construct a homogeneous equation in three unknowns whose solutions are the linear combinations of the vectors $(1,1,2)$ and $(1,2,3)$. This is the reverse of the previous exercise, but the two problems are really the same.

11. The fundamental theorem is often stated in the form of *Fredholm's alternative*: For any A and b, one and only one of the following systems has a solution:

(i) $Ax = b$.

(ii) $A^{\mathrm{T}}y = 0$, $y^{\mathrm{T}}b \neq 0$.

Either b is in the column space $\boldsymbol{C}(A)$ or there is a y in $\boldsymbol{N}(A^{\mathrm{T}})$ such that $y^{\mathrm{T}}b \neq 0$. Show that it is contradictory for (i) and (ii) both to have solutions.

12. Find a basis for the orthogonal complement of the row space of A:

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & 4 \end{bmatrix}.$$

Split $x = (3,3,3)$ into a row space component x_r and a nullspace component x_n.

13. Illustrate the action of A^{T} by a picture corresponding to Figure 3.4, sending $\boldsymbol{C}(A)$ back to the row space and the left nullspace to zero.

14. Show that $x - y$ is orthogonal to $x + y$ if and only if $\|x\| = \|y\|$.

15. Find a matrix whose row space contains $(1,2,1)$ and whose nullspace contains $(1,-2,1)$, or prove that there is no such matrix.

16. Find all vectors that are perpendicular to $(1,4,4,1)$ and $(2,9,8,2)$.

17. If $\mathbf{V}$ is the orthogonal complement of $\mathbf{W}$ in $\mathbf{R}^n$, is there a matrix with row space $\mathbf{V}$ and nullspace $\mathbf{W}$? Starting with a basis for $\mathbf{V}$, construct such a matrix.

18. If $\mathbf{S} = \{0\}$ is the subspace of $\mathbf{R}^4$ containing only the zero vector, what is $\mathbf{S}^{\perp}$? If $\mathbf{S}$ is spanned by $(0,0,0,1)$, what is $\mathbf{S}^{\perp}$? What is $(\mathbf{S}^{\perp})^{\perp}$?

19. *Why are these statements false*?

(a) If $\mathbf{V}$ is orthogonal to $\mathbf{W}$, then $\mathbf{V}^{\perp}$ is orthogonal to $\mathbf{W}^{\perp}$.

(b) $\mathbf{V}$ orthogonal to $\mathbf{W}$ and $\mathbf{W}$ orthogonal to $\mathbf{Z}$ makes $\mathbf{V}$ orthogonal to $\mathbf{Z}$.

20. Let $\mathbf{S}$ be a subspace of $\mathbf{R}^n$. Explain what $(\mathbf{S}^{\perp})^{\perp} = \mathbf{S}$ means and why it is true.

21. Let $\mathbf{P}$ be the plane in $\mathbf{R}^2$ with equation $x + 2y - z = 0$. Find a vector perpendicular to $\mathbf{P}$. What matrix has the plane $\mathbf{P}$ as its nullspace, and what matrix has $\mathbf{P}$ as its row space?

22. Let $\mathbf{S}$ be the subspace of $\mathbf{R}^4$ containing all vectors with $x_1 + x_2 + x_3 + x_4 = 0$. Find a basis for the space $\mathbf{S}^{\perp}$, containing all vectors orthogonal to $\mathbf{S}$.

23. Construct an unsymmetric 2 by 2 matrix of rank 1. Copy Figure 3.4 and put one vector in each subspace. Which vectors are orthogonal?

24. Redraw Figure 3.4 for a 3 by 2 matrix of rank $r = 2$. Which subspace is $\mathbf{Z}$ (zero vector only)? The nullspace part of any vector x in $\mathbf{R}^2$ is $x_n =$ ____.

25. Construct a matrix with the required property or say why that is impossible.

(a) Column space contains $\begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix}$, nullspace contains $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

(b) Row space contains $\begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix}$, nullspace contains $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

(c) $Ax = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ has a solution and $A^{\mathrm{T}} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

(d) Every row is orthogonal to every column (A is not the zero matrix).

(e) The columns add up to a column of 0s, the rows add to a row of 1s.

26. If $AB = 0$ then the columns of B are in the ____ of A. The rows of A are in the ____ of B. Why can't A and B be 3 by 3 matrices of rank 2?

27. (a) If $Ax = b$ has a solution and $A^{\mathrm{T}}y = 0$, then y is perpendicular to ____.

(b) If $A^{\mathrm{T}}y = c$ has a solution and $Ax = 0$, then x is perpendicular to ____.

28. This is a system of equations $Ax = b$ with *no solution*:

$$\begin{aligned} x + 2y + 2z &= 5 \\ 2x + 2y + 3z &= 5 \\ 3x + 4y + 5z &= 9. \end{aligned}$$

Find numbers y_1, y_2, y_3 to multiply the equations so they add to $0 = 1$. You have found a vector y in which subspace? The inner product $y^{\mathrm{T}}b$ is 1.

29. In Figure 3.4, how do we know that Ax_r is equal to Ax? How do we know that this vector is in the column space? If $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ what is x_r?

30. If Ax is in the nullspace of A^{T} then $Ax = 0$. Reason: Ax is also in the ____ of A and the spaces are ____. *Conclusion:* $A^{\mathrm{T}}A$ *has the same nullspace as* A.

31. Suppose A is a symmetric matrix ($A^{\mathrm{T}} = A$).

(a) Why is its column space perpendicular to its nullspace?

(b) If $Ax = 0$ and $Az = 5z$, which subspaces contain these "eigenvectors" x and z? **Symmetric matrices have perpendicular eigenvectors** (see Section 5.5).

32. (Recommended) Draw Figure 3.4 to show each subspace for

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \qquad \text{and} \qquad B = \begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix}.$$

33. Find the pieces x_r and x_n, and draw Figure 3.4 properly, if

$$A = \begin{bmatrix} 1 & -1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \qquad \text{and} \qquad x = \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

Problems 34–44 are about orthogonal subspaces.

34. Put bases for the orthogonal subspaces **V** and **W** into the columns of matrices V and W. *Why does* $V^{\mathrm{T}}W = \textit{zero matrix}$? This matches $v^{\mathrm{T}}w = 0$ for vectors.

35. The floor and the wall are not orthogonal subspaces because they share a nonzero vector (along the line where they meet). Two planes in $\mathbf{R}^3$ cannot be orthogonal! Find a vector in both column spaces $C(A)$ and $C(B)$:

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 1 & 2 \end{bmatrix} \qquad \text{and} \qquad B = \begin{bmatrix} 5 & 4 \\ 6 & 3 \\ 5 & 1 \end{bmatrix}.$$

This will be a vector Ax and also $B\widehat{x}$. Think 3 by 4 with the matrix $[A \;\; B]$.

36. Extend Problem 35 to a p-dimensional subspace **V** and a q-dimensional subspace **W** of $\mathbf{R}^n$. What inequality on $p+q$ guarantees that **V** intersects **W** in a nonzero vector? These subspaces cannot be orthogonal.

37. Prove that every y in $N(A^{\mathrm{T}})$ is perpendicular to every Ax in the column space, using the matrix shorthand of equation (8). Start from $A^{\mathrm{T}}y = 0$.

38. If **S** is the subspace of $\mathbf{R}^3$ containing only the zero vector, what is $\mathbf{S}^{\perp}$? If **S** is spanned by $(1,1,1)$, what is $\mathbf{S}^{\perp}$? If **S** is spanned by $(2,0,0)$ and $(0,0,3)$, what is $\mathbf{S}^{\perp}$?

39. Suppose **S** only contains $(1,5,1)$ and $(2,2,2)$ (not a subspace). Then $\mathbf{S}^{\perp}$ is the nullspace of the matrix $A =$ ____. $\mathbf{S}^{\perp}$ is a subspace even if **S** is not.

40. Suppose **L** is a one-dimensional subspace (a line) in $\mathbf{R}^3$. Its orthogonal complement $\mathbf{L}^{\perp}$ is the ____ perpendicular to **L**. Then $(\mathbf{L}^{\perp})^{\perp}$ is a ____ perpendicular to $\mathbf{L}^{\perp}$. In fact $(\mathbf{L}^{\perp})^{\perp}$ is the same as ____.

41. Suppose **V** is the whole space $\mathbf{R}^4$. Then $\mathbf{V}^{\perp}$ contains only the vector ____. Then $(\mathbf{V}^{\perp})^{\perp}$ is ____. So $(\mathbf{V}^{\perp})^{\perp}$ is the same as ____.

42. Suppose **S** is spanned by the vectors $(1,2,2,3)$ and $(1,3,3,2)$. Find two vectors that span $\mathbf{S}^{\perp}$. This is the same as solving $Ax = 0$ for which A?

43. If **P** is the plane of vectors in $\mathbf{R}^4$ satisfying $x_1 + x_2 + x_3 + x_4 = 0$, write a basis for $\mathbf{P}^{\perp}$. Construct a matrix that has **P** as its nullspace.

44. If a subspace **S** is contained in a subspace **V**, prove that $\mathbf{S}^{\perp}$ contains $\mathbf{V}^{\perp}$.

Problems 45–50 are about perpendicular columns and rows.

45. Suppose an n by n matrix is invertible: $AA^{-1} = I$. Then the first column of A^{-1} is orthogonal to the space spanned by which rows of A?

46. Find $A^{\mathrm{T}}A$ if the columns of A are unit vectors, all mutually perpendicular.

47. Construct a 3 by 3 matrix A with no zero entries whose columns are mutually perpendicular. Compute $A^{\mathrm{T}}A$. Why is it a diagonal matrix?

48. The lines $3x+y=b_1$ and $6x+2y=b_2$ are ____. They are the same line if ____. In that case (b_1,b_2) is perpendicular to the vector ____. The nullspace of the matrix is the line $3x+y=$ ____. One particular vector in that nullspace is ____.

49. Why is each of these statements false?

(a) $(1,1,1)$ is perpendicular to $(1,1,-2)$, so the planes $x+y+z=0$ and $x+y-2z=0$ are orthogonal subspaces.

(b) The subspace spanned by $(1,1,0,0,0)$ and $(0,0,0,1,1)$ is the orthogonal complement of the subspace spanned by $(1,-1,0,0,0)$ and $(2,-2,3,4,-4)$.

(c) Two subspaces that meet only in the zero vector are orthogonal.

50. Find a matrix with $v=(1,2,3)$ in the row space and column space. Find another matrix with v in the nullspace and column space. Which pairs of subspaces can v *not* be in?

51. Suppose A is 3 by 4, B is 4 by 5, and $AB=0$. Prove $\operatorname{rank}(A)+\operatorname{rank}(B)\le 4$.

52. The command `N = null(A)` will produce a basis for the nullspace of A. Then the command `B = null(N')` will produce a basis for the ____ of A.

3.2 Cosines and Projections onto Lines

Vectors with $x^{\mathrm{T}}y=0$ are orthogonal. Now we allow inner products that are ***not zero***, and angles that are ***not right angles***. We want to connect inner products to angles, and also to transposes. In Chapter 1 the transpose was constructed by flipping over a matrix as if it were some kind of pancake. We have to do better than that.

One fact is unavoidable: *The orthogonal case is the most important*. Suppose we want to find the distance from a point b to the line in the direction of the vector a. We are looking along that line for the point p closest to b. The key is in the geometry: ***The line connecting*** b ***to*** p (the dotted line in Figure 3.5) ***is perpendicular to*** a. *This fact will allow us to find the projection* p. Even though a and b are not orthogonal, the distance problem automatically brings in orthogonality.

The situation is the same when we are given a plane (or any subspace **S**) instead of a line. Again the problem is to find the point p on that subspace that is closest to b. ***This***

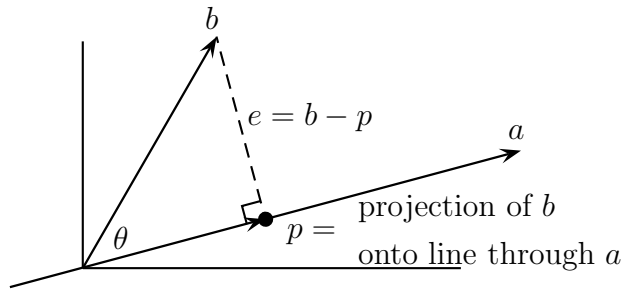


Figure 3.5: The projection p is the point (on the line through a) closest to b.

point p is the projection of b onto the subspace. A perpendicular line from b to $\mathbf{S}$ meets the subspace at p. Geometrically, that gives the distance between points b and subspaces $\mathbf{S}$. But there are two questions that need to be asked:

1. Does this projection actually arise in practical applications?

2. If we have a basis for the subspace $\mathbf{S}$, is there a formula for the projection p?

The answers are certainly yes. This is exactly the problem of the ***least-squares solution to an overdetermined system***. The vector b represents the data from experiments or questionnaires, and it contains too many errors to be found in the subspace $\mathbf{S}$. When we try to write b as a combination of the basis vectors for $\mathbf{S}$, it cannot be done—the equations are inconsistent, and $Ax = b$ has no solution.

The least-squares method selects p as the best choice to replace b. There can be no doubt of the importance of this application. In economics and statistics, least squares enters *regression analysis*. In geodesy, the U.S. mapping survey tackled 2.5 million equations in 400,000 unknowns.

A formula for p is easy when the subspace is a line. We will project b onto a in several different ways, and relate the projection p to inner products and angles. Projection onto a higher dimensional subspace is by far the most important case; it corresponds to a least-squares problem with several parameters, and it is solved in Section 3.3. The formulas are even simpler when we produce an orthogonal basis for $\mathbf{S}$.

inner products and cosines

We pick up the discussion of inner products and angles. You will soon see that it is not the angle, but ***the cosine of the angle***, that is directly related to inner products. We look back to trigonometry in the two-dimensional case to find that relationship. Suppose the vectors a and b make angles α and β with the x-axis (Figure 3.6). The length $\|a\|$ is the hypotenuse in the triangle OaQ. So the sine and cosine of α are

$$\sin\alpha = \frac{a_2}{\|a\|}, \qquad \cos\alpha = \frac{a_1}{\|a\|}.$$

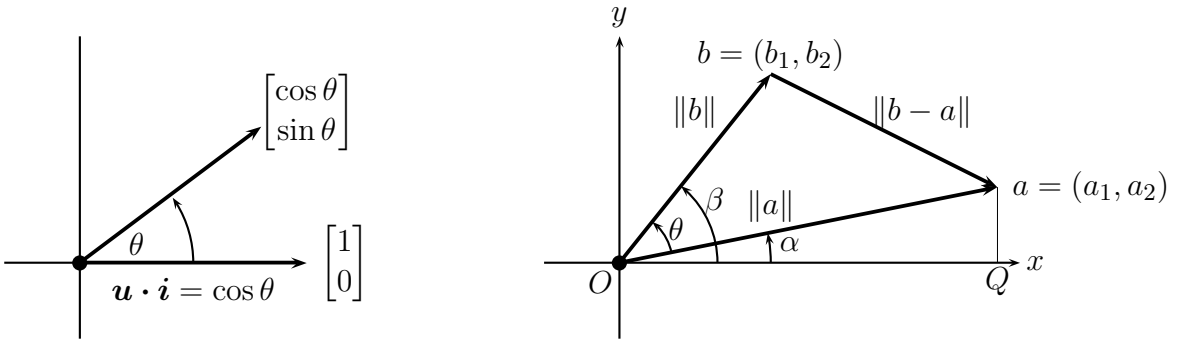


Figure 3.6: The cosine of the angle $\theta = \beta - \alpha$ using inner products.

For the angle β, the sine is $b_2/\|b\|$ and the cosine is $b_1/\|b\|$. The cosine of $\theta = \beta - \alpha$ comes from an identity that no one could forget:

$$\textbf{Cosine formula} \qquad \cos\theta = \cos\beta\cos\alpha + \sin\beta\sin\alpha = \frac{a_1b_1 + a_2b_2}{\|a\|\|b\|}. \tag{1}$$

The numerator in this formula is exactly the inner product of a and b. It gives the relationship between $a^{\mathrm{T}}b$ and $\cos\theta$:

3G The cosine of the angle between any nonzero vectors a and b is

$$\textbf{Cosine of } \theta \qquad \cos\theta = \frac{a^{\mathrm{T}}b}{\|a\|\|b\|}. \tag{2}$$

This formula is dimensionally correct; if we double the length of b, then both numerator and denominator are doubled, and the cosine is unchanged. Reversing the sign of b, on the other hand, reverses the sign of $\cos\theta$—and changes the angle by 180°.

There is another law of trigonometry that leads directly to the same result. It is not so unforgettable as the formula in equation (1), but it relates the lengths of the sides of any triangle:

$$\textbf{Law of Cosines} \qquad \|b-a\|^2 = \|b\|^2 + \|a\|^2 - 2\|b\|\|a\|\cos\theta. \tag{3}$$

When θ is a right angle, we are back to Pythagoras: $\|b-a\|^2 = \|b\|^2 + \|a\|^2$. For any angle θ, the expression $\|b-a\|^2$ is $(b-a)^{\mathrm{T}}(b-a)$, and equation (3) becomes

$$b^{\mathrm{T}}b - 2a^{\mathrm{T}}b + a^{\mathrm{T}}a = b^{\mathrm{T}}b + a^{\mathrm{T}}a - 2\|b\|\|a\|\cos\theta.$$

Canceling $b^{\mathrm{T}}b$ and $a^{\mathrm{T}}a$ on both sides of this equation, you recognize formula (2) for the cosine: $a^{\mathrm{T}}b = \|a\|\|b\|\cos\theta$. In fact, this proves the cosine formula in n dimensions, since we only have to worry about the plane triangle Oab.

Projection onto a Line

Now we want to find the projection point p. This point must be some multiple $p = \widehat{x}a$ of the given vector a—every point on the line is a multiple of a. The problem is to compute

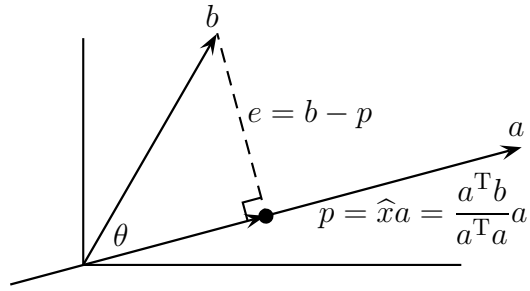


Figure 3.7: The projection p of b onto a, with $\cos\theta = \dfrac{Op}{Ob} = \dfrac{a^{\mathrm{T}}b}{\|a\|\|b\|}$.

the coefficient $\widehat{x}$. All we need is the geometrical fact that ***the line from*** b ***to the closest point*** $p = \widehat{x}a$ ***is perpendicular to the vector*** a:

$$(b-\widehat{a})\perp a, \quad \text{or} \quad a^{\mathrm{T}}(b-\widehat{a}) = 0, \quad \text{or} \quad \widehat{x} = \frac{a^{\mathrm{T}}b}{a^{\mathrm{T}}a}. \tag{4}$$

That gives the formula for the number $\widehat{x}$ and the projection p:

3H The projection of the vector b onto the line in the direction of a is $p = \widehat{x}a$:

$$\textbf{Projection onto a line} \qquad p = \widehat{x}a = \frac{a^{\mathrm{T}}b}{a^{\mathrm{T}}a}a. \tag{5}$$

This allows us to redraw Figure 3.5 with a correct formula for p (Figure 3.7).

This leads to the Schwarz inequality in equation (6), which is the most important inequality in mathematics. A special case is the fact that arithmetic means $\frac{1}{2}(x+y)$ are larger than geometric means $\sqrt{xy}$. (It is also equivalent—see Problem 1 at the end of this section—to the triangle inequality for vectors.) The Schwarz inequality seems to come almost accidentally from the statement that $\|e\|^2 = \|b-p\|^2$ in Figure 3.7 cannot be negative:

$$\left\| b - \frac{a^{\mathrm{T}}b}{a^{\mathrm{T}}a}a \right\|^2 = b^{\mathrm{T}}b - 2\frac{(a^{\mathrm{T}}b)^2}{a^{\mathrm{T}}a} + \left(\frac{a^{\mathrm{T}}b}{a^{\mathrm{T}}a}\right)^2 a^{\mathrm{T}}a = \frac{(b^{\mathrm{T}}b)(a^{\mathrm{T}}a) - (a^{\mathrm{T}}b)^2}{(a^{\mathrm{T}}a)} \geq 0.$$

This tells us that $(b^{\mathrm{T}}b)(a^{\mathrm{T}}a) \geq (a^{\mathrm{T}}b)^2$—and then we take square roots:

3I All vectors a and b satisfy the ***Schwarz inequality***, which is $|\cos\theta| \leq 1$ in $\mathbf{R}^n$:

$$\textbf{Schwarz inequality} \qquad |a^{\mathrm{T}}b| \leq \|a\|\|b\|. \tag{6}$$

According to formula (2), the ratio between $a^{\mathrm{T}}b$ and $\|a\|\|b\|$ is exactly $|\cos\theta|$. Since all cosines lie in the interval $-1 \leq \cos\theta \leq 1$, this gives another proof of equation (6): *the Schwarz inequality is the same as* $|\cos\theta| \leq 1$. In some ways that is a more easily understood proof, because cosines are so familiar. Either proof is all right in $\mathbf{R}^n$, but

notice that ours came directly from the calculation of $\|b-p\|^2$. This stays nonnegative when we introduce new possibilities for the lengths and inner products. The name of Cauchy is also attached to this inequality $|a^{\mathrm{T}}b| \le \|a\|\|b\|$, and the Russians refer to it as the Cauchy-Schwarz-Buniakowsky inequality! Mathematical historians seem to agree that Buniakowsky's claim is genuine.

One final observation about $|a^{\mathrm{T}}b| \le \|a\|\|b\|$. *Equality holds if and only if b is a multiple of a.* The angle is $\theta = 0^\circ$ or $\theta = 180^\circ$ and the cosine is 1 or -1. In this case b is identical with its projection p, and the distance between b and the line is zero.

Example 1. Project $b = (1,2,3)$ onto the line through $a = (1,1,1)$ to get $\widehat{x}$ and p:

$$\widehat{x} = \frac{a^{\mathrm{T}}b}{a^{\mathrm{T}}a} = \frac{6}{3} = 2.$$

The projection is $p = \widehat{x}a = (2,2,2)$. The angle between a and b has

$$\cos\theta = \frac{\|p\|}{\|b\|} = \frac{\sqrt{12}}{\sqrt{14}} \qquad \text{and also} \quad \cos\theta = \frac{a^{\mathrm{T}}b}{\|a\|\|b\|} = \frac{6}{\sqrt{3}\sqrt{14}}.$$

The Schwarz inequality $|a^{\mathrm{T}}b| \le \|a\|\|b\|$ is $6 \le \sqrt{3}\sqrt{14}$. If we write 6 as $\sqrt{36}$, that is the same as $\sqrt{36} \le \sqrt{42}$. The cosine is less than 1, because b is not parallel to a.

Projection Matrix of Rank 1

The projection of b onto the line through a lies at $p = a(a^{\mathrm{T}}b/a^{\mathrm{T}}a)$. That is our formula $p = \widehat{x}a$, but it is written with a slight twist: The vector a is put before the number $\widehat{x} = a^{\mathrm{T}}b/a^{\mathrm{T}}a$. There is a reason behind that apparently trivial change. Projection onto a line is carried out by a ***projection matrix*** P, and written in this new order we can see what it is. *P is the matrix that multiplies b and produces p*:

$$P = a\frac{a^{\mathrm{T}}b}{a^{\mathrm{T}}a} \quad \text{so the projection matrix is} \quad P = \frac{aa^{\mathrm{T}}}{a^{\mathrm{T}}a}. \tag{7}$$

That is a column times a row—a square matrix—divided by the number $a^{\mathrm{T}}a$.

Example 2. The matrix that projects onto the line through $a = (1,1,1)$ is

$$P = \frac{aa^{\mathrm{T}}}{a^{\mathrm{T}}a} = \frac{1}{3}\begin{bmatrix}1\\1\\1\end{bmatrix}\begin{bmatrix}1 & 1 & 1\end{bmatrix} = \begin{bmatrix}\frac{1}{3} & \frac{1}{3} & \frac{1}{3}\\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3}\\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3}\end{bmatrix}.$$

This matrix has two properties that we will see as typical of projections:

1. P is a symmetric matrix.
2. Its square is itself: $P^2 = P$.

P^2b is the projection of Pb—and Pb is already on the line! So $P^2b = Pb$. This matrix P also gives a great example of the four fundamental subspaces:

The column space consists of the line through $a = (1,1,1)$.

The nullspace consists of the plane perpendicular to a.

The rank is $r = 1$.

Every column is a multiple of a, and so is $Pb = \widehat{x}a$. The vectors that project to $p = 0$ are especially important. They satisfy $a^{\mathrm{T}}b = 0$—they are perpendicular to a and their component along the line is zero. They lie in the nullspace = perpendicular plane.

Actually that example is too perfect. It has the nullspace orthogonal to the column space, which is haywire. The nullspace should be orthogonal to the *row space*. But because P is symmetric, its row and column spaces are the same.

Remark on scaling The projection matrix $aa^{\mathrm{T}}/a^{\mathrm{T}}a$ is the same if a is doubled:

$$a = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} \quad \text{gives} \quad P = \frac{1}{12}\begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}\begin{bmatrix} 2 & 2 & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix} \quad \text{as before.}$$

The line through a is the same, and that's all the projection matrix cares about. If a has unit length, the denominator is $a^{\mathrm{T}}a = 1$ and the matrix is just $P = aa^{\mathrm{T}}$.

Example 3. Project onto the "θ-direction" in the x-y plane. The line goes through $a = (\cos\theta, \sin\theta)$ and the matrix is symmetric with $P^2 = P$:

$$P = \frac{aa^{\mathrm{T}}}{a^{\mathrm{T}}a} = \frac{\begin{bmatrix} c \\ s \end{bmatrix}\begin{bmatrix} c & s \end{bmatrix}}{\begin{bmatrix} c & s \end{bmatrix}\begin{bmatrix} c \\ s \end{bmatrix}} = \begin{bmatrix} c^2 & cs \\ cs & s^2 \end{bmatrix}.$$

Here c is $\cos\theta$, s is $\sin\theta$, and $c^2 + s^2 = 1$ in the denominator. This matrix P was discovered in Section 2.6 on linear transformations. Now we know P in any number of dimensions. We emphasize that it produces the projection p:

> ***To project b onto a, multiply by the projection matrix P: $p = Pb$.***

Transposes from Inner Products

Finally we connect inner products to A^{T}. Up to now, A^{T} is simply the reflection of A across its main diagonal; the rows of A become the columns of A^{T}, and vice versa. The entry in row i, column j of A^{T} is the (j,i) entry of A:

$$\textbf{Transpose by reflection} \qquad A^{\mathrm{T}}_{ij} = (A)_{ji}.$$

There is a deeper significance to A^{T}, Its close connection to inner products gives a new and much more "abstract" definition of the transpose:

3J The transpose A^{T} can be defined by the following property: The inner product of Ax with y equals the inner product of x with $A^{\mathrm{T}}y$. Formally, this simply means that

$$(Ax)^{\mathrm{T}}y = x^{\mathrm{T}}A^{\mathrm{T}}y = x^{\mathrm{T}}(A^{\mathrm{T}}y). \tag{8}$$

This definition gives us another (better) way to verify the formula $(AB)^{\mathrm{T}} = B^{\mathrm{T}}A^{\mathrm{T}}$, Use equation (8) twice:

Move A then move B $\qquad (ABx)^{\mathrm{T}}y = (Bx)^{\mathrm{T}}(A^{\mathrm{T}}Y) = x^{\mathrm{T}}(B^{\mathrm{T}}A^{\mathrm{T}}y).$

The transposes turn up in reverse order on the right side, just as the inverses do in the formula $(AB)^{-1} = B^{-1}A^{-1}$. We mention again that these two formulas meet to give the remarkable combination $(A^{-1})^{\mathrm{T}} = (A^{\mathrm{T}})^{-1}$.

Problem Set 3.2

1. (a) Given any two positive numbers x and y, choose the vector b equal to $(\sqrt{x}, \sqrt{y})$, and choose $a = (\sqrt{y}, \sqrt{x})$. Apply the Schwarz inequality to compare the arithmetic mean $\frac{1}{2}(x+y)$ with the geometric mean $\sqrt{xy}$.

(b) Suppose we start with a vector from the origin to the point x, and then add a vector of length $\|y\|$ connecting x to $x+y$. The third side of the triangle goes from the origin to $x+y$. *The triangle inequality asserts that this distance cannot be greater than the sum of the first two*:

$$\|x+y\| \le \|x\| + \|y\|.$$

After squaring both sides, and expanding $(x+y)^{\mathrm{T}}(x+y)$, reduce this to the Schwarz inequality.

2. Verify that the length of the projection in Figure 3.7 is $\|p\| = \|b\| \cos\theta$, using formula (5).

3. What multiple of $a = (1,1,1)$ is closest to the point $b = (2,4,4)$? Find also the point closest to a on the line through b.

4. Explain why the Schwarz inequality becomes an equality in the case that a and b lie on the same line through the origin, and only in that case. What if they lie on opposite sides of the origin?

5. In n dimensions, what angle does the vector $(1,1,\ldots,1)$ make with the coordinate axes? What is the projection matrix P onto that vector?

6. The Schwarz inequality has a one-line proof if a and b are normalized ahead of time to be unit vectors:

$$|a^{\mathrm{T}}b| = \left|\sum a_j b_j\right| \le \sum |a_j||b_j| \le \sum \frac{|a_j|^2 + |b_j|^2}{2} = \frac{1}{2} + \frac{1}{2} = \|a\|\|b\|.$$

Which previous problem justifies the middle step?

7. By choosing the correct vector b in the Schwarz inequality, prove that

$$(a_1 + \cdots + a_n)^2 \le n(a_1^2 + \cdots + a_n^2).$$

When does equality hold?

8. The methane molecule CH_4 is arranged as if the carbon atom were at the center of a regular tetrahedron with four hydrogen atoms at the vertices. If vertices are placed at $(0,0,0)$, $(1,1,0)$, $(1,0,1)$, and $(0,1,1)$—note that all six edges have length $\sqrt{2}$, so the tetrahedron is regular—what is the cosine of the angle between the rays going from the center $(\frac{1}{2},\frac{1}{2},\frac{1}{2})$ to the vertices? (The bond angle itself is about 109.5°, an old friend of chemists.)

9. Square the matrix $P = aa^{\mathrm{T}}/a^{\mathrm{T}}a$, which projects onto a line, and show that $P^2 = P$. (Note the number $a^{\mathrm{T}}a$ in the middle of the matrix $aa^{\mathrm{T}}aa^{\mathrm{T}}$!)

10. Is the projection matrix P invertible? Why or why not?

11. (a) Find the projection matrix P_1 onto the line through $a = [\begin{smallmatrix}1\\3\end{smallmatrix}]$ and also the matrix P_2 that projects onto the line perpendicular to a.

(b) Compute $P_1 + P_2$ and P_1P_2 and explain.

12. Find the matrix that projects every point in the plane onto the line $x + 2y = 0$.

13. Prove that the *trace* of $P = aa^{\mathrm{T}}/a^{\mathrm{T}}a$—which is the sum of its diagonal entries—always equals 1.

14. What matrix P projects every point in $\mathbf{R}^3$ onto the line of intersection of the planes $x + y + t = 0$ and $x - t = 0$?

15. Show that the length of Ax equals the length of $A^{\mathrm{T}}x$ if $AA^{\mathrm{T}} = A^{\mathrm{T}}A$.

16. Suppose P is the projection matrix onto the line through a.

(a) Why is the inner product of x with Py equal to the inner product of Px with y?

(b) Are the two angles the same? Find their cosines if $a = (1,1,-1)$, $x = (2,0,1)$, $y = (2,1,2)$.

(c) Why is the inner product of Px with Py again the same? What is the angle between those two?

Problems 17–26 ask for projections onto lines. Also errors $e = b - p$ and matrices P.

17. Project the vector b onto the line through a. Check that e is perpendicular to a:

(a) $b=\begin{bmatrix}1\\2\\2\end{bmatrix}$ and $a=\begin{bmatrix}1\\1\\1\end{bmatrix}$. (b) $b=\begin{bmatrix}1\\3\\1\end{bmatrix}$ and $a=\begin{bmatrix}-1\\-3\\-1\end{bmatrix}$.

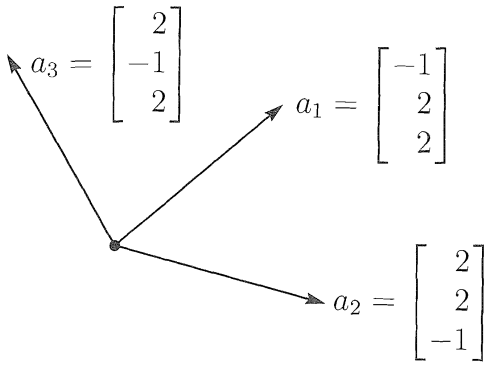
18. *Draw* the projection of b onto a and also compute it from $p=\widehat{x}a$:

(a) $b=\begin{bmatrix}\cos\theta\\\sin\theta\end{bmatrix}$ and $a=\begin{bmatrix}1\\0\end{bmatrix}$. (b) $b=\begin{bmatrix}1\\1\end{bmatrix}$ and $a=\begin{bmatrix}1\\-1\end{bmatrix}$.

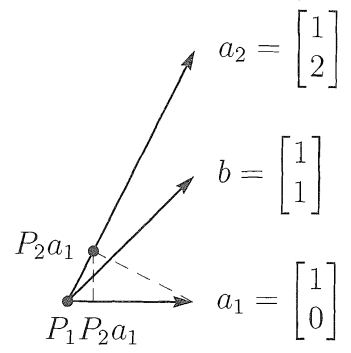
19. In Problem 17, find the projection matrix $P=aa^{\mathrm{T}}/a^{\mathrm{T}}a$ onto the line through each vector a. Verify in both cases that $P^2=P$. Multiply Pb in each case to compute the projection p.

20. Construct the projection matrices P_1 and P_2 onto the lines through the a's in Problem 18. Is it true that $(P_1+P_2)^2=P_1+P_2$? This *would* be true if $P_1P_2=0$.

For Problems 21–26, consult the accompanying figures.



Problems 21–23



Problems 24–26

21. Compute the projection matrices $aa^{\mathrm{T}}/a^{\mathrm{T}}a$ onto the lines through $a_1=(-1,2,2)$ and $a_2=(2,2,-1)$, Multiply those projection matrices and explain why their product P_1P_2 is what it is.

22. Project $b=(1,0,0)$ onto the lines through a_1 and a_2 in Problem 21 and also onto $a_3=(2,-1,2)$. Add the three projections $p_1+p_2+p_3$.

23. Continuing Problems 21–22, find the projection matrix P_3 onto $a_3=(2,-1,2)$. Verify that $P_1+P_2+P_3=I$. The basis a_1, a_2, a_3 is orthogonal!

24. Project the vector $b=(1,1)$ onto the lines through $a_1=(1,0)$ and $a_2=(1,2)$. Draw the projections p_1 and p_2 and add p_1+p_2. The projections do not add to b because the a's are not orthogonal.

25. In Problem 24, the projection of b onto the *plane* of a_1 and a_2 will equal b. Find $P=A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$ for $A=[a_1\ \ a_2]\left[\begin{smallmatrix}1&1\\0&2\end{smallmatrix}\right]$.

26. Project $a_1 = (1,0)$ onto $a_2 = (1,2)$. Then project the result back onto a_1. Draw these projections and multiply the projection matrices P_1P_2: Is this a projection?

3.3 Projections and Least Squares

Up to this point, $Ax = b$ either has a solution or not. If b is not in the column space $\boldsymbol{C}(A)$, the system is inconsistent and Gaussian elimination fails. This failure is almost certain when there are several equations and only one unknown:

$$\begin{array}{rcl} \textbf{More equations} & 2x & = & b_1 \\ \textbf{than unknowns—} & 3x & = & b_2 \\ \textbf{no solution?} & 4x & = & b_3. \end{array}$$

This is solvable when b_1, b_2, b_3 are in the ratio 2:3:4. The solution x will exist only if b is on the same line as the column $a = (2,3,4)$.

In spite of their unsolvability, inconsistent equations arise all the time in practice. They have to be solved! One possibility is to determine x from part of the system, and ignore the rest; this is hard to justify if all m equations come from the same source. Rather than expecting no error in some equations and large errors in the others, it is much better *to choose the x that minimizes an average error E in the m equations.*

The most convenient "average" comes from the *sum of squares*:

$$\textbf{Squared error} \qquad E^2 = (2x - b_1)^2 + (3x - b_2)^2 + (4x - b_3)^2.$$

If there is an exact solution, the minimum error is $E = 0$. In the more likely case that b is not proportional to a, the graph of E^2 will be a parabola. The minimum error is at the lowest point, where the derivative is zero:

$$\frac{dE^2}{dx} = 2\big[(2x - b_1)2 + (3x - b_2)3 + (4x - b_3)4\big] = 0.$$

Solving for x, the least-squares solution of this model system $ax = b$ is denoted by $\widehat{x}$:

$$\textbf{Leastsquares solution} \qquad \widetilde{x} = \frac{2b_1 + 3b_2 + 4b_3}{2^2 + 3^2 + 4^2} = \frac{a^{\mathrm{T}}b}{a^{\mathrm{T}}a}.$$

You recognize $a^{\mathrm{T}}b$ in the numerator and $a^{\mathrm{T}}a$ in the denominator.

The general case is the same. We "solve" $ax = b$ by minimizing

$$E^2 = \|ax - b\|^2 = (a_1x - b_1)^2 + \cdots + (a_mx - b_m)^2.$$

The derivative of E^2 is zero at the point $\widehat{x}$, if

$$(a_1\widehat{x} - b_1)a_1 + \cdots + (a_m\widehat{x} - b_m)a_m = 0.$$

We are minimizing the distance from b to the line through a, and calculus gives the same answer, $\widehat{x} = (a_1b_1 + \cdots + a_mb_m)/(a_1^2 + \cdots + a_m^2)$, that geometry did earlier:

3K The least-squares solution to a problem $ax = b$ in one unknown is $\widehat{x} = \dfrac{a^{\mathrm{T}}b}{a^{\mathrm{T}}a}$.

You see that we keep coming back to the geometrical interpretation of a least-squares problem—to minimize a distance. By setting the derivative of E^2 to zero, calculus confirms the geometry of the previous section. *The error vector e connecting b to p must be perpendicular to a*:

Orthogonality of a and e $$a^{\mathrm{T}}(b - \widehat{x}a) = a^{\mathrm{T}}b - \frac{a^{\mathrm{T}}b}{a^{\mathrm{T}}a}a^{\mathrm{T}}a = 0.$$

As a side remark, notice the degenerate case $a = 0$. All multiples of a are zero, and the line is only a point. Therefore $p = 0$ is the only candidate for the projection. But the formula for $\widehat{x}$ becomes a meaningless $0/0$, and correctly reflects the fact that $\widehat{x}$ is completely undetermined. All values of x give the same error $E = \|0x - b\|$, so E^2 is a horizontal line instead of a parabola. The "pseudoinverse" assigns the definite value $\widehat{x} = 0$, which is a more "symmetric" choice than any other number.

Least Squares Problems with Several Variables

Now we are ready for the serious step, ***to project b onto a subspace***—rather than just onto a line. This problem arises from $Ax = b$ when A is an m by n matrix. Instead of one column and one unknown x, the matrix now has n columns. The number m of observations is still larger than the number n of unknowns, so it must be expected that $Ax = b$ will be inconsistent. *Probably, there will not exist a choice of x that perfectly fits the data b*. In other words, the vector b probably will not be a combination of the columns of A; it will be outside the column space.

Again the problem is to choose $\widehat{x}$ so as to minimize the error, and again this minimization will be done in the least-squares sense. The error is $E = \|Ax - b\|$, and ***this is exactly the distance from b to the point Ax in the column space***. Searching for the least-squares solution $\widehat{x}$, which minimizes E, is the same as locating the point $p = A\widehat{x}$ that is closer to b than any other point in the column space.

We may use geometry or calculus to determine $\widehat{x}$. In n dimensions, we prefer the appeal of geometry; p must be the "projection of b onto the column space." ***The error vector $e = b - A\widehat{x}$ must be perpendicular to that space*** (Figure 3.8). Finding $\widehat{x}$ and the projection $p = A\widehat{x}$ is so fundamental that we do it in two ways:

1. All vectors perpendicular to the column space lie in the *left nullspace*. Thus the error vector $e = b - A\widehat{x}$ must be in the nullspace of A^{T}:

$$A^{\mathrm{T}}(b - A\widehat{x}) = 0 \qquad \text{or} \qquad A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}b.$$

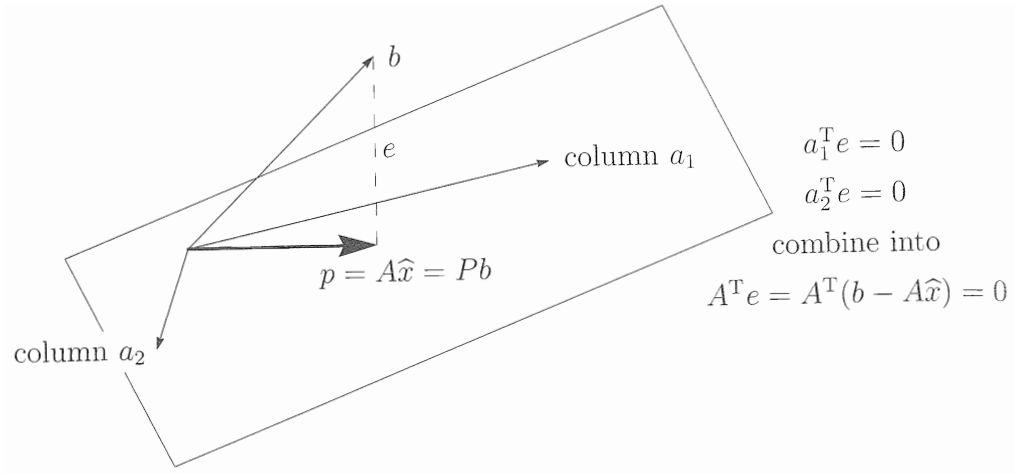


Figure 3.8: Projection onto the column space of a 3 by 2 matrix.

2. The error vector must be perpendicular to *each column* $a_1, \ldots, a_n$ of A:

$$\begin{matrix} a_1^{\mathrm{T}}(b - A\widehat{x}) = 0 \\ \vdots \\ a_n^{\mathrm{T}}(b - A\widehat{x}) = 0 \end{matrix} \qquad \text{or} \qquad \begin{bmatrix} a_1^{\mathrm{T}} \\ \vdots \\ a_n^{\mathrm{T}} \end{bmatrix} \begin{bmatrix} \\ b - A\widehat{x} \\ \\ \end{bmatrix} = 0.$$

This is again $A^{\mathrm{T}}(b - A\widehat{x}) = 0$ and $A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}b$, The calculus way is to take partial derivatives of $E^2 = (Ax - b)^{\mathrm{T}}(Ax - b)$. That gives the same $2A^{\mathrm{T}}Ax - 2A^{\mathrm{T}}b = 0$. The fastest way is just ***to multiply the unsolvable equation*** $Ax = b$ ***by*** A^{T}. All these equivalent methods produce a square coefficient matrix $A^{\mathrm{T}}A$. It is symmetric (its transpose is not AA^{T}!) and it is the fundamental matrix of this chapter.

The equations $A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}b$ are known in statistics as the ***normal equations***.

3L When $Ax = b$ is inconsistent, its least-squares solution minimizes $\|Ax - b\|^2$:

$$\textbf{Normal equations} \qquad A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}b. \tag{1}$$

$A^{\mathrm{T}}A$ is invertible exactly when the columns of A are linearly independent! Then,

$$\textbf{Best estimate } \widehat{x} \qquad \widehat{x} = (A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}b. \tag{2}$$

The projection of b onto the column space is the nearest point $A\widehat{x}$:

$$\textbf{Projection} \qquad p = A\widehat{x} = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}b. \tag{3}$$

We choose an example in which our intuition is as good as the formulas:

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 0 & 0 \end{bmatrix}, \qquad b = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \qquad \begin{matrix} Ax = b \quad \text{has no solution} \\ A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}b \quad \text{gives the best } x. \end{matrix}$$

Both columns end with a zero, so $C(A)$ is the x-y plane within three-dimensional space The projection of $b = (4,5,6)$ is $p = (4,5,0)$—the x and y components stay the same but $z = 6$ will disappear. That is confirmed by solving the normal equations:

$$A^{\mathrm{T}}A = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 3 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ 5 & 13 \end{bmatrix}.$$

$$\widehat{x} = (A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}b = \begin{bmatrix} 13 & -5 \\ -5 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 2 & 3 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

$$\textbf{Projection} \qquad p = A\widehat{x} = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 0 \end{bmatrix}.$$

In this special case, the best we can do is to solve the first two equations of $Ax = b$. Then $\widehat{x}_1 = 2$ and $\widehat{x}_2 = 1$. The error in the equation $0x_1 + 0x_2 = 6$ is sure to be 6.

***Remark* 4.** Suppose b is actually in the column space of A—it is a combination $b = Ax$ of the columns. Then the projection of b is still b:

$$b \textbf{ in column space} \qquad p = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}Ax = Ax = b.$$

The closest point p is just b itself—which is obvious.

***Remark* 5.** At the other extreme, suppose b is *perpendicular* to every column, so $A^{\mathrm{T}}b = 0$. In this case b projects to the zero vector:

$$b \textbf{ in left nullspace} \qquad p = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}b = A(A^{\mathrm{T}}A)^{-1}0 = 0.$$

***Remark* 6.** When A is square and invertible, the column space is the whole space. Every vector projects to itself, p equals b, and $\widehat{x} = x$:

$$\textbf{If } A \textbf{ is invertible} \qquad p = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}b = AA^{-1}(A^{\mathrm{T}})^{-1}A^{\mathrm{T}}b = b.$$

This is the only case when we can take apart $(A^{\mathrm{T}}A)^{-1}$*, and write it as* $A^{-1}(A^{\mathrm{T}})^{-1}$. When A is rectangular that is not possible.

***Remark* 7.** Suppose A has only one column, containing a. Then the matrix $A^{\mathrm{T}}A$ is the number $a^{\mathrm{T}}a$ and $\widehat{x}$ is $a^{\mathrm{T}}b/a^{\mathrm{T}}a$. We return to the earlier formula.

The Cross-Product Matrix $A^{\mathrm{T}}A$

The matrix $A^{\mathrm{T}}A$ is certainly symmetric. Its transpose is $(A^{\mathrm{T}}A)^{\mathrm{T}} = A^{\mathrm{T}}A^{\mathrm{TT}}$, which is $A^{\mathrm{T}}A$ again. Its i, j entry (and j, i entry) is the inner product of column i of A with column j of A. The key question is the invertibility of $A^{\mathrm{T}}A$, and fortunately

$A^{\mathrm{T}}A$ has the same nullspace as A.

Certainly if $Ax = 0$ then $A^{\mathrm{T}}Ax = 0$. Vectors x in the nullspace of A are also in the nullspace of $A^{\mathrm{T}}A$. To go in the other direction, start by supposing that $A^{\mathrm{T}}Ax = 0$, and take the inner product with x to show that $Ax = 0$:

$$x^{\mathrm{T}}A^{\mathrm{T}}Ax = 0, \quad \text{or} \quad \|Ax\|^2 = 0, \quad \text{or} \quad Ax = 0.$$

The two nullspaces are identical. In particular, if A has independent columns (and only $x = 0$ is in its nullspace), then the same is true for $A^{\mathrm{T}}A$:

3M If A has independent columns, then $A^{\mathrm{T}}A$ is *square, symmetric*, and *invertible*.

We show later that $A^{\mathrm{T}}A$ is also positive definite (all pivots and eigenvalues are positive).

This case is by far the most common and most important. Independence is not so hard in m-dimensional space if $m > n$. We assume it in what follows.

Projection Matrices

We have shown that the closest point to b is $p = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}b$. *This formula expresses in matrix terms the construction of a perpendicular line from b to the column space of A*. The matrix that gives p is a projection matrix, denoted by P:

$$\textbf{Projection matrix} \qquad P = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}. \tag{4}$$

This matrix projects any vector b onto the column space of A.[1] In other words, $p = Pb$ is the component of b in the column space, and the error $e = b - Pb$ is the component in the orthogonal complement. ($I - P$ is also a projection matrix! It projects b onto the orthogonal complement, and the projection is $b - Pb$.)

In short, we have a matrix formula for splitting any b into two perpendicular components. Pb is in the column space $\boldsymbol{C}(A)$, and the other component $(I - P)b$ is in the left nullspace $\boldsymbol{N}(A^{\mathrm{T}})$—which is orthogonal to the column space.

These projection matrices can be understood geometrically and algebraically.

3N The projection matrix $P = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$ has two basic properties:

(i) It equals its square: $P^2 = P$.

(ii) It equals its transpose: $P^{\mathrm{T}} = P$.

Conversely, any symmetric matrix with $P^2 = P$ represents a projection.

[1] There may be a risk of confusion with permutation matrices, also denoted by P, but the risk should be small, and we try never to let both appear on the same page.

Proof. It is easy to see why $P^2 = P$. If we start with any b, then Pb lies in the subspace we are projecting onto. ***When we project again nothing is changed***. The vector Pb is already in the subspace, and $P(Pb)$ is still Pb. In other words $P^2 = P$. Two or three or fifty projections give the same point p as the first projection:

$$P^2 = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}} = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}} = P.$$

To prove that P is also symmetric, take its transpose. Multiply the transposes in reverse order, and use symmetry of $(A^{\mathrm{T}}A)^{-1}$, to come back to P:

$$P^{\mathrm{T}} = (A^{\mathrm{T}})^{\mathrm{T}}\left((A^{\mathrm{T}}A)^{-1}\right)^{\mathrm{T}}A^{\mathrm{T}} = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}} = P.$$

For the converse, we have to deduce from $P^2 = P$ and $P^{\mathrm{T}} = P$ that Pb **is the projection of** b **onto the column space of** P. The error vector $b - Pb$ is *orthogonal to the space*. For any vector Pc in the space, the inner product is zero:

$$(b - Pb)^{\mathrm{T}}Pc = b^{\mathrm{T}}(I - P)^{\mathrm{T}}Pc = b^{\mathrm{T}}(P - P^2)c = 0.$$

Thus $b - Pb$ is orthogonal to the space, and Pb is the projection onto the column space. □

Example 1. Suppose A is actually invertible. If it is 4 by 4, then its four columns are independent and its column space is all of $\mathbf{R}^4$. What is the projection *onto the whole space*? It is the identity matrix.

$$P = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}} = AA^{-1}(A^{\mathrm{T}})^{-1}A^{\mathrm{T}} = I. \tag{5}$$

The identity matrix is symmetric, $I^2 = I$, and the error $b - Ib$ is zero.

The point of all other examples is that what happened in equation (5) is *not allowed*. To repeat: We cannot invert the separate parts A^{T} and A when those matrices are rectangular. It is the square matrix $A^{\mathrm{T}}A$ that is invertible.

Least-Squares Fitting of Data

Suppose we do a series of experiments, and expect the output b to be a linear function of the input t. We look for a ***straight line*** $b = C + Dt$. For example:

1. At different times we measure the distance to a satellite on its way to Mars. In this case t is the time and b is the distance. Unless the motor was left on or gravity is strong, the satellite should move with nearly constant velocity v: $b = b_0 + vt$.
2. We vary the load on a structure, and measure the movement it produces. In this experiment t is the load and b is the reading from the strain gauge. Unless the load is so great that the material becomes plastic, a linear relation $b = C + Dt$ is normal in the theory of elasticity.

3. The cost of producing t books like this one is nearly linear, $b = C + Dt$, with editing and typesetting in C and then printing and binding in D. C is the set-up cost and D is the cost for each additional book.

How to compute C and D? If there is no experimental error, then two measurements of b will determine the line $b = C + Dt$. But if there is error, we must be prepared to "average" the experiments and find an optimal line. That line is not to be confused with the line through a on which b was projected in the previous section! In fact, since there are two unknowns C and D to be determined, we now project onto a *two-dimensional* subspace. A perfect experiment would give a perfect C and D:

$$\begin{array}{rcrcl} C & + & Dt_1 & = & b_1 \\ C & + & Dt_2 & = & b_2 \\ & & \vdots & & \\ C & + & Dt_m & = & b_m. \end{array} \tag{6}$$

This is an *overdetermined* system, with m equations and only two unknowns. If errors are present, it will have no solution. A has two columns, and $x = (C, D)$:

$$\begin{bmatrix} 1 & t_1 \\ 1 & t_2 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}, \qquad \text{or} \qquad Ax = b. \tag{7}$$

The best solution $(\widehat{C}, \widehat{D})$ is the $\widehat{x}$ that minimizes the squared error E^2:

$$\textbf{Minimize} \qquad E^2 = \|b - Ax\|^2 = (b_1 - C - Dt_1)^2 + \cdots + (b_m - C - Dt_m)^2.$$

The vector $p = A\widehat{x}$ is as close as possible to b. Of all straight lines $b = C + Dt$, we are choosing the one that best fits the data (Figure 3.9). On the graph, the errors are the ***vertical distances*** $b - C - Dt$ to the straight line (not perpendicular distances!). It is the vertical distances that are squared, summed, and minimized.

Example 2. Three measurements b_1, b_2, b_3 are marked on Figure 3.9a:

$$b = 1 \quad \text{at} \quad t = -1, \qquad b = 1 \quad \text{at} \quad t = 1, \qquad b = 3 \quad \text{at} \quad t = 2.$$

Note that the values $t = -1, 1, 2$ are not required to be equally spaced. The first step is *to write the equations that* ***would*** *hold if a line could go through all three points.* Then every $C + Dt$ would agree exactly with b:

$$Ax = b \quad \text{is} \quad \begin{array}{rcrcl} C & - & D & = & 1 \\ C & + & D & = & 1 \\ C & + & 2D & = & 3 \end{array} \quad \text{or} \quad \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}.$$

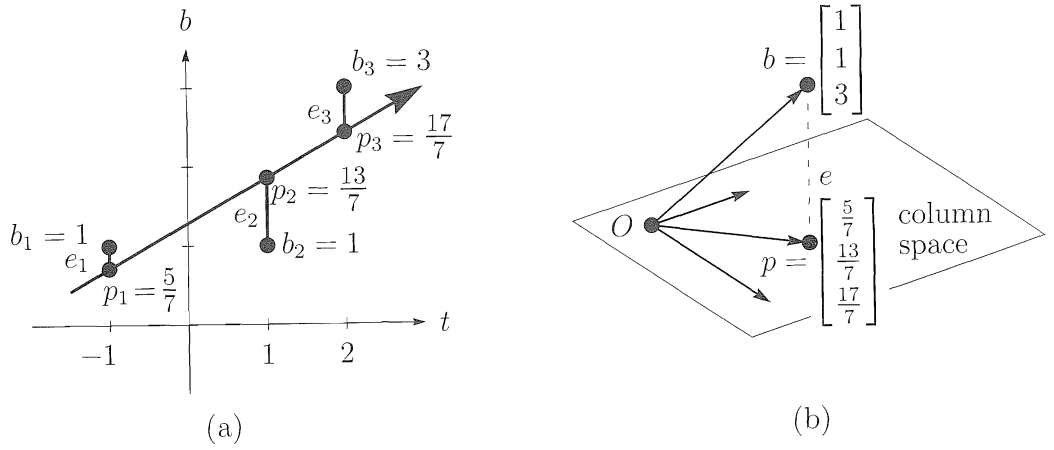


Figure 3.9: Straight-line approximation matches the projection p of b.

If those equations $Ax = b$ could be solved, there would be no errors. They can't be solved because the points are not on a line. Therefore they are solved by least squares:

$$A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}b \quad \text{is} \quad \begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} \widehat{C} \\ \widehat{D} \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

The best solution is $\widehat{C} = \frac{9}{7}$, $\widehat{D} = \frac{4}{7}$ and the best line is $\frac{9}{7} + \frac{4}{7}t$.

Note the beautiful connections between the two figures. The problem is the same but the art shows it differently. In Figure 3.9b, b is not a combination of the columns $(1,1,1)$ and $(-1,1,2)$. In Figure 3.9, the three points are not on a line. Least squares replaces points b that are not on a line by points p that are! Unable to solve $Ax = b$, we solve $A\widehat{x} = p$.

The line $\frac{9}{7} + \frac{4}{7}t$ has heights $\frac{5}{7}$, $\frac{13}{7}$, $\frac{17}{7}$ at the measurement times -1, 1, 2. ***Those points do lie on a line***. Therefore the vector $p = (\frac{5}{7}, \frac{13}{7}, \frac{17}{7})$ is in the column space. *This vector is the projection*. Figure 3.9b is in three dimensions (or m dimensions if there are m points) and Figure 3.9a is in two dimensions (or n dimensions if there are n parameters).

Subtracting p from b, the errors are $e = (\frac{2}{7}, -\frac{6}{7}, \frac{4}{7})$. Those are the vertical errors in Figure 3.9a, and they are the components of the dashed vector in Figure 3.9b. This error vector is orthogonal to the first column $(1,1,1)$, since $-\frac{2}{7} - \frac{6}{7} + \frac{4}{7} = 0$. It is orthogonal to the second column $(-1,1,2)$, because $-\frac{2}{7} - \frac{6}{7} + \frac{8}{7} = 0$. It is orthogonal to the column space, and it is in the left nullspace.

Question: If the measurements $b = (\frac{2}{7}, -\frac{6}{7}, \frac{4}{7})$ were those errors, what would be the best line and the best $\widehat{x}$? Answer: The zero line—which is the horizontal axis—and $\widehat{x} = 0$. Projection to zero.

We can quickly summarize the equations for fitting by a straight line. The first column of A contains 1s, and the second column contains the times t_i. Therefore $A^{\mathrm{T}}A$ contains the sum of the 1s and the t_i and the t_i^2:

3O The measurements $b_1,\dots,b_m$ are given at distinct points $t_1,\dots,t_m$. Then the straight line $\widehat{C}+\widehat{D}t$ which minimizes E^2 comes from least squares:

$$A^{\mathrm{T}}A\begin{bmatrix}\widehat{C}\\ \widehat{D}\end{bmatrix}=A^{\mathrm{T}}b \qquad \text{or} \qquad \begin{bmatrix}m & \sum t_i\\ \sum t_i & \sum t_i^2\end{bmatrix}\begin{bmatrix}\widehat{C}\\ \widehat{D}\end{bmatrix}=\begin{bmatrix}\sum b_i\\ \sum t_i b_i\end{bmatrix}.$$

Remark. The mathematics of least squares is not limited to fitting the data by straight lines. In many experiments there is no reason to expect a linear relationship, and it would be crazy to look for one. Suppose we are handed some radioactive material, The output b will be the reading on a Geiger counter at various times t. We may know that we are holding a mixture of two chemicals, and we may know their half-lives (or rates of decay), but we do not know how much of each is in our hands. If these two unknown amounts are C and D, then the Geiger counter readings would behave like the sum of two exponentials (and not like a straight line):

$$b=Ce^{-\lambda t}+De^{-\mu t}. \tag{8}$$

In practice, the Geiger counter is not exact. Instead, we make readings $b_1,\dots,b_m$ at times $t_1,\dots,t_m$, and equation (8) is approximately satisfied:

$$Ax=b \quad \text{is} \quad \begin{array}{rcl} Ce^{-\lambda t_1} + De^{-\mu t_1} &\approx& b_1\\ \vdots & & \\ Ce^{-\lambda t_m} + De^{-\mu t_m} &\approx& b_m. \end{array}$$

If there are more than two readings, $m>2$, then in all likelihood we cannot solve for C and D. But the least-squares principle will give optimal values $\widehat{C}$ and $\widehat{D}$.

The situation would be completely different if we knew the amounts C and D, and were trying to discover the decay rates λ and μ. This is a problem in *nonlinear least squares*, and it is harder. We would still form E^2, the sum of the squares of the errors, and minimize it. But setting its derivatives to zero will not give linear equations for the optimal λ and μ. In the exercises, we stay with linear least squares.

Weighted Least Squares

A simple least-squares problem is the estimate $\widehat{x}$ of a patient's weight from two observations $x=b_1$ and $x=b_2$. Unless $b_1=b_2$, we are faced with an inconsistent system of two equations in one unknown:

$$\begin{bmatrix}1\\1\end{bmatrix}\begin{bmatrix}x\end{bmatrix}=\begin{bmatrix}b_1\\b_2\end{bmatrix}.$$

Up to now, we accepted b_1 and b_2 as equally reliable. We looked for the value $\widehat{x}$ that minimized $E^2=(x-b_1)^2+(x-b_2)^2$:

$$\frac{dE^2}{dx}=0 \qquad \text{at} \qquad \widehat{x}=\frac{b_1+b_2}{2}.$$

The optimal $\widehat{x}$ is the average. The same conclusion comes from $A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}b$. In fact $A^{\mathrm{T}}A$ is a 1 by 1 matrix, and the normal equation is $2\widehat{x} = b_1 + b_2$.

Now suppose the two observations are not trusted to the same degree. The value $x = b_1$ may be obtained from a more accurate scale—or, in a statistical problem, from a larger sample—than $x = b_2$. Nevertheless, if b_2 contains some information, we are not willing to rely totally on b_1. The simplest compromise is to attach different weights w_1^2 and w_2^2, and choose the $\widehat{x}_W$ that minimizes the *weighted sum of squares*:

$$\textbf{Weighted error} \qquad E^2 = w_1^2(x-b_1)^2 + w_2^2(x-b_2)^2.$$

If $w_1 > w_2$, more importance is attached to b_1. The minimizing process (derivative $= 0$) tries harder to make $(x-b_1)^2$ small:

$$\frac{dE^2}{dx} = 2\left[w_1^2(x-b_1) + w_2^2(x-b_2)\right] = 0 \qquad \text{at} \qquad \widehat{x}_W = \frac{w_1^2 b_1 + w_2^2 b_2}{w_1^2 + w_2^2}. \tag{9}$$

Instead of the average of b_1 and b_2 (for $w_1 = w_2 = 1$), $\widehat{x}_W$ is a ***weighted average*** of the data. This average is closer to b_1 than to b_2.

The ordinary least-squares problem leading to $\widehat{x}_W$ comes from changing $Ax = b$ to the new system $WAx = Wb$. ***This changes the solution from $\widehat{x}$ to $\widehat{x}_W$.*** The matrix $W^{\mathrm{T}}W$ turns up on both sides of the weighted normal equations:

The least squares solution to $WAx = Wb$ is $\widehat{x}_W$:

$$\textbf{Weighted normal equations} \qquad (A^{\mathrm{T}}W^{\mathrm{T}}WA)\widehat{x}_W = A^{\mathrm{T}}W^{\mathrm{T}}Wb.$$

What happens to the picture of b projected to $A\widehat{x}$? The projection $A\widehat{x}_W$ is still the point in the column space that is closest to b. But the word "closest" has a new meaning when the length involves W. The *weighted length* of x equals the ordinary length of Wx. Perpendicularity no longer means $y^{\mathrm{T}}x = 0$; in the new system the test is $(Wy)^{\mathrm{T}}(Wx) = 0$. The matrix $W^{\mathrm{T}}W$ appears in the middle. In this new sense, the projection $A\widehat{x}_W$ and the error $b - A\widehat{x}_W$ are again perpendicular.

That last paragraph describes ***all inner products***: They come from invertible matrices W. They involve only the symmetric combination $C = W^{\mathrm{T}}W$. ***The inner product of x and y is $y^{\mathrm{T}}Cx$.*** For an orthogonal matrix $W = Q$, when this combination is $C = Q^{\mathrm{T}}Q = I$, the inner product is not new or different. Rotating the space leaves the inner product unchanged. Every other W changes the length and inner product.

For any invertible matrix W, these rules define a new inner product and length:

$$\textbf{Weighted by } W \qquad (x,y)_W = (Wy)^{\mathrm{T}}(Wx) \quad \textit{and} \quad \|x\|_W = \|Wx\|. \tag{10}$$

Since W is invertible, no vector is assigned length zero (except the zero vector). All possible inner products—which depend linearly on x and y and are positive when $x = y \neq 0$—are found in this way, from some matrix $C = W^{\mathrm{T}}W$.

In practice, the important question is the choice of C. The best answer comes from statisticians, and originally from Gauss. We may know that the average error is zero. That is the "expected value" of the error in b—although the error is not really expected to be zero! We may also know the ***average of the square*** of the error; that is the ***variance***. If the errors in the b_i are independent of each other, and their variances are σ_i^2, then ***the right weights are*** $w_i = 1/\sigma_i$. A more accurate measurement, which means a smaller variance, gets a heavier weight.

In addition to unequal reliability, *the observations may not be independent*. If the errors are coupled—the polls for President are not independent of those for Senator, and certainly not of those for Vice-President—then W has off-diagonal terms. The best unbiased matrix $C = W^{\mathrm{T}}W$ is the ***inverse of the covariance matrix***—whose i, j entry is the expected value of (error in b_i) times (error in b_j). Then the main diagonal of C^{-1} contains the variances σ_i^2, which are the average of (error in $b_i)^2$.

Example 3. Suppose two bridge partners both guess (after the bidding) the total number of spades they hold. For each guess, the errors -1, 0, 1 might have equal probability $\frac{1}{3}$. Then the expected error is zero and the variance is $\frac{2}{3}$:

$$E(e) = \tfrac{1}{3}(-1) + \tfrac{1}{3}(0) + \tfrac{1}{3}(1) = 0$$
$$E(e^2) = \tfrac{1}{3}(-1)^2 + \tfrac{1}{3}(0)^2 + \tfrac{1}{3}(1)^2 = \tfrac{2}{3}.$$

The two guesses are dependent, because they are based on the same bidding—but not identical, because they are looking at different hands. Say the chance that they are both too high or both too low is zero, but the chance of opposite errors is $\frac{1}{3}$. Then $E(e_1e_2) = \frac{1}{3}(-1)$, and the inverse of the covariance matrix is $W^{\mathrm{T}}W$:

$$\begin{bmatrix} E(e_1^2) & E(e_1e_2) \\ E(e_1e_2) & E(e_2^2) \end{bmatrix}^{-1} = \begin{bmatrix} \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} \end{bmatrix}^{-1} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = C = W^{\mathrm{T}}W.$$

This matrix goes into the middle of the weighted normal equations.

Problem Set 3.3

1. Find the best least-squares solution $\widehat{x}$ to $3x = 10$, $4x = 5$. What error E^2 is minimized? Check that the error vector $(10 - 3\widehat{x}, 5 - 4\widehat{x})$ is perpendicular to the column $(3,4)$.

2. Suppose the values $b_1 = 1$ and $b_2 = 7$ at times $t_1 = 1$ and $t_2 = 2$ are fitted by a line $b = Dt$ *through the origin*. Solve $D = 1$ and $2D = 7$ by least squares, and sketch the best line.

3. Solve $Ax = b$ by least squares, and find $p = A\widehat{x}$ if

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

Verify that the error $b - p$ is perpendicular to the columns of A.

4. Write out $E^2 = \|Ax - b\|^2$ and set to zero its derivatives with respect to u and v, if

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \qquad x = \begin{bmatrix} u \\ v \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}.$$

Compare the resulting equations with $A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}b$, confirming that calculus as well as geometry gives the normal equations. Find the solution $\widehat{x}$ and the projection $p = A\widehat{x}$. Why is $p = b$?

5. The following system has no solution:

$$Ax = \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 9 \end{bmatrix} = b.$$

Sketch and solve a straight-line fit that leads to the minimization of the quadratic $(C - D - 4)^2 + (C - 5)^2 + (C + D - 9)^2$? What is the projection of b onto the column space of A?

6. Find the projection of b onto the column space of A:

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ -2 & 4 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix}.$$

Split b into $p + q$, with p in the column space and q perpendicular to that space. Which of the four subspaces contains q?

7. Find the projection matrix P onto the space spanned by $a_1 = (1,0,1)$ and $a_2 = (1,1,-1)$.

8. If P is the projection matrix onto a k-dimensional subspace $\mathbf{S}$ of the whole space $\mathbf{R}^n$, what is the column space of P and what is its rank?

9. (a) If $P = P^{\mathrm{T}}P$, show that P is a projection matrix.

(b) What subspace does the matrix $P = 0$ project onto?

10. If the vectors a_1, a_2, and b are orthogonal, what are $A^{\mathrm{T}}A$ and $A^{\mathrm{T}}b$? What is the projection of b onto the plane of a_1 and a_2?

11. Suppose P is the projection matrix onto the subspace $\mathbf{S}$ and Q is the projection onto the orthogonal complement $\mathbf{S}^{\perp}$. What are $P+Q$ and PQ? Show that $P-Q$ is its own inverse.

12. If $\mathbf{V}$ is the subspace spanned by $(1,1,0,1)$ and $(0,0,1,0)$, find

(a) a basis for the orthogonal complement $\mathbf{V}^{\perp}$.

(b) the projection matrix P onto $\mathbf{V}$.

(c) the vector in $\mathbf{V}$ closest to the vector $b=(0,1,0,-1)$ in $\mathbf{V}^{\perp}$.

13. Find the best straight-line fit (least squares) to the measurements

$$\begin{array}{llll} b=4 & \text{at} & t=-2, & \qquad b=3 \quad \text{at} \quad t=-1, \\ b=1 & \text{at} & t=0, & \qquad b=0 \quad \text{at} \quad t=2. \end{array}$$

Then find the projection of $b=(4,3,1,0)$ onto the column space of

$$A=\begin{bmatrix} 1 & -2 \\ 1 & -1 \\ 1 & 0 \\ 1 & 2 \end{bmatrix}.$$

14. The vectors $a_1=(1,1,0)$ and $a_2=(1,1,1)$ span a plane in $\mathbf{R}^3$. Find the projection matrix P onto the plane, and find a nonzero vector b that is projected to zero.

15. If P is the projection matrix onto a line in the x-y plane, draw a figure to describe the effect of the "reflection matrix" $H=I-2P$. Explain both geometrically and algebraically why $H^2=I$.

16. Show that if u has unit length, then the rank-1 matrix $P=uu^{\mathrm{T}}$ is a projection matrix: It has properties (i) and (ii) in 3N. By choosing $u=a/\|a\|$, P becomes the projection onto the line through a, and Pb is the point $p=\widehat{x}a$. Rank-1 projections correspond exactly to least-squares problems in one unknown.

17. What 2 by 2 matrix projects the x-y plane onto the -45° line $x+y=0$?

18. We want to fit a plane $y=C+Dt+Ez$ to the four points

$$\begin{array}{llll} y=3 & \text{at} & t=1,z=1 & \qquad y=6 \quad \text{at} \quad t=0,z=3 \\ y=5 & \text{at} & t=2,z=1 & \qquad y=0 \quad \text{at} \quad t=0,z=0. \end{array}$$

(a) Find 4 equations in 3 unknowns to pass a plane through the points (if there is such a plane).

(b) Find 3 equations in 3 unknowns for the best least-squares solution.

19. If $P_C=A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$ is the projection onto the column space of A, what is the projection P_R onto the row space? (It is not P_C^{T}!)

20. If P is the projection onto the column space of A, what is the projection onto the left nullspace?

21. Suppose L_1 is the line through the origin in the direction of a_1 and L_2 is the line through b in the direction of a_2. To find the closest points x_1a_1 and $b+x_2a_2$ on the two lines, write the two equations for the x_1 and x_2 that minimize $\|x_1a_1 - x_2a_2 - b\|$. Solve for x if $a_1 = (1,1,0)$, $a_2 = (0,1,0)$, $b = (2,1,4)$.

22. Find the best line $C+Dt$ to fit $b = 4,2,-1,0,0$ at times $t = -2,-1,0,1,2$.

23. Show that the best least-squares fit to a set of measurements $y_1,\dots,y_m$ by a *horizontal line* (a constant function $y = C$) is their average

$$C = \frac{y_1 + \cdots + y_m}{m}.$$

24. Find the best straight-line fit to the following measurements, and sketch your solution:

$$\begin{array}{llll} y = 2 & \text{at} & t = -1, & y = 0 \quad \text{at} \quad t = 0, \\ y = -3 & \text{at} & t = 1, & y = -5 \quad \text{at} \quad t = 2. \end{array}$$

25. Suppose that instead of a straight line, we fit the data in Problem 24 by a parabola: $y = C + Dt + Et^2$. In the inconsistent system $Ax = b$ that comes from the four measurements, what are the coefficient matrix A, the unknown vector x, and the data vector b? You need not compute $\widehat{x}$.

26. A Middle-Aged man was stretched on a rack to lengths $L = 5$, 6, and 7 feet under applied forces of $F = 1$, 2, and 4 tons. Assuming Hooke's law $L = a + bF$, find his normal length a by least squares.

Problems 27–31 introduce basic ideas of statistics—the foundation for least squares.

27. (Recommended) This problem projects $b = (b_1,\dots,b_m)$ onto the line through $a = (1,\dots,1)$. We solve m equations $ax = b$ in 1 unknown (by least squares).

(a) Solve $a^{\mathrm{T}}a\widehat{x} = a^{\mathrm{T}}b$ to show that is the *mean* (the average) of the b's,

(b) Find $e = b - a\widehat{x}$, the *variance* $\|e\|^2$, and the *standard deviation* $\|e\|$.

(c) The horizontal line $\widehat{b} = 3$ is closest to $b = (1,2,6)$, Check that $p = (3,3,3)$ is perpendicular to e and find the projection matrix P.

28. First assumption behind least squares: Each measurement error has ***mean zero***. Multiply the 8 error vectors $b - Ax = (\pm 1, \pm 1, \pm 1)$ by $(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$ to show that the 8 vectors $\widehat{x} - x$ also average to zero. The estimate $\widehat{x}$ is *unbiased*.

29. Second assumption behind least squares: The m errors e_i are independent with variance σ^2, so the average of $(b-Ax)(b-Ax)^{\mathrm{T}}$ is $\sigma^2 I$. Multiply on the left by $(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$ and on the right by $A(A^{\mathrm{T}}A)^{-1}$ to show that the average of $(\widehat{x}-x)(\widehat{x}-x)^{\mathrm{T}}$ is $\sigma^2(A^{\mathrm{T}}A)^{-1}$. This is the all-important ***covariance matrix*** for the error in $\widehat{x}$.

30. A doctor takes four readings of your heart rate. The best solution to $x=b_1,\dots,x=b_4$ is the average $\widehat{x}$ of $b_1,\dots,b_4$. The matrix A is a column of 1s. Problem 29 gives the expected error $(\widehat{x}-x)^2$ as $\sigma^2(A^{\mathrm{T}}A)^{-1}=$ ____. By averaging, the variance drops from σ^2 to $\sigma^2/4$.

31. If you know the average $\widehat{x}_9$ of 9 numbers $b_1,\dots,b_9$, how can you quickly find the average $\widehat{x}_{10}$ with one more number b_{10}? The idea of *recursive* least squares is to avoid adding 10 numbers. What coefficient of $\widehat{x}_9$ correctly gives $\widehat{x}_{10}$?

$$\widehat{x}_{10}=\tfrac{1}{10}\widehat{b}_{10}+\underline{\quad\quad}\widehat{x}_9=\tfrac{1}{10}(b_1+\cdots+b_{10}).$$

Problems 32–37 use four points $b=(0,8,8,20)$ to bring out more ideas.

32. With $b=0,8,8,20$ at $t=0,1,3,4$, set up and solve the normal equations $A^{\mathrm{T}}A\widehat{x}=A^{\mathrm{T}}b$. For the best straight line as in Figure 3.9a, find its four heights p_i and four errors e_i. What is the minimum value $E^2=e_1^2+e_2^2+e_3^2+e_4^2$?

33. (Line $C+Dt$ does go through p's) With $b=0,8,8,20$ at times $t=0,1,3,4$, write the four equations $Ax=b$ (unsolvable). Change the measurements to $p=1,5,13,17$ and find an exact solution to $A\widehat{x}=p$.

34. Check that $e=b-p=(-1,3,-5,3)$ is perpendicular to both columns of A. What is the shortest distance $\|e\|$ from b to the column space of A?

35. For the closest parabola $b=C+Dt+Et^2$ to the same four points, write the unsolvable equations $Ax=b$ in three unknowns $x=(C,D,E)$. Set up the three normal equations $A^{\mathrm{T}}A\widehat{x}=A^{\mathrm{T}}b$ (solution not required). You are now fitting a parabola to four points—what is happening in Figure 3.9b?

36. For the closest cubic $b=C+Dt+Et^2+Ft^3$ to the same four points, write the four equations $Ax=b$. Solve them by elimination, This cubic now goes exactly through the points. What are p and e?

37. The average of the four times is $\widehat{t}=\frac{1}{4}(0+1+3+4)=2$. The average of the four b's is $\widehat{b}=\frac{1}{4}(0+8+8+20)=9$.

(a) Verify that the best line goes *through the center point* $(\widehat{t},\widehat{b})=(2,9)$.

(b) Explain why $C+D\widehat{t}=\widehat{b}$ comes from the first equation in $A^{\mathrm{T}}A\widehat{x}=A^{\mathrm{T}}b$.

38. What happens to the weighted average $\widehat{x}_W=(w_1^2b_1+w_2^2b_2)/(w_1^2+w_2^2)$ if the first weight w_1 approaches zero? The measurement b_1 is totally unreliable.

39. From m independent measurements $b_1,\dots,b_m$ of your pulse rate, weighted by $w_1,\dots,w_m$, what is the weighted average that replaces equation (9)? It is the best estimate when the statistical variances are $\sigma_i^2 \equiv 1/w_i^2$.

40. If $W = \left[\begin{smallmatrix} 2 & 0 \\ 0 & 1 \end{smallmatrix}\right]$, find the W-inner product of $x = (2,3)$ and $y = (1,1)$, and the W-length of x. What line of vectors is W-perpendicular to y?

41. Find the weighted least-squares solution $\widehat{x}_W$ to $Ax = b$:

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \qquad b = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \qquad W = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Check that the projection $A\widehat{x}_W$ is still perpendicular (in the W-inner product!) to the error $b - A\widehat{x}_W$.

42. (a) Suppose you guess your professor's age, making errors $e = -2, -1, 5$ with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{4}$. Check that the expected error $E(e)$ is zero and find the variance $E(e^2)$.

(b) If the professor guesses too (or tries to remember), making errors -1, 0, 1 with probabilities $\frac{1}{8}, \frac{6}{8}, \frac{1}{8}$, what weights w_1 and w_2 give the reliability of your guess and the professor's guess?

3.4 Orthogonal Bases and Gram-Schmidt

In an orthogonal basis, every vector is perpendicular to every other vector. The coordinate axes are mutually orthogonal. That is just about optimal, and the one possible improvement is easy: Divide each vector by its length, to make it a *unit vector*. That changes an ***orthogonal*** basis into an ***orthonormal*** basis of q's:

3P The vectors $q_1,\dots,q_n$ are ***orthonormal*** if

$$q_i^{\mathrm{T}} q_j = \begin{cases} 0 & \text{whenever } i \neq j, \quad \text{giving the orthogonality;} \\ 1 & \text{whenever } i = j, \quad \text{giving the normalization.} \end{cases}$$

A matrix with orthonormal columns will be called Q.

The most important example is the *standard basis*. For the x-y plane, the best-known axes $e_1 = (1,0)$ and $e_2 = (0,1)$ are not only perpendicular but horizontal and vertical. Q is the 2 by 2 identity matrix. In n dimensions the standard basis $e_1,\dots,e_n$ again consists

of *the columns of* $Q = I$:

Standard basis

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \cdots, \quad e_n = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}.$$

That is not the only orthonormal basis! We can rotate the axes without changing the right angles at which they meet. These rotation matrices will be examples of Q.

If we have a subspace of $\mathbf{R}^n$, the standard vectors e_i might not lie in that subspace. But the subspace always has an orthonormal basis, and it can be constructed in a simple way out of any basis whatsoever. This construction, which converts a skewed set of axes into a perpendicular set, is known as ***Gram-Schmidt orthogonalization***.

To summarize, the three topics basic to this section are:

1. The definition and properties of orthogonal matrices Q.
2. The solution of $Qx = b$, either n by n or rectangular (least squares).
3. The Gram-Schmidt process and its interpretation as a new factorization $A = QR$.

Orthogonal Matrices

3Q If Q (square or rectangular) has orthonormal columns, then $Q^{\mathrm{T}}Q = I$:

Orthonormal columns

$$\begin{bmatrix} — & q_1^{\mathrm{T}} & — \\ — & q_2^{\mathrm{T}} & — \\ & \vdots & \\ — & q_n^{\mathrm{T}} & — \end{bmatrix} \begin{bmatrix} | & | & & | \\ q_1 & q_2 & \cdots & q_n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdot & 0 \\ 0 & 1 & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & \cdot & 1 \end{bmatrix} = I. \tag{1}$$

An orthogonal matrix is a square matrix with orthonormal columns.[2] Then Q^{T} is Q^{-1}. For square orthogonal matrices, ***the transpose is the inverse***.

When row i of Q^{T} multiplies column j of Q, the result is $q_j^{\mathrm{T}} q_j = 0$. On the diagonal where $i = j$, we have $q_i^{\mathrm{T}} q_i = 1$. That is the normalization to unit vectors of length 1.

Note that $Q^{\mathrm{T}}Q = I$ even if Q is rectangular. But then Q^{T} is only a left-inverse.

Example 1.

$$Q = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \qquad Q^{\mathrm{T}} = Q^{-1} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}.$$

[2]Orthonormal matrix would have been a better name, but it is too late to change. Also, there is no accepted word for a rectangular matrix with orthonormal columns. We still write Q, but we won't call it an "orthogonal matrix" unless it is square.

Q rotates every vector through the angle θ, and Q^{T} rotates it back through $-\theta$. The columns are clearly orthogonal, and they are orthonormal because $sin^2\theta + cos^2\theta = 1$. The matrix Q^{T} is just as much an orthogonal matrix as Q.

Example 2. Any permutation matrix P is an orthogonal matrix. The columns are certainly unit vectors and certainly orthogonal—because the 1 appears in a different place in each column: The transpose is the inverse.

$$\text{If} \quad P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad \text{then} \quad P^{-1} = P^{\mathrm{T}} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

An anti-diagonal P, with $P_{13} = P_{22} = P_{31} = I$, takes the x-y-z axes into the z-y-x axes—a "right-handed" system into a "left-handed" system. So we were wrong if we suggested that every orthogonal Q represents a rotation. *A reflection is also allowed.* $P = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ reflects every point (x,y) into (y,x), its mirror image across the 45° line. Geometrically, an orthogonal Q is the product of a rotation and a reflection.

There does remain one property that is shared by rotations and reflections, and in fact by every orthogonal matrix. It is not shared by projections, which are not orthogonal or even invertible. Projections reduce the length of a vector, whereas orthogonal matrices have a property that is the most important and most characteristic of all:

3R Multiplication by any Q preserves lengths:

$$\textbf{Lengths unchanged} \qquad \|Qx\| = \|x\| \quad \text{for every vector } x. \tag{2}$$

It also preserves inner products and angles, since $(Qx)^{\mathrm{T}}(Qy) = x^{\mathrm{T}}Q^{\mathrm{T}}Qy = x^{\mathrm{T}}y$.

The preservation of lengths comes directly from $Q^{\mathrm{T}}Q = I$:

$$\|Qx\|^2 = \|x\|^2 \quad \text{because} \quad (Qx)^{\mathrm{T}}(Qx) = x^{\mathrm{T}}Q^{\mathrm{T}}Qx = x^{\mathrm{T}}x. \tag{3}$$

All inner products and lengths are preserved, when the space is rotated or reflected.

We come now to the calculation that uses the special property $Q^{\mathrm{T}} = Q^{-1}$. If we have a basis, then any vector is a combination of the basis vectors. This is exceptionally simple for an orthonormal basis, which will be a key idea behind Fourier series. The problem is *to find the coefficients of the basis vectors*:

Write b ***as a combination*** $b = x_1q_1 + x_2q_2 + \cdots + x_nq_n$.

To compute x_1 there is a neat trick. *Multiply both sides of the equation by* q_1^{T}. On the left-hand side is $q_1^{\mathrm{T}}b$. On the right-hand side all terms disappear (because $q_1^{\mathrm{T}}q_j = 0$) except the first term. We are left with

$$q_1^{\mathrm{T}}b = x_1q_1^{\mathrm{T}}q_1.$$

Since $q_1^{\mathrm{T}} q_1 = 1$, we have found $x_1 = q_1^{\mathrm{T}} b$. Similarly the second coefficient is $x_2 = q_2^{\mathrm{T}} b$; that term survives when we multiply by q_2^{T}. The other terms die of orthogonality. Each piece of b has a simple formula, and recombining the pieces gives back b:

$$\textbf{\textit{Every vector } b \textit{ is equal to }} (q_1^{\mathrm{T}} b)q_1 + (q_2^{\mathrm{T}} b)q_2 + \cdots + (q_n^{\mathrm{T}} b)q_n. \tag{4}$$

I can't resist putting this orthonormal basis into a square matrix Q. The vector equation $x_1 q_1 + \cdots + x_n q_n = b$ is identical to $Qx = b$. (The columns of Q multiply the components of x.) Its solution is $x = Q^{-1} b$. But since $Q^{-1} = Q^{\mathrm{T}}$—this is where orthonormality enters—the solution is also $x = Q^{\mathrm{T}} b$:

$$x = Q^{\mathrm{T}} b = \begin{bmatrix} — & q_1^{\mathrm{T}} & — \\ & \vdots & \\ — & q_n^{\mathrm{T}} & — \end{bmatrix} \begin{bmatrix} \\ b \\ \\ \end{bmatrix} = \begin{bmatrix} q_1^{\mathrm{T}} b \\ \vdots \\ q_n^{\mathrm{T}} b \end{bmatrix} \tag{5}$$

The components of x are the inner products $q_i^{\mathrm{T}} b$, as in equation (4).

The matrix form also shows what happens when the columns are *not* orthonormal. Expressing b as a combination $x_1 a_1 + \cdots + x_n a_n$ is the same as solving $Ax = b$. The basis vectors go into the columns of A. In that case we need A^{-1}, which takes work. In the orthonormal case we only need Q^{T}.

***Remark* 1.** The ratio $a^{\mathrm{T}} b / a^{\mathrm{T}} a$ appeared earlier, when we projected b onto a line. Here a is q_1, the denominator is 1, and the projection is $(q_1^{\mathrm{T}} b)q_1$. Thus we have a new interpretation for formula (4): *Every vector b is the sum of its one-dimensional projections onto the lines through the q's.*

Since those projections are orthogonal, Pythagoras should still be correct. The square of the hypotenuse should still be the sum of squares of the components:

$$\|b\|^2 = (q_1^{\mathrm{T}} b)^2 + (q_2^{\mathrm{T}} b)^2 + \cdots + (q_n^{\mathrm{T}} b)^2 \quad \text{which is} \quad \|Q^{\mathrm{T}} b\|^2. \tag{6}$$

Remark* 2.** Since $Q^{\mathrm{T}} = Q^{-1}$, we also have $QQ^{\mathrm{T}} = I$. When Q comes before Q^{T}, multiplication takes the inner products of the *rows* of Q. (For $Q^{\mathrm{T}} Q$ it was the columns.) Since the result is again the identity matrix, we come to a surprising conclusion: ***The rows of a square matrix are orthonormal whenever the columns are. The rows point in completely different directions from the columns, and I don't see geometrically why they are forced to be orthonormal—but they are.

Orthonormal columns
Orthonormal rows

$$Q = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 0 & -2/\sqrt{6} \\ 1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \end{bmatrix}.$$

Rectangular Matrices with Orthogonal Columns

This chapter is about $Ax = b$, when A is not necessarily square. For $Qx = b$ we now admit the same possibility—there may be more rows than columns. The n orthonormal

vectors q_i in the columns of Q have $m > n$ components. Then Q is an m by n matrix and we cannot expect to solve $Qx = b$ exactly. *We solve it by least squares*.

If there is any justice, orthonormal columns should make the problem simple. It worked for square matrices, and now it will work for rectangular matrices. The key is to notice that *we still have* $Q^{\mathrm{T}}Q = I$. So Q^{T} is still the ***left-inverse*** of Q.

For least squares that is all we need. The normal equations came from multiplying $Ax = b$ by the transpose matrix, to give $A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}b$. Now the normal equations are $Q^{\mathrm{T}}Q = Q^{\mathrm{T}}b$. But $Q^{\mathrm{T}}Q$ is the identity matrix! Therefore $\widehat{x} = Q^{\mathrm{T}}b$, whether Q is square and $\widehat{x}$ is an exact solution, or Q is rectangular and we need least squares.

3S If Q has orthonormal columns, the least-squares problem becomes easy: rectangular system with no solution for most b.

$$
\begin{array}{rcll}
Qx & = & b & \text{rectangular system with no solution for most } b. \\
Q^{\mathrm{T}}Q\widehat{x} & = & Q^{\mathrm{T}}b & \text{normal equation for the best } \widehat{x}\text{—in which } Q^{\mathrm{T}}Q = I. \\
\widehat{x} & = & Q^{\mathrm{T}}b & \widehat{x}_i \text{ is } q_i^{\mathrm{T}}b. \\
p & = & Q\widehat{x} & \text{the projection of } b \text{ is } (q_1^{\mathrm{T}}b)q_1 + \cdots + (q_n^{\mathrm{T}}b)q_n. \\
p & = & QQ^{\mathrm{T}}b & \text{the projection matrix is } P = QQ^{\mathrm{T}}.
\end{array}
$$

The last formulas are like $p = A\widehat{x}$ and $P = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$. When the columns are orthonormal, the "cross-product matrix" $A^{\mathrm{T}}A$ becomes $Q^{\mathrm{T}}Q = I$. The hard part of least squares disappears when vectors are orthonormal. The projections onto the axes are uncoupled, and p is the sum $p = (q_1^{\mathrm{T}}b)q_1 + \cdots + (q_n^{\mathrm{T}}b)q_n$.

We emphasize that those projections do not reconstruct b. In the square case $m = n$, they did. In the rectangular case $m > n$, they don't. They give the projection p and not the original vector b—which is all we can expect when there are more equations than unknowns, and the q's are no longer a basis. The projection matrix is usually $A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$, and here it simplifies to

$$
P = Q(Q^{\mathrm{T}}Q)^{-1}Q^{\mathrm{T}} \qquad \text{or} \qquad P = QQ^{\mathrm{T}}. \tag{7}
$$

Notice that $Q^{\mathrm{T}}Q$ is the n by n identity matrix, whereas QQ^{T} is an m by m projection P. It is the identity matrix on the columns of Q (P leaves them alone), But QQ^{T} is the zero matrix on the orthogonal complement (the nullspace of Q^{T}).

Example 3. The following case is simple but typical. Suppose we project a point $b = (x, y, z)$ onto the x-y plane. Its projection is $p = (x, y, 0)$, and this is the sum of the separate projections onto the x- and y-axes:

$$
q_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \text{and} \quad (q_1^{\mathrm{T}}b)q_1 = \begin{bmatrix} x \\ 0 \\ 0 \end{bmatrix}; \qquad q_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad (q_2^{\mathrm{T}}b)q_2 = \begin{bmatrix} 0 \\ y \\ 0 \end{bmatrix}.
$$

The overall projection matrix is

$$P = q_1 q_1^{\mathrm{T}} + q_2 q_2^{\mathrm{T}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \text{and} \quad P \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ 0 \end{bmatrix}.$$

Projection onto a plane = sum of projections onto orthonormal q_1 and q_2.

Example 4. When the measurement times average to zero, fitting a straight line leads to orthogonal columns. Take $t_1 = -3$, $t_2 = 0$, and $t_3 = 3$. Then the attempt to fit $y = C + Dt$ leads to three equations in two unknowns:

$$\begin{array}{rcrcl} C & + & Dt_1 & = & y_1 \\ C & + & Dt_2 & = & y_2 \\ C & + & Dt_3 & = & y_3 \end{array}, \qquad \text{or} \qquad \begin{bmatrix} 1 & -3 \\ 1 & 0 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}.$$

The columns $(1,1,1)$ *and* $(-3,0,3)$ *are orthogonal.* We can project y separately onto each column, and the best coefficients $\widehat{C}$ and $\widehat{D}$ can be found separately:

$$\widehat{C} = \frac{\begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix}^{\mathrm{T}}}{1^2 + 1^2 + 1^2}, \qquad \widehat{D} = \frac{\begin{bmatrix} -3 & 0 & 3 \end{bmatrix} \begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix}^{\mathrm{T}}}{(-3)^2 + 0^2 + 3^2}.$$

Notice that $\widehat{C} = (y_1 + y_2 + y_3)/3$ is the *mean* of the data. $\widehat{C}$ gives the best fit by a horizontal line, whereas $\widehat{D}t$ is the best fit by a straight line through the origin. *The columns are orthogonal, so the sum of these two separate pieces is the best fit by any straight line whatsoever.* The columns are not unit vectors, so $\widehat{C}$ and $\widehat{D}$ have the length squared in the denominator.

Orthogonal columns are so much better that it is worth changing to that case. if the average of the observation times is not zero—it is $\bar{t} = (t_1 + \cdots + t_m)/m$—then the time origin can be shifted by $\bar{t}$. Instead of $y = C + Dt$ we work with $y = c + d(t - \bar{t})$. The best line is the same! As in the example, we find

$$\begin{aligned} \widehat{c} &= \frac{\begin{bmatrix} 1 & \cdots & 1 \end{bmatrix} \begin{bmatrix} y_1 & \cdots & y_m \end{bmatrix}^{\mathrm{T}}}{1^2 + 1^2 + \cdots + 1^2} = \frac{y_1 + \cdots + y_m}{m} \\ \widehat{d} &= \frac{\begin{bmatrix} (t_1 - \bar{t}) & \cdots & (t_m - \bar{t}) \end{bmatrix} \begin{bmatrix} y_1 & \cdots & y_m \end{bmatrix}^{\mathrm{T}}}{(t_1 - \bar{t})^2 + \cdots + (t_m - \bar{t})^2} = \frac{\sum (t_i - \bar{t}) y_i}{\sum (t_i - \bar{t})^2}. \end{aligned} \tag{8}$$

The best $\widehat{c}$ is the mean, and we also get a convenient formula for $\widehat{d}$. The earlier $A^{\mathrm{T}}A$ had the off-diagonal entries $\sum t_i$, and shifting the time by $\bar{t}$ made these entries zero. This shift is an example of the Gram-Schmidt process, ***which orthogonalizes the situation in advance***.

Orthogonal matrices are crucial to numerical linear algebra, because they introduce no instability. While lengths stay the same, roundoff is under control. Orthogonalizing vectors has become an essential technique. Probably it comes second only to elimination. And it leads to a factorization $A = QR$ that is nearly as famous as $A = LU$.

The Gram-Schmidt Process

Suppose you are given three independent vectors a, b, c. If they are orthonormal, life is easy. To project a vector v onto the first one, you compute $(a^{\mathrm{T}}v)a$. To project the same vector v onto the plane of the first two, you just add $(a^{\mathrm{T}}v)a+(b^{\mathrm{T}}v)b$. To project onto the span of a, b, c, you add three projections. All calculations require only the inner products $a^{\mathrm{T}}v$, $b^{\mathrm{T}}v$, and $c^{\mathrm{T}}v$. But to make this true, we are forced to say, "***If*** they are orthonormal." Now we propose to find a way to ***make*** them orthonormal.

The method is simple. We are given a, b, c and we want q_1, q_2, q_3. There is no problem with q_1: it can go in the direction of a. We divide by the length, so that $q_1 = a/\|a\|$ is a unit vector. The real problem begins with q_2—which has to be orthogonal to q_1. If the second vector b has any component in the direction of q_1 (which is the direction of a), ***that component has to be subtracted***:

$$\textbf{Second vector} \qquad B = b-(q_1^{\mathrm{T}}b)q_1 \quad \text{and} \quad q_2 = B/\|B\|. \tag{9}$$

B is orthogonal to q_1. It is the part of b that goes in a new direction, and not in the a. In Figure 3.10, B is perpendicular to q_1. It sets the direction for q_2.

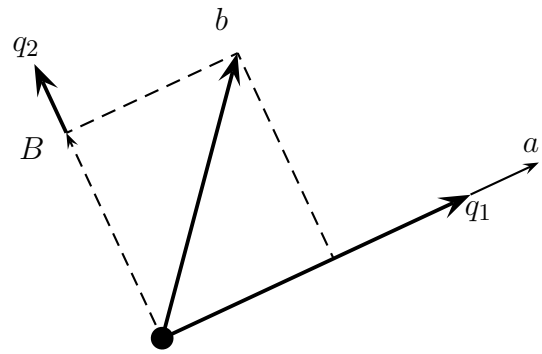


Figure 3.10: The q_i component of b is removed; a and B normalized to q_1 and q_2.

At this point q_1 and q_2 are set. The third orthogonal direction starts with c. It will not be in the plane of q_1 and q_2, which is the plane of a and b. However, it may have a component in that plane, and that has to be subtracted. (If the result is $C=0$, this signals that a, b, c were not independent in the first place) What is left is the component C we want, the part that is in a new direction perpendicular to the plane:

$$\textbf{Third vector} \qquad C = c-(q_1^{\mathrm{T}}c)q_1-(q_2^{\mathrm{T}}c)q_2 \quad \text{and} \quad q_3 = C/\|C\|. \tag{10}$$

This is the one idea of the whole Gram-Schmidt process, ***to subtract from every new vector its components in the directions that are already settled***. That idea is used over and over again.[3] When there is a fourth vector, we subtract away its components in the directions of q_1, q_2, q_3.

[3]If Gram thought of it first, what was left for Schmidt?

Example 5. Gram-Schmidt Suppose the independent vectors are a, b, c:

$$a = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \qquad c = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}.$$

To find q_1, make the first vector into a unit vector: $q_1 = a/\sqrt{2}$. To find q_2, subtract from the second vector its component in the first direction:

$$B = b - (q_1^{\mathrm{T}} b)q_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - \frac{1}{\sqrt{2}} \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}.$$

The normalized q_2 is B divided by its length, to produce a unit vector:

$$q_2 = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{bmatrix}.$$

To find q_3, subtract from c its components along q_1 and q_2:

$$\begin{aligned} C &= c - (q_1^{\mathrm{T}} c)q_1 - (q_2^{\mathrm{T}} c)q_2 \\ &= \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} - \sqrt{2} \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix} - \sqrt{2} \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}. \end{aligned}$$

This is already a unit vector, so it is q_3. I went to desperate lengths to cut down the number of square roots (the painful part of Gram-Schmidt). The result is a set of orthonormal vectors q_1, q_2, q_3, which go into the columns of an orthogonal matrix Q:

Orthonormal basis
$$Q = \begin{bmatrix} q_1 & q_2 & q_3 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \end{bmatrix}.$$

3T The Gram-Schmidt process starts with independent vectors $a_1, \ldots, a_n$ and ends with orthonormal vectors $q_1, \ldots, q_n$. At step j it subtracts from a_j its components in the directions $q_1, \ldots, q_{j-1}$ that are already settled:

$$A_j = a_j - (q_1^{\mathrm{T}} a_j)q_1 - \cdots - (q_{j-1}^{\mathrm{T}} a_j)q_{j-1}. \tag{11}$$

Then q_j is the unit vector $A_j/\|A_j\|$.

Remark on the calculations I think it is easier to compute the orthogonal a, B, C, without forcing their lengths to equal one. Then square roots enter only at the end, when

dividing by those lengths. The example above would have the same B and C, without using square roots. Notice the $\frac{1}{2}$ from $a^{\mathrm{T}}b/a^{\mathrm{T}}a$ instead of $\frac{1}{\sqrt{2}}$ from $q^{\mathrm{T}}b$:

$$B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - \frac{1}{2}\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad \text{and then} \quad C = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - 2\begin{bmatrix} \frac{1}{2} \\ 0 \\ -\frac{1}{2} \end{bmatrix}.$$

The Factorization $A = QR$

We started with a matrix A, whose columns were a, b, c. We ended with a matrix Q, whose columns are q_1, q_2, q_3. What is the relation between those matrices? The matrices A and Q are m by n when the n vectors are in m-dimensional space, and there has to be a third matrix that connects them.

The idea is to write the a's as combinations of the q's. The vector b in Figure 3.10 is a combination of the orthonormal q_1 and q_2, and we know what combination it is:

$$b = (q_1^{\mathrm{T}}b)q_1 + (q_2^{\mathrm{T}}b)q_2.$$

Every vector in the plane is the sum of its q_1 and q_2 components. Similarly c is the sum of its q_1, q_2, q_3 components: $c = (q_1^{\mathrm{T}}c)q_1 + (q_2^{\mathrm{T}}c)q_2 + (q_3^{\mathrm{T}}c)q_3$. If we express that in matrix form we have ***the new factorization*** $A = QR$:

$$QR \textbf{ factors} \qquad A = \begin{bmatrix} a & b & c \end{bmatrix} = \begin{bmatrix} q_1 & q_2 & q_3 \end{bmatrix} \begin{bmatrix} q_1^{\mathrm{T}}a & q_1^{\mathrm{T}}b & q_1^{\mathrm{T}}c \\ & q_2^{\mathrm{T}}b & q_2^{\mathrm{T}}c \\ & & q_3^{\mathrm{T}}c \end{bmatrix} = QR \tag{12}$$

Notice the zeros in the last matrix! R is *upper triangular* because of the way Gram-Schmidt was done. The first vectors a and q_1 fell on the same line. Then q_1, q_2 were in the same plane as a, b. The third vectors c and q_3 were not involved until step 3.

The QR factorization is like $A = LU$, except that the first factor Q has orthonormal columns. The second factor is called R, because the nonzeros are to the *right* of the diagonal (and the letter U is already taken). The off-diagonal entries of R are the numbers $q_1^{\mathrm{T}}b = 1/\sqrt{2}$ and $q_1^{\mathrm{T}}c = q_2^{\mathrm{T}}c = \sqrt{2}$, found above. The whole factorization is

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} \sqrt{2} & 1/\sqrt{2} & \sqrt{2} \\ & 1/\sqrt{2} & \sqrt{2} \\ & & 1 \end{bmatrix} = QR.$$

You see the lengths of a, B, C on the diagonal of R. The orthonormal vectors q_1, q_2, q_3, which are the whole object of orthogonalization, are in the first factor Q.

Maybe QR is not as beautiful as LU (because of the square roots). Both factorizations are vitally important to the theory of linear algebra, and absolutely central to the calculations. If LU is Hertz, then QR is Avis.

The entries $r_{ij} = q_i^{\mathrm{T}} a_j$ appear in formula (11), when $\|A_j\| q_j$ is substituted for A_j:

$$a_j = (q_1^{\mathrm{T}} a_j) q_1 + \cdots + (q_{j-1}^{\mathrm{T}} a_j) q_{j-1} + \|A_j\| q_j = Q \text{ times column } j \text{ of } R. \tag{13}$$

3U Every m by n matrix with independent columns can be factored into $A = QR$. The columns of Q are orthonormal, and R is upper triangular and invertible. When $m = n$ and all matrices are square, Q becomes an orthogonal matrix.

I must not forget the main point of orthogonalization. It simplifies the least-squares problem $Ax = b$. The normal equations are still correct, but $A^{\mathrm{T}}A$ becomes easier:

$$A^{\mathrm{T}}A = R^{\mathrm{T}}Q^{\mathrm{T}}QR = R^{\mathrm{T}}R. \tag{14}$$

The fundamental equation $A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}b$ simplifies to a triangular system:

$$R^{\mathrm{T}}R\widehat{x} = R^{\mathrm{T}}Q^{\mathrm{T}}b \qquad \text{or} \qquad R\widehat{x} = Q^{\mathrm{T}}b. \tag{15}$$

Instead of solving $QRx = b$, which can't be done, we solve $R\widehat{x} = Q^{\mathrm{T}}b$ which is just back-substitution because R is triangular. The real cost is the mn^2 operations of Gram-Schmidt, which are needed to find Q and R in the first place.

The same idea of orthogonality applies to functions, The sines and cosines are orthogonal; the powers $1, x, x^2$ are not. When $f(x)$ is written as a combination of sines and cosines, that is a ***Fourier series***. Each term is a projection onto a line—the line in function space containing multiples of $\cos nx$ or $\sin nx$. It is completely parallel to the vector case, and very important. And finally we have a job for Schmidt: To orthogonalize the powers of x and produce the Legendre polynomials.

Function Spaces and Fourier Series

This is a brief and optional section, but it has a number of good intentions:

1. to introduce the most famous infinite-dimensional vector space (*Hilbert space*);
2. to extend the ideas of length and inner product from vectors v to functions $f(x)$:
3. to recognize the Fourier series as a sum of one-dimensional projections (the orthogonal "columns" are the sines and cosines);
4. to apply Gram-Schmidt orthogonalization to the polynomials $1, x, x^2, \ldots$; and
5. to find the best approximation to $f(x)$ by a straight line.

We will try to follow this outline, which opens up a range of new applications for linear algebra, in a systematic way.

1. Hilbert Space. After studying $\mathbf{R}^n$, it is natural to think of the space $\mathbf{R}^\infty$. It contains all vectors $v = (v_1, v_2, v_3, \ldots)$ with an infinite sequence of components. This space

is actually too big when there is no control on the size of components v_j. A much better idea is to keep the familiar definition of length, using a sum of squares, and *to include only those vectors that have a* ***finite length***:

$$\textbf{Length squared} \qquad \|v\|^2 = v_1^2 + v_2^2 + v_3^2 + \cdots \tag{16}$$

The infinite series must converge to a finite sum. This leaves $(1, \frac{1}{2}, \frac{1}{3}, \ldots)$ but not $(1,1,1,\ldots)$. Vectors with finite length can be added ($\|v+w\| \le \|v\| + \|w\|$) and multiplied by scalars, so they form a vector space. It is the celebrated ***Hilbert space***.

Hilbert space is the natural way to let the number of dimensions become infinite, and at the same time to keep the geometry of ordinary Euclidean space. Ellipses become infinite-dimensional ellipsoids, and perpendicular lines are recognized exactly as before. The vectors v and w are orthogonal when their inner product is zero:

$$\textbf{Orthogonality} \qquad v^{\mathrm{T}} w = v_1 w_1 + v_2 w_2 + v_3 w_3 + \cdots = 0.$$

This sum is guaranteed to converge, and for any two vectors it still obeys the Schwarz inequality $|v^{\mathrm{T}} w| \le \|v\| \|w\|$. The cosine, even in Hilbert space, is never larger than 1.

There is another remarkable thing about this space: It is found under a great many different disguises. Its "vectors" can turn into functions, which is the second point.

2. Lengths and Inner Products. Suppose $f(x) = \sin x$ on the interval $0 \le x \le 2\pi$. This f is like a vector with a whole continuum of components, the values of $\sin x$ along the whole interval. To find the length of such a vector, the usual rule of adding the squares of the components becomes impossible. This summation is replaced, in a natural and inevitable way, by *integration*:

$$\textbf{Length } \|f\| \textbf{ of function} \qquad \|f\|^2 = \int_0^{2\pi} (f(x))^2 dx = \int_0^{2\pi} (\sin x)^2 dx = \pi \tag{17}$$

Our Hilbert space has become a ***function space***. The vectors are functions, we have a way to measure their length, and the space contains all those functions that have a finite length—just as in equation (16). It does not contain the function $F(x) = 1/x$, because the integral of $1/x^2$ is infinite.

The same idea of replacing summation by integration produces the ***inner product of two functions***: If $f(x) = \sin x$ and $g(x) = \cos x$, then their inner product is

$$(f,g) = \int_0^{2\pi} f(x) g(x) dx = \int_0^{2\pi} \sin x \cos x dx = 0. \tag{18}$$

This is exactly like the vector inner product $f^{\mathrm{T}} g$. It is still related to the length by $(f,f) = \|f\|^2$. The Schwarz inequality is still satisfied: $|(f,g)| \le \|f\| \|g\|$. Of course, two functions like $\sin x$ and $\cos x$—whose inner product is zero—will be called orthogonal. They are even orthonormal after division by their length $\sqrt{\pi}$.

3. The ***Fourier series*** of a function is an expansion into sines and cosines:

$$f(x) = a_0 + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x + \cdots .$$

To compute a coefficient like b_1, *multiply* both sides by the corresponding function $\sin x$ and *integrate* from 0 to 2π. (The function $f(x)$ is given on that interval.) In other words, take the inner product of both sides with $\sin x$:

$$\int_0^{2\pi} f(x) \sin x dx = a_0 \int_0^{2\pi} \sin x dx + a_1 \int_0^{2\pi} \cos x \sin x dx + b_1 \int_0^{2\pi} (\sin x)^2 dx + \cdots .$$

On the right-hand side, every integral is zero except one—the one in which $\sin x$ multiplies itself. *The sines and cosines are mutually orthogonal* as in equation (18) Therefore b_1 is the left-hand side divided by that one nonzero integral:

$$b_1 = \frac{\int_0^{2\pi} f(x) \sin x dx}{\int_0^{2\pi} (\sin x)^2 dx} = \frac{(f, \sin x)}{(\sin x, \sin x)}.$$

The Fourier coefficient a_1 would have $\cos x$ in place of $\sin x$, and a_2 would use $\cos 2x$.

The whole point is to see the analogy with projections. The component of the vector b along the line spanned by a is $b^{\mathrm{T}}a/a^{\mathrm{T}}a$. A ***Fourier series is projecting*** $f(x)$ ***onto*** $\sin x$. Its component p in this direction is exactly $b_1 \sin x$.

The coefficient b_1 is the least squares solution of the inconsistent equation $b_1 \sin x = f(x)$. This brings $b_1 \sin x$ as close as possible to $f(x)$. All the terms in the series are projections onto a sine or cosine. Since the sines and cosines are orthogonal, *the Fourier series gives the coordinates of the "vector" $f(x)$ with respect to a set of* (infinitely many) *perpendicular axes*.

4. Gram-Schmidt for Functions. There are plenty of useful functions other than sines and cosines, and they are not always orthogonal. The simplest are the powers of x, and unfortunately there is no interval on which even 1 and x^2 are perpendicular. (Their inner product is always positive, because it is the integral of x^2.) Therefore the closest parabola to $f(x)$ is *not* the sum of its projections onto 1, x, and x^2. There will be a matrix like $(A^{\mathrm{T}}A)^{-1}$, and this coupling is given by the ill-conditioned **Hilbert matrix**. On the interval $0 \le x \le 1$,

$$A^{\mathrm{T}}A = \begin{bmatrix} (1,1) & (1,x) & (1,x^2) \\ (x,1) & (x,x) & (x,x^2) \\ (x^2,1) & (x^2,x) & (x^2,x^2) \end{bmatrix} = \begin{bmatrix} \int 1 & \int x & \int x^2 \\ \int x & \int x^2 & \int x^3 \\ \int x^2 & \int x^3 & \int x^4 \end{bmatrix} = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{bmatrix}.$$

This matrix has a large inverse, because the axes 1, x, x^2 are far from perpendicular. The situation becomes impossible if we add a few more axes. *It is virtually hopeless to solve $A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}b$ for the closest polynomial of degree ten.*

More precisely, it is hopeless to solve this by Gaussian elimination; every roundoff error would be amplified by more than 10^{13}. On the other hand, we cannot just give

up; approximation by polynomials has to be possible. The right idea is to switch to orthogonal axes (by Gram-Schmidt). We look for combinations of 1, x, and x^2 that *are* orthogonal.

It is convenient to work with a symmetrically placed interval like $-1 \le x \le 1$, because this makes all the odd powers of x orthogonal to all the even powers:

$$(1,x) = \int_{-1}^{1} x\,dx = 0, \qquad (x,x^2) = \int_{-1}^{1} x^3\,dx = 0.$$

Therefore the Gram-Schmidt process can begin by accepting $v_1 = 1$ and $v_2 = x$ as the first two perpendicular axes. Since $(x,x^2) = 0$, it only has to correct the angle between 1 and x^2. By formula (10), the third orthogonal polynomial is

Orthogonalize $$v_3 = x^2 - \frac{(1,x^2)}{(1,1)}1 - \frac{(x,x^2)}{(x,x)}x = x^2 - \frac{\int_{-1}^{1} x^2\,dx}{\int_{-1}^{1} 1\,dx} = x^2 - \frac{1}{3}.$$

The polynomials constructed in this way are called the ***Legendre polynomials*** and they are orthogonal to each other over the interval $-1 \le x \le 1$.

Check $$\left(1, x^2 - \frac{1}{3}\right) = \int_{-1}^{1} \left(x^2 - \frac{1}{3}\right) dx = \left[\frac{x^3}{3} - \frac{x}{3}\right]_{-1}^{1} = 0.$$

The closest polynomial of degree ten is now computable, without disaster, by projecting onto each of the first 10 (or 11) Legendre polynomials.

5. Best Straight Line. Suppose we want to approximate $y = x^5$ by a straight line $C + Dx$ between $x = 0$ and $x = 1$. There are at least three ways of finding that line, and if you compare them the whole chapter might become clear!

1. Solve $[1 \;\; x]\begin{bmatrix} C \\ D \end{bmatrix} = x^5$ by least squares. The equation $A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}b$ is

$$\begin{bmatrix} (1,1) & (1,x) \\ (x,1) & (x,x) \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} (1,x^5) \\ (x,x^5) \end{bmatrix} \qquad \text{or} \qquad \begin{bmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} \frac{1}{6} \\ \frac{1}{17} \end{bmatrix}.$$

2. Minimize $E^2 = \int_0^1 (x^5 - C - Dx)^2 dx = \frac{1}{11} - \frac{2}{6}C - \frac{2}{7}D + C^2 + CD + \frac{1}{3}D^2$. The derivatives with respect to C and D, after dividing by 2, bring back the normal equations of method 1 (and the solution is $\widehat{C} = \frac{1}{6} - \frac{5}{14}$, $\widehat{D} = \frac{5}{17}$):

$$-\frac{1}{6} + C + \frac{1}{2}D = 0 \qquad \text{and} \qquad -\frac{1}{7} + \frac{1}{2}C + \frac{1}{3}D = 0.$$

3. Apply Gram-Schmidt to replace x by $x - (1,x)/(1,1)$. That is $x - \frac{1}{2}$, which is orthogonal to 1. Now the one-dimensional projections add to the best line:

$$C + Dx = \frac{(x^5,1)}{(1,1)}1 + \frac{(x^5, x-\frac{1}{2})}{(x-\frac{1}{2}, x-\frac{1}{2})}(x - \tfrac{1}{2}) = \frac{1}{6} + \frac{5}{7}\left(x - \frac{1}{2}\right).$$

Problem Set 3.4

1. (a) Write the four equations for fitting $y = C + Dt$ to the data

$$\begin{array}{llll} y=-4 & \text{at} & t=-2, & y=-3 \quad \text{at} \quad t=-1 \\ y=-1 & \text{at} & t=1, & y=0 \quad \text{at} \quad t=2. \end{array}$$

Show that the columns are orthogonal.

(b) Find the optimal straight line, draw its graph, and write E^2.

(c) Interpret the zero error in terms of the original system of four equations in two unknowns: The right-hand side $(-4,-3,-1,0)$ is in the ____ space.

2. Project $b = (0,3,0)$ onto each of the orthonormal vectors $a_1 = (\frac{2}{3}, \frac{2}{3}, -\frac{1}{3})$ and $a_2 = (-\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$, and then find its projection p onto the plane of a_1 and a_2.

3. Find also the projection of $b = (0,3,0)$ onto $a_3 = (\frac{2}{3}, -\frac{1}{3}, \frac{2}{3})$, and add the three projections. Why is $P = a_1a_1^{\mathrm{T}} + a_2a_2^{\mathrm{T}} + a_3a_3^{\mathrm{T}}$ equal to I?

4. If Q_1 and Q_2 are orthogonal matrices, so that $Q^{\mathrm{T}}Q = I$, show that Q_1Q_2 is also orthogonal. If Q_1 is rotation through θ, and Q_2 is rotation through ϕ, what is Q_1Q_2? Can you find the trigonometric identities for $\sin(\theta+\phi)$ and $\cos(\theta+\phi)$ in the matrix multiplication Q_1Q_2?

5. If u is a unit vector, show that $Q = I - 2uu^{\mathrm{T}}$ is a symmetric orthogonal matrix. (It is a reflection, also known as a Householder transformation.) Compute Q when $u^{\mathrm{T}} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \end{bmatrix}$.

6. Find a third column so that the matrix

$$Q = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{14} & \\ 1/\sqrt{3} & 2/\sqrt{14} & \\ 1/\sqrt{3} & -3/\sqrt{14} & \end{bmatrix}$$

is orthogonal. It must be a unit vector that is orthogonal to the other columns; how much freedom does this leave? Verify that the rows automatically become orthonormal at the same time.

7. Show, by forming $b^{\mathrm{T}}b$ directly, that Pythagoras's law holds for any combination $b = x_1q_1 + \cdots + x_nq_n$ of orthonormal vectors: $\|b\|^2 = x_1^2 + \cdots + x_n^2$. In matrix terms, $b = Qx$, so this again proves that lengths are preserved: $\|Qx\|^2 = \|x\|^2$.

8. Project the vector $b = (1,2)$ onto two vectors that are not orthogonal, $a_1 = (1,0)$ and $a_2 = (1,1)$. Show that, unlike the orthogonal case, the sum of the two one-dimensional projections does not equal b.

9. If the vectors q_1, q_2, q_3 are orthonormal, what combination of q_1 and q_2 is closest to q_3?

10. If q_1 and q_2 are the outputs from Gram-Schmidt, what were the possible input vectors a and b?

11. Show that an orthogonal matrix that is upper triangular must be diagonal.

12. What multiple of $a_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ should be subtracted from $a2 = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$ to make the result orthogonal to a_1? Factor $\begin{bmatrix} 1 & 4 \\ 1 & 0 \end{bmatrix}$ into QR with orthonormal vectors in Q.

13. Apply the Gram-Schmidt process to

$$a = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \qquad b = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \qquad c = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

and write the result in the form $A = QR$.

14. From the nonorthogonal a, b, c, find orthonormal vectors q_1, q_2, q_3:

$$a = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \qquad c = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

15. Find an orthonormal set q_1, q_2, q_3 for which q_1, q_2 span the column space of

$$A = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -2 & 4 \end{bmatrix}.$$

Which fundamental subspace contains q_3? What is the least-squares solution of $Ax = b$ if $b = [1 \;\; 2 \;\; 7]^{\mathrm{T}}$?

16. Express the Gram-Schmidt orthogonalization of a_1, a_2 as $A = QR$:

$$a_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \qquad a_2 = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}.$$

Given n vectors a_i with m components, what are the shapes of A, Q, and R?

17. With the same matrix A as in Problem 16, and with $b = [1 \;\; 1 \;\; 1]^{\mathrm{T}}$, use $A = QR$ to solve the least-squares problem $Ax = b$.

18. If $A = QR$, find a simple formula for the projection matrix P onto the column space of A.

19. Show that these *modified Gram-Schmidt* steps produce the same C as in equation (10):

$$C^* = c - (q_1^{\mathrm{T}} c)q_1 \qquad \text{and} \qquad C = C^* - (q_2^{\mathrm{T}} C^*)q_2.$$

This is much more stable, to subtract the projections one at a time.

20. In Hilbert space, find the length of the vector $v = (1/\sqrt{2}, 1/\sqrt{4}, 1/\sqrt{8}, \ldots)$ and the length of the function $f(x) = e^x$ (over the interval $0 \le x \le 1$). What is the inner product over this interval of e^x and e^{-x}?

21. What is the closest function $a\cos x + b\sin x$ to the function $f(x) = \sin 2x$ on the interval from $-\pi$ to π? What is the closest straight line $c + dx$?

22. By setting the derivative to zero, find the value of b_1 that minimizes

$$\|b_1 \sin x - \cos x\|^2 = \int_0^{2\pi} (b_1 \sin x - \cos x)^2 dx.$$

Compare with the Fourier coefficient b_1.

23. Find the Fourier coefficients a_0, a_1, b_1 of the step function $y(x)$, which equals 1 on the interval $0 \le x \le \pi$ and 0 on the remaining interval $\pi < x < 2\pi$:

$$a_0 = \frac{(y,1)}{(1,1)} \qquad a_1 = \frac{(y,\cos x)}{(\cos x, \cos x)} \qquad b_1 = \frac{(y,\sin x)}{(\sin x, \sin x)}.$$

24. Find the fourth Legendre polynomial. It is a cubic $x^3 + ax^2 + bx + c$ that is orthogonal to 1, x, and $x^2 - \frac{1}{3}$ over the interval $-1 \le x \le 1$.

25. What is the closest straight line to the parabola $y = x^2$ over $-1 \le x \le 1$?

26. In the Gram-Schmidt formula (10), verify that C is orthogonal to q_1 and q_2.

27. Find an orthonormal basis for the subspace spanned by $a_1 = (1,-1,0,0)$, $a_2 = (0,1,-1,0)$, $a_3 = (0,0,1,-1)$.

28. Apply Gram-Schmidt to $(1,-1,0)$, $(0,1,-1)$, and $(1,0,-1)$, to find an orthonormal basis on the plane $x_1 + x_2 + x_3 = 0$. What is the dimension of this subspace, and how many nonzero vectors come out of Gram-Schmidt?

29. (Recommended) Find orthogonal vectors A, B, C by Gram-Schmidt from a, b, c:

$$a = (1,-1,0,0) \qquad b = (0,1,-1,0) \qquad c = (0,0,1,-1).$$

A, B, C and a, b, c are bases for the vectors perpendicular to $d = (1,1,1,1)$.

30. If $A = QR$ then $A^{\mathrm{T}}A = R^{\mathrm{T}}R =$ ____ triangular times ____ triangular. *Gram-Schmidt on A corresponds to elimination on $A^{\mathrm{T}}A$.* Compare

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \qquad \text{with} \qquad A^{\mathrm{T}}A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}.$$

For $A^{\mathrm{T}}A$, the pivots are 2, $\frac{3}{2}$, $\frac{4}{3}$ and the multipliers are $-\frac{1}{2}$ and $-\frac{2}{3}$.

SPECTRAL THEORY

Strang Chapter 5, Sections 5.1-5.2

Chapter 5

Eigenvalues and Eigenvectors

5.1 Introduction

This chapter begins the "second half" of linear algebra. The first half was about $Ax = b$. The new problem $Ax = \lambda x$ will still be solved by simplifying a matrix—making it diagonal if possible. *The basic step is no longer to subtract a multiple of one row from another*: Elimination changes the eigenvalues, which we don't want.

Determinants give a transition from $Ax = b$ to $Ax = \lambda x$. In both cases the determinant leads to a "formal solution": to Cramer's rule for $x = A^{-1}b$, and to the polynomial $\det(A - \lambda I)$, whose roots will be the eigenvalues. (All matrices are now square; the eigenvalues of a rectangular matrix make no more sense than its determinant.) The determinant can actually be used if $n = 2$ or 3. For large n, computing λ is more difficult than solving $Ax = b$.

The first step is to understand how eigenvalues can be useful, One of their applications is to ordinary differential equations. We shall not assume that the reader is an expert on differential equations! If you can differentiate x^n, $\sin x$, and e^x, you know enough. As a specific example, consider the coupled pair of equations

$$\begin{aligned} \frac{dv}{dt} &= 4v - 5w, \quad v = 8 \quad \text{at} \quad t = 0, \\ \frac{dw}{dt} &= 2v - 3w, \quad w = 5 \quad \text{at} \quad t = 0. \end{aligned} \tag{1}$$

This is an ***initial-value problem***. The unknown is specified at time $t = 0$ by the given initial values 8 and 5. The problem is to find $v(t)$ and $w(t)$ for later times $t > 0$.

It is easy to write the system in matrix form. Let the unknown vector be $u(t)$, with initial value $u(0)$. The coefficient matrix is A:

$$\textbf{Vector unknown} \qquad u(t) = \begin{bmatrix} v(t) \\ w(t) \end{bmatrix}, \quad u(0) = \begin{bmatrix} 8 \\ 5 \end{bmatrix}, \quad A = \begin{bmatrix} 4 & -5 \\ 2 & -3 \end{bmatrix}.$$

The two coupled equations become the vector equation we want:

$$\textbf{Matrix form} \qquad \frac{du}{dt} = Au \quad \text{with} \quad u = u(0) \ \text{ at } \ t = 0. \tag{2}$$

This is the basic statement of the problem. Note that it is a first-order equation—no higher derivatives appear—and it is *linear* in the unknowns, It also has *constant coefficients*; the matrix A is independent of time.

How do we find $u(t)$? If there were only one unknown instead of two, that question would be easy to answer. We would have a scalar instead of a vector equation:

$$\textbf{Single equation} \qquad \frac{du}{dt} = au \quad \text{with} \quad u = u(0) \ \text{ at } \ t = 0. \tag{3}$$

The solution to this equation is the one thing you need to know:

$$\textbf{Pure exponential} \qquad u(t) = e^{at}u(0). \tag{4}$$

At the initial time $t = 0$, u equals $u(0)$ because $e^0 = 1$. The derivative of e^{at} has the required factor a, so that $du/dt = au$. Thus the initial condition and the equation are both satisfied.

Notice the behavior of u for large times. The equation is unstable if $a > 0$, neutrally stable if $a = 0$, or stable if $a < 0$; the factor e^{at} approaches infinity, remains bounded, or goes to zero. If a were a complex number, $a = \alpha + i\beta$, then the same tests would be applied to the real part α. The complex part produces oscillations $e^{i\beta t} = \cos\beta t + i\sin\beta t$. Decay or growth is governed by the factor $e^{\alpha t}$.

So much for a single equation. We shall take a direct approach to systems, and look for solutions with the *same exponential dependence on t* just found in the scalar case:

$$\begin{aligned} v(t) &= e^{\lambda t}y \\ w(t) &= e^{\lambda t}z \end{aligned} \tag{5}$$

or in vector notation

$$u(t) = e^{\lambda t}x. \tag{6}$$

This is the whole key to differential equations $du/dt = Au$: ***Look for pure exponential solutions***. Substituting $v = e^{\lambda t}y$ and $w = e^{\lambda t}z$ into the equation, we find

$$\begin{aligned} \lambda e^{\lambda t}y &= 4e^{\lambda t}y - 5e^{\lambda t}z \\ \lambda e^{\lambda t}z &= 2e^{\lambda t}y - 3e^{\lambda t}z. \end{aligned}$$

The factor $e^{\lambda t}$ is common to every term, and can be removed. This cancellation is the reason for assuming the same exponent λ for both unknowns; it leaves

$$\textbf{Eigenvalue problem} \qquad \begin{aligned} 4y - 5z &= \lambda y \\ 2y - 3z &= \lambda z. \end{aligned} \tag{7}$$

That is the eigenvalue equation. In matrix form it is $Ax = \lambda x$. You can see it again if we use $u = e^{\lambda t}x$—a number $e^{\lambda t}$ that grows or decays times a fixed vector x. ***Substituting into*** $du/dt = Au$ ***gives*** $\lambda e^{\lambda t}x = Ae^{\lambda t}x$***. The cancellation of*** $e^{\lambda t}$ ***produces***

$$\textbf{Eigenvalue equation} \qquad Ax = \lambda x. \tag{8}$$

Now we have the fundamental equation of this chapter. It involves two unknowns λ and x. It is an algebra problem, and differential equations can be forgotten! The number λ (lambda) is an ***eigenvalue*** of the matrix A, and the vector x is the associated ***eigenvector***. Our goal is to find the eigenvalues and eigenvectors, λ's and x's, and to use them.

The Solution of $Ax = \lambda x$

Notice that $Ax = \lambda x$ is a nonlinear equation; λ multiplies x. If we could discover λ, then the equation for x would be linear. In fact we could write λIx in place of λx, and bring this term over to the left side:

$$(A - \lambda I)x = 0. \tag{9}$$

The identity matrix keeps matrices and vectors straight; the equation $(A - \lambda)x = 0$ is shorter, but mixed up. This is the key to the problem:

> ***The vector x is in the nullspace of $A - \lambda I$.***
> ***The number λ is chosen so that $A - \lambda I$ has a nullspace.***

Of course every matrix has a nullspace. It was ridiculous to suggest otherwise, but you see the point. We want a *nonzero* eigenvector x, The vector $x = 0$ always satisfies $Ax = \lambda x$, but it is useless in solving differential equations. The goal is to build $u(t)$ out of exponentials $e^{\lambda t}x$, and *we are interested only in those particular values λ for which there is a nonzero eigenvector x*. To be of any use, the nullspace of $A - \lambda I$ must contain vectors other than zero. In short, $A - \lambda I$ ***must be singular***.

For this, the determinant gives a conclusive test.

5A The number λ is an eigenvalue of A if and only if $A - \lambda I$ is singular:

$$\det(A - \lambda I) = 0. \tag{10}$$

This is the characteristic equation. Each λ is associated with eigenvectors x:

$$(A - \lambda I)x = 0 \qquad \text{or} \qquad Ax = \lambda x. \tag{11}$$

In our example, we shift A by λI to make it singular:

$$\textbf{Subtract } \lambda I \qquad A - \lambda I = \begin{bmatrix} 4 - \lambda & -5 \\ 2 & -3 - \lambda \end{bmatrix}.$$

Note that λ is subtracted only from the main diagonal (because it multiplies I).

$$\textbf{Determinant} \qquad |A - \lambda I| = (4 - \lambda)(-3 - \lambda) + 10 \qquad \text{or} \qquad \lambda^2 - \lambda - 2.$$

This is the *characteristic polynomial*. Its roots, where the determinant is zero, are the eigenvalues. They come from the general formula for the roots of a quadratic, or from

factoring into $\lambda^2-\lambda-2=(\lambda+1)(\lambda-2)$. That is zero if $\lambda=-1$ or $\lambda=2$, as the general formula confirms:

$$\textbf{Eigenvalues} \qquad \lambda=\frac{-b\pm\sqrt{b^2-4ac}}{2a}=\frac{1\pm\sqrt{9}}{2}=-1 \text{ and } 2.$$

There are two eigenvalues, because a quadratic has two roots. Every 2 by 2 matrix $A-\lambda I$ has λ^2 (and no higher power of λ) in its determinant.

The values $\lambda=-1$ and $\lambda=2$ lead to a solution of $Ax=\lambda x$ or $(A-\lambda I)x=0$. A matrix with zero determinant is singular, so there must be nonzero vectors x in its nullspace. In fact the nullspace contains a whole *line* of eigenvectors; it is a subspace!

$$\lambda_1=-1: \qquad (A-\lambda_1 I)x=\begin{bmatrix}5 & -5\\ 2 & -2\end{bmatrix}\begin{bmatrix}y\\ z\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}.$$

The solution (the first eigenvector) is any nonzero multiple of x_1:

$$\textbf{Eigenvector for } \lambda_1 \qquad x_1=\begin{bmatrix}1\\ 1\end{bmatrix}.$$

The computation for λ_2 is done separately:

$$\lambda_2=2: \qquad (A-\lambda_2 I)x=\begin{bmatrix}2 & -5\\ 2 & -5\end{bmatrix}\begin{bmatrix}y\\ z\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}.$$

The second eigenvector is any nonzero multiple of x_2:

$$\textbf{Eigenvector for } \lambda_2 \qquad x_2=\begin{bmatrix}5\\ 2\end{bmatrix}.$$

You might notice that the columns of $A-\lambda_1 I$ give x_2, and the columns of $A-\lambda_2 I$ are multiples of x_1. This is special (and useful) for 2 by 2 matrices.

In the 3 by 3 case, I often set a component of x equal to 1 and solve $(A-\lambda I)x=0$ for the other components. Of course if x is an eigenvector then so is $7x$ and so is $-x$. All vectors in the nullspace of $A-\lambda I$ (which we call the *eigenspace*) will satisfy $Ax=\lambda x$. In our example the eigenspaces are the lines through $x_1=(1,1)$ and $x_2=(5,2)$.

Before going back to the application (the differential equation), we emphasize the steps in solving $Ax=\lambda x$:

1. ***Compute the determinant of*** $A-\lambda I$. With λ subtracted along the diagonal, this determinant is a polynomial of degree n. It starts with $(-\lambda)^n$.

2. ***Find the roots of this polynomial***. The n roots are the eigenvalues of A.

3. ***For each eigenvalue solve the equation*** $(A-\lambda I)x=0$. Since the determinant is zero, there are solutions other than $x=0$. Those are the eigenvectors.

In the differential equation, this produces the special solutions $u = e^{\lambda t}x$. They are the *pure exponential solutions* to $du/dt = Au$. Notice e^{-t} and e^{2t}:

$$u(t) = e^{\lambda_1 t}x_1 = e^{-t}\begin{bmatrix}1\\1\end{bmatrix} \qquad \text{and} \qquad u(t) = e^{\lambda_2 t}x_2 = e^{2t}\begin{bmatrix}5\\2\end{bmatrix}.$$

These two special solutions give the complete solution. They can be multiplied by any numbers c_1 and c_2, and they can be added together. When u_1 and u_2 satisfy the linear equation $du/dt = Au$, so does their sum $u_1 + u_2$:

$$\textbf{Complete solution} \qquad u(t) = c_1 e^{\lambda_1 t}x_1 + c_2 e^{\lambda_2 t}x_2 \tag{12}$$

This is ***superposition***, and it applies to differential equations (homogeneous and linear) just as it applied to matrix equations $Ax = 0$. The nullspace is always a subspace, and combinations of solutions are still solutions.

Now we have two free parameters c_1 and c_2, and it is reasonable to hope that they can be chosen to satisfy the initial condition $u = u(0)$ at $t = 0$:

$$\textbf{Initial condition} \qquad c_1x_1 + c_2x_2 = u(0) \qquad \text{or} \qquad \begin{bmatrix}1 & 5\\1 & 2\end{bmatrix}\begin{bmatrix}c_1\\c_2\end{bmatrix} = \begin{bmatrix}8\\5\end{bmatrix}. \tag{13}$$

The constants are $c_1 = 3$ and $c_2 = 1$, and ***the solution to the original equation is***

$$u(t) = 3e^{-t}\begin{bmatrix}1\\1\end{bmatrix} + e^{2t}\begin{bmatrix}5\\2\end{bmatrix}. \tag{14}$$

Writing the two components separately, we have $v(0) = 8$ and $w(0) = 5$:

$$\textbf{Solution} \qquad v(t) = 3e^{-t} + 5e^{2t}, \qquad w(t) = 3e^{-t} + 2e^{2t}.$$

The key was in the eigenvalues λ and eigenvectors x. Eigenvalues are important in themselves, and not just part of a trick for finding u. Probably the homeliest example is that of soldiers going over a bridge.[1] Traditionally, they stop marching and just walk across. If they happen to march at a frequency equal to one of the eigenvalues of the bridge, it would begin to oscillate. (Just as a child's swing does; you soon notice the natural frequency of a swing, and by matching it you make the swing go higher.) An engineer tries to keep the natural frequencies of his bridge or rocket away from those of the wind or the sloshing of fuel. And at the other extreme, a stockbroker spends his life trying to get in line with the natural frequencies of the market. The eigenvalues are the most important feature of practically any dynamical system.

Summary and Examples

To summarize, this introduction has shown how λ and x appear naturally and automatically when solving $du/dt = Au$. Such an equation has *pure exponential solutions*

[1] One which I never really believed—but a bridge did crash this way in 1831.

$u = e^{\lambda t}x$; the eigenvalue gives the rate of growth or decay, and the eigenvector x develops at this rate. The other solutions will be *mixtures* of these pure solutions, and the mixture is adjusted to fit the initial conditions.

The key equation was $Ax = \lambda x$. Most vectors x will not satisfy such an equation. They change direction when multiplied by A, so that Ax is not a multiple of x. This means that ***only certain special numbers are eigenvalues, and only certain special vectors x are eigenvectors***. We can watch the behavior of each eigenvector, and then combine these "normal modes" to find the solution. To say the same thing in another way, *the underlying matrix can be diagonalized*.

The diagonalization in Section 5.2 will be applied to difference equations, Fibonacci numbers, and Markov processes, and also to differential equations. In every example, we start by computing the eigenvalues and eigenvectors; there is no shortcut to avoid that. Symmetric matrices are especially easy. "Defective matrices" lack a full set of eigenvectors, so they are not diagonalizable. Certainly they have to be discussed, but we will not allow them to take over the book.

We start with examples of particularly good matrices.

Example 1. Everything is clear when A is a ***diagonal matrix***:

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \quad \text{has} \quad \lambda_1 = 3 \quad \text{with} \quad x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \lambda_2 = 2 \quad \text{with} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

On each eigenvector A acts like a multiple of the identity: $Ax_1 = 3x_1$ and $Ax_2 = 2x_2$. Other vectors like $x = (1,5)$ are mixtures $x_1 + 5x_2$ of the two eigenvectors, and when A multiplies x_1 and x_2 it produces the eigenvalues $\lambda_1 = 3$ and $\lambda_2 = 2$:

$$A \quad \text{times} \quad x_1 + 5x_2 \quad \text{is} \quad 3x_1 + 10x_2 = \begin{bmatrix} 3 \\ 10 \end{bmatrix}.$$

This is Ax for a typical vector x—not an eigenvector. But the action of A is determined by its eigenvectors and eigenvalues.

Example 2. The eigenvalues of a ***projection matrix*** are 1 or 0!

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \quad \text{has} \quad \lambda_1 = 1 \quad \text{with} \quad x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \lambda_2 = 0 \quad \text{with} \quad x_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

We have $\lambda = 1$ when x projects to itself, and $\lambda = 0$ when x projects to the zero vector. The column space of P is filled with eigenvectors, and so is the nullspace. If those spaces have dimension r and $n-r$, then $\lambda = 1$ is repeated r times and $\lambda = 0$ is repeated $n-r$ times (*always* n λ's):

Four eigenvalues allowing repeats

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{has} \quad \lambda = 1, 1, 0, 0.$$

There is nothing exceptional about $\lambda = 0$. Like every other number, zero might be an eigenvalue and it might not. If it is, then its eigenvectors satisfy $Ax = 0x$. Thus x is in the nullspace of A. A zero eigenvalue signals that A is singular (not invertible); its determinant is zero. Invertible matrices have all $\lambda \neq 0$.

Example 3. The eigenvalues are on the main diagonal when A is ***triangular***:

$$\det(A-\lambda I) = \begin{vmatrix} 1-\lambda & 4 & 5 \\ 0 & \frac{3}{4}-\lambda & 6 \\ 0 & 0 & \frac{1}{2}-\lambda \end{vmatrix} = (1-\lambda)(\tfrac{3}{4}-\lambda)(\tfrac{1}{2}-\lambda).$$

The determinant is just the product of the diagonal entries. It is zero if $\lambda = 1$, $\lambda = \frac{3}{4}$, or $\lambda = \frac{1}{2}$; the eigenvalues were already sitting along the main diagonal.

This example, in which the eigenvalues can be found by inspection, points to one main theme of the chapter: To transform A into a diagonal or triangular matrix *without changing its eigenvalues*. We emphasize once more that the Gaussian factorization $A = LU$ is not suited to this purpose. The eigenvalues of U may be visible on the diagonal, but they are ***not*** the eigenvalues of A.

For most matrices, there is no doubt that the eigenvalue problem is computationally more difficult than $Ax = b$. With linear systems, a finite number of elimination steps produced the exact answer in a finite time. (Or equivalently, Cramer's rule gave an exact formula for the solution.) No such formula can give the eigenvalues, or Galois would turn in his grave. For a 5 by 5 matrix, $\det(A-\lambda I)$ involves λ^5. Galois and Abel proved that there can be no algebraic formula for the roots of a fifth-degree polynomial.

All they will allow is a few simple checks on the eigenvalues, *after* they have been computed, and we mention two good ones: ***sum and product***.

5B The ***sum*** of the n eigenvalues equals the sum of the n diagonal entries:

$$\textbf{Trace of} \quad A = \lambda_1 + \cdots + \lambda_n = a_{11} + \cdots + a_{nn}. \tag{15}$$

Furthermore, the ***product*** of the n eigenvalues equals the ***determinant*** of A.

The projection matrix P had diagonal entries $\frac{1}{2}$, $\frac{1}{2}$ and eigenvalues 1, 0. Then $\frac{1}{2}+\frac{1}{2}$ agrees with $1+0$ as it should. So does the determinant, which is $0 \cdot 1 = 0$. A singular matrix, with zero determinant, has one or more of its eigenvalues equal to zero.

There should be no confusion between the diagonal entries and the eigenvalues. For a triangular matrix they are the same—but that is exceptional. Normally the pivots, diagonal entries, and eigenvalues are completely different, And for a 2 by 2 matrix, the trace and determinant tell us everything:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{has trace } a+d, \text{ and determinant } ad-bc$$

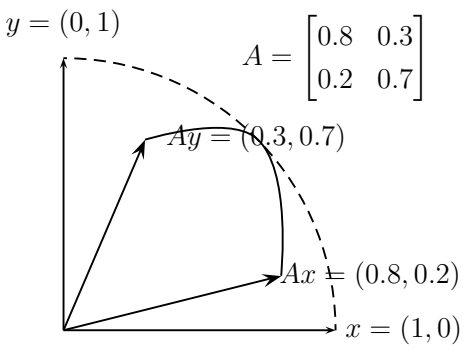
$$\det(A-\lambda I)=\det\begin{vmatrix} a-\lambda & b \\ c & d-\lambda \end{vmatrix}=\lambda^2-(\text{trace})\lambda+\text{determinant}$$

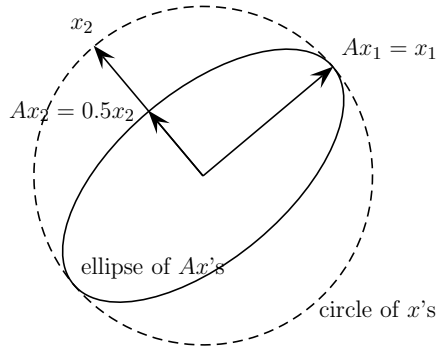
$$\text{The eigenvalues are } \lambda=\frac{\text{trace}\pm\left[(\text{trace})^2-4\det\right]^{1/2}}{2}.$$

Those two λ's add up to the trace; Exercise 9 gives $\sum\lambda_i=$ trace for all matrices.

Eigshow

There is a MATLAB demo (just type eigshow), displaying the eigenvalue problem for a 2 by 2 matrix. It starts with the unit vector $x=(1,0)$. *The mouse makes this vector move around the unit circle.* At the same time the screen shows Ax, in color and also moving. Possibly Ax is ahead of x. Possibly Ax is behind x. *Sometimes Ax is parallel to x.* At that parallel moment, $Ax=\lambda x$ (twice in the second figure).





The eigenvalue λ is the length of Ax, when the unit eigenvector x is parallel. The built-in choices for A illustrate three possibilities: 0, 1, or 2 real eigenvectors.

1. There are *no real eigenvectors. Ax stays behind or ahead of x.* This means the eigenvalues and eigenvectors are complex, as they are for the rotation Q.
2. There is only *one* line of eigenvectors (unusual). The moving directions Ax and x meet but don't cross. This happens for the last 2 by 2 matrix below.
3. There are eigenvectors in *two* independent directions. This is typical! Ax crosses x at the first eigenvector x_1, and it crosses back at the second eigenvector x_2.

Suppose A is singular (rank 1). Its column space is a line. The vector Ax has to stay on that line while x circles around. One eigenvector x is along the line. Another eigenvector appears when $Ax_2=0$. Zero is an eigenvalue of a singular matrix.

You can mentally follow x and Ax for these six matrices. How many eigenvectors and where? When does Ax go clockwise, instead of counterclockwise with x?

$$A=\begin{bmatrix}2&0\\0&1\end{bmatrix}\quad\begin{bmatrix}2&0\\0&-1\end{bmatrix}\quad\begin{bmatrix}0&1\\1&0\end{bmatrix}\quad\begin{bmatrix}0&1\\-1&0\end{bmatrix}\quad\begin{bmatrix}1&1\\1&1\end{bmatrix}\quad\begin{bmatrix}1&1\\0&1\end{bmatrix}$$

Problem Set 5.1

1. Find the eigenvalues and eigenvectors of the matrix $A = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$. Verify that the trace equals the sum of the eigenvalues, and the determinant equals their product.

2. With the same matrix A, solve the differential equation $du/dt = Au$, $u(0) = \begin{bmatrix} 0 \\ 6 \end{bmatrix}$. What are the two pure exponential solutions?

3. If we shift to $A - 7I$, what are the eigenvalues and eigenvectors and how are they related to those of A?

$$B = A - 7I = \begin{bmatrix} -6 & -1 \\ 2 & -3 \end{bmatrix}.$$

4. Solve $du/dt = Pu$, when P is a projection:

$$\frac{du}{dt} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} u \qquad \text{with} \qquad u(0) = \begin{bmatrix} 5 \\ 3 \end{bmatrix}.$$

Part of $u(0)$ increases exponentially while the nullspace part stays fixed.

5. Find the eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 3 & 4 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \qquad \text{and} \qquad B = \begin{bmatrix} 0 & 0 & 2 \\ 0 & 2 & 0 \\ 2 & 0 & 0 \end{bmatrix}.$$

Check that $\lambda_1 + \lambda_2 + \lambda_3$ equals the trace and $\lambda_1\lambda_2\lambda_3$ equals the determinant.

6. Give an example to show that the eigenvalues can be changed when a multiple of one row is subtracted from another. Why is a zero eigenvalue *not* changed by the steps of elimination?

7. Suppose that λ is an eigenvalue of A, and x is its eigenvector: $Ax = \lambda x$.

(a) Show that this same x is an eigenvector of $B = A - 7I$, and find the eigenvalue. This should confirm Exercise 3.

(b) Assuming $\lambda \neq 0$, show that x is also an eigenvector of A^{-1}—and find the eigenvalue.

8. Show that the determinant equals the product of the eigenvalues by imagining that the characteristic polynomial is factored into

$$\det(A - \lambda I) = (\lambda_1 - \lambda)(\lambda_2 - \lambda)\cdots(\lambda_n - \lambda), \tag{16}$$

and making a clever choice of λ.

9. Show that the trace equals the sum of the eigenvalues, in two steps. First, find the coefficient of $(-\lambda)^{n-1}$ on the right side of equation (16). Next, find all the terms in

$$\det(A-\lambda I)=\det\begin{bmatrix} a_{11}-\lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22}-\lambda & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn}-\lambda \end{bmatrix}$$

that involve $(-\lambda)^{n-1}$. They all come from the main diagonal! Find that coefficient of $(-\lambda)^{n-1}$ and compare.

10. (a) Construct 2 by 2 matrices such that the eigenvalues of AB are not the products of the eigenvalues of A and B, and the eigenvalues of $A+B$ are not the sums of the individual eigenvalues.

(b) Verify, however, that the sum of the eigenvalues of $A+B$ equals the sum of all the individual eigenvalues of A and B, and similarly for products. Why is this true?

11. ***The eigenvalues of*** A ***equal the eigenvalues of*** A^{T}. This is because $\det(A-\lambda I)$ equals $\det(A^{\mathrm{T}}-\lambda I)$. That is true because ____. Show by an example that the eigenvectors of A and A^{T} are *not* the same.

12. Find the eigenvalues and eigenvectors of

$$A=\begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix} \qquad \text{and} \qquad A=\begin{bmatrix} a & b \\ b & a \end{bmatrix}.$$

13. If B has eigenvalues 1, 2, 3, C has eigenvalues 4, 5, 6, and D has eigenvalues 7, 8, 9, what are the eigenvalues of the 6 by 6 matrix $A=\left[\begin{smallmatrix} B & C \\ 0 & D \end{smallmatrix}\right]$?

14. Find the rank and all four eigenvalues for both the matrix of ones and the checker board matrix:

$$A=\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \qquad \text{and} \qquad C=\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}.$$

Which eigenvectors correspond to nonzero eigenvalues?

15. What are the rank and eigenvalues when A and C in the previous exercise are n by n? Remember that the eigenvalue $\lambda=0$ is repeated $n-r$ times.

16. If A is the 4 by 4 matrix of ones, find the eigenvalues and the determinant of $A-I$.

17. Choose the third row of the "companion matrix"

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ . & . & . \end{bmatrix}$$

so that its characteristic polynomial $|A - \lambda I|$ is $-\lambda^3 + 4\lambda^2 + 5\lambda + 6$.

18. Suppose A has eigenvalues 0, 3, 5 with independent eigenvectors u, v, w.

(a) Give a basis for the nullspace and a basis for the column space.

(b) Find a particular solution to $Ax = v + w$. Find all solutions.

(c) Show that $Ax = u$ has no solution. (If it had a solution, then ____ would be in the column space.)

19. The powers A^k of this matrix A approaches a limit as $k \to \infty$:

$$A = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix}, \qquad A^2 = \begin{bmatrix} .70 & .45 \\ .30 & .55 \end{bmatrix}, \quad \text{and} \quad A^\infty = \begin{bmatrix} .6 & .6 \\ .4 & .4 \end{bmatrix}.$$

The matrix A^2 is halfway between A and A^∞. Explain why $A^2 = \frac{1}{2}(A + A^\infty)$ from the eigenvalues and eigenvectors of these three matrices.

20. Find the eigenvalues and the eigenvectors of these two matrices:

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \qquad \text{and} \qquad A + I = \begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix}.$$

$A + I$ has the ____ eigenvectors as A. Its eigenvalues are ____ by 1.

21. Compute the eigenvalues and eigenvectors of A and A^{-1}:

$$A = \begin{bmatrix} 0 & 2 \\ 2 & 3 \end{bmatrix} \qquad \text{and} \qquad A^{-1} = \begin{bmatrix} -3/4 & 1/2 \\ 1/2 & 0 \end{bmatrix}.$$

A^{-1} has the ____ eigenvectors as A. When A has eigenvalues λ_1 and λ_2, its inverse has eigenvalues ____.

22. Compute the eigenvalues and eigenvectors of A and A^2:

$$A = \begin{bmatrix} -1 & 3 \\ 2 & 0 \end{bmatrix} \qquad \text{and} \qquad A^2 = \begin{bmatrix} 7 & -3 \\ -2 & 6 \end{bmatrix}.$$

A^2 has the same ____ as A. When A has eigenvalues λ_1 and λ_2, A^2 has eigenvalues ____.

23. (a) If you know x is an eigenvector, the way to find λ is to ____.

(b) If you know λ is an eigenvalue, the way to find x is to ____.

24. What do you do to $Ax = \lambda x$, in order to prove (a), (b), and (c)?

(a) λ^2 is an eigenvalue of A^2, as in Problem 22.

(b) λ^{-1} is an eigenvalue of A^{-1}, as in Problem 21.

(c) $\lambda + 1$ is an eigenvalue of $A + I$, as in Problem 20.

25. From the unit vector $u = \left(\frac{1}{6}, \frac{1}{6}, \frac{3}{6}, \frac{5}{6}\right)$, construct the rank-1 projection matrix $P = uu^{\mathrm{T}}$.

(a) Show that $Pu = u$. Then u is an eigenvector with $\lambda = 1$.

(b) If v is perpendicular to u show that $Pv =$ zero vector. Then $\lambda = 0$.

(c) Find three independent eigenvectors of P all with eigenvalue $\lambda = 0$.

26. Solve $\det(Q - \lambda I) = 0$ by the quadratic formula, to reach $\lambda = \cos\theta \pm i\sin\theta$:

$$Q = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \quad \text{rotates the } xy\text{-plane by the angle } \theta.$$

Find the eigenvectors of Q by solving $(Q - \lambda I)x = 0$. Use $i^2 = -1$.

27. Every permutation matrix leaves $x = (1, 1, \ldots, 1)$ unchanged. Then $\lambda = 1$. Find two more λ's for these permutations:

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \qquad \text{and} \qquad P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

28. If A has $\lambda_1 = 4$ and $\lambda_2 = 5$, then $\det(A - \lambda I) = (\lambda - 4)(\lambda - 5) = \lambda^2 - 9\lambda + 20$. Find three matrices that have trace $a + d = 9$, determinant 20, and $\lambda = 4, 5$.

29. A 3 by 3 matrix B is known to have eigenvalues 0, 1, 2, This information is enough to find three of these:

(a) the rank of B,

(b) the determinant of $B^{\mathrm{T}}B$,

(c) the eigenvalues of $B^{\mathrm{T}}B$, and

(d) the eigenvalues of $(B + I)^{-1}$.

30. Choose the second row of $A = \begin{bmatrix} 0 & 1 \\ * & * \end{bmatrix}$ so that A has eigenvalues 4 and 7.

31. Choose a, b, c, so that $\det(A - \lambda I) = 9\lambda - \lambda^3$. Then the eigenvalues are -3, 0, 3:

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ a & b & c \end{bmatrix}.$$

32. Construct any 3 by 3 Markov matrix M: positive entries down each column add to 1. If $e = (1,1,1)$, verify that $M^{\mathrm{T}}e = e$. By Problem 11, $\lambda = 1$ is also an eigenvalue of M. Challenge: A 3 by 3 singular Markov matrix with trace $\frac{1}{2}$ has eigenvalues $\lambda =$ ____.

33. Find three 2 by 2 matrices that have $\lambda_1 = \lambda_2 = 0$. The trace is zero and the determinant is zero. The matrix A might not be 0 but check that $A^2 = 0$.

34. This matrix is singular with rank 1. Find three λ's and three eigenvectors:

$$A = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 2 \\ 4 & 2 & 4 \\ 2 & 1 & 2 \end{bmatrix}.$$

35. Suppose A and B have the same eigenvalues $\lambda_1, \ldots, \lambda_n$ with the same independent eigenvectors $x_1, \ldots, x_n$. Then $A = B$. *Reason*: Any vector x is a combination $c_1x_1 + \cdots + c_nx_n$. What is Ax? What is Bx?

36. (Review) Find the eigenvalues of A, B, and C:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 0 \end{bmatrix}, \quad \text{and} \quad C = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix}.$$

37. When $a + b = c + d$, show that $(1,1)$ is an eigenvector and find both eigenvalues:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

38. When P exchanges rows 1 and 2 *and* columns 1 and 2, the eigenvalues don't change. Find eigenvectors of A and PAP for $\lambda = 11$:

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 6 & 3 \\ 4 & 8 & 4 \end{bmatrix} \qquad \text{and} \qquad PAP = \begin{bmatrix} 6 & 3 & 3 \\ 2 & 1 & 1 \\ 8 & 4 & 4 \end{bmatrix}.$$

39. Challenge problem: *Is there a real* 2 *by* 2 *matrix* (other than I) *with* $A^3 = I$? Its eigenvalues must satisfy $\lambda^3 = I$. They can be $e^{2\pi i/3}$ and $e^{-2\pi i/3}$. What trace and determinant would this give? Construct A.

40. There are six 3 by 3 permutation matrices P. What numbers can be the *determinants* of P? What numbers can be *pivots*? What numbers can be the *trace* of P? What *four numbers* can be eigenvalues of P?

5.2 Diagonalization of a Matrix

We start right off with the one essential computation. It is perfectly simple and will be used in every section of this chapter. ***The eigenvectors diagonalize a matrix***:

5C Suppose the n by n matrix A has n linearly independent eigenvectors. If these eigenvectors are the columns of a matrix S, then $S^{-1}AS$ is a diagonal matrix Λ. The eigenvalues of A are on the diagonal of Λ:

$$\textbf{Diagonalization} \qquad S^{-1}AS = \Lambda = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix}. \tag{1}$$

We call S the "eigenvector matrix" and Λ the "eigenvalue matrix"—using a capital lambda because of the small lambdas for the eigenvalues on its diagonal.

Proof. Put the eigenvectors x_i in the columns of S, and compute AS by columns:

$$AS = A\begin{bmatrix} | & | & & | \\ x_1 & x_2 & \cdots & x_n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ \lambda_1 x_1 & \lambda_2 x_2 & \cdots & \lambda_n x_n \\ | & | & & | \end{bmatrix}.$$

Then the trick is to split this last matrix into a quite different product $S\Lambda$:

$$\begin{bmatrix} & & & \\ \lambda_1 x_1 & \lambda_2 x_2 & \cdots & \lambda_n x_n \\ & & & \end{bmatrix} = \begin{bmatrix} & & & \\ x_1 & x_2 & \cdots & x_n \\ & & & \end{bmatrix} \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix}.$$

It is crucial to keep these matrices in the right order. If Λ came before S (instead of after), then λ_1 would multiply the entries in the first row. We want λ_1 to appear in the first column. As it is, $S\Lambda$ is correct. Therefore,

$$AS = S\Lambda, \quad \text{or} \quad S^{-1}AS = \Lambda, \quad \text{or} \quad A = S\Lambda S^{-1}. \tag{2}$$

S is invertible, because its columns (the eigenvectors) were assumed to be independent.

We add four remarks before giving any examples or applications. □

Remark* 1.** If the matrix A has no repeated eigenvalues—the numbers $\lambda_1, \ldots, \lambda_n$ are distinct—then its n eigenvectors are automatically independent (see 5D below). Therefore ***any matrix with distinct eigenvalues can be diagonalized.

***Remark* 2.** The diagonalizing matrix S is *not unique*. An eigenvector x can be multiplied by a constant, and remains an eigenvector. We can multiply the columns of S by any nonzero constants, and produce a new diagonalizing S. Repeated eigenvalues leave even more freedom in S. For the trivial example $A = I$, any invertible S will do: $S^{-1}IS$ is is always diagonal (Λ is just I). All vectors are eigenvectors of the identity.

***Remark* 3.** *Other matrices S will not produce a diagonal Λ.* Suppose the first column of S is y. Then the first column of $S\Lambda$ is $\lambda_1 y$. If this is to agree with the first column of AS, which by matrix multiplication is Ay, then y must be an eigenvector: $Ay = \lambda_1 y$. The *order* of the eigenvectors in S and the eigenvalues in Λ is automatically the same.

Remark* 4.** Not all matrices possess n linearly independent eigenvectors, so ***not all matrices are diagonalizable. The standard example of a "defective matrix" is

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Its eigenvalues are $\lambda_1 = \lambda_2 = 0$, since it is triangular with zeros on the diagonal:

$$\det(A - \lambda I) = \det \begin{bmatrix} -\lambda & 1 \\ 0 & -\lambda \end{bmatrix} = \lambda^2.$$

All eigenvectors of this A are multiples of the vector $(1,0)$:

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad \text{or} \qquad x = \begin{bmatrix} c \\ 0 \end{bmatrix}.$$

$\lambda = 0$ is a double eigenvalue—its *algebraic multiplicity* is 2. But the *geometric multiplicity* is 1—there is only one independent eigenvector. We can't construct S.

Here is a more direct proof that this A is not diagonalizable. Since $\lambda_1 = \lambda_2 = 0$, Λ would have to be the zero matrix, But if $\Lambda = S^{-1}AS = 0$, then we premultiply by S and postmultiply by S^{-1}, to deduce falsely that $A = 0$. There is no invertible S.

That failure of diagonalization was ***not*** a result of $\lambda = 0$. It came from $\lambda_1 = \lambda_2$:

Repeated eigenvalues $\qquad A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$ and $A = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$.

Their eigenvalues are 3, 3 and 1, 1. They are not singular! The problem is the shortage of eigenvectors—which are needed for S. That needs to be emphasized:

> ***Diagonalizability of A depends on enough eigenvectors.***
> ***Invertibility of A depends on nonzero eigenvalues.***

There is no connection between diagonalizability (n independent eigenvector) and invertibility (no zero eigenvalues). The only indication given by the eigenvalues is this: *Diagonalization can fail only if there are repeated eigenvalues.* Even then, it does not always fail. $A = I$ has repeated eigenvalues $1, 1, \ldots, 1$ but it is already diagonal! There is no shortage of eigenvectors in that case.

The test is to check, for an eigenvalue that is repeated p times, whether there are p independent eigenvectors—in other words, whether $A - \lambda I$ has rank $n - p$. To complete that circle of ideas, we have to show that *distinct* eigenvalues present no problem.

5D If eigenvectors $x_1,\dots,x_k$ correspond to *different eigenvalues* $\lambda_1,\dots,\lambda_k$, then those eigenvectors are linearly independent.

Suppose first that $k=2$, and that some combination of x_1 and x_2 produces zero: $c_1x_1+c_2x_2=0$. Multiplying by A, we find $c_1\lambda_1x_1+c_2\lambda_2x_2=0$. Subtracting λ_2 times the previous equation, the vector x_2 disappears:

$$c_1(\lambda_1-\lambda_2)x_1=0.$$

Since $\lambda_1\neq\lambda_2$ and $x_1\neq 0$, we are forced into $c_1=0$. Similarly $c_2=0$, and the two vectors are independent; only the trivial combination gives zero.

This same argument extends to any number of eigenvectors: If some combination produces zero, multiply by A, subtract λ_k times the original combination, and x_k disappears—leaving a combination of $x_1,\dots,x_{k-1}$, which produces zero. By repeating the same steps (this is really *mathematical induction*) we end up with a multiple of x_1 that produces zero. This forces $c_1=0$, and ultimately every $c_i=0$. Therefore eigenvectors that come from distinct eigenvalues are automatically independent.

A matrix with n distinct eigenvalues can be diagonalized. This is the typical case.

Examples of Diagonalization

The main point of this section is $S^{-1}AS=A$. The eigenvector matrix S converts A into its eigenvalue matrix Λ (diagonal). We see this for projections and rotations.

Example 1. The projection $A=\begin{bmatrix}\frac{1}{2}&\frac{1}{2}\\ \frac{1}{2}&\frac{1}{2}\end{bmatrix}$ has eigenvalue matrix $\Lambda=\begin{bmatrix}1&0\\0&0\end{bmatrix}$. The eigenvectors go into the columns of S:

$$S=\begin{bmatrix}1&1\\1&-1\end{bmatrix}\qquad\text{and}\qquad AS=S\Lambda=\begin{bmatrix}1&0\\1&0\end{bmatrix}.$$

That last equation can be verified at a glance. Therefore $S^{-1}AS=\Lambda$.

Example 2. The eigenvalues themselves are not so clear for a ***rotation***:

$$\textbf{90° rotation}\qquad K=\begin{bmatrix}0&-1\\1&0\end{bmatrix}\quad\text{has}\quad\det(K-\lambda I)=\lambda^2+1.$$

How can a vector be rotated and still have its direction unchanged? Apparently it can't—except for the zero vector, which is useless. But there must be eigenvalues, and we must be able to solve $du/dt=Ku$. The characteristic polynomial λ^2+1 should still have two roots—but those roots are *not real*.

You see the way out. The eigenvalues of K are *imaginary numbers*, $\lambda_1=i$ and $\lambda_2=-i$. The eigenvectors are also not real. Somehow, in turning through 90°, they are

multiplied by i or $-i$:

$$(K-\lambda_1 I)x_1 = \begin{bmatrix} -i & -1 \\ 1 & -i \end{bmatrix}\begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \qquad \text{and} \qquad x_1 = \begin{bmatrix} 1 \\ -i \end{bmatrix}$$

$$(K-\lambda_2 I)x_2 = \begin{bmatrix} i & -1 \\ 1 & i \end{bmatrix}\begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \qquad \text{and} \qquad x_2 = \begin{bmatrix} 1 \\ i \end{bmatrix}.$$

The eigenvalues are distinct, even if imaginary, and the eigenvectors are independent. They go into the columns of S:

$$S = \begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix} \qquad \text{and} \qquad S^{-1}KS = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}.$$

We are faced with an inescapable fact, that ***complex numbers are needed even for real matrices***. If there are too few real eigenvalues, there are always n complex eigenvalues. (Complex includes real, when the imaginary part is zero.) If there are too few eigenvectors in the real world $\mathbf{R}^3$, or in $\mathbf{R}^n$, we look in $\mathbf{C}^3$ or $\mathbf{C}^n$. The space $\mathbf{C}^n$ contains all column vectors with complex components, and it has new definitions of length and inner product and orthogonality. But it is not more difficult than $\mathbf{R}^n$, and in Section 5.5 we make an easy conversion to the complex case.

Powers and Products: A^k and AB

There is one more situation in which the calculations are easy. ***The eigenvalue of A^2 are exactly $\lambda_1^2,\dots,\lambda_n^2$, and every eigenvector of A is also an eigenvector of A^2.*** We start from $Ax = \lambda x$, and multiply again by A:

$$A^2x = A\lambda x = \lambda Ax = \lambda^2 x. \tag{3}$$

Thus λ^2 is an eigenvalue of A^2, with the same eigenvector x. If the first multiplication by A leaves the direction of x unchanged, then so does the second.

The same result comes from diagonalization, by squaring $S^{-1}AS = \Lambda$:

$$\textbf{Eigenvalues of } A^2 \qquad (S^{-1}AS)(S^{-1}AS) = \Lambda^2 \quad \text{or} \quad S^{-1}A^2S = \Lambda^2.$$

The matrix A^2 is diagonalized by the same S, so the eigenvectors are unchanged. The eigenvalues are squared. This continues to hold for any power of A:

5E The eigenvalues of A^k are $\lambda_1^k,\dots,\lambda_n^k$, and each eigenvector of A is still an eigenvector of A^k. When S diagonalizes A, it also diagonalizes A^k:

$$\Lambda^k = (S^{-1}AS)(S^{-1}AS)\cdots(S^{-1}AS) = S^{-1}A^kS. \tag{4}$$

Each S^{-1} cancels an S, except for the first S^{-1} and the last S.

If A is invertible this rule also applies to its inverse (the power $k=-1$). ***The eigenvalues of*** A^{-1} ***are*** $1/\lambda_i$. That can be seen even without diagonalizing:

$$\textit{if} \quad Ax=\lambda x \quad \textit{then} \quad x=\lambda A^{-1}x \quad \textit{and} \quad \frac{1}{\lambda}x=A^{-1}x.$$

Example 3. If K is rotation through 90°, then K^2 is rotation through 180° (which means $-I$) and K^{-1} is rotation through $-90°$:

$$K=\begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}, \qquad K^2=\begin{bmatrix}-1 & 0\\ 0 & -1\end{bmatrix}, \quad \text{and} \quad K^{-1}=\begin{bmatrix}0 & 1\\ -1 & 0\end{bmatrix}.$$

The eigenvalues of K are i and $-i$; their squares are -1 and -1; their reciprocals are $1/i=-i$ and $1/(-i)=i$. Then K^4 is a complete rotation through 360°:

$$K^4=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} \quad \text{and also} \quad \Lambda^4=\begin{bmatrix}i^4 & 0\\ 0 & (-i)^4\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}.$$

For a ***product of two matrices***, we can ask about the eigenvalues of AB—but we won't get a good answer. It is very tempting to try the same reasoning, hoping to prove what is *not in general true*. If λ is an eigenvalue of A and μ is an eigenvalue of B, here is the false proof that AB has the eigenvalue $\mu\lambda$:

$$\textbf{False proof} \qquad ABx=A\mu x=\mu Ax=\mu\lambda x.$$

The mistake lies in assuming that A and B share the *same* eigenvector x. In general, they do not, We could have two matrices with zero eigenvalues, while AB has $\lambda=1$:

$$AB=\begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}\begin{bmatrix}0 & 0\\ 1 & 0\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}.$$

The eigenvectors of this A and B are completely different, which is typical. For the same reason, the eigenvalues of $A+B$ generally have nothing to do with $\lambda+\mu$.

This false proof does suggest what *is* true. If the eigenvector is the same for A and B, then the eigenvalues multiply and AB has the eigenvalue $\mu\lambda$. But there is something more important. There is an easy way to recognize when A and B share a full set of eigenvectors, and that is a key question in quantum mechanics:

5F Diagonalizable matrices share the same eigenvector matrix S if and only if $AB=BA$.

Proof. If the same S diagonalizes both $A=S\Lambda_1S^{-1}$ and $B=S\Lambda_2S^{-1}$, we can multiply in either order:

$$AB=S\Lambda_1S^{-1}S\Lambda_2S^{-1}=S\Lambda_1\Lambda_2S^{-1} \quad \text{and} \quad BA=S\Lambda_2S^{-1}S\Lambda_1S^{-1}=S\Lambda_2\Lambda_1S^{-1}.$$

Since $\Lambda_1\Lambda_2=\Lambda_2\Lambda_1$ (diagonal matrices always commute) we have $AB=BA$.

In the opposite direction, suppose $AB = BA$. Starting from $Ax = \lambda x$, we have

$$ABx = BAx = B\lambda x = \lambda Bx.$$

Thus x and Bx are both eigenvectors of A, sharing the same λ (or else $Bx = 0$). If we assume for convenience that the eigenvalues of A are distinct—the eigenspaces are all one-dimensional—then *Bx must be a multiple of x.* in other words x is an eigenvector of B as well as A. The proof with repeated eigenvalues is a little longer. □

Heisenberg's uncertainty principle comes from noncommuting matrices, like position P and momentum Q. Position is symmetric, momentum is skew-symmetric, and together they satisfy $QP - PQ = I$. The uncertainty principle follows directly from the Schwarz inequality $(Qx)^{\mathrm{T}}(Px) \le \|Qx\|\|Px\|$ of Section 3.2:

$$\|x\|^2 = x^{\mathrm{T}}x = x^{\mathrm{T}}(QP - PQ)x \le 2\|Qx\|\|Px\|.$$

The product of $\|Qx\|/\|x\|$ and $\|Px\|/\|x\|$—momentum and position errors, when the wave function is x—is at least $\frac{1}{2}$. It is impossible to get both errors small, because when you try to measure the position of a particle you change its momentum.

At the end we come back to $A = S\Lambda S^{-1}$. That factorization is particularly suited to take powers of A, and the simplest case A^2 makes the point. The LU factorization is hopeless when squared, but $S\Lambda S^{-1}$ is perfect. The square is $S\Lambda^2 S^{-1}$, and the eigenvectors are unchanged. By following those eigenvectors we will solve difference equations and differential equations.

Problem Set 5.2

1. Factor the following matrices into $S\Lambda S^{-1}$:

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \qquad \text{and} \qquad A = \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}.$$

2. Find the matrix A whose eigenvalues are 1 and 4, and whose eigenvectors are $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$, respectively. (*Hint*: $A = S\Lambda S^{-1}$.)

3. Find *all* the eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

and write two different diagonalizing matrices S.

4. If a 3 by 3 upper triangular matrix has diagonal entries 1, 2, 7, how do you know it can be diagonalized? What is Λ?

5. Which of these matrices cannot be diagonalized?

$$A_1 = \begin{bmatrix} 2 & -2 \\ 2 & -2 \end{bmatrix} \qquad A_2 = \begin{bmatrix} 2 & 0 \\ 2 & -2 \end{bmatrix} \qquad A_3 = \begin{bmatrix} 2 & 0 \\ 2 & 2 \end{bmatrix}.$$

6. (a) If $A^2 = I$, what are the possible eigenvalues of A?

(b) If this A is 2 by 2, and not I or $-I$, find its trace and determinant.

(c) If the first row is $(3,-1)$, what is the second row?

7. If $A = \left[\begin{smallmatrix} 4 & 3 \\ 1 & 2 \end{smallmatrix}\right]$, find A^{100} by diagonalizing A.

8. Suppose $A = uv^{\mathrm{T}}$ is a column times a row (a rank-1 matrix).

(a) By multiplying A times u, show that u is an eigenvector. What is λ?

(b) What are the other eigenvalues of A (and why)?

(c) Compute trace(A) from the sum on the diagonal and the sum of λ's.

9. Show by direct calculation that AB and BA have the same trace when

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \qquad \text{and} \qquad B = \begin{bmatrix} q & r \\ s & t \end{bmatrix}.$$

Deduce that $AB - BA = I$ is impossible (except in infinite dimensions).

10. Suppose A has eigenvalues 1, 2, 4. What is the trace of A^2? What is the determinant of $(A^{-1})^{\mathrm{T}}$?

11. If the eigenvalues of A are 1, 1, 2, which of the following are certain to be true? Give a reason if true or a counterexample if false:

(a) A is invertible.

(b) A is diagonalizable.

(c) A is not diagonalizable.

12. Suppose the only eigenvectors of A are multiples of $x = (1,0,0)$. True or false:

(a) A is not invertible.

(b) A has a repeated eigenvalue.

(c) A is not diagonalizable.

13. Diagonalize the matrix $A = \left[\begin{smallmatrix} 5 & 4 \\ 4 & 5 \end{smallmatrix}\right]$ and find one of its square roots—a matrix such that $R^2 = A$. How many square roots will there be?

14. Suppose the eigenvector matrix S has $S^{\mathrm{T}} = S^{-1}$. Show that $A = S\Lambda S^{-1}$ is symmetric and has orthogonal eigenvectors.

Problems 15–24 are about the eigenvalue and eigenvector matrices.

15. Factor these two matrices into $A = S\Lambda S^{-1}$:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \qquad \text{and} \qquad A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}.$$

16. If $A = S\Lambda S^{-1}$ then $A^3 = (\quad)(\quad)(\quad)$ and $A^{-1} = (\quad)(\quad)(\quad)$.

17. If A has $\lambda_1 = 2$ with eigenvector $x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\lambda_2 = 5$ with $x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, use $S\Lambda S^{-1}$ to find A. No other matrix has the same λ's and x's.

18. Suppose $A = S\Lambda S^{-1}$. What is the eigenvalue matrix for $A + 2I$? What is the eigenvector matrix? Check that $A + 2I = (\quad)(\quad)(\quad)^{-1}$.

19. True or false: If the n columns of S (eigenvectors of A) are independent, then

(a) A is invertible.

(b) A is diagonalizable.

(c) S is invertible.

(d) S is diagonalizable.

20. If the eigenvectors of A are the columns of I, then A is a _____ matrix. If the eigenvector matrix S is triangular, then S^{-1} is triangular and A is triangular.

21. Describe all matrices S that diagonalize this matrix A:

$$A = \begin{bmatrix} 4 & 0 \\ 1 & 2 \end{bmatrix}.$$

Then describe all matrices that diagonalize A^{-1}.

22. Write the most general matrix that has eigenvectors $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

23. Find the eigenvalues of A and B and $A + B$:

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad A + B = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Eigenvalues of $A + B$ (are equal to)(are not equal to) eigenvalues of A plus eigenvalues of B.

24. Find the eigenvalues of A, B, AB, and BA:

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad AB = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad \text{and} \quad BA = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}.$$

Eigenvalues of AB (are equal to)(are not equal to) eigenvalues of A times eigenvalues of B. Eigenvalues of AB (are)(are not) equal to eigenvalues of BA.

Problems 25–28 are about the diagonalizability of A.

25. True or false: If the eigenvalues of A are 2, 2, 5, then the matrix is certainly

(a) invertible.

(b) diagonalizable.

(c) not diagonalizable.

26. If the eigenvalues of A are 1 and 0, write everything you know about the matrices A and A^2.

27. Complete these matrices so that $\det A = 25$. Then trace $= 10$, and $\lambda = 5$ is repeated! Find an eigcnvector with $Ax = 5x$. These matrices will nothe diagonalizabie because there is no second line of eigenvectors.

$$A = \begin{bmatrix} 8 & \\ & 2 \end{bmatrix}, \qquad A = \begin{bmatrix} 9 & 4 \\ & 1 \end{bmatrix}, \quad \text{and} \quad A = \begin{bmatrix} 10 & 5 \\ -5 & \end{bmatrix}.$$

28. The matrix $A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$ is not diagonalizable because the rank of $A - 3I$ is ____. Change one entry to make A diagonalizable. Which entries could you change?

Problems 29–33 are about powers of matrices.

29. $A^k = S\Lambda^k S^{-1}$ approaches the zero matrix as $k \to \infty$ if and only if every λ has absolute value less than ____. Does $A^k \to 0$ or $B^k \to 0$?

$$A = \begin{bmatrix} .6 & .4 \\ .4 & .6 \end{bmatrix} \qquad \text{and} \qquad B = \begin{bmatrix} .6 & .9 \\ .1 & .6 \end{bmatrix}.$$

30. (Recommended) Find Λ and S to diagonalize A in Problem 29. What is the limit of Λ^k as $k \to \infty$? What is the limit of $S\Lambda^k S^{-1}$? In the columns of this limiting matrix you see the ____.

31. Find Λ and S to diagonalize B in Problem 29. What is $B^{10}u_0$ for these u_0?

$$u_0 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \qquad u_0 = \begin{bmatrix} 3 \\ -1 \end{bmatrix}, \quad \text{and} \quad u_0 = \begin{bmatrix} 6 \\ 0 \end{bmatrix}.$$

32. Diagonalize A and compute $S\Lambda^k S^{-1}$ to prove this formula for A^k:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \qquad \text{has} \qquad A^k = \frac{1}{2}\begin{bmatrix} 3^k + 1 & 3^k - 1 \\ 3^k - 1 & 3^k + 1 \end{bmatrix}.$$

33. Diagonalize B and compute $S\Lambda^k S^{-1}$ to prove this formula for B^k:

$$B = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \qquad \text{has} \qquad B^k = \begin{bmatrix} 3^k & 3^k - 2^k \\ 0 & 2^k \end{bmatrix}.$$

Problems 34–44 are new applications of $A = S\Lambda S^{-1}$.

34. Suppose that $A = S\Lambda S^{-1}$. Take determinants to prove that $\det A = \lambda_1\lambda_2\cdots\lambda_n =$ product of λ's. This quick proof only works when A is ____.

35. The trace of S times ΛS^{-1} equals the trace of ΛS^{-1} times S. So the trace of a diagonalizable A equals the trace of Λ, which is ____.

36. If $A = S\Lambda S^{-1}$, diagonalize the block matrix $B = \begin{bmatrix} A & 0 \\ 0 & 2A \end{bmatrix}$. Find its eigenvalue and eigenvector matrices.

37. Consider all 4 by 4 matrices A that are diagonalized by the same fixed eigenvector matrix S. Show that the A's form a subspace (cA and $A_1 + A_2$ have this same S). What is this subspace when $S = I$? What is its dimension?

38. Suppose $A^2 = A$. On the left side A multiplies each column of A. Which of our four subspaces contains eigenvectors with $\lambda = 1$? Which subspace contains eigenvectors with $\lambda = 0$? From the dimensions of those subspaces, A has a full set of independent eigenvectors and can be diagonalized.

39. Suppose $Ax = \lambda x$. If $\lambda = 0$, then x is in the nullspace. If $\lambda \neq 0$, then x is in the column space. Those spaces have dimensions $(n-r) + r = n$. So why doesn't every square matrix have n linearly independent eigenvectors?

40. Substitute $A = S\Lambda S^{-1}$ into the product $(A-\lambda_1 I)(A-\lambda_2 I)\cdots(A-\lambda_n I)$ and explain why this produces the *zero matrix*. We are substituting the matrix A for the number λ in the polynomial $p(\lambda) = \det(A - \lambda I)$. The ***Cayley-Hamilton Theorem*** says that this product is always $p(A) =$ *zero matrix*, even if A is not diagonalizable.

41. Test the Cayley-Hamilton Theorem on Fibonacci's matrix $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$. The theorem predicts that $A^2 - A - I = 0$, since $\det(A - \lambda I)$ is $\lambda^2 - \lambda - 1$.

42. If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $\det(A - \lambda I)$ is $(\lambda - a)(\lambda - d)$. Check the Cayley-Hamilton statement that $(A - aI)(A - dI) =$ *zero matrix*.

43. If $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ and $AB = BA$, show that $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is also diagonal. B has the same eigen____ as A, but different eigen____. These diagonal matrices B form a two-dimensional subspace of matrix space. $AB - BA = 0$ gives four equations for the unknowns **a**, **b**, **c**, **d**—find the rank of the 4 by 4 matrix.

44. If A is 5 by 5. then $AB - BA =$ zero matrix gives 25 equations for the 25 entries in B. Show that the 25 by 25 matrix is singular by noticing a simple nonzero solution B.

45. Find the eigenvalues and eigenvectors for both of these Markov matrices A and A^∞. Explain why A^{100} is close to A^∞:

$$A = \begin{bmatrix} .6 & .2 \\ .4 & .8 \end{bmatrix} \qquad \text{and} \qquad A^\infty = \begin{bmatrix} 1/3 & 1/3 \\ 2/3 & 2/3 \end{bmatrix}.$$

QUADRATIC FORMS

Strang Chapter 6, Sections 6.1-6.2

Chapter 6

Positive Definite Matrices

6.1 Minima, Maxima, and Saddle Points

Up to now, we have hardly thought about **the signs of the eigenvalues**. We couldn't ask whether λ was positive before it was known to be real. Chapter 5 established that every symmetric matrix has real eigenvalues. Now we will find a test that can be applied directly to A, without computing its eigenvalues, which will ***guarantee that all those eigenvalues are positive***. The test brings together three of the most basic ideas in the book—*pivots*, *determinants*, and *eigenvalues*.

The signs of the eigenvalues are often crucial. For stability in differential equations, we needed negative eigenvalues so that $e^{\lambda t}$ would decay. The new and highly important problem is to recognize a ***minimum point***. This arises throughout science and engineering and every problem of optimization. The mathematical problem is to move the second derivative test $F'' > 0$ into n dimensions. Here are two examples:

$$F(x,y) = 7 + 2(x+y)^2 - y\sin y - x^3 \qquad f(x,y) = 2x^2 + 4xy + y^2.$$

Does either $F(x,y)$ or $f(x,y)$ have a minimum at the point $x = y = 0$?

Remark 3. The *zero-order terms* $F(0,0) = 7$ and $f(0,0) = 0$ have no effect on the answer. They simply raise or lower the graphs of F and f.

Remark 4. The *linear terms* give a necessary condition: To have any chance of a minimum, the first derivatives must vanish at $x = y = 0$:

$$\frac{\partial F}{\partial x} = 4(x+y) - 3x^2 = 0 \qquad \text{and} \qquad \frac{\partial F}{\partial y} = 4(x+y) - y\cos y - \sin y = 0$$

$$\frac{\partial f}{\partial x} = 4x + 4y = 0 \qquad \text{and} \qquad \frac{\partial f}{\partial y} = 4x + 2y = 0. \quad \textit{All zero.}$$

Thus $(x,y) = (0,0)$ is a stationary point for both functions. The surface $z = F(x,y)$ is tangent to the horizontal plane $z = 7$, and the surface $z = f(x,y)$ is tangent to the plane $z = 0$. The question is whether the graphs go *above those planes or not*, as we move away from the tangency point $x = y = 0$.

Remark* 5. *The second derivatives at* $(0,0)$ *are decisive:

$$\frac{\partial^2 F}{\partial x^2} = 4 - 6x = 4 \qquad \frac{\partial^2 f}{\partial x^2} = 4$$
$$\frac{\partial^2 F}{\partial x \partial y} = \frac{\partial^2 F}{\partial y \partial x} = 4 \qquad \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = 4$$
$$\frac{\partial^2 F}{\partial y^2} = 4 + y\sin y - 2\cos y = 2 \qquad \frac{\partial^2 f}{\partial y^2} = 2.$$

These second derivatives 4, 4, 2 contain the answer. Since they are the same for F and f, they must contain the same answer. The two functions behave in exactly the same way near the origin. F ***has a minimum if and only if*** f ***has a minimum***. I am going to show that those functions don't!

***Remark* 6.** The *higher-degree terms* in F have no effect on the question of a *local* minimum, but they can prevent it from being a *global* minimum. In our example the term $-x^3$ must sooner or later pull F toward $-\infty$. For $f(x,y)$, with no higher terms, all the action is at $(0,0)$.

Every quadratic form $f = ax^2 + 2bxy + cy^2$ *has a stationary point at the origin, where* $\partial f/\partial x = \partial f/\partial y = 0$. A local minimum would also be a global minimum, The surface $z = f(x,y)$ will then be shaped like a bowl, resting on the origin (Figure 6.1). If the stationary point of F is at $x = \alpha$, $y = \beta$, the only change would be to use the second derivatives at α, β:

Quadratic part of F
$$f(x,y) = \frac{x^2}{2}\frac{\partial^2 F}{\partial x^2}(\alpha,\beta) + xy\frac{\partial^2 F}{\partial x \partial y}(\alpha,\beta) + \frac{y^2}{2}\frac{\partial^2 F}{\partial y^2}(\alpha,\beta). \quad (1)$$

This $f(x,y)$ behaves near $(0,0)$ in the same way that $F(x,y)$ behaves near (α,β).

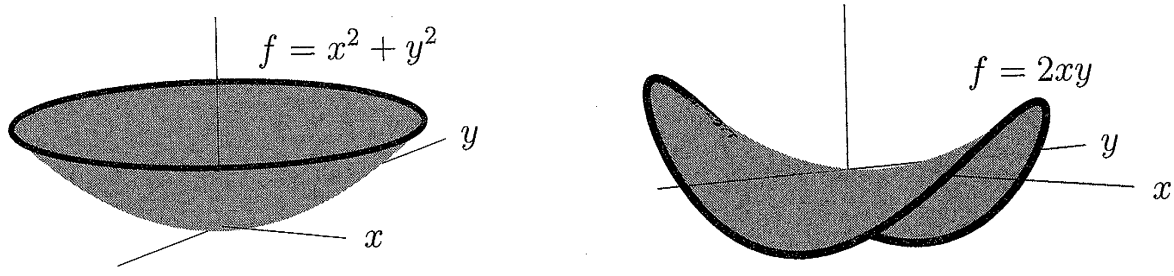


Figure 6.1: A bowl and a saddle: Definite $A = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right]$ and indefinite $A = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$.

The third derivatives are drawn into the problem when the second derivatives fail to give a definite decision. That happens when the quadratic part is singular. For a true minimum, f is allowed to vanish *only* at $x = y = 0$. When $f(x,y)$ is strictly positive at all other points (the bowl goes up), it is called ***positive definite***.

Definite versus Indefinite: Bowl versus Saddle

The problem comes down to this: For a function of two variables x and y, what is the correct replacement for the condition $\partial^2 F/\partial x^2 > 0$? With only one variable, the sign of the second derivative decides between a minimum or a maximum. Now we have three second derivatives: F_{xx}, $F_{xy} = F_{yx}$, and F_{yy}. These three numbers (like 4, 4, 2) must determine whether or not F (as well as f) has a minimum.

What conditions on a, b, and c ensure that the quadratic $f(x,y) = ax^2 + 2bxy + cy^2$ is positive definite? One necessary condition is easy:

(i) *If $ax^2 + 2bxy + cy^2$ is positive definite, then necessarily $a > 0$.*

We look at $x = 1$, $y = 0$, where $ax^2 + 2bxy + cy^2$ is equal to a. This must be positive. Translating back to F, that means that $\partial^2 F/\partial x^2 > 0$. The graph must go up in the x direction. Similarly, fix $x = 0$ and look in the y direction where $f(0,y) = cy^2$:

(ii) *If $f(x,y)$ is positive definite, then necessarily $c > 0$.*

Do these conditions $a > 0$ and $c > 0$ guarantee that $f(x,y)$ is always positive? The answer is **no**. A large cross term $2bxy$ can pull the graph below zero.

Example 1. $f(x,y) = x^2 - 10xy + y^2$. Here $a = 1$ and $c = 1$ are both positive. But f is not positive definite, because $f(1,1) = -8$. The conditions $a > 0$ and $c > 0$ ensure that $f(x,y)$ is positive on the x and y axes. But this function is negative on the line $x = y$, because $b = -10$ overwhelms a and c.

Example 2. In our original f the coefficient $2b = 4$ was positive. Does this ensure a minimum? Again the answer is **no**; the sign of b is of no importance! *Even though its second derivatives are positive, $2x^2 + 4xy + y^2$ is not positive definite. **Neither F nor f has a minimum at** $(0,0)$ **because** $f(1,-1) = 2 - 4 + 1 = -1$.*

It is the size of b, compared to a and c, that must be controlled. We now want a necessary and sufficient condition for positive definiteness. The simplest technique is to complete the square:

Express $f(x,y)$ using squares

$$f = ax^2 + 2bxy + cy^2 = a\left(x + \frac{b}{a}y\right)^2 + \left(c - \frac{b^2}{a}\right)y^2. \quad (2)$$

The first term on the right is never negative, when the square is multiplied by $a > 0$. But this square can be zero, and the second term must then be positive. That term has coefficient $(ac - b^2)/a$. The last requirement for positive definiteness is that this coefficient must be positive:

(iii) If $ax^2 + 2bxy + cy^2$ stays positive, then necessarily $ac > b^2$.

Test for a minimum: The conditions $a > 0$ and $ac > b^2$ are just right. They guarantee $c > 0$. The right side of (2) is positive, and we have found a minimum:

6A $ax^2+2bxy+cy^2$ is positive definite if and only if $a>0$ and $ac>b^2$. Any $f(x,y)$ has a minimum at a point where $\partial F/\partial x=\partial F/\partial y=0$ with

$$\frac{\partial F^2}{\partial x^2}>0 \qquad \text{and} \qquad \left[\frac{\partial F^2}{\partial x^2}\right]\left[\frac{\partial F^2}{\partial y^2}\right]>\left[\frac{\partial F^2}{\partial x\partial y}\right]^2. \tag{3}$$

Test for a maximum: Since f has a maximum whenever $-f$ has a minimum, we just reverse the signs of a, b, and c. This actually leaves $ac>b^2$ unchanged: The quadratic form is ***negative definite*** if and only if $a<0$ and $ac>b^2$. The same change applies for a maximum of $F(x,y)$.

Singular case $ac=b^2$**:** The second term in equation (2) disappears to leave only the first square—which is either ***positive semidefinite***, when $a>0$, or ***negative semidefinite***, when $a<0$. The prefix *semi* allows the possibility that f can equal zero, as it will at the point $x=b$, $y=-a$. The surface $z=f(x,y)$ degenerates from a bowl into a valley. For $f=(x+y)^2$, the valley runs along the line $x+y=0$.

Saddle Point $ac<b^2$**:** In one dimension, $F(x)$ has a minimum or a maximum, or $F''=0$. In two dimensions, a very important possibility still remains: *The combination* $ac-b^2$ *may be negative*. This occurred in both examples, when b dominated a and c. It also occurs if a and c have opposite signs. Then two directions give opposite results—in one direction f increases, in the other it decreases. It is useful to consider two special cases:

Saddle points at $(0,0)$ $\qquad f_1=2xy$ and $f_2=x^2-y^2$ and $ac-b^2=-1$.

In the first, $b=1$ dominates $a=c=0$. In the second, $a=1$ and $c=-1$ have opposite sign. The saddles $2xy$ and x^2-y^2 are practically the same; if we turn one through 45° we get the other. They are also hard to draw.

These quadratic forms are ***indefinite***, because they can take either sign. So we have a stationary point that is neither a maximum or a minimum. It is called a ***saddle point***. The surface $z=x^2-y^2$ goes down in the direction of the y axis, where the legs fit (if you still ride a horse). In case you switched to a car, think of a road going over a mountain pass. The top of the pass is a minimum as you look along the range of mountains, but it is a maximum as you go along the road.

Higher Dimensions: Linear Algebra

Calculus would be enough to find our conditions $F_{xx}>0$ and $F_{xx}F_{yy}>F_{xy}^2$ for a minimum. But linear algebra is ready to do more, because the second derivatives fit into a symmetric matrix A. The terms ax^2 and cy^2 appear *on the diagonal*. The cross derivative $2bxy$ is

split between the same entry b above and below. A quadratic $f(x,y)$ comes directly from a symmetric 2 by 2 matrix!

$$x^{\mathrm{T}}Ax \textbf{ in } \mathbf{R}^2 \qquad ax^2+2bxy+cy^2=\begin{bmatrix}x & y\end{bmatrix}\begin{bmatrix}a & b\\ b & c\end{bmatrix}\begin{bmatrix}x\\ y\end{bmatrix}. \tag{4}$$

This identity (please multiply it out) is the key to the whole chapter. It generalizes immediately to n dimensions, and it is a perfect shorthand for studying maxima and minima. When the variables are $x_1,\dots,x_n$, they go into a column vector x. ***For any symmetric matrix A, the product $x^{\mathrm{T}}Ax$ is a pure quadratic form*** $f(x_1,\dots,x_n)$:

$$x^{\mathrm{T}}Ax \textbf{ in } \mathbf{R}^n \qquad \begin{bmatrix}x_1 & x_2 & \cdot & x_n\end{bmatrix}\begin{bmatrix}a_{11} & a_{12} & \cdot & a_{1n}\\ a_{21} & a_{22} & \cdot & a_{2n}\\ \cdot & \cdot & \cdot & \cdot\\ a_{n1} & a_{n2} & \cdot & a_{nn}\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ \cdot\\ x_n\end{bmatrix}=\sum_{i=1}^{n}\sum_{j=1}^{n}a_{ij}x_ix_j. \tag{5}$$

The diagonal entries a_{11} to a_{nn} multiply x_1^2 to x_n^2. The pair $a_{ij}=a_{ji}$ combines into $2a_{ij}x_ix_j$. Then $f=a_{11}x_1^2+2a_{12}x_1x_2+\cdots+a_{nn}x_n^2$.

There are no higher-order terms or lower-order terms—only second-order. The function is zero at $x=(0,\dots,0)$, and its first derivatives are zero. The tangent is flat; this is a stationary point. We have to decide if $x=0$ is a minimum or a maximum or a saddle point of the function $f=x^{\mathrm{T}}Ax$.

Example 3. $f=2x^2+4xy+y^2$ and $A=\begin{bmatrix}2 & 2\\ 2 & 1\end{bmatrix}\rightarrow$ *saddle point.*

Example 4. $f=2xy$ and $A=\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\rightarrow$ *saddle point.*

Example 5. A is 3 by 3 for $2x_1^2-2x_1x_2+2x_2^2-2x_2x_3+2x_3^2$:

$$f=\begin{bmatrix}x_1 & x_2 & x_3\end{bmatrix}\begin{bmatrix}2 & -1 & 0\\ -1 & 2 & -1\\ 0 & -1 & 2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}\rightarrow \textit{minimum at } (0,0,0).$$

Any function $F(x_1,\dots,x_n)$ is approached in the same way. At a stationary point all first derivatives are zero. A is the **"second derivative matrix"** with entries $a_{ij}=\partial^2F/\partial x_i\partial x_j$. This automatically equals $a_{ji}=\partial^2F/\partial x_j\partial x_i$, so A is symmetric. **Then** F ***has a minimum when the pure quadratic $x^{\mathrm{T}}Ax$ is positive definite***. These second-order terms control F near the stationary point:

$$\textbf{Taylor series} \qquad F(x)=F(0)+x^{\mathrm{T}}(\text{grad } F)+\frac{1}{2}x^{\mathrm{T}}Ax+\text{higher order terms}. \tag{6}$$

At a stationary point, grad $F=(\partial F/\partial x_1,\dots,\partial F/\partial x_n)$ is a vector of zeros. The second derivatives in $x^{\mathrm{T}}Ax$ take the graph up or down (or saddle). If the stationary point is at x_0

instead of 0, $F(x)$ and all derivatives are computed at x_0. Then x changes to $x - x_0$ on the right-hand side.

The next section contains the tests to decide whether $x^{\mathrm{T}}Ax$ is positive (the bowl goes up from $x = 0$). Equivalently, **the tests decide whether the matrix A is positive definite**—which is the main goal of the chapter.

Problem Set 6.1

1. The quadratic $f = x^2 + 4xy + 2y^2$ has a saddle point at the origin, despite the fact that its coefficients are positive. Write f as a *difference of two squares*.

2. Decide for or against the positive definiteness of these matrices, and write out the corresponding $f = x^{\mathrm{T}}Ax$:

(a) $\begin{bmatrix} 1 & 3 \\ 3 & 5 \end{bmatrix}$. (b) $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$. (c) $\begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$. (d) $\begin{bmatrix} -1 & 2 \\ 2 & -8 \end{bmatrix}$.

The determinant in (b) is zero; along what line is $f(x,y) = 0$?

3. If a 2 by 2 symmetric matrix passes the tests $a > 0$, $ac > b^2$, solve the quadratic equation $\det(A - \lambda I) = 0$ and show that both eigenvalues are positive.

4. Decide between a minimum, maximum, or saddle point for the following functions.

(a) $F = -1 + 4(e^x - x) - 5x\sin y + 6y^2$ at the point $x = y = 0$.

(b) $F = (x^2 - 2x)\cos y$, with stationary point at $x = 1$, $y = \pi$.

5. (a) For which numbers b is the matrix $A = \left[\begin{smallmatrix} 1 & b \\ b & 9 \end{smallmatrix}\right]$ positive definite?

(b) Factor $A = LDL^{\mathrm{T}}$ when b is in the range for positive definiteness.

(c) Find the minimum value of $\frac{1}{2}(x^2 + 2bxy + 9y^2) - y$ for b in this range.

(d) What is the minimum if $b = 3$?

6. Suppose the positive coefficients a and c dominate b in the sense that $a + c > 2b$. Find an example that has $ac < b^2$, so the matrix is not positive definite.

7. (a) What 3 by 3 symmetric matrices A_1 and A_2 correspond to f_1 and f_2?

$$f_1 = x_1^2 + x_2^2 + x_3^2 - 2x_1x_2 - 2x_1x_3 + 2x_2x_3$$
$$f_2 = x_1^2 + 2x_2^2 + 11x_3^2 - 2x_1x_2 - 2x_1x_3 - 4x_2x_3.$$

(b) Show that f_1 is a *single* perfect square and not positive definite. Where is f_1 equal to 0?

(c) Factor A_2 into LL^{T}, Write $f_2 = x^{\mathrm{T}}A_2x$ as a sum of three squares.

8. If $A = \left[\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right]$ is positive definite, test $A^{-1} = \left[\begin{smallmatrix} p & q \\ q & r \end{smallmatrix}\right]$ for positive definiteness.

9. The quadratic $f(x_1,x_2) = 3(x_1+2x_2)^2 + 4x_2^2$ is positive. Find its matrix A, factor it into LDL^{T}, and connect the entries in D and L to 3, 2, 4 in f.

10. If $R = \left[\begin{smallmatrix} p & q \\ q & r \end{smallmatrix}\right]$, write out R^2 and check that it is positive definite unless R is singular.

11. (a) If $A = \left[\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right]$ is Hermitian (*complex b*), find its pivots and determinant.

(b) Complete the square for $x^{\mathrm{H}}Ax$. Now $x^{\mathrm{H}} = [\bar{x}_1 \;\; \bar{x}_2]$ can be complex

$$a|x_1|^2 + 2\mathrm{Re}\, b\bar{x}_1 x_2 + c|x_2|^2 = a|x_1 + (b/a)x_2|^2 + ____|x_2|^2.$$

(c) Show that $a > 0$ and $ac > |b|^2$ ensure that A is positive definite.

(d) Are the matrices $\left[\begin{smallmatrix} 1 & 1+i \\ 1-i & 2 \end{smallmatrix}\right]$ and $\left[\begin{smallmatrix} 3 & 4+i \\ 4-i & 6 \end{smallmatrix}\right]$ positive definite?

12. Decide whether $F = x^2y^2 - 2x - 2y$ has a minimum at the point $x = y = 1$ (after showing that the first derivatives are zero at that point).

13. Under what conditions on a, b, c is $ax^2 + 2bxy + cy^2 > x^2 + y^2$ for all x, y?

Problems 14–18 are about tests for positive definiteness.

14. Which of A_1, A_2, A_3, A_4 has two positive eigenvalues? Test $a > 0$ and $ac > b^2$, don't compute the eigenvalues. Find an x so that $x^{\mathrm{T}}A_1x < 0$.

$$A_1 = \begin{bmatrix} 5 & 6 \\ 6 & 7 \end{bmatrix} \qquad A_2 = \begin{bmatrix} -1 & -2 \\ -2 & -5 \end{bmatrix} \qquad A_3 = \begin{bmatrix} 1 & 10 \\ 10 & 100 \end{bmatrix} \qquad A_4 = \begin{bmatrix} 1 & 10 \\ 10 & 101 \end{bmatrix}.$$

15. What is the quadratic $f = ax^2 + 2bxy + cy^2$ for each of these matrices? Complete the square to write f as a sum of one or two squares $d_1(\quad)^2 + d_2(\quad)^2$.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 9 \end{bmatrix} \qquad \text{and} \qquad A = \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix}.$$

16. Show that $f(x,y) = x^2 + 4xy + 3y^2$ does not have a minimum at $(0,0)$ even though it has positive coefficients. Write f as a *difference* of squares and find a point (x,y) where f is negative.

17. (*Important*) If A has *independent columns*, then $A^{\mathrm{T}}A$ is square and symmetric and invertible (Section 4.2). ***Rewrite*** $x^{\mathrm{T}}A^{\mathrm{T}}Ax$ ***to show why it is positive except when*** $x = 0$. Then $A^{\mathrm{T}}A$ is positive definite.

18. Test to see if $A^{\mathrm{T}}A$ is positive definite in each case:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}, \qquad A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 2 & 1 \end{bmatrix}, \quad \text{and} \quad A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix}.$$

19. Find the 3 by 3 matrix A and its pivots, rank, eigenvalues, and determinant:

$$\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} & A & \\ & & \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 4(x_1 - x_2 + 2x_3)^2.$$

20. For $F_1(x,y) = \frac{1}{4}x^4 + x^2y + y^2$ and $F_2(x,y) = x^3 + xy - x$, find the second derivative matrices A_1 and A_2:

$$A = \begin{bmatrix} \partial^2 F/\partial x^2 & \partial^2 F/\partial x \partial y \\ \partial^2 F/\partial y \partial x & \partial^2 F/\partial y^2 \end{bmatrix}.$$

A_1 is positive definite, so F_1 is concave up (= convex). Find the minimum point of F_1 and the saddle point of F_2 (look where first derivatives are zero).

21. The graph of $z = x^2 + y^2$ is a bowl opening upward. *The graph* of $z = x^2 - y^2$ is *a saddle*. The graph of $z = -x^2 - y^2$ is a bowl opening downward. What is a test on $F(x,y)$ to have a saddle at $(0,0)$?

22. Which values of c give a bowl and which give a saddle point for the graph of $z = 4x^2 + 12xy + cy^2$? Describe this graph at the borderline value of c.

6.2 Tests for Positive Definiteness

Which symmetric matrices have the property that $x^{\mathrm{T}}Ax > 0$ for all nonzero vectors x? There are four or five different ways to answer this question, and we hope to find all of them. The previous section began with some hints about the signs of eigenvalues. but that gave place to the tests on a, b, c:

$$b = \begin{bmatrix} a & b \\ b & c \end{bmatrix} \quad \text{is positive definite when} \quad a > 0 \quad \text{and} \quad ac - b^2 > 0.$$

From those conditions, ***both eigenvalues are positive***. Their product $\lambda_1\lambda_2$ is determinant $ac - b^2 > 0$, so the eigenvalues are either both positive or both negative. They must be positive because their sum is the trace $a + c > 0$.

Looking at a and $ac - b^2$, it is even possible to spot the appearance of the ***pivots***. They turned up when we decomposed $x^{\mathrm{T}}Ax$ into a sum of squares:

$$\textbf{Sum of squares} \qquad ax^2 + 2bxy + cy^2 = a\left(x + \frac{b}{a}y\right)^2 + \frac{ac - b^2}{a}y^2. \tag{1}$$

Those coefficients a and $(ac - b^2)/a$ are the pivots for a 2 by 2 matrix. For larger matrices the pivots still give a simple test for positive definiteness: $x^{\mathrm{T}}Ax$ stays positive when n independent squares are multiplied by ***positive pivots***.

One more preliminary remark. The two parts of this hook were linked by the chapter on determinants. Therefore we ask what part determinants play. ***It is not enough to require that the determinant of A is positive***. If $a = c = -1$ and $b = 0$. then $\det A = 1$ but $A = -I =$ negative definite. The determinant test is applied not only to A itself, giving $ac - b^2 > 0$, but also to the 1 by 1 submatrix a in the upper left-hand corner.

The natural generalization will involve all n of the *upper left submatrices* of A:

$$A_1 = \begin{bmatrix} a_{11} \end{bmatrix}, \qquad A_2 = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \qquad A_3 = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \cdots, \qquad A_n = A.$$

Here is the main theorem on positive definiteness, and a reasonably detailed proof:

6B Each of the following tests is a necessary and sufficient condition for the real symmetric matrix A to be ***positive definite***:

(I) $x^{\mathrm{T}} k x > 0$ for all nonzero real vectors x.

(II) All the eigenvalues of A satisfy $\lambda_i > 0$.

(III) All the upper left submatrices A_k have positive determinants.

(IV) All the pivots (without row exchanges) satisfy $d_k > 0$.

Proof. Condition I defines a positive definite matrix. Our first step shows that each eigenvalue will be positive:

$$\text{If} \quad Ax = \lambda x, \quad \text{then} \quad x^{\mathrm{T}} A x = x^{\mathrm{T}} \lambda x = \lambda \|x\|^2.$$

A positive definite matrix has positive eigenvalues, since $x^{\mathrm{T}} A x > 0$.

Now we go in the other direction. If all $\lambda_i > 0$, we have to prove $x^{\mathrm{T}} A x > 0$ for every vector x (not just the eigenvectors). Since symmetric matrices have a full set of orthonormal eigenvectors, any x is a combination $c_1 x_1 + \cdots + c_n x_n$. Then

$$Ax = c_1 A x_1 + \cdots + c_n A x_n = c_1 \lambda_1 x_1 + \cdots + c_n \lambda_n x_n.$$

Because of the orthogonality $x_i^{\mathrm{T}} x_i = 0$, and the normalization $x_i^{\mathrm{T}} x_i = 1$,

$$\begin{aligned} x^{\mathrm{T}} A x &= \left(c_1 x_1^{\mathrm{T}} + \cdots + c_n x_n^{\mathrm{T}} \right) \left(c_1 \lambda_1 x_1 + \cdots + c_n \lambda_n x_n \right) \\ &= c_1^2 \lambda_1 + \cdots + c_n^2 \lambda_n. \end{aligned} \tag{2}$$

If every $\lambda_i > 0$, then equation (2) shows that $x^{\mathrm{T}} A x > 0$. Thus condition II implies condition I.

If condition I holds, so does condition III: The determinant of A is the product of the eigenvalues. And if condition I holds, we already know that these eigenvalues are positive. But we also have to deal with every upper left submatrix A_k. The trick is to look at all nonzero vectors whose last $n - k$ components are zero:

$$x^{\mathrm{T}} A x = \begin{bmatrix} x_k^{\mathrm{T}} & 0 \end{bmatrix} \begin{bmatrix} A_k & * \\ * & * \end{bmatrix} \begin{bmatrix} x_k \\ 0 \end{bmatrix} = x_k^{\mathrm{T}} A_k x_k > 0.$$

Thus A_k is positive definite. Its eigenvalues (not the same λ_1!) must be positive. Its determinant is their product, so all upper left determinants are positive.

If condition III holds, so does condition IV: According to Section 4.4, the kth pivot d_k is the ratio of $\det A_k$ to $\det A_{k-1}$. If the determinants are all positive, so are the pivots.

If condition IV holds, so does condition I: We are given positive pivots, and must deduce that $x^{\mathrm{T}}Ax > 0$. This is what we did in the 2 by 2 case, by completing the square. The pivots were the numbers outside the squares. To see how that happens for symmetric matrices of any size, we go back to *elimination on a symmetric matrix*: $A = LDL^{\mathrm{T}}$.

Example 1. Positive pivots 2, $\frac{3}{2}$, and $\frac{4}{3}$:

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 0 & -\frac{2}{3} & 1 \end{bmatrix} \begin{bmatrix} 2 & & \\ & \frac{3}{2} & \\ & & \frac{4}{3} \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & 1 & -\frac{2}{3} \\ 0 & 0 & 1 \end{bmatrix} = LDL^{\mathrm{T}}.$$

I want to split $x^{\mathrm{T}}Ax$ into $x^{\mathrm{T}}LDL^{\mathrm{T}}x$:

$$\text{If} \quad x = \begin{bmatrix} u \\ v \\ w \end{bmatrix}, \quad \text{then} \quad L^{\mathrm{T}}x = \begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & 1 & -\frac{2}{3} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} u - \frac{1}{2}v \\ v - \frac{2}{3}w \\ w \end{bmatrix}.$$

So $x^{\mathrm{T}}Ax$ is a sum of squares with the pivots 2, $\frac{3}{2}$, and $\frac{4}{3}$ as coefficients:

$$x^{\mathrm{T}}Ax = (L^{\mathrm{T}}x)^{\mathrm{T}}D(L^{\mathrm{T}}x) = \mathbf{2}\left(u - \frac{\mathbf{1}}{\mathbf{2}}v\right)^2 + \frac{\mathbf{3}}{\mathbf{2}}\left(v - \frac{\mathbf{2}}{\mathbf{3}}w\right)^2 + \frac{\mathbf{4}}{\mathbf{3}}(w)^2.$$

Those positive pivots in D multiply perfect squares to make $x^{\mathrm{T}}Ax$ positive. Thus condition IV implies condition I, and the proof is complete.

□

It is beautiful that elimination and completing the square are actually the same. Elimination removes x_1 from all later equations. Similarly, the first square accounts for all terms in $x^{\mathrm{T}}Ax$ involving x_1. The sum of squares has the pivots outside. *The multipliers ℓ_{ij} are inside*! You can see the numbers $-\frac{1}{2}$ and $-\frac{2}{3}$ inside the squares in the example.

Every diagonal entry a_{ii} must be positive. As we know from the examples, however, it is far from sufficient to look only at the diagonal entries.

The pivots d_i are not to be confused with the eigenvalues. For a typical positive definite matrix, they are two completely different sets of positive numbers, In our 3 by 3 example, probably the determinant test is the easiest:

Determinant test $\det A_1 = 2, \quad \det A_2 = 3, \quad \det A_3 = \det A = 4.$

The pivots are the ratios $d_1 = 2$, $d_2 = \frac{3}{2}$, $d_3 = \frac{4}{3}$. Ordinarily the eigenvalue test is the longest computation. For this A we know the λ's are all positive:

Eigenvalue test $\lambda_1 = 2 - \sqrt{2}, \quad \lambda_2 = 2, \quad \lambda_3 = 2 + \sqrt{2}.$

Even though it is the hardest to apply to a single matrix, eigenvalues can be the most useful test for theoretical purposes. ***Each test is enough by itself***.

Positive Definite Matrices and Least Squares

I hope you will allow one more test for positive definiteness. It is already close. We connected positive definite matrices to pivots (Chapter 1), determinants (Chapter 4), and eigenvalues (Chapter 5). Now we see them in the least-squares problems in Chapter 3, coming from the rectangular matrices of Chapter 2.

The rectangular matrix will be R and the least-squares problem will be $Rx = b$. It has m equations with $m \geq n$ (square systems are included). *The least-square choice $\widehat{x}$ is the solution of $R^{\mathrm{T}}R\widehat{x} = R^{\mathrm{T}}b$.* That matrix $AR^{\mathrm{T}}R$ is not only symmetric but positive definite, as we now show—provided that the n columns of R are linearly independent:

6C The symmetric matrix A is positive definite if and only if

(V) There is a matrix R with independent columns such that $A = R^{\mathrm{T}}R$.

The key is to recognize $x^{\mathrm{T}}Ax$ as $x^{\mathrm{T}}R^{\mathrm{T}}Rx = (Rx)^{\mathrm{T}}(Rx)$. This squared length $\|Rx\|^2$ is positive (unless $x = 0$), because R has independent columns. (If x is nonzero then Rx is nonzero.) Thus $x^{\mathrm{T}}R^{\mathrm{T}}Rx > 0$ and $R^{\mathrm{T}}R$ is positive definite.

It remains to find an R For which $A = R^{\mathrm{T}}R$. We have almost done this twice already:

$$\textbf{Elimination} \qquad A = LDL^{\mathrm{T}} = (L\sqrt{D})(\sqrt{D}L^{\mathrm{T}}). \quad \text{So take } R = \sqrt{D}L^{\mathrm{T}}.$$

This ***Cholesky decomposition*** has the pivots split evenly between L and L^{T}.

$$\textbf{Eigenvalues} \qquad A = Q\Lambda Q^{\mathrm{T}} = (Q\sqrt{\Lambda})(\sqrt{\Lambda}Q^{\mathrm{T}}). \quad \text{So take } R = \sqrt{\Lambda}Q^{\mathrm{T}}. \tag{3}$$

A third possibility is $R = Q\sqrt{\Lambda}Q^{\mathrm{T}}$, the ***symmetric positive definite square root*** of A. There are many other choices, square or rectangular, and we can see why. If you multiply any R by *a matrix Q with orthonormal columns*, then $(QR)^{\mathrm{T}}(QR) = R^{\mathrm{T}}Q^{\mathrm{T}}QR = R^{\mathrm{T}}IR = A$. Therefore QR is another choice.

Applications of positive definite matrices are developed in my earlier book *Introduction to Applied Mathematics* and also the new *Applied Mathematics and Scientific Computing* (see `www.wellesleycambridge.com`). We mention that $Ax = \lambda Mx$ arises constantly in engineering analysis. If A and M are positive definite, this generalized problem is parallel to the familiar $Ax = \lambda x$, and $\lambda > 0$. M is a ***mass matrix*** for the *finite element method* in Section 6.4.

Semidefinite Matrices

The tests for semidefiniteness will relax $x^{\mathrm{T}}Ax > 0$, $\lambda > 0$, $d > 0$, and det > 0, to allow zeros to appear. The main point is to see the analogies with the positive definite case.

6D Each of the following tests is a necessary and sufficient condition for a symmetric matrix A to be ***positive semidefinite***:

(I′) $x^{\mathrm{T}}Ax \geq 0$ for all vectors x (this defines positive semidefinite).

(II′) All the eigenvalues of A satisfy $\lambda_i \geq 0$.

(III′) No principal submatrices have negative determinants.

(IV′) No pivots are negative.

(V′) There is a matrix R, possibly with dependent columns, such that $A = R^{\mathrm{T}}R$.

The diagonalization $A = Q\Lambda Q^{\mathrm{T}}$ leads to $x^{\mathrm{T}}Ax = x^{\mathrm{T}}Q\Lambda Q^{\mathrm{T}}x = y^{\mathrm{T}}\Lambda y$. If A has rank r, there are r nonzero λ's and r perfect squares in $y^{\mathrm{T}}\Lambda y = \lambda_1 y_1^2 + \cdots + \lambda_r y_r^2$.

Note. The novelty is that condition III′ applies to all the principal submatrices, not only those in the upper left-hand corner. Otherwise, we could not distinguish between two matrices whose upper left determinants were all zero:

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \text{ is positive semidefinite, and } \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix} \text{ is negative semidefinite.}$$

A row exchange comes with the same column exchange to maintain symmetry.

Example 2.

$$A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \quad \text{is positive } \textit{semi}\text{definite, by all five tests:}$$

(I′) $x^{\mathrm{T}}Ax = (x_1 - x_2)^2 + (x_1 - x_3)^2 + (x_2 - x_3)^2 \geq 0$ (zero if $x_1 = x_2 = x_3$).

(II′) The eigenvalues are $\lambda_1 = 0$, $\lambda_2 = \lambda_3 = 3$ (a zero eigenvalue).

(III′) $\det A = 0$ and smaller determinants are positive.

(IV′) $$A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \to \begin{bmatrix} 2 & 0 & 0 \\ 0 & \frac{3}{2} & -\frac{3}{2} \\ 0 & -\frac{3}{2} & \frac{3}{2} \end{bmatrix} \to \begin{bmatrix} 2 & 0 & 0 \\ 0 & \frac{3}{2} & 0 \\ 0 & 0 & \mathbf{0} \end{bmatrix} \quad \text{(missing pivot).}$$

(V′) $A = R^{\mathrm{T}}R$ with dependent columns in R:

$$\begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \quad (1,1,1) \text{ in the nullspace.}$$

Remark. The conditions for semidefiniteness could also be deduced from the origin conditions I-V for definiteness by the following trick: Add a small multiple of the identity giving a positive definite matrix $A + \varepsilon I$. Then let ε approach zero. Since the determinants and eigenvalues depend continuously on ε, they will be positive until the very last moment. At $\varepsilon = 0$ they must still be nonnegative.

My class often asks about *unsymmetric* positive definite matrices. I never use that term. One reasonable definition is that the symmetric part $\frac{1}{2}(A+A^{\mathrm{T}})$ should be positive definite. That guarantees that *the real parts of the eigenvalues are positive*. But it is not necessary: $A = \left[\begin{smallmatrix} 1 & 4 \\ 0 & 1 \end{smallmatrix}\right]$ has $\lambda > 0$ but $\frac{1}{2}(A+A^{\mathrm{T}}) = \left[\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right]$ is indefinite.

If $Ax = \lambda x$, then $x^{\mathrm{H}}Ax = \lambda x^{\mathrm{H}}x$ and $x^{\mathrm{H}}A^{\mathrm{H}}x = \overline{\lambda}x^{\mathrm{H}}x$.

Adding, $\frac{1}{2}x^{\mathrm{H}}(A+A^{\mathrm{H}})x = (\mathrm{Re}\lambda)x^{\mathrm{H}}x > 0$, so that $\mathrm{Re}\lambda > 0$.

Ellipsoids in n Dimensions

Throughout this book, geometry has helped the matrix algebra. A linear equation produced a plane. The system $Ax = b$ gives an intersection of planes. Least squares gives a perpendicular projection. The determinant is the volume of a box. Now, for a positive definite matrix and its $x^{\mathrm{T}}Ax$, we finally get a figure that is curved. It is an *ellipse* in two dimensions, and an *ellipsoid* in n dimensions.

The equation to consider is $x^{\mathrm{T}}Ax = 1$. If A is the identity matrix, this simplifies to $x_1^2 + x_2^2 + \cdots + x_n^2 = 1$. This is the equation of the "unit sphere" in $\mathbf{R}^n$. If $A = 4I$, the sphere gets smaller. The equation changes to $4x_1^2 + \cdots + 4x_n^2 = 1$. Instead of $(1,0,\ldots,0)$, it goes through $(\frac{1}{2},0,\ldots,0)$. The center is at the origin, because if x satisfies $x^{\mathrm{T}}Ax = 1$, so does the opposite vector $-x$. The important step is to go from the identity matrix to a *diagonal matrix*:

Ellipsoid For $A = \begin{bmatrix} 4 & & \\ & 1 & \\ & & \frac{1}{9} \end{bmatrix}$, the equation is $x^{\mathrm{T}}Ax = 4x_1^2 + x_2^2 + \frac{1}{9}x_3^2 = 1$.

Since the entries are unequal (and positive!) the sphere changes to an ellipsoid.

One solution is $x = (\frac{1}{2},0,0)$ along the first axis. Another is $x = (0,1,0)$. The major axis has the farthest point $x = (0,0,3)$. It is like a football or a rugby ball, but not quite—those are closer to $x_1^2 + x_2^2 + \frac{1}{2}x_3^2 = 1$. The two equal coefficients make them circular in the x_1-x_2 plane, and much easier to throw!

Now comes the final step, to allow nonzeros away from the diagonal of A.

Example 3. $A = \left[\begin{smallmatrix} 5 & 4 \\ 4 & 5 \end{smallmatrix}\right]$ and $x^{\mathrm{T}}Ax = 5u^2 + 8uv + 5v^2 = 1$. That ellipse is centered at $u = v = 0$, but the axes are not so clear. The off-diagonal 4s leave the matrix positive definite, but they rotate the ellipse—its axes no longer line up with the coordinate axes (Figure 6.2). We will show that ***the axes of the ellipse point toward the eigenvector of*** A. Because $A = A^{\mathrm{T}}$, those eigenvectors and axes are orthogonal. The *major* axis of the ellipse corresponds to the *smallest* eigenvalue of A.

To locate the ellipse we compute $\lambda_1 = 1$ and $\lambda_2 = 9$. The unit eigenvectors are $(1,-1)/\sqrt{2}$ and $(1,1)/\sqrt{2}$. Those are at 45° angles with the u-v axes, and they are lined up with the axes of the ellipse. The way to see the ellipse properly is to *rewrite*

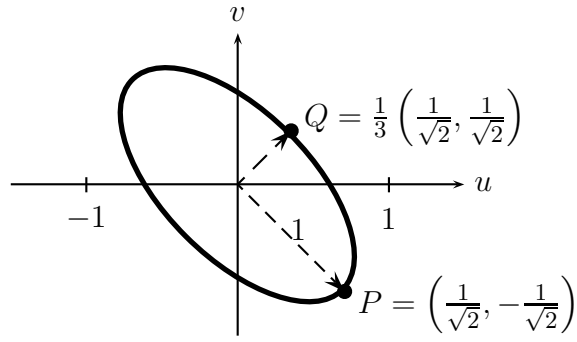


Figure 6.2: The ellipse $x^{\mathrm{T}}Ax = 5u^2 + 8uv + 5v^2 = 1$ and its principal axes.

$x^{\mathrm{T}}Ax = 1$:

$$\textbf{New squares} \qquad 5u^2 + 8uv + v^2 = \left(\frac{u}{\sqrt{2}} - \frac{v}{\sqrt{2}}\right)^2 + 9\left(\frac{u}{\sqrt{2}} + \frac{v}{\sqrt{2}}\right)^2 = 1. \tag{4}$$

$\lambda = 1$ and $\lambda = 9$ are outside the squares. The eigenvectors are inside. This is different from completing the square to $5(u + \frac{4}{5}v)^2 + \frac{9}{5}v^2$, with the *pivots* outside.

The first square equals 1 at $(1/\sqrt{2}, -1/\sqrt{2})$ at the end of the major axis. The minor axis is one-third as long, since we need $(\frac{1}{3})^2$ to cancel the 9.

Any ellipsoid $x^{\mathrm{T}}Ax = 1$ can be simplified in the same way. *The key step is to diagonalize* $A = Q\Lambda Q^{\mathrm{T}}$. We straightened the picture by rotating the axes. Algebraically, the change to $y = Q^{\mathrm{T}}x$ produces a sum of squares:

$$x^{\mathrm{T}}Ax = (x^{\mathrm{T}}Q)\Lambda(Q^{\mathrm{T}}x) = y^{\mathrm{T}}\Lambda y = \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2 = 1. \tag{5}$$

The *major axis* has $y_1 = 1/\sqrt{\lambda_1}$ along the eigenvector with the smallest eigenvalue.

The other axes are along the other eigenvectors. Their lengths are $1/\sqrt{\lambda_2}, \ldots, 1/\sqrt{\lambda_n}$. Notice that the λ's must be positive—*the matrix must be positive definite*—or these square roots are in trouble. An indefinite equation $y_1^2 - 9y_2^2 = 1$ describes a hyperbola and not an ellipse. A hyperbola is a cross-section through a saddle, and an ellipse is a cross-section through a bowl.

The change from x to $y = Q^{\mathrm{T}}x$ rotates the axes of the space, to match the axes of the ellipsoid. In the y variables we can see that it is an ellipsoid, because the equation becomes so manageable:

6E Suppose $A = Q\Lambda Q^{\mathrm{T}}$ with $\lambda_i > 0$. Rotating $y = Q^{\mathrm{T}}x$ simplifies $x^{\mathrm{T}}Ax = 1$:

$$x^{\mathrm{T}}Q\Lambda Q^{\mathrm{T}}x = 1, \qquad y^{\mathrm{T}}\Lambda y = 1, \quad \text{and} \quad \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2 = 1.$$

This is the equation of an ellipsoid. Its axes have lengths $1/\sqrt{\lambda_1}, \ldots, 1/\sqrt{\lambda_n}$ from the center. In the original x-space they point along the eigenvectors of A.

The Law of Inertia

For elimination and eigenvalues, matrices become simpler by elementary operations The essential thing is to know which properties of the matrix stay unchanged. When a multiple of one row is subtracted from another, the row space, nullspace. rant and determinant all remain the same. For eigenvalues, the basic operation was a similarity transformation $A \to S^{-1}AS$ (or $A \to M^{-1}AM$). The eigenvalues are unchanged (and also the Jordan form). Now we ask the same question for symmetric matrices: *What are the elementary operations and their invariants for* $x^{\mathrm{T}}Ax$?

The basic operation on a quadratic form is to change variables. A new vector y is related to x by some nonsingular matrix, $x = Cy$. The quadratic form becomes $y^{\mathrm{T}}C^{\mathrm{T}}ACy$. This shows the fundamental operation on A:

$$\textbf{Congruence transformation} \qquad A \to C^{\mathrm{T}}AC \quad \text{for some nonsingular } C. \tag{6}$$

The symmetry of A is preserved, since $C^{\mathrm{T}}AC$ remains symmetric. The real question is, What other properties are shared by A and $C^{\mathrm{T}}AC$? The answer is given by Sylvester's ***law of inertia***.

6F $C^{\mathrm{T}}AC$ has the same number of positive eigenvalues, negative eigenvalues, and zero eigenvalues as A.

The *signs* of the eigenvalues (and not the eigenvalues themselves) are preserved by a congruence transformation. In the proof, we will suppose that A is nonsingular. Then $C^{\mathrm{T}}AC$ is also nonsingular, and there are no zero eigenvalues to worry about. (Otherwise we can work with the nonsingular $A + \varepsilon I$ and $A - \varepsilon I$, and at the end let $\varepsilon \to 0$.)

Proof. We want to borrow a trick from topology. Suppose C is linked to an orthogonal matrix Q by a continuous chain of nonsingular matrices $C(t)$. At $t = 0$ and $t = 1$, $C(0) = C$ and $C(1) = Q$. Then the eigenvalues of $C(t)^{\mathrm{T}}AC(t)$ will change gradually, as t goes from 0 to 1, from the eigenvalues of $C^{\mathrm{T}}AC$ to the eigenvalues of $Q^{\mathrm{T}}AQ$. Because $C(t)$ is never singular, *none of these eigenvalues can touch zero* (not to mention cross over it!). Therefore the number of eigenvalues to the right of zero, and the number to the left, is the same for $C^{\mathrm{T}}AC$ as for $Q^{\mathrm{T}}AQ$. And A has exactly the same eigenvalues as the similar matrix $Q^{-1}AQ = Q^{\mathrm{T}}AQ$.

One good choice for Q is to apply Gram-Schmidt to the columns of C. Then $C = QR$, and the chain of matrices is $C(t) = tQ + (1-t)QR$. The family $C(t)$ goes slowly through Gram-Schmidt, from QR to Q. It is invertible, because Q is invertible and the triangular factor $tI + (1-t)R$ has positive diagonal. That ends the proof. □

Example 4. Suppose $A = I$. Then $C^{\mathrm{T}}AC = C^{\mathrm{T}}C$ is positive definite. Both I and $C^{\mathrm{T}}C$ have n positive eigenvalues, confirming the law of inertia.

Example 5. If $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, then $C^{\mathrm{T}}AC$ has a negative determinant:

$$\det C^{\mathrm{T}}AC = (\det C^{\mathrm{T}})(\det A)(\det C) = -(\det C)^2 < 0.$$

Then $C^{\mathrm{T}}AC$ must have one positive and one negative eigenvalue, like A.

Example 6. This application is the important one:

> **6G** For any symmetric matrix A, ***the signs of the pivots agree with the signs of the eigenvalues***. The eigenvalue matrix Λ and the pivot matrix D have the same number of positive entries, negative entries, and zero entries.

We will assume that A allows the symmetric factorization $A = LDL^{\mathrm{T}}$ (without row exchanges). By the law of inertia, A has the same number of positive eigenvalues as D. But the eigenvalues of D are just its diagonal entries (the pivots). Thus the number of positive pivots matches the number of positive eigenvalues of A.

That is both beautiful and practical. It is beautiful because it brings together (for symmetric matrices) two parts of this book that were previously separate: *pivots* and *eigenvalues*. It is also practical, because the pivots can locate the eigenvalues:

A **has positive pivots**
$A - 2I$ **has a negative pivot**

$$A = \begin{bmatrix} 3 & 3 & 0 \\ 3 & 10 & 7 \\ 0 & 7 & 8 \end{bmatrix} \qquad A - 2I = \begin{bmatrix} 1 & 3 & 0 \\ 3 & 8 & 7 \\ 0 & 7 & 6 \end{bmatrix}.$$

A has positive eigenvalues, by our test. But we know that $\lambda_{\min}$ *is smaller than* 2, because subtracting 2 dropped it below zero. The next step looks at $A - I$, to see if $\lambda_{\min} < 1$. (It is, because $A - I$ has a negative pivot.) That interval containing λ is cut in half at every step by checking the signs of the pivots.

This was almost the first practical method of computing eigenvalues. It was dominant about 1960, after one important improvement—to make A tridiagonal first. Then the pivots are computed in $2n$ steps instead of $\frac{1}{6}n^3$. Elimination becomes fast, and the search for eigenvalues (by halving the intervals) becomes simple. The current favorite is the QR method in Chapter 7.

The Generalized Eigenvalue Problem

Physics, engineering, and statistics are usually kind enough to produce symmetric matrices in their eigenvalue problems. ***But sometimes*** $Ax = \lambda x$ ***is replaced by*** $Ax = \lambda Mx$. *There are two matrices rather than one.*

An example is the motion of two unequal masses in a line of springs:

$$\begin{aligned} m_1 \frac{d^2v}{dt^2} + 2v - w &= 0 \\ m_2 \frac{d^2w}{dt^2} - v + 2w &= 0 \end{aligned} \quad \text{or} \quad \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \frac{d^2u}{dt^2} + \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} u = 0. \tag{7}$$

When the masses were equal, $m_1 = m_2 = 1$, this was the old system $u'' + Au = 0$. Now it is $Mu'' + Au = 0$, with a *mass matrix* M. The eigenvalue problem arises when we look

for exponential solutions $e^{i\omega t}x$:

$$Mu'' + Au = 0 \qquad \text{becomes} \qquad M(i\omega)^2 e^{i\omega t}x + Ae^{i\omega t}x = 0. \tag{8}$$

Canceling $e^{i\omega t}$, and writing λ for ω^2, this is an eigenvalue problem:

$$\textbf{Generalized problem } Ax = \lambda Mx \qquad \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} x = \lambda \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} x. \tag{9}$$

There is a solution when $A - \lambda M$ is singular. The special choice $M = I$ brings back the usual $\det(A - \lambda I) = 0$. We work out $\det(A - \lambda M)$ with $m_1 = 1$ and $m_2 = 2$:

$$\det \begin{bmatrix} 2-\lambda & -1 \\ -1 & 2-2\lambda \end{bmatrix} = 2\lambda^2 - 6\lambda + 3 = 0 \quad \text{gives} \quad \lambda = \frac{3 \pm \sqrt{3}}{2}.$$

For the eigenvector $x_1(\sqrt{3}-1, 1)$, the two masses oscillate together—but the first mass only moves as far as $\sqrt{3}-1 \approx .73$. In the fastest mode, the components of $x_2 = (1+\sqrt{3}, -1)$ have opposite signs and the masses move in opposite directions. This time the smaller mass goes much further.

The underlying theory is easier to explain if M is split into $R^{\mathrm{T}}R$. (*M is assumed to be positive definite.*) Then the substitution $y = Rx$ changes

$$Ax = \lambda Mx = \lambda R^{\mathrm{T}}Rx \quad \text{into} \quad AR^{-1}y = \lambda R^{\mathrm{T}}y.$$

Writing C for R^{-1}, and multiplying through by $(R^{\mathrm{T}})^{-1} = C^{\mathrm{T}}$, this becomes a standard eigenvalue problem for the *single* symmetric matrix $C^{\mathrm{T}}AC$:

$$\textbf{Equivalent problem} \qquad C^{\mathrm{T}}ACy = \lambda y. \tag{10}$$

The eigenvalues λ_j are the same as for the original $Ax = \lambda Mx$. and the eigenvectors are related by $y_j = Rx_j$. The properties of $C^{\mathrm{T}}AC$ lead directly to thc properties of $Ax = \lambda Mx$, when $A = A^{\mathrm{T}}$ and M is positive definite:

1. The eigenvalues for $Ax = \lambda Mx$ are real, because $C^{\mathrm{T}}AC$ is symmetric.
2. The λ's have the same signs as the eigenvalues of A, by the law of inertia.
3. $C^{\mathrm{T}}AC$ has orthogonal eigenvectors y_j. So the eigenvectors of $Ax = \lambda Mx$ have

$$\textbf{“}M\textbf{-orthogonality”} \qquad x_i^{\mathrm{T}}Mx_j = x_i^{\mathrm{T}}R^{\mathrm{T}}Rx_j = y_i^{\mathrm{T}}y_j = 0. \tag{11}$$

A and M are being *simultaneously diagonalized*. If S has the x_j in its columns, then $S^{\mathrm{T}}AS = \Lambda$ and $S^{\mathrm{T}}MS = I$. This is a *congruence* transformation, with S^{T} on the left, and not a similarity transformation with S^{-1}. The main point is easy to summarize: As long as M is positive definite, the generalized eigenvalue problem $Ax = -\lambda Mx$ behaves exactly like $Ax = \lambda x$.

Problem Set 6.2

1. For what range of numbers a and b are the matrices A and B positive definite?

$$A = \begin{bmatrix} a & 2 & 2 \\ 2 & a & 2 \\ 2 & 2 & a \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 2 & 4 \\ 2 & b & 8 \\ 4 & 8 & 7 \end{bmatrix}.$$

2. Decide for or against the positive definiteness of

$$A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix}, \qquad C = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}^2.$$

3. Construct an indefinite matrix with its largest entries on the main diagonal:

$$A = \begin{bmatrix} 1 & b & -b \\ b & 1 & b \\ -b & b & 1 \end{bmatrix} \quad \text{with } |b| < 1 \text{ can have } \det A < 0.$$

4. Show from the eigenvalues that if A is positive definite, so is A^2 and so is A^{-1}.

5. *If A and B are positive definite, then $A+B$ is positive definite.* Pivots and eigenvalues are not convenient for $A+B$. Much better to prove $x^{\mathrm{T}}(A+B)x > 0$.

6. From the pivots, eigenvalues, and eigenvectors of $A = \left[\begin{smallmatrix} 5 & 4 \\ 4 & 5 \end{smallmatrix}\right]$, write A as $R^{\mathrm{T}}R$ in three ways: $(L\sqrt{D})(\sqrt{D}L^{\mathrm{T}})$, $(Q\sqrt{\Lambda})(\sqrt{\Lambda}Q^{\mathrm{T}})$, and $(Q\sqrt{\Lambda}Q^{\mathrm{T}})(Q\sqrt{\Lambda}Q^{\mathrm{T}})$.

7. If $A = Q\Lambda Q^{\mathrm{T}}$ is symmetric positive definite, then $R = Q\sqrt{\Lambda}Q^{\mathrm{T}}$ is its *symmetric positive definite square root*. Why does R have positive eigenvalues? Compute R and verify $R^2 = A$ for

$$A = \begin{bmatrix} 10 & 6 \\ 6 & 10 \end{bmatrix} \qquad \text{and} \qquad A = \begin{bmatrix} 10 & -6 \\ -6 & 10 \end{bmatrix}.$$

8. If A is symmetric positive definite and C is nonsingular, prove that $B = C^{\mathrm{T}}AC$ is also symmetric positive definite.

9. If $A = R^{\mathrm{T}}R$ prove the generalized Schwarz inequality $|x^{\mathrm{T}}Ay|^2 \le (x^{\mathrm{T}}Ax)(y^{\mathrm{T}}Ay)$.

10. The ellipse $u^2 + 4v^2 = 1$ corresponds to $A = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 4 \end{smallmatrix}\right]$. Write the eigenvalues and eigenvectors, and sketch the ellipse.

11. Reduce the equation $3u^2 - 2\sqrt{2}uv + 2v^2 = 1$ to a sum of squares by finding the eigenvalues of the corresponding A, and sketch the ellipse.

12. In three dimensions, $\lambda_1 y_1^2 + \lambda_2 y_2^2 + \lambda_3 y_3^2 = 1$ represents an ellipsoid when all $\lambda_i > 0$. Describe all the different kinds of surfaces that appear in the positive semidefinite case when one or more of the eigenvalues is zero.

13. Write down the five conditions for a 3 by 3 matrix to be *negative definite* ($-A$ is positive definite) with special attention to condition III: How is $\det(-A)$ related to $\det A$?

14. Decide whether the following matrices are positive definite, negative definite, semidefinite, or indefinite:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 4 \\ 3 & 4 & 9 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 2 & 6 & -2 & 0 \\ 0 & -2 & 5 & -2 \\ 0 & 0 & -2 & 3 \end{bmatrix}, \qquad C = -B, \qquad D = A^{-1}.$$

Is there a real solution to $-x^2 - 5y^2 - 9z^2 - 4xy - 6xz - 8yz = 1$?

15. Suppose A is symmetric positive definite and Q is an orthogonal matrix. True or false:

(a) $Q^{\mathrm{T}}AQ$ is a diagonal matrix.

(b) $Q^{\mathrm{T}}AQ$ is symmetric positive definite.

(c) $Q^{\mathrm{T}}AQ$ has the same eigenvalues as A.

(d) e^{-A} is symmetric positive definite.

16. If A is positive definite and a_{11} is increased, prove from cofactors that the determinant is increased. Show by example that this can fail if A is indefinite.

17. From $A = R^{\mathrm{T}}R$. show for positive definite matrices that $\det A \le a_{11}a_{22}\cdots a_{nn}$. (The length squared of column j of R is a_{jj}. Use determinant = volume.)

18. (Lyapunov test for stability of M) Suppose $AM + M^{\mathrm{H}}A = -I$ with positive definite A. If $Mx = \lambda x$ show that $\mathrm{Re}\lambda < 0$. (*Hint*: Multiply the first equation by x^{H} and x.)

19. Which 3 by 3 symmetric matrices A produce these functions $f = x^{\mathrm{T}}Ax$? Why is the first matrix positive definite but not the second one?

(a) $f = 2(x_1^2 + x_2^2 + x_3^2 - x_1x_2 - x_2x_3)$.

(b) $f = 2(x_1^2 + x_2^2 + x_3^2 - x_1x_2 - x_1x_3 - x_2x_3)$.

20. Compute the three upper left determinants to establish positive definiteness. Verify that their ratios give the second and third pivots.

$$A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 3 \\ 0 & 3 & 8 \end{bmatrix}.$$

21. A positive definite matrix cannot have a zero (or even worse, a negative number) on its diagonal. Show that this matrix fails to have $x^{\mathrm{T}}Ax > 0$:

$$\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 4 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad \text{is not positive when} \quad (x_1, x_2, x_3) = (\quad , \quad , \quad).$$

22. A diagonal entry a_{jj} of a symmetric matrix cannot be smaller than all λ's. If it were, then $A - a_{jj}I$ would have ____ eigenvalues and would be positive definite. But $A - a_{jj}I$ has a ____ on the main diagonal.

23. Give a quick reason why each of these statements is true:

(a) Every positive definite matrix is invertible.

(b) The only positive definite projection matrix is $P = I$.

(c) A diagonal matrix with positive diagonal entries is positive definite.

(d) A symmetric matrix with a positive determinant might not be positive definite!

24. For which s and t do A and B have all $\lambda > 0$ (and are therefore positive definite)?

$$A = \begin{bmatrix} s & -4 & -4 \\ -4 & s & -4 \\ -4 & -4 & s \end{bmatrix} \qquad \text{and} \qquad B = \begin{bmatrix} t & 3 & 0 \\ 3 & t & 4 \\ 0 & 4 & t \end{bmatrix}.$$

25. You may have seen the equation for an ellipse as $(\frac{x}{a})^2 + (\frac{y}{b})^2 = 1$. What are a and b when the equation is written as $\lambda_1 x^2 + \lambda_2 y^2 = 1$? The ellipse $9x^2 + 16y^2 = 1$ has half-axes with lengths $a =$ ____, and $b =$ ____.

26. Draw the tilted ellipse $x^2 + xy + y^2 = 1$ and find the half-lengths of its axes from the eigenvalues of the corresponding A.

27. With positive pivots in D, the factorization $A = LDL^{\mathrm{T}}$ becomes $L\sqrt{D}\sqrt{D}L^{\mathrm{T}}$. (Square roots of the pivots give $D = \sqrt{D}\sqrt{D}$.) Then $C = L\sqrt{D}$ yields the ***Cholesky factorization*** $A = CC^{\mathrm{T}}$, which is "symmetrized LU":

$$\text{From} \quad C = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix} \quad \text{find } A. \qquad \text{From} \quad A = \begin{bmatrix} 4 & 8 \\ 8 & 25 \end{bmatrix} \quad \text{find } C.$$

28. In the Cholesky factorization $A = CC^{\mathrm{T}}$, with $C = L\sqrt{D}$, the square roots of the pivots are on the diagonal of C. Find C (lower triangular) for

$$A = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 8 \end{bmatrix} \qquad \text{and} \qquad A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 7 \end{bmatrix}.$$

29. The symmetric factorization $A = LDL^{\mathrm{T}}$ means that $x^{\mathrm{T}}Ax = x^{\mathrm{T}}LDL^{\mathrm{T}}x$:

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 1 & 0 \\ b/a & 1 \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & (ac-b^2)/a \end{bmatrix} \begin{bmatrix} 1 & b/a \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

The left-hand side is $ax^2 + 2bxy + cy^2$. The right-hand side is $a(x + \frac{b}{a}y)^2 + ____y^2$. The second pivot completes the square! Test with $a = 2$, $b = 4$, $c = 10$.

30. Without multiplying $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$, find

(a) the determinant of A. (b) the eigenvalues of A.
(c) the eigenvectors of A. (d) a reason why A is symmetric positive definite.

31. For the semidefinite matrices

$$A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \text{ (rank 2)} \qquad \text{and} \qquad B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \text{ (rank 1)},$$

write $x^{\mathrm{T}}Ax$ as a sum of two squares and $x^{\mathrm{T}}Bx$ as one square.

32. Apply any three tests to each of the matrices

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} \qquad \text{and} \qquad B = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 1 & 1 \\ 2 & 1 & 2 \end{bmatrix},$$

to decide whether they are positive definite, positive semidefinite, or indefinite.

33. For $C = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$ and $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, confirm that $C^{\mathrm{T}}AC$ has eigenvalues of the same signs as A. Construct a chain of nonsingular matrices $C(t)$ linking C to an orthogonal Q. Why is it impossible to construct a nonsingular chain linking C to the identity matrix?

34. If the pivots of a matrix are all greater than 1, are the eigenvalues all greater than 1? Test on the tridiagonal $-1, 2, -1$ matrices.

35. Use the pivots of $A - \frac{1}{2}I$ to decide whether A has an eigenvalue smaller than $\frac{1}{2}$:

$$A - \frac{1}{2}I = \begin{bmatrix} 2.5 & 3 & 0 \\ 3 & 9.5 & 7 \\ 0 & 7 & 7.5 \end{bmatrix}.$$

36. An algebraic proof of the law of inertia starts with the orthonormal eigenvectors $x_1, \ldots, x_p$ of A corresponding to eigenvalues $\lambda_i > 0$. and the orthonormal eigenvectors $y_1, \ldots, y_q$ of $C^{\mathrm{T}}AC$ corresponding to eigenvalues $\mu_i < 0$.

(a) To prove that the $p+q$ vectors $x_1,\dots,x_p$, $Cy_1,\dots,Cy_q$ are independent, assume that some combination gives zero:

$$a_1x_1+\cdots+a_px_p = b_1Cy_1+\cdots+b_qCy_q \quad (=z,\text{ say}).$$

Show that $z^{\mathrm{T}}Az = \lambda_1 a_1^2+\cdots+\lambda_p a_p^2 \geq 0$ and $z^{\mathrm{T}}Az = \mu_1 b_1^2+\cdots+\mu_q b_q^2 \leq 0$.

(b) Deduce that the a's and b's are zero (proving linear independence). From that deduce $p+q \leq n$.

(c) The same argument for the $n-p$ negative λ's and the $n-q$ positive μ's gives $n-p+n-q \leq n$. (We again assume no zero eigenvalues—which are handled separately). Show that $p+q=n$, so the number p of positive λ's equals the number $n-q$ of positive μ's—which is the law of inertia.

37. If C is nonsingular, show that A and $C^{\mathrm{T}}AC$ have the same rank. Thus they have the same number of zero eigenvalues.

38. Find by experiment the number of positive, negative, and zero eigenvalues of

$$A = \begin{bmatrix} I & B \\ B^{\mathrm{T}} & 0 \end{bmatrix}$$

when the block B (of order $\frac{1}{2}n$) is nonsingular.

39. Do A and $C^{\mathrm{T}}AC$ always satisfy the law of inertia when C is not square?

40. In equation (9) with $m_1 = 1$ and $m_2 = 2$, verify that the normal modes are M-orthogonal: $x_1^{\mathrm{T}}Mx_2 = 0$.

41. Find the eigenvalues and eigenvectors of $Ax = \lambda Mx$:

$$\begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix} x = \frac{\lambda}{18} \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} x.$$

42. If the symmetric matrices A and M are indefinite, $Ax = \lambda Mx$ might not have real eigenvalues. Construct a 2 by 2 example.

43. A *group* of nonsingular matrices includes AB and A^{-1} if it includes A and B. "Products and inverses stay in the group." Which of these sets are groups? *Positive definite symmetric matrices A, orthogonal matrices Q, all exponentials e^{tA} of a fixed matrix A, matrices P with positive eigenvalues, matrices D with determinant* 1. Invent a group containing only positive definite matrices.

SOLUTIONS

Strang Solutions to Selected Exercises

Solutions to Selected Exercises

Problem Set 1.2

1. The lines intersect at $(x, y) = (3, 1)$. Then 3(column 1) + 1(column 2) = (4, 4).

3. These "planes" intersect in a line in four-dimensional space. The fourth plane normally intersects that line in a point. An inconsistent equation like $u + w = 5$ leaves no solution (no intersection).

5. The two points on the plane are (1, 0, 0, 0) and (0, 1, 0, 0).

7. Solvable for (3, 5, 8) and (1, 2, 3); not solvable for $b = (3, 5, 7)$ or $b = (1, 2, 2)$.

9. Column 3 = 2(column 2) − column 1. If $b = (0, 0, 0)$, then $(u, v, w) = (c, -2c, c)$

11. Both $a = 2$ and $a = -2$ give a line of solutions. All other a give $x = 0$, $y = 0$.

13. The row picture has two lines meeting at (4, 2). The column picture has $4(1, 1) + 2(-2, 1) = 4$(column 1) + 2(column 2) = right-hand side (0, 6).

15. The row picture shows four *lines*. The column picture is in *four*-dimensional space. No solution unless the right-hand side is a combination of *the two columns*.

17. If x, y, z satisfy the first two equations, they also satisfy the third equation. The line **L** of solutions contains $v = (1, 1, 0)$, $w = \left(\frac{1}{2}, 1, \frac{1}{2}\right)$, and $u = \frac{1}{2}v + \frac{1}{2}w$, and all combinations $cv + dw$ with $c + d = 1$.

19. Column 3 = column 1; solutions $(x, y, z) = (1, 1, 0)$ or (0, 1, 1) and you can add any multiple of (−1, 0, 1); $b = (4, 6, c)$ needs $c = 10$ for solvability.

21. The second plane and row 2 of the matrix and all columns of the matrix are changed. The solution is not changed.

23. $u = 0$, $v = 0$, $w = 1$, because 1(column 3) = b.

Problem Set 1.3

1. Multiply by $\ell = \frac{10}{2} = 5$, and subtract to find $2x + 3y = 1$ and $-6y = 6$. Pivots 2, −6.

3. Subtract $-\frac{1}{2}$ times equation 1 (or add $\frac{1}{2}$ times equation 1). The new second equation is $3y = 3$. Then $y = 1$ and $x = 5$. If the right-hand side changes sign, so does the solution: $(x, y) = (-5, -1)$.

5. $6x + 4y$ is 2 times $3x + 2y$. There is no solution unless the right-hand side is $2 \cdot 10 = 20$. Then all points on the line $3x + 2y = 10$ are solutions, including (0, 5) and (4, −1).

7. If $a = 2$, elimination must fail. The equations have no solution. If $a = 0$, elimination stops for a row exchange. Then $3y = -3$ gives $y = -1$ and $4x + 6y = 6$ gives $x = 3$.

9. $6x - 4y$ is 2 times $(3x - 2y)$. Therefore, we need $b_2 = 2b_1$. Then there will be infinitely many solutions. The columns (3, 6) and (−2, −4) are on the same line.

11.
$$\begin{array}{rrrrrrl} 2x & - & 3y & = 3 & 2x - 3y = 3 & x = 3 & \text{Subtract } 2\times \text{row 1 from row 2} \\ y & + & z & = 1 \text{ gives} & y + z = 1 \text{ and} & y = 1 & \text{Subtract } 1\times \text{row 1 from row 3} \\ 2y & - & 3z & = 2 & -5z = 0 & z = 0 & \text{Subtract } 2\times \text{row 2 from row 3} \end{array}$$

13. The second pivot position will contain $-2 - b$. If $b = -2$, we exchange with row 3. If $b = -1$ (singular case), the second equation is $-y - z = 0$. A solution is $(1, 1, -1)$.

15. If row 1 = row 2, then row 2 is zero after the first step; exchange the zero row with row 3 and there is no *third* pivot. If column 1 = column 2, there is no *second* pivot.

17. Row 2 becomes $3y - 4z = 5$, then row 3 becomes $(q + 4)z = t - 5$. If $q = -4$, the system is singular — no third pivot. Then, if $t = 5$, the third equation is $0 = 0$. Choosing $z = 1$, the equation $3y - 4z = 5$ gives $y = 3$ and equation 1 gives $x = -9$.

19. The system is singular if row 3 is a *combination of rows* 1 *and* 2. From the end view, the three planes form a triangle. This happens if rows 1 + 2 = row 3 on the left-hand side but not the right-hand side: for example, $x + y + z = 0$, $x - 2y - z = 1$, $2x - y = 9$. No two planes are parallel, but still no solution.

21. The fifth pivot is $\frac{6}{5}$. The nth pivot is $\frac{(n+1)}{n}$.

23. Triangular system
$$\begin{cases} u + v + w = 2 \\ 2v + 2w = -2 \\ 2w = 2 \end{cases} \quad \text{Solution} \quad \begin{array}{l} u = 3 \\ v = -2. \\ w = 1 \end{array}$$

25. $(u, v, w) = (3/2, 1/2, -3)$. Change to $+1$ would make the system singular (2 equal columns).

27. $a = 0$ requires a row exchange, but the system is nonsingular: $a = 2$ makes it singular (one pivot, infinity of solutions); $a = -2$ makes it singular (one pivot, no solution).

29. The second term $bc + ad$ is $(a + b)(c + d) - ac - bd$ (only 1 additional multiplication).

31. Elimination fails for $a = 2$ (equal columns), $a = 4$ (equal rows), $a = 0$ (zero column).

Problem Set 1.4

1. $\begin{bmatrix} 17 \\ 4 \\ 17 \end{bmatrix}, \begin{bmatrix} 5 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \end{bmatrix}$. With sides to (2, 1) and (0, 3), the parallelogram goes to (2, 4).

3. Inner products 54 and 0, column times row gives $\begin{bmatrix} 3 & 5 & 1 \\ -6 & -10 & -2 \\ 21 & 35 & 7 \end{bmatrix}$.

5. $Ax = (0, 0, 0)$, so $x = (2, 1, 1)$ is a solution; other solutions are $cx = (2c, c, c)$.

7. Examples: Diagonal $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 7 \end{bmatrix}$, symmetric $\begin{bmatrix} 1 & 3 & 4 \\ 3 & 2 & 0 \\ 4 & 0 & 7 \end{bmatrix}$, triangular $\begin{bmatrix} 1 & 3 & 4 \\ 0 & 2 & 0 \\ 0 & 0 & 7 \end{bmatrix}$, skew-symmetric $\begin{bmatrix} 0 & 3 & 4 \\ -3 & 0 & 0 \\ -4 & 0 & 0 \end{bmatrix}$.

9. (a) a_{11} (b) $\ell_{i1} = a_{i1}/a_{11}$ (c) new a_{ij} is $a_{ij} - \dfrac{a_{i1}}{a_{11}}a_{1j}$

(d) second pivot $a_{22} - \dfrac{a_{21}}{a_{11}}a_{12}$.

11. The coefficients of rows of B are 2, 1, 4 from A. The first row of AB is [6 3].

13. $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $C = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $D = A$, $E = F = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$.

15. $AB_1 = B_1A$ gives $b = c = 0$. $AB_2 = B_2A$ gives $a = d$. So $A = aI$.

17. $A(A + B) + B(A + B)$, $(A + B)(B + A)$, $A^2 + AB + BA + B^2$ always equal $(A + B)^2$.

19. $\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} p & q \\ r & s \end{bmatrix} = \begin{bmatrix} a \\ c \end{bmatrix}[p \;\; q] + \begin{bmatrix} b \\ d \end{bmatrix}[r \;\; s] = \begin{bmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{bmatrix}$.

21. $A^n = A$; $B^n = \begin{bmatrix} 1 & 0 \\ 0 & (-1)^n \end{bmatrix}$, $C = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ = zero matrix.

23. $E_{32}E_{21}b = (1, -5, -35)$ but $E_{21}E_{32}b = (1, -5, \mathbf{0})$. Then row 3 feels no effect from row 1.

25. Changing a_{33} from 7 to 11 will change the third pivot from 5 to 9. Changing a_{33} from 7 to 2 will change the pivot from 5 to *no pivot.*

27. To reverse E_{31}, *add* 7 times row 1 to row 3. The matrix is $R_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 7 & 0 & 1 \end{bmatrix}$.

29. $E_{13} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$; $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$; $E_{31}E_{13} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$. Test on the identity matrix!

31. E_{21} has $\ell_{21} = -\frac{1}{2}$, E_{32} has $\ell_{32} = -\frac{2}{3}$, E_{43} has $\ell_{43} = -\frac{3}{4}$. Otherwise the E's match I.

33. $\begin{matrix} a + b + c = 4 \\ a + 2b + 4c = 8 \\ a + 3b + 9c = 14 \end{matrix}$ gives $\begin{matrix} a = 2 \\ b = 1. \\ c = 1 \end{matrix}$

35. (a) Each column is E times a column of B.

(b) $EB = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & 4 \\ 1 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 4 & 8 \end{bmatrix}$.

Rows of EB are combinations of rows of B, so they are multiples of [1 2 4].

37. (row 3) $\cdot\, x$ is $\sum a_{3j}x_j$, and $(A^2)_{11}$ = (row 1) $\cdot$ (column 1) $= \sum a_{1j}a_{j1}$.

39. $BA = 3I$ is 5 by 5, $AB = 5I$ is 3 by 3, $ABD = 5D$ is 3 by 1, ABD: No, $A(B+C)$: No.

41. (a) $B = 4I$. (b) $B = 0$. (c) $B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$.

(d) Every row of B is 1, 0, 0.

43. (a) mn (every entry). (b) mnp. (c) n^3 (this is n^2 dot products).

45. $\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}[3 \;\; 3 \;\; 0] + \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix}[1 \;\; 2 \;\; 1] = \begin{bmatrix} 3 & 3 & 0 \\ 6 & 6 & 0 \\ 6 & 6 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 4 & 8 & 4 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 3 & 0 \\ 10 & 14 & 4 \\ 7 & 8 & 1 \end{bmatrix}$.

47. A times B is $A\begin{bmatrix}|&|&|&|\\|&|&|&|\\|&|&|&|\end{bmatrix}, \begin{bmatrix}=\\=\\=\end{bmatrix}B, \begin{bmatrix}=\\=\\=\end{bmatrix}\begin{bmatrix}|&|&|&|\\|&|&|&|\\|&|&|&|\end{bmatrix}, \begin{bmatrix}|&|&|&|\\|&|&|&|\\|&|&|&|\end{bmatrix}\begin{bmatrix}=\\=\\=\end{bmatrix}$.

49. The (2, 2) block $S = D - CA^{-1}B$ is the **Schur complement**: blocks in $d - (cb/a)$.

51. A times $X = [x_1 \;\; x_2 \;\; x_3]$ will be the identity matrix $I = [Ax_1 \;\; Ax_2 \;\; Ax_3]$.

53. $\begin{bmatrix} a+b & a+b \\ c+d & c+d \end{bmatrix}$ agrees with $\begin{bmatrix} a+c & b+d \\ a+c & b+d \end{bmatrix}$ when $b = c$ and $a = d$.

55. $2x + 3y + z + 5t = 8$ is $Ax = b$ with the 1 by 4 matrix $A = [2 \;\; 3 \;\; 1 \;\; 5]$. The solutions x fill a 3D "plane" in four dimensions.

57. The dot product $[1 \;\; 4 \;\; 5]\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ = (1 by 3)(3 by 1) is zero for points (x, y, z) on a plane $x + 4y + 5z = 0$ in three dimensions. The columns of A are one-dimensional vectors.

59. A * v = [3 4 5]′ and v′ * v = 50; v * A gives an error message.

61. $M = \begin{bmatrix} 8 & 3 & 4 \\ 1 & 5 & 9 \\ 6 & 7 & 2 \end{bmatrix} = \begin{bmatrix} 5+u & 5-u+v & 5-v \\ 5-u-v & 5 & 5+u+v \\ 5+v & 5+u-v & 5-u \end{bmatrix}$;

$M_3(1, 1, 1) = (15, 15, 15)$; $M_4(1, 1, 1, 1) = (34, 34, 34, 34)$ because the numbers 1 to 16 add to 136, which is 4(34).

Problem Set 1.5

1. U is nonsingular when no entry on the main diagonal is zero.

3. $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$; $\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix} = I$ also.

$(E^{-1}F^{-1}G^{-1})(GFE) = E^{-1}F^{-1}FE = E^{-1}E = I$; also $(GFE)(E^{-1}F^{-1}G^{-1}) = I$.

5. $LU = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}\begin{bmatrix} 2 & 3 & 3 \\ 0 & 5 & 7 \\ 0 & 0 & -1 \end{bmatrix}$; after elimination, $\begin{bmatrix} 2 & 3 & 3 \\ 0 & 5 & 7 \\ 0 & 0 & -1 \end{bmatrix}\begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}$.

7. $FGH = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 2 & 1 \end{bmatrix}$; $HGF = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 4 & 2 & 1 & 0 \\ 8 & 4 & 2 & 1 \end{bmatrix}$.

9. (a) Nonsingular when $d_1d_2d_3 \neq 0$. (b) Suppose $d_3 \neq 0$: Solve $Lc = b$ going downward: $Lc = b$ gives $c = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. Then $\begin{bmatrix} d_1 & -d_1 & 0 \\ 0 & d_2 & -d_2 \\ 0 & 0 & d_3 \end{bmatrix}\begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ gives

$x = \begin{bmatrix} 1/d_3 \\ 1/d_3 \\ 1/d_3 \end{bmatrix}$.

11. $Lc = b$ going downward gives $c = \begin{bmatrix} 2 \\ -2 \\ 0 \end{bmatrix}$; $Ux = c$ upward gives $x = \begin{bmatrix} 5 \\ -2 \\ 0 \end{bmatrix}$.

13. Permutation rows 2 and 3 $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}$;

permutation rows 1 and 2 $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$.

15. $PA = LDU$ is $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$;

$PA = LDU$ is $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

17. L becomes $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$. MATLAB and other codes use $PA = LU$.

19. $a = 4$ leads to a row exchange; $3b + 10a = 40$ leads to a singular matrix; $c = 0$ leads to a row exchange; $c = 3$ leads to a singular matrix.

21. $\ell_{31} = 1$ and $\ell_{32} = 2$ ($\ell_{33} = 1$): reverse steps to recover $x + 3y + 6z = 11$ from $Ux = c$: 1 times $(x + y + z = 5) + 2$ times $(y + 2z = 2) + 1$ times $(z = 2)$ gives $x + 3y + 6z = 11$.

23. $\begin{bmatrix} 1 & & \\ 0 & 1 & \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & & \\ -2 & 1 & \\ 0 & 0 & 1 \end{bmatrix} A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & -6 \end{bmatrix} = U$.

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix} U = E_{21}^{-1} E_{32}^{-1} U = LU.$$

25. 2 by 2: $d = 0$ not allowed;

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & & \\ \ell & 1 & \\ m & n & 1 \end{bmatrix} \begin{bmatrix} d & e & g \\ & f & h \\ & & i \end{bmatrix}$$

$d = 1$, $e = 1$, then $\ell = 1$
$f = 0$ is not allowed
no pivot in row 2.

27. $A = \begin{bmatrix} 2 & 4 & 8 \\ 0 & 3 & 9 \\ 0 & 0 & 7 \end{bmatrix}$ has $L = I$ and $D = \begin{bmatrix} 2 & & \\ & 3 & \\ & & 7 \end{bmatrix}$; $A = LU$ has $U = A$ (pivots on the diagonal); $A = LDU$ has $U = D^{-1}A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$ with 1s on the diagonal.

29. $\begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{bmatrix} = \begin{bmatrix} 1 & & & \\ 1 & 1 & & \\ 1 & 1 & 1 & \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} a & a & a & a \\ & b-a & b-a & b-a \\ & & c-b & c-b \\ & & & d-c \end{bmatrix}$. Need $\begin{matrix} a \neq 0 \\ b \neq a \\ c \neq b \\ d \neq c. \end{matrix}$

31. $\begin{bmatrix} 1 & & \\ 1 & 1 & \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ & 1 & 1 \\ & & 1 \end{bmatrix} = LIU$; $\begin{bmatrix} a & a & 0 \\ a & a+b & b \\ 0 & b & b+c \end{bmatrix} =$ (same L) $\begin{bmatrix} a & & \\ & b & \\ & & c \end{bmatrix}$ (same U).

33. $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} c = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$ gives $c = \begin{bmatrix} 4 \\ 1 \\ 1 \end{bmatrix}$. $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} x = \begin{bmatrix} 4 \\ 1 \\ 1 \end{bmatrix}$ gives $x = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$. $A = LU$.

35. The 2 by 2 upper submatrix B has the first two pivots 2, 7. Reason: Elimination on A starts in the upper left corner with elimination on B.

37.
$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 6 & 10 & 15 \\ 1 & 4 & 10 & 20 & 35 \\ 1 & 5 & 15 & 35 & 70 \end{bmatrix} = \begin{bmatrix} 1 & & & & \\ 1 & 1 & & & \\ 1 & 2 & 1 & & \\ 1 & 3 & 3 & 1 & \\ 1 & 4 & 6 & 4 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ & 1 & 2 & 3 & 4 \\ & & 1 & 3 & 6 \\ & & & 1 & 4 \\ & & & & 1 \end{bmatrix}.$$

Pascal's triangle in L and U. MATLAB's lu code will wreck the pattern. chol does no row exchanges for symmetric matrices with positive pivots.

39. Each new *right-hand side* costs only n^2 steps compared to $n^3/3$ for full elimination $A\backslash b$.

41. 2 exchanges; 3 exchanges; 50 exchanges and then 51.

43. $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$; $P_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ and $P_2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ (P_2 gives a column exchange).

45. There are $n!$ permutation matrices of order n. Eventually two powers of P *must be the same*: If $P^r = P^s$ then $P^{r-s} = I$. Certainly $r - s \leq n!$

$$P = \begin{bmatrix} P_2 & \\ & P_3 \end{bmatrix} \text{ is 5 by 5 with } P_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, P_3 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \text{ and } P^6 = I.$$

47. The solution is $x = (1, 1, \ldots, 1)$. Then $x = Px$.

Problem Set 1.6

1. $A_1^{-1} = \begin{bmatrix} 0 & \frac{1}{3} \\ \frac{1}{2} & 0 \end{bmatrix}$; $A_2^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ -1 & \frac{1}{2} \end{bmatrix}$; $A_3^{-1} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$.

3. $A^{-1} = BC^{-1}$; $A^{-1} = U^{-1}L^{-1}P$.

5. $A(AB) = (\text{move parentheses}) = (A^2)(B) = I$.

7. $\begin{bmatrix} \sqrt{3}/2 & 1/2 \\ 1/2 & -\sqrt{3}/2 \end{bmatrix}, \begin{bmatrix} -\sqrt{3}/2 & 1/2 \\ 1/2 & \sqrt{3}/2 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ all have $A^2 = I$.

9. If row 3 of A^{-1} were (a, b, c, d), then $A^{-1}A = I$ would give $2a = 0$, $a + 3b = 0$, $4a + 8b = 1$. This has no solution.

11. (a) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$. (b) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

(c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$; $(B^{-1} + A^{-1})^{-1} = B(A + B)^{-1}A$.

13. $A^{\mathrm{T}}B = 8$; $B^{\mathrm{T}}A = 8$; $AB^{\mathrm{T}} = \begin{bmatrix} 6 & 6 \\ 2 & 2 \end{bmatrix}$; $BA^{\mathrm{T}} = \begin{bmatrix} 6 & 2 \\ 6 & 2 \end{bmatrix}$.

15. (a) $n(n + 1)/2$ entries on and above diagonal. (b) $(n - 1)n/2$ entries above diagonal.

17. (a) The inverse of a lower (upper) triangular matrix is still lower (upper) triangular. Multiplying lower (upper) triangular matrices gives a lower (upper) triangular matrix. (b) The main diagonals of $L_1^{-1}L_2D_2$ and $D_1U_1U_2^{-1}$ are the same as those of D_2 and D_1 respectively. $L_1^{-1}L_2D_2 = D_1U_1U_2^{-1}$, so we have $D_1 = D_2$. By comparing the off-diagonals of $L_1^{-1}L_2D_2 = D_1U_1U_2^{-1}$, both matrices must be diagonal. $L_1^{-1}L_2D_2 = D_2$, $D_1U_1U_2^{-1} = D_1$, D_1 is invertible so $L_1^{-1}L_2 = I$, $U_1U_2^{-1} = I$. Then $L_1 = L_2$, $U_1 = U_2$.

19. $\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 5 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 3 & 5 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$;

$\begin{bmatrix} 1 & 0 \\ b/a & 1 \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & d - (b^2/a) \end{bmatrix} \begin{bmatrix} 1 & b/a \\ 0 & 1 \end{bmatrix} = LDL^{\mathrm{T}}$.

21. From $B(I - AB) = (I - BA)B$ we get $(I - BA)^{-1} = B(I - AB)^{-1}B^{-1}$, an explicit inverse provided B and $I - AB$ are invertible. *Second approach*: If $I - BA$ is not invertible, then $BAx = x$ for some nonzero x. Therefore $ABAx = Ax$, or $ABy = y$, and $I - AB$ could not be invertible. (Note that $y = Ax$ is nonzero, from $BAx = x$.)

23. $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} .5 \\ -.2 \end{bmatrix}$, $\begin{bmatrix} t \\ z \end{bmatrix} = \begin{bmatrix} -.2 \\ .1 \end{bmatrix}$ so $A^{-1} = \frac{1}{10}\begin{bmatrix} 5 & -2 \\ -2 & 1 \end{bmatrix}$.

25. (a) In $Ax = (1, 0, 0)$, equation 1 + equation 2 − equation 3 is $0 = 1$. (b) The right-hand sides must satisfy $b_1 + b_2 = b_3$. (c) Row 3 becomes a row of zeros—no third pivot.

27. If B exchanges rows 1 and 2 of A, then B^{-1} exchanges *columns* 1 and 2 of A^{-1}.

29. If A has a column of zeros, so does BA. So $BA = I$ is impossible. There is no A^{-1}

31. $\begin{bmatrix} 1 & & \\ & 1 & \\ -1 & & 1 \end{bmatrix} \begin{bmatrix} 1 & & \\ & 1 & \\ -1 & & 1 \end{bmatrix} \begin{bmatrix} 1 & & \\ -1 & 1 & \\ & & 1 \end{bmatrix} = \begin{bmatrix} 1 & & \\ -1 & 1 & \\ 0 & -1 & 1 \end{bmatrix} = E$;

then $\begin{bmatrix} 1 & & \\ 1 & 1 & \\ 1 & 1 & 1 \end{bmatrix}$ is $L = E^{-1}$, after reversing the order of these 3 elementary matrice and changing -1 to $+1$.

33. A * ones(4,1) gives the zero vector, so A cannot be invertible.

35. $\begin{bmatrix}1&3&1&0\\2&7&0&1\end{bmatrix} \to \begin{bmatrix}1&3&1&0\\0&1&-2&1\end{bmatrix} \to \begin{bmatrix}1&0&7&-3\\0&1&-2&1\end{bmatrix} = [I \quad A^{-1}]$;

$\begin{bmatrix}1&3&1&0\\3&8&0&1\end{bmatrix} \to \begin{bmatrix}1&0&-8&3\\0&1&3&-1\end{bmatrix} = [I \quad A^{-1}]$.

37. $\begin{bmatrix}1&a&b&1&0&0\\0&1&c&0&1&0\\0&0&1&0&0&1\end{bmatrix} \to \begin{bmatrix}1&a&0&1&0&-b\\0&1&0&0&1&-c\\0&0&1&0&0&1\end{bmatrix}$

$$\to \begin{bmatrix}1&0&0&1&-a&ac-b\\0&1&0&0&1&-c\\0&0&1&0&0&1\end{bmatrix}.$$

39. $\begin{bmatrix}2&2&0&1\\0&2&1&0\end{bmatrix} \to \begin{bmatrix}2&0&-1&1\\0&2&1&0\end{bmatrix} \to \begin{bmatrix}1&0&-1/2&1/2\\0&1&1/2&0\end{bmatrix} = [I \quad A^{-1}]$.

41. Not invertible for $c = 7$ (equal columns), $c = 2$ (equal rows), $c = 0$ (zero column).

43. $A^{-1} = \begin{bmatrix}1&1&0&0\\0&1&1&0\\0&0&1&1\\0&0&0&1\end{bmatrix}$. The 5 by 5 A^{-1} also has 1s on the diagonal and super-diagonal.

45. $\begin{bmatrix}I&0\\-C&I\end{bmatrix}$, $\begin{bmatrix}A^{-1}&0\\-D^{-1}CA^{-1}&D^{-1}\end{bmatrix}$, and $\begin{bmatrix}-D&I\\I&0\end{bmatrix}$.

47. For $Ax = b$ with A = ones(4, 4) = singular matrix and b = ones(4, 1), A\b will pick $x = (1, 0, 0, 0)$ and pinv(A) * b will pick the shortest solution $x = (1, 1, 1, 1)/4$.

49. $A^{\mathrm{T}} = \begin{bmatrix}1&9\\0&3\end{bmatrix}$, $A^{-1} = \begin{bmatrix}1&0\\-3&1/3\end{bmatrix}$, $(A^{-1})^{\mathrm{T}} = (A^{\mathrm{T}})^{-1} = \begin{bmatrix}1&-3\\0&1/3\end{bmatrix}$; $A^{\mathrm{T}} = A$ and then $A^{-1} = \dfrac{1}{c^2}\begin{bmatrix}0&c\\c&-1\end{bmatrix} = (A^{-1})^{\mathrm{T}} = (A^{\mathrm{T}})^{-1}$.

51. $\left((AB)^{-1}\right)^{\mathrm{T}} = (B^{-1}A^{-1})^{\mathrm{T}} = (A^{-1})^{\mathrm{T}}(B^{-1})^{\mathrm{T}}$; $(U^{-1})^{\mathrm{T}}$ is *lower* triangular.

53. (a) $x^{\mathrm{T}}Ay = a_{22} = 5$. (b) $x^{\mathrm{T}}A = [4 \quad 5 \quad 6]$. (c) $Ay = \begin{bmatrix}2\\5\end{bmatrix}$.

55. $(Px)^{\mathrm{T}}(Py) = x^{\mathrm{T}}P^{\mathrm{T}}Py = x^{\mathrm{T}}y$ because $P^{\mathrm{T}}P = I$; usually $Px \cdot y = x \cdot P^{\mathrm{T}}y \neq x \cdot Py$:

$$\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix}\begin{bmatrix}1\\2\\3\end{bmatrix} \cdot \begin{bmatrix}1\\1\\2\end{bmatrix} \neq \begin{bmatrix}1\\2\\3\end{bmatrix} \cdot \begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix}\begin{bmatrix}1\\1\\2\end{bmatrix}.$$

57. PAP^{T} recovers the symmetry.

59. (a) The transpose of $R^{\mathrm{T}}AR$ is $R^{\mathrm{T}}A^{\mathrm{T}}R^{\mathrm{TT}} = R^{\mathrm{T}}AR = n$ by n.
(b) $(R^{\mathrm{T}}R)_{jj}$ = (column j of R) · (column j of R) = length squared of column j.

61. Total currents are $A^{\mathrm{T}}y = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 0 & -1 & -1 \end{bmatrix} \begin{bmatrix} y_{BC} \\ y_{CS} \\ y_{BS} \end{bmatrix} = \begin{bmatrix} y_{BC} + y_{BS} \\ -y_{BC} + y_{CS} \\ -y_{CS} - y_{BS} \end{bmatrix}$.

Either way $(Ax)^{\mathrm{T}}y = x^{\mathrm{T}}(A^{\mathrm{T}}y) = x_B y_{BC} + x_B y_{BS} - x_C y_{BC} + x_C y_{CS} - x_S y_{CS} - x_S y_{BS}$.

63. $Ax \cdot y$ is the *cost* of inputs, whereas $x \cdot A^{\mathrm{T}}y$ is the *value* of outputs.

65. These are groups: Lower triangular with diagonal 1s, diagonal invertible D, and permutations P. Two more: Even permutations, all nonsingular matrices.

67. Reordering the rows and/or columns of $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ will move entry a, not giving $\begin{bmatrix} a & c \\ b & d \end{bmatrix}$.

69. Random matrices are almost surely invertible.

71. The $-1, 2, -1$ matrix in Section 1.7 has $A = LDL^{\mathrm{T}}$ with $\ell_{i,i-1} = 1 - \frac{1}{i}$.

Problem Set 1.7

1.
$$\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix}$$
$$= \begin{bmatrix} 1 & & & \\ -\frac{1}{2} & 1 & & \\ & -\frac{2}{3} & 1 & \\ & & -\frac{3}{4} & 1 \end{bmatrix} \begin{bmatrix} 2 & & & \\ & \frac{3}{2} & & \\ & & \frac{4}{3} & \\ & & & \frac{5}{4} \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{2} & & \\ & 1 & -\frac{2}{3} & \\ & & 1 & -\frac{3}{4} \\ & & & 1 \end{bmatrix} = LDL^{\mathrm{T}}$$
$\det = 5.$

3. $A_0 = \begin{bmatrix} 1 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & -1 & \\ & & -1 & 2 & -1 \\ & & & -1 & 1 \end{bmatrix}$. Each row adds to 1, so $A_0 \begin{bmatrix} c \\ c \\ c \\ c \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$.

5. $(u_1, u_2, u_3) = (\pi^2/8, 0, -\pi^2/8)$ instead of the true values $(1, 0, -1)$.

7. $H^{-1} = \begin{bmatrix} 9 & -36 & 30 \\ -36 & 192 & -180 \\ 30 & -180 & 180 \end{bmatrix}$.

9. The 10 by 10 Hilbert matrix is very ill-conditioned.

11. A large pivot is multiplied by less than 1 in eliminating each entry below it. An extreme case, with multipliers $= 1$ and pivots $= \frac{1}{2}, \frac{1}{2}, 4$, is $A = \begin{bmatrix} 1/2 & 1/2 & 1 \\ -1/2 & 0 & 1 \\ -1/2 & -1 & 1 \end{bmatrix}$.

Problem Set 2.1

1. (a) The set of all (u, v), where u and v are ratios p/q of integers. (b) The set of all (u, v), where $u = 0$ or $v = 0$.

3. $\boldsymbol{C}(A)$ is the x-axis; $\boldsymbol{N}(A)$ is the line through $(1, 1)$; $\boldsymbol{C}(B)$ is $\mathbf{R}^2$; $\boldsymbol{N}(B)$ is the line through $(-2, 1, 0)$; $\boldsymbol{C}(C)$ is the point $(0, 0)$ in $\mathbf{R}^2$; the nullspace $\boldsymbol{N}(C)$ is $\mathbf{R}^3$.

5. Broken rules: (a) 7, 8 (b) 1 (c) 1, 2, 8.

7. (b), (d), (e) are subspaces. Can't multiply by -1 in (a) and (c). Can't add in (f).

9. The sum of two nonsingular matrices may be singular ($A + (-A)$). The sum of two singular matrices may be nonsingular.

11. (a) One possibility: The matrices cA form a subspace not containing B.
(b) Yes: the subspace must contain $A - B = I$.
(c) The subspace of matrices whose main diagonal is all zero.

13. If $(f + g)(x)$ is the usual $f(g(x))$, then $(g + f)x$ is $g(f(x))$, which is different. In rule 2 both sides are $f(g(h(x)))$. Rule 4 is broken because there might be no inverse function $f^{-1}(x)$ such that $f(f^{-1}(x)) = x$. If the inverse function exists, it will be the vector $-f$.

15. The sum of (4, 0, 0) and (0, 4, 0) is not on the plane; it has $x + y - 2z = 8$.

17. (a) The subspaces of $\mathbf{R}^2$ are $\mathbf{R}^2$ itself, lines through (0, 0), and the point (0, 0).
(b) The subspaces of $\mathbf{R}^4$ are $\mathbf{R}^4$ itself, three-dimensional planes $n \cdot v = 0$, two-dimensional subspaces ($n_1 \cdot v = 0$ and $n_2 \cdot v = 0$), one-dimensional lines through (0, 0, 0, 0), and (0, 0, 0, 0) alone.

19. The smallest subspace containing **P** and **L** is either **P** or $\mathbf{R}^3$.

21. The column space of A is the x-axis = all vectors $(x, 0, 0)$. The column space of B is the x-y plane = all vectors $(x, y, 0)$. The column space of C is the line of vectors $(x, 2x, 0)$.

23. A combination of the columns of C is also a combination of the columns of A (same column space; B has a different column space).

25. The extra column b enlarges the column space, unless b is *already in* that space:

$$[A \;\; b] = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{matrix} \text{(larger column space)} \\ \text{(no solution to } Ax = b\text{)}. \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \quad \begin{matrix} (b \text{ already in column space}) \\ (Ax = b \text{ has a solution}). \end{matrix}$$

27. Column space = $\mathbf{R}^8$. Every b is a combination of the columns, since $Ax = b$ is solvable.

29. $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ or $\begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$; $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \\ 3 & 6 & 0 \end{bmatrix}$ (columns on 1 line).

31. $\mathbf{R}^2$ contains vectors with *two* components—they don't belong to $\mathbf{R}^3$.

Problem Set 2.2

1. $x + y + z = 1, x + y + z = 0$. Changing 1 to 0, $(x, y, z) = c(-1, 1, 0) + d(-1, 0, 1)$.

3. Echelon form $U = \begin{bmatrix} 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$; free variables x_1, x_3, x_4; special solutions (1, 0, 0, 0), (0, 0, 1, 0), and (0, −3, 0, 1). Consistent when $b_2 = 2b_1$. Complete solution $(0, b_1, 0, 0)$ plus any combination of special solutions.

5. $\begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} -2v-3 \\ v \\ 2 \end{bmatrix} = v\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} -3 \\ 0 \\ 2 \end{bmatrix}$; no solution!

7. $c = 7$ allows $u = 1$, $v = 1$, $w = 0$. The column space is a plane.

9. (a) $x = x_2\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4\begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$, for any x_2, x_4. Row-reduced $R = \begin{bmatrix} 1 & 2 & 0 & -2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

(b) Complete solution $x = \begin{bmatrix} a-3b \\ 0 \\ b \\ 0 \end{bmatrix} + x_2\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4\begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$, for any x_2, x_4.

11. $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ has nullspace = line through $(-1, 1)$ but no solution. Any $b = \begin{bmatrix} c \\ c \end{bmatrix}$ has many particular solutions to $Ax_p = b$.

13. $R = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}; R = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}; R = \begin{bmatrix} 1 & -1 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. (a) $r = 1$. (b) $r = 2$. (c) $r = 1$.

15. A nullspace matrix $N = \begin{bmatrix} -F \\ I \end{bmatrix}$ is n by $n - r$.

17. I think this is true.

19. The special solutions are the columns of $N = \begin{bmatrix} -2 & -3 \\ -4 & -5 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $N = \begin{bmatrix} 1 & 0 \\ 0 & -2 \\ 0 & 1 \end{bmatrix}$.

21. The r pivot columns of A form an m by r submatrix of rank r, so that matrix A^* has r independent pivot rows, giving an r by r invertible submatrix of A. (The pivot rows of A^* and A are the same, since elimination is done in the same order—we just don't see for A^* the "free" columns of zeros that appear for A.)

23. $(uv^{\mathrm{T}})(wz^{\mathrm{T}}) = u(v^{\mathrm{T}}w)z^{\mathrm{T}}$ has rank 1 unless $v^{\mathrm{T}}w = 0$.

25. We are given $AB = I$ which has rank n. Then $\mathrm{rank}(AB) \leq \mathrm{rank}(A)$ forces $\mathrm{rank}(A) = n$.

27. If $R = EA$ and the same $R = E^*B$, then $B = (E^*)^{-1}EA$. (To get B, reduce A to R and then invert steps back to B.) B is an *invertible* matrix times A, when they share the same R.

29. Since R starts with r independent rows, R^{T} starts with r independent columns (and then zeros). So *its* reduced echelon form is $\begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix}$ where I is r by r.

31. If $c = 1$, $R = \begin{bmatrix} 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ has x_2, x_3, x_4 free.

If $c \neq 1$, $R = \begin{bmatrix} 1 & 0 & 2 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ has x_3, x_4 free.

Special solutions in $N = \begin{bmatrix} -1 & -2 & -2 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $(c = 1)$ and $N = \begin{bmatrix} -2 & -2 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$ $(c \neq 1)$.

If $c = 1$, $R = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ has x_1 free; if $c = 2$, $R = \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix}$ has x_2 free; $R = I$ if $c \neq 1, 2$.

Special solutions in $N = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $(c = 1)$ or $N = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ $(c = 2)$ or $N = 2$ by 0 *empty matrix*.

33. $x_{\text{complete}} = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}$; $x_{\text{complete}} = \begin{bmatrix} 1/2 \\ 0 \\ 1/2 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix}$.

35. (a) Solvable if $b_2 = 2b_1$ and $3b_1 - 3b_3 + b_4 = 0$. Then $x = \begin{bmatrix} 5b_1 - 2b_3 \\ b_3 - 2b_1 \end{bmatrix}$ (no free variables). (b) Solvable if $b_2 = 2b_1$ and $3b_1 - 3b_3 + b_4 = 0$. Then $x = \begin{bmatrix} 5b_1 - 2b_3 \\ b_3 - 2b_1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$.

37. A 1 by 3 system has at least two free variables.

39. (a) The particular solution x_p is always multiplied by 1. (b) Any solution can be x_p. (c) $\begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 6 \end{bmatrix}$. Then $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is shorter (length $\sqrt{2}$) than $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$. (d) The "homogeneous" solution in the nullspace is $x_n = 0$ when A is invertible.

41. Multiply x_p by 2, same x_n; $\begin{bmatrix} x \\ X \end{bmatrix}_p$ is $\begin{bmatrix} x_p \\ 0 \end{bmatrix}$, special solutions also include the columns of $\begin{bmatrix} -I \\ I \end{bmatrix}$; x_p and the special solutions are not changed.

43. For A, $q = 3$ gives rank 1, every other q gives rank 2. For B, $q = 6$ gives rank 1, every other q gives rank 2.

45. (a) $r < m$, always $r \leq n$. (b) $r = m, r < n$. (c) $r < m, r = n$. (d) $r = m = n$.

47. $R = \begin{bmatrix} 1 & 0 & 0 & \mathbf{0} \\ 0 & 0 & 1 & \mathbf{0} \\ 0 & 0 & 0 & \mathbf{0} \end{bmatrix}$ and $x_n = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$; $\begin{bmatrix} 1 & 0 & 0 & \mathbf{-1} \\ 0 & 0 & 1 & \mathbf{2} \\ 0 & 0 & 0 & \mathbf{5} \end{bmatrix}$: no solution because of row 3.

49. $A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 0 & 3 \end{bmatrix}$; B can't exist since two equations in three unknowns can't have one solution.

51. A has rank $4 - 1 = 3$; the complete solution to $Ax = 0$ is $x = (2, 3, 1, 0)$.

$$R = \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{ with } -2, -3 \text{ in the free column.}$$

53. (a) False. (b) True. (c) True (only n columns). (d) True (only m rows).

55. $U = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ and $R = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ (R doesn't come from this U).

57. If column 1 = column 5, then x_5 is a free variable. Its special solution is $(-1, 0, 0, 0, 1)$.

59. Column 5 is sure to have no pivot since it is a combination of earlier columns, and x_5 is free. With four pivots in the other columns, the special solution is $(1, 0, 1, 0, 1)$. The nullspace contains all multiples of $(1, 0, 1, 0, 1)$ (a line in $\mathbf{R}^5$).

61. $A = \begin{bmatrix} 1 & 0 & 0 & -4 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & -2 \end{bmatrix}$.

63. This construction is impossible: two pivot columns, two free variables, only three columns.

65. $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$.

67. R is most likely to be I; R is most likely to be I with fourth row of zeros.

69. Any zero rows come after these rows: $R = [1 \quad -2 \quad -3]$, $R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, $R = I$.

Problem Set 2.3

1. $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = 0$ gives $c_3 = c_2 = c_1 = 0$. But $v_1 + v_2 - 4v_3 + v_4 = 0$ (dependent).

3. If $a = 0$ then column 1 $= 0$; if $d = 0$ then b(column 1) $- a$(column 2) $= 0$; if $f = 0$ then all columns end in zero (all are perpendicular to $(0, 0, 1)$, all in the xy plane, must be dependent).

5. (a) $\begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -7 \\ 0 & -1 & -5 \end{bmatrix}$

$\to \begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -7 \\ 0 & 0 & -18/5 \end{bmatrix}$: invertible $\Rightarrow$ independent columns (could have used rows).

(b) $\begin{bmatrix} 1 & 2 & -3 \\ -3 & 1 & 2 \\ 2 & -3 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & -3 \\ 0 & 7 & -7 \\ 0 & 0 & 0 \end{bmatrix}$; $A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$, columns add to 0 (could use rows).

7. The sum $v_1 - v_2 + v_3 = 0$ because $(w_2 - w_3) - (w_1 - w_3) + (w_1 - w_2) = 0$.

9. (a) The four vectors are the columns of a 3 by 4 matrix A with at least one free variable, so $Ax = 0$. (b) Dependent if $[v_1 \quad v_2]$ has rank 0 or 1.
(c) $0v_1 + c(0, 0, 0) = 0$ has a nonzero solution (take any $c \neq 0$).

11. (a) Line in $\mathbf{R}^3$. (b) Plane in $\mathbf{R}^3$. (c) Plane in $\mathbf{R}^3$. (d) All of $\mathbf{R}^3$.

13. All dimensions are 2. The row spaces of A and U are the same.

15. $v = \frac{1}{2}(v + w) + \frac{1}{2}(v - w)$ and $w = \frac{1}{2}(v + w) - \frac{1}{2}(v - w)$. The two pairs *span* the same space. They are a basis when v and w are *independent.*

17. If elimination produces one or more zero rows, the rows of A are linearly dependent;

for example in Problem 16 $\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

$\to \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

19. The n independent vectors span a space of dimension n. They are a *basis* for that space. If they are the columns of A then m is *not less* than n ($m \geq n$).

21. $C(U)$: Any bases for $\mathbf{R}^2$; $N(U)$: (row 1 and row 2) or (row 1 and row 1 + row 2).

23. Independent columns $\Rightarrow$ rank n. Columns span $\mathbf{R}^m \Rightarrow$ rank m. Columns are basis for $\mathbf{R}^m \Rightarrow$ rank $= m = n$.

25. (a) The only solution is $x = 0$ because *the columns are independent.* (b) $Ax = b$ is solvable because *the columns span* $\mathbf{R}^5$.

27. Columns 1 and 2 are bases for the (different) column spaces of A and U; rows 1 and 2 are bases for the (equal) row spaces; $(1, -1, 1)$ is a basis for the (equal) nullspaces.

29. rank$(A) = 2$ if $c = 0$ and $d = 2$; rank$(B) = 2$ except when $c = d$ or $c = -d$.

31. Let $v_1 = (1, 0, 0, 0), \ldots, v_4 = (0, 0, 0, 1)$ be the coordinate vectors. If $\mathbf{W}$ is the line through $(1, 2, 3, 4)$, none of the v's are in $\mathbf{W}$.

33. (a) If it were not a basis, we could add more independent vectors, which would exceed the given dimension k. (b) If it were not a basis, we could delete some vectors, leaving less than the given dimension k.

35. (a) False, might be no solution. (b) True, 7 vectors in $\mathbf{R}^5$ are dependent.

37. (a) $\begin{bmatrix}1&0&0\\0&0&0\\0&0&0\end{bmatrix}, \begin{bmatrix}0&0&0\\0&1&0\\0&0&0\end{bmatrix}, \begin{bmatrix}0&0&0\\0&0&0\\0&0&1\end{bmatrix}$. (b) Add $\begin{bmatrix}0&1&0\\1&0&0\\0&0&0\end{bmatrix}, \begin{bmatrix}0&0&1\\0&0&0\\1&0&0\end{bmatrix}, \begin{bmatrix}0&0&0\\0&0&1\\0&1&0\end{bmatrix}$. (c) $\begin{bmatrix}0&1&0\\-1&0&0\\0&0&0\end{bmatrix}, \begin{bmatrix}0&0&1\\0&0&0\\-1&0&0\end{bmatrix}, \begin{bmatrix}0&0&0\\0&0&1\\0&-1&0\end{bmatrix}$ are a basis for all $A = -A^{\mathrm{T}}$.

39. $y(0) = 0$ requires $A + B + C = 0$. One basis is $\cos x - \cos 2x$ and $\cos x - \cos 3x$.

41. $y_1(x), y_2(x), y_3(x)$ can be $x, 2x, 3x$ (dim 1) or $x, 2x, x^2$ (dim 2) or x, x^2, x^3 (dim 3).

43. $$\begin{bmatrix}\mathbf{1}&&\\&\mathbf{1}&\\&&\mathbf{1}\end{bmatrix} = \begin{bmatrix}&\mathbf{1}&\\\mathbf{1}&&\\&&\mathbf{1}\end{bmatrix} - \begin{bmatrix}&\mathbf{1}&\\&&\mathbf{1}\\\mathbf{1}&&\end{bmatrix} + \begin{bmatrix}&&\mathbf{1}\\&\mathbf{1}&\\\mathbf{1}&&\end{bmatrix} + \begin{bmatrix}\mathbf{1}&&\\&&\mathbf{1}\\&\mathbf{1}&\end{bmatrix} - \begin{bmatrix}&&\mathbf{1}\\\mathbf{1}&&\\&\mathbf{1}&\end{bmatrix}.$$

Check the (1, 1) entry, then (3, 2), then (3, 3), then (1, 2) to show that those five P's are independent. Four conditions on the nine entries make all row sums and column sums equal: row sum 1 = row sum 2 = row sum 3 = column sum 1 = column sum 2 (= column sum 3 is automatic because sum of all rows = sum of all columns).

45. If the 5 by 5 matrix $[A \;\; b]$ is invertible, b is not a combination of the columns of A. If $[A \;\; b]$ is singular, and A has independent columns, b *is* a combination of those columns.

Problem Set 2.4

1. False, we only know dimensions are equal. Left nullspace has smaller dim $= m - r$.

3. $\boldsymbol{C}(A)$: $r = 2$, (1, 0, 1) (0, 1, 0); $\boldsymbol{N}(A)$: $n - r = 2$, (2, −1, 1, 0), (−1, 0, 0, 1);
$\boldsymbol{C}(A^{\mathrm{T}})$: $r = 2$, (1, 2, 0, 1), (0, 1, 1, 0); $\boldsymbol{N}(A^{\mathrm{T}})$: $m - r = 1$, (−1, 0, 1);
$\boldsymbol{C}(U)$: (1, 0, 0), (0, 1, 0); $\boldsymbol{N}(U)$: (2, −1, 1, 0), (−1, 0, 0, 0);
$\boldsymbol{C}(U^{\mathrm{T}})$: (1, 2, 0, 1), (0, 1, 1, 0); $\boldsymbol{N}(A^{\mathrm{T}})$: (0, 0, 1).

5. A times every column of B is zero, so $\boldsymbol{C}(B)$ is contained in the nullspace $\boldsymbol{N}(A)$.

7. From $Ax = 0$, the row space and the nullspace must be orthogonal. See Chapter 3.

9. [1 2 4]; $\begin{bmatrix}1&2&4\\2&4&8\\3&6&12\end{bmatrix}$ has the same nullspace.

11. If $Ax = 0$ has a nonzero solution, then $r < n$ and $\boldsymbol{C}(A^{\mathrm{T}})$ is smaller than $\mathbf{R}^n$. So $A^{\mathrm{T}}y = f$ is not solvable for some f. Example: $A = [1 \;\; 1]$ and $f = (1, 2)$.

13. $d = bc/a$; the only pivot is a.

15. With independent columns: rank n; nullspace $= \{0\}$; row space is $\mathbf{R}^n$; *left* inverse.

17. $A = [1 \;\; 1 \;\; 0]$; $B = [0 \;\; 0 \;\; 1]$.

19. No—for example, all invertible n by n matrices have the same four subspaces.

21. (a) $\begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$. (b) Impossible: dimensions $1 + 1 \neq 3$. (c) $[1 \quad 1]$.

(d) $\begin{bmatrix} -9 & -3 \\ 3 & 1 \end{bmatrix}$. (e) Impossible: Row space = column space requires $m = n$. Then $m - r = n - r$.

23. Invertible A: row space basis = column space basis = $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$; nullspace and left nullspace bases are empty. B: row space basis $(1, 0, 0, 1, 0, 0)$, $(0, 1, 0, 0, 1, 0)$, and $(0, 0, 1, 0, 0, 1)$; column space basis $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$; nullspace basis $(-1, 0, 0, 1, 0, 0)$, $(0, -1, 0, 0, 1, 0)$, and $(0, 0, -1, 0, 0, 1)$; left nullspace basis is empty.

25. (a) Same row space and nullspace. Therefore rank (dimension of row space) is the same. (b) Same column space and left nullspace. Same rank (dimension of column space).

27. (a) No solution means that $r < m$. Always $r \leq n$. Can't compare m and n.
(b) If $m - r > 0$, the nullspace of A^T contains a nonzero vector.

29. Row space basis $(1, 2, 3, 4)$, $(0, 1, 2, 3)$, $(0, 0, 1, 2)$; nullspace basis $(0, 1, -2, 1)$; column space basis $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$; left nullspace has empty basis.

31. If $Av = 0$ and v is a row of A then $v \cdot v = 0$. Only $v = 0$ is in both spaces.

33. Row 3 $-$ 2 (row 2) $+$ row 1 $=$ zero row, so the vectors $c(1, -2, 1)$ are in the left nullspace. The same vectors happen to be in the nullspace.

35. (a) u and w span $\boldsymbol{C}(A)$. (b) v and z span $\boldsymbol{C}(A^T)$. (c) rank < 2 if u and w are dependent or v and z are dependent. (d) The rank of $uv^T + wz^T$ is 2.

37. (a) True (same rank). (b) False ($A = [1 \quad 0]$). (c) False (A can be invertible and also unsymmetric). (d) True.

39. $a_{11} = 1, a_{12} = 0, a_{13} = 1, a_{22} = 0, a_{32} = 1, a_{31} = 0, a_{23} = 1, a_{33} = 0, a_{21} = 1$ (not unique).

41. Rank $r = n$ means nullspace = zero vector and $x_n = 0$.

Problem Set 2.5

1. $A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & -1 \end{bmatrix}$; $\boldsymbol{N}(A)$ contains multiples of $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$; $\boldsymbol{N}(A^T)$ contains multiples of $\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$.

3. The entries in each row add to zero. Therefore, any combination will have that same property: $f_1 + f_2 + f_3 = 0$; $A^T y = f \Rightarrow y_1 + y_3 = f_1, -y_1 + y_2 = f_2, -y_2 - y_3 = f_3 \Rightarrow f_1 + f_2 + f_3 = 0$. It means that the total current entering from outside is zero.

5. $\begin{bmatrix} c_1+c_3 & -c_1 & -c_3 \\ -c_1 & c_1+c_2 & -c_2 \\ -c_3 & -c_2 & c_2+c_3 \end{bmatrix}$; $\begin{bmatrix} c_1+c_3 & -c_1 \\ -c_1 & c_1+c_2 \end{bmatrix}$ has pivots c_1+c_3, $\dfrac{c_1c_3+c_1c_2+c_2c_3}{c_1+c_3}$.

7. Conditions on b are $b_1+b_4-b_5=0$, $b_3-b_4+b_6=0$, $b_2-b_5+b_6=0$.

9. $\begin{bmatrix} 3 & -1 & -1 & -1 \\ -1 & 3 & -1 & -1 \\ -1 & -1 & 3 & -1 \\ -1 & -1 & -1 & 3 \end{bmatrix}$, $\begin{bmatrix} c_1+c_2+c_5 & -c_1 & -c_2 & -c_5 \\ -c_1 & c_1+c_3+c_4 & -c_3 & -c_4 \\ -c_2 & -c_3 & c_2+c_3+c_6 & -c_6 \\ -c_5 & -c_4 & -c_6 & c_4+c_5+c_6 \end{bmatrix}$.

Those c's that connect to node j will appear in row j.

11. $\begin{bmatrix} 1 & 0 & 0 & 0 & -1 & 1 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & -1 & 0 & 1 \\ 0 & 0 & \frac{1}{2} & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 \\ -1 & -1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ f_1 \\ f_2 \\ f_3 \end{bmatrix}$; $x = \begin{bmatrix} -4 \\ -\frac{5}{3} \\ -\frac{14}{3} \\ 0 \end{bmatrix}$; $y = \begin{bmatrix} -\frac{7}{3} \\ \frac{4}{3} \\ \frac{10}{3} \\ \frac{14}{3} \end{bmatrix}$.

13. There are 20 choices of 3 edges out of 6 because "6 choose 3" $= \dfrac{6!}{3!3!} = 20$. Four choices give triangles, leaving 16 spanning trees.

15. I think it is already built in.

17. 9 nodes $-$ 12 edges $+$ 4 loops $= 1$; 7 nodes $-$ 12 edges $+$ 6 loops $= 1$.

19. $x = (1, 1, 1, 1)$ gives $Ax = 0$; then $A^{\mathrm{T}}Ax = 0$; rank is again $n - 1$.

21. $M = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$ and

$M^2 = \begin{bmatrix} 3 & 2 & 2 & 2 \\ 2 & 3 & 2 & 2 \\ 2 & 2 & 3 & 2 \\ 2 & 2 & 2 & 3 \end{bmatrix}$.

$(M^2)_{ij} = a_{i1}a_{1j} + \cdots + a_{in}a_{nj}$ and we get $a_{ik}a_{kj} = 1$ when there is a 2-step path i to k to j. Notice 3 paths from a node to itself.

Problem Set 2.6

1. Rotation $\begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix}$; $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

3. $\|Ax\|^2 = 1$ always produces an ellipse.

5. They are transformed to $(1, 3)$, $(2, 6)$, $(-1, -3)$. The x-axis turns; vertical lin shift up/down but stay vertical.

7. Second derivative matrix $\begin{bmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. The nullspace is spanned by $(1,0,0,0)$ and $(0,1,0,0)$, which gives linear P_1. Second derivatives of linear functions are zero. The column space is accidentally the same as the nullspace, because second derivatives of cubics are linear.

9. e^t and e^{-t} are a basis for the solutions of $u'' = u$.

11. $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ so $H^2 = I$.

13. (a) Yes. (b) Yes. We don't need parentheses $(AB)C$ or $A(BC)$ for ABC!

15. $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ and $A^2 = I$; the double transpose of a matrix gives the matrix itself.
Note $A_{23} = 1$ because transpose of matrix 2 is matrix 3.

17. $A = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$; $B = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$; $AB = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$; $BA = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

19. (a) is invertible with $T^{-1}(y) = y^{1/3}$; (c) is invertible with $T^{-1}(y) = y - 11$.

21. With $w = 0$, linearity gives $T(v+0) = T(v) + T(0)$. Thus $T(0) = 0$. With $c = -1$, linearity gives $T(-0) = -T(0)$. Certainly $T(-0) = T(0)$. Thus $T(0) = 0$.

23. $S(T(v)) = S(v) = v$.

25. (b) and (c) are linear, (a) fails $T(2v) = 2T(v)$, (d) fails $T(v+w) = T(v) + T(w)$.

27. $T(T(v)) = (v_3, v_1, v_2)$; $T^3(v) = v$; $T^{100}(v) = T(T^{99}(v)) = T(v)$.

29. (a) $T(1,0) = 0$. (b) $(0,0,1)$ is not in the range. (c) $T(0,1) = 0$.

31. Associative law gives $A(M_1 + M_2) = AM_1 + AM_2$. Distributive law over c's gives $A(cM) = c(AM)$.

33. No matrix A gives $A \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. To professors: The matrix space has dimension 4. Linear transformations on that space must come from 4 by 4 matrices (16 parameters). Those multiplications by A in Problems 31 and 32 were special transformations with only 4 parameters.

35. $T(I) = 0$ but $M = \begin{bmatrix} 0 & b \\ 0 & 0 \end{bmatrix} = T(M)$; these fill the range. $M = \begin{bmatrix} a & 0 \\ c & d \end{bmatrix}$ in the kernel.

37. (a) $M = \begin{bmatrix} r & s \\ t & u \end{bmatrix}$. (b) $N = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1}$. (c) $ad = bc$.

39. Reorder basis by *permutation matrix*; change lengths by *positive diagonal matrix.*

41. $\begin{bmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{bmatrix} \begin{bmatrix} A \\ B \\ C \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$; Vandermonde determinant $= (b-a)(c-a)(c-b)$; the points a, b, c must be different, and then determinant $\neq 0$ (interpolation is possible).

43. If T is not invertible, then $T(v_1), \ldots, T(v_n)$ will not be a basis. Then we couldn't choose $w_i = T(v_i)$ as output basis.

45. $S(T(v)) = (-1, 2)$ but $S(v) = (-2, 1)$ and $T(S(v)) = (1, -2)$. So $TS \neq ST$.

47. The Hadamard matrix H has orthogonal columns of length 2. So the inverse of H is $H^{\mathrm{T}}/4 = H/4$.

49. False: the n nonzero vectors would have to be independent.

Problem Set 3.1

1. $\|x\| = \sqrt{21}$; $\|y\| = 3\sqrt{2}$; $x^{\mathrm{T}}y = 0$.

3. $(x_2/x_1)(y_2/y_1) = -1$ means that $x_1y_1 + x_2y_2 = 0$, so $x^{\mathrm{T}}y = 0$.

5. v_1 and v_3 are orthogonal, also v_2 and v_3.

7. $x = (-2, 1, 0)$; $y = (-1, -1, 1)$; the row $z = (1, 2, 1)$ is orthogonal to the nullspace.

9. The orthogonal complement is the line through $(-1, -1, 1)$ and $(0, 0, 0)$.

11. If $A^{\mathrm{T}}y = 0$, then $y^{\mathrm{T}}b = y^{\mathrm{T}}Ax = (y^{\mathrm{T}}A)x = 0$, which contradicts $y^{\mathrm{T}}b \neq 0$.

13. The figure splits any y in $\mathbf{R}^m$ into column space part + left nullspace part.

15. No such matrix, because $(1, 2, 1)^{\mathrm{T}}(1, -2, 1) \neq 0$.

17. The matrix with the basis for $\mathbf{V}$ as its rows. Then the nullspace is $\mathbf{V}^{\perp} = \mathbf{W}$.

19. (a) If $\mathbf{V}$ and $\mathbf{W}$ are lines in $\mathbf{R}^3$, $\mathbf{V}^{\perp}$ and $\mathbf{W}^{\perp}$ are intersecting planes. (b) $\mathbf{V}$.

21. $(1, 2, -1)$ is perpendicular to $\mathbf{P}$. $A = \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 2 \end{bmatrix}$ has $N(A) = \mathbf{P}$; $B = [1 \quad 2 \quad -1]$ has row space $= \mathbf{P}$.

23. $A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$ has subspaces = four lines; $(1, 1)$ orthogonal to $(-1, 1)$, $(1, 2)$ orthogonal to $(-2, 1)$. Always row space $\perp$ nullspace.

25. (a) $\begin{bmatrix} 1 & 2 & -3 \\ 2 & -3 & 1 \\ -3 & 5 & -2 \end{bmatrix}$. (b) $\begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix}$ is not orthogonal to $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. (c) $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ in $C(A)$ and $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ in $N(A^{\mathrm{T}})$ is impossible: not perpendicular. (d) $A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$ has $A^2 = 0$. (e) $(1, 1, 1)$ will be in the nullspace and row space; no such matrix.

27. (a) If $Ax = b$ has a solution and $A^{\mathrm{T}}y = 0$, then $b^{\mathrm{T}}y = (Ax)^{\mathrm{T}}y = 0$.
(b) b is not in the column space; so, not perpendicular to all y in the left nullspace.

29. $x = x_r + x_n$, where x_r is in the row space and x_n is in the nullspace. Then $Ax_n = 0$ and $Ax = Ax_r + Ax_n = Ax_r$. All vectors Ax are combinations of the columns of A. If $x = (1, 0)$, then $x_r = (1/2, 1/2)$.

31. (a) For a symmetric matrix, the column space and row space are the same.
(b) x is in the nullspace and z is in the column space = row space: so these "eigenvectors" have $x^{\mathrm{T}}z = 0$.

33. x splits into $x_r + x_n = (1, -1) + (1, 1) = (2, 0)$.

35. $Ax = B\widehat{x}$ means that $[\,A \quad B\,]\begin{bmatrix} x \\ -\widehat{x} \end{bmatrix} = 0$. Three homogeneous equations in four unknowns always have a nonzero solution. Here $x = (3, 1)$ and $\widehat{x} = (1, 0)$, and $Ax = B\widehat{x} = (5, 6, 5)$ is in both column spaces. Two planes in $\mathbf{R}^3$ (through zero) must intersect in a line at least!

37. $A^{\mathrm{T}}y = 0$ gives $(Ax)^{\mathrm{T}}y = x^{\mathrm{T}}A^{\mathrm{T}}y = 0$. Then $y \perp Ax$ and $N(A^{\mathrm{T}}) \perp C(A)$.

39. $\mathbf{S}^{\perp}$ is the nullspace of $A = \begin{bmatrix} 1 & 5 & 1 \\ 2 & 2 & 2 \end{bmatrix}$. Therefore $\mathbf{S}^{\perp}$ is a *subspace* even if $\mathbf{S}$ is not.

41. If $\mathbf{V}$ is all of $\mathbf{R}^4$, then $\mathbf{V}^{\perp}$ contains only the *zero vector*. Then $(\mathbf{V}^{\perp})^{\perp} = \mathbf{R}^4 = \mathbf{V}$.

43. $(1, 1, 1, 1)$ is a basis for $\mathbf{P}^{\perp}$. $A = [1 \quad 1 \quad 1 \quad 1]$ has the plane $\mathbf{P}$ as its nullspace.

45. Column 1 of A^{-1} is orthogonal to the space spanned by the 2nd, ..., nth rows of A.

47. $A = \begin{bmatrix} 2 & 2 & -1 \\ -1 & 2 & 2 \\ 2 & -1 & 2 \end{bmatrix}$,

$A^{\mathrm{T}}A = 9I$ is *diagonal*: $(A^{\mathrm{T}}A)_{ij} = (\text{column } i \text{ of } A) \cdot (\text{column } j)$.

49. (a) $(1, -1, 0)$ is in both planes. Normal vectors are perpendicular, planes still intersect! (b) Need *three* orthogonal vectors to span the whole orthogonal complement in $\mathbf{R}^5$. (c) Lines can meet without being orthogonal.

51. When $AB = 0$, the column space of B is contained in the nullspace of A. Therefore the dimension of $C(B) \le$ dimension of $N(A)$. This means $\operatorname{rank}(B) \le 4 - \operatorname{rank}(A)$.

Problem Set 3.2

1. (a) $(x + y)/2 \ge \sqrt{xy}$ (arithmetic mean $\ge$ geometric mean of x and y).
(b) $\|x+y\|^2 \le (\|x\|+\|y\|)^2$ means that $(x+y)^{\mathrm{T}}(x+y) \le \|x\|^2+2\|x\|\|y\|+\|y\|^2$. The left-hand side is $x^{\mathrm{T}}x + 2x^{\mathrm{T}}y + y^{\mathrm{T}}y$. After cancelling this is $x^{\mathrm{T}}y \le \|x\|\|y\|$.

3. $p = (10/3, 10/3, 10/3)$; $(5/9, 10/9, 10/9)$.

5. $\cos\theta = 1/\sqrt{n}$ so $\theta = \arccos(1/\sqrt{n})$; $P = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}[1/n \quad \cdots \quad 1/n] =$ all entries $\dfrac{1}{n}$.

7. Choose $b = (1, \ldots, 1)$; equality if $a_1 = \cdots = a_n$ (then a is parallel to b).

9. $P^2 = \dfrac{aa^{\mathrm{T}}aa^{\mathrm{T}}}{a^{\mathrm{T}}aa^{\mathrm{T}}a} = \dfrac{a(a^{\mathrm{T}}a)a^{\mathrm{T}}}{(a^{\mathrm{T}}a)(a^{\mathrm{T}}a)} = \dfrac{aa^{\mathrm{T}}}{a^{\mathrm{T}}a} = P.$

11. (a) $P = \begin{bmatrix} \frac{1}{10} & \frac{3}{10} \\ \frac{3}{10} & \frac{9}{10} \end{bmatrix}$; $P_2 = I - P_1 = \begin{bmatrix} \frac{9}{10} & -\frac{3}{10} \\ -\frac{3}{10} & \frac{1}{10} \end{bmatrix}$. (b) $P_1 + P_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$; $P_1P_2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$. The sum of the projections onto two perpendicular lines gives the vector itself. The projection onto one line and then a perpendicular line gives the zero vector.

13. Trace $= \dfrac{a_1a_1}{a^{\mathrm{T}}a} + \cdots + \dfrac{a_na_n}{a^{\mathrm{T}}a} = \dfrac{a^{\mathrm{T}}a}{a^{\mathrm{T}}a} = 1.$

15. $\|Ax\|^2 = (Ax)^{\mathrm{T}}(Ax) = xA^{\mathrm{T}}Ax$, $\|A^{\mathrm{T}}x\|^2 = (A^{\mathrm{T}}x)^{\mathrm{T}}(A^{\mathrm{T}}x) = xAA^{\mathrm{T}}x$. If $A^{\mathrm{T}}A = AA^{\mathrm{T}}$, then $\|Ax\| = \|A^{\mathrm{T}}x\|$. (These matrices are called *normal*.)

17. (a) $a^{\mathrm{T}}b/a^{\mathrm{T}}a = 5/3$; $p = (5/3, 5/3, 5/3)$; $e = (-2/3, 1/3, 1/3)$ has $e^{\mathrm{T}}a = 0$. (b) $a^{\mathrm{T}}b/a^{\mathrm{T}}a = -1$; $p = (1, 3, 1) = b$ and $e = (0, 0, 0)$.

19. $P_1 = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = P_1^2$ and $P_1 b = \frac{1}{3}\begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix}$. $P_2 = \frac{1}{11}\begin{bmatrix} 1 & 3 & 1 \\ 3 & 9 & 3 \\ 1 & 3 & 1 \end{bmatrix}$ and $P_2 b = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$.

21. $P_1 = \frac{1}{9}\begin{bmatrix} 1 & -2 & -2 \\ -2 & 4 & 4 \\ -2 & 4 & 4 \end{bmatrix}$, $P_2 = \frac{1}{9}\begin{bmatrix} 4 & 4 & -2 \\ 4 & 4 & -2 \\ -2 & -2 & 1 \end{bmatrix}$.
$P_1 P_2 =$ zero matrix because $a_1 \perp a_2$.

23. $P_1 + P_2 + P_3$
$$= \frac{1}{9}\begin{bmatrix} 1 & -2 & -2 \\ -2 & 4 & 4 \\ -2 & 4 & 4 \end{bmatrix} + \frac{1}{9}\begin{bmatrix} 4 & 4 & -2 \\ 4 & 4 & -2 \\ -2 & -2 & 1 \end{bmatrix} + \frac{1}{9}\begin{bmatrix} 4 & -2 & 4 \\ -2 & 1 & -2 \\ 4 & -2 & 4 \end{bmatrix} = I.$$

25. Since A is invertible, $P = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}} = AA^{-1}(A^{\mathrm{T}})^{-1}A^{\mathrm{T}} = I$: project onto all of $\mathbf{R}^2$.

Problem Set 3.3

1. $\widehat{x} = 2$; $E^2 = (10 - 3x)^2 + (5 - 4x)^2$ is minimized; $(4, -3)^{\mathrm{T}}(3, 4) = 0$.

3. $\widehat{x} = \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \end{bmatrix}$; $p = \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{2}{3} \end{bmatrix}$; $b - p = \begin{bmatrix} \frac{2}{3} \\ \frac{2}{3} \\ -\frac{2}{3} \end{bmatrix}$ is perpendicular to both columns.

5. $b = 4, 5, 9$ at $t = -1, 0, 1$; the best line is $6 + (5/2)t$; $p = (7/2, 6, 17/2)$.

7. $P = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}} = \begin{bmatrix} 1 & 1/2 & 0 \\ 1/2 & 1/2 & -1/2 \\ 0 & -1/2 & 1 \end{bmatrix}$.

9. (a) $P^{\mathrm{T}} = (P^{\mathrm{T}}P)^{\mathrm{T}} = P$. Then $P = P^{\mathrm{T}}P = P^2$. (b) P projects onto the space $\mathbf{Z} = \{0\}$.

11. $P + Q = I$, $PQ = 0$, transpose to $QP = 0$, so $(P - Q)(P - Q) = P - 0 - 0 + Q = I$.

13. Best line $61/35 - (36/35)t$; $p = (133/35, 95/35, 61/35, -11/35)$ from $C + Dt$.

15. $H^2 = (I - 2P)^2 = I - 4P + 4P^2 = I - 4P + 4P = I$. Two reflections give I.

17. Projection onto $x + y = 0$ = Projection onto $(-1, 1) = \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix}$.

19. Projection matrix onto row space would be $A^{\mathrm{T}}(AA^{\mathrm{T}})^{-1}A$ if the rows were independent.

21. Best line is $\begin{bmatrix} a_1^Ta_1 & -a_1^Ta_2 \\ -a_1^Ta_2 & a_2^Ta_2 \end{bmatrix} \begin{bmatrix} \widehat{x}_1 \\ \widehat{x}_2 \end{bmatrix} = \begin{bmatrix} a_1^Tb \\ -a_2^Tb \end{bmatrix}; \widehat{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

23. $C = (A^TA)^{-1}A^Tb$, $A^T = [1 \cdots 1]$, $b = (y_1, \ldots, y_m)^T$ then

$$C = \frac{A^Tb}{A^TA} = \frac{y_1 + \cdots + y_m}{m}.$$

25. $A = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix}; x = \begin{bmatrix} C \\ D \\ E \end{bmatrix}; b = \begin{bmatrix} 2 \\ 0 \\ -3 \\ -5 \end{bmatrix}$.

27. (a) $a^Ta = m$, $a^Tb = b_1 + \cdots + b_m$. Therefore $\widehat{x}$ is the mean of the b's. (b) The variance is $\|e\|^2 = \sum_{i=1}^m (b_i - \widehat{x})^2$. (c) $p = (3, 3, 3)$, $e = (-2, -1, 3)$, $p^Te = 0$.

$$P = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

29. $(\widehat{x} - x)(\widehat{x} - x)^T = (A^TA)^{-1}A^T[(b - Ax)(b - Ax)^T]A(A^TA)^{-1}$. For independent errors, substituting $(b - Ax)(b - Ax)^T = \sigma^2 I$ gives the *covariance matrix* $(A^TA)^{-1}A^T\sigma^2A(A^TA)^{-1}$. This simplifies to $\sigma^2(A^TA)^{-1}$: neat formula for the covariance matrix.

31. $\frac{1}{10}b_{10} + \frac{\mathbf{9}}{\mathbf{10}}\widehat{x}_9 = \frac{1}{10}(b_1 + \cdots + b_{10})$.

33. $\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 0 \\ 8 \\ 8 \\ 20 \end{bmatrix}$. Change right side to $p = \begin{bmatrix} 1 \\ 5 \\ 13 \\ 17 \end{bmatrix}$; $\widehat{x} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$ solves $A\widehat{x} = p$.

35. Closest parabola: $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{bmatrix} \begin{bmatrix} C \\ D \\ E \end{bmatrix} = \begin{bmatrix} 0 \\ 8 \\ 8 \\ 20 \end{bmatrix}$.

$$A^TA\widehat{x} = \begin{bmatrix} 4 & 8 & 26 \\ 8 & 26 & 92 \\ 26 & 92 & 338 \end{bmatrix} \begin{bmatrix} \widehat{C} \\ \widehat{D} \\ \widehat{E} \end{bmatrix} = \begin{bmatrix} 36 \\ 112 \\ 400 \end{bmatrix}.$$

37. (a) The best line is $x = 1 + 4t$, which goes through the center point $(\widehat{t}, \widehat{b}) = (2, 9)$.
(b) From the first equation: $Cm + D\sum t_i = \sum b_i$. Divide by m to get $C + D\widehat{t} = \widehat{b}$.

39. $\widehat{x}_W = \dfrac{w_1^2b_1 + \cdots + w_m^2b_m}{w_1^2 + \cdots + w_m^2}$.

41. $\widehat{x}_W = (1/21, 4/7)$; $A\widehat{x}_W = (1/21, 13/21, 25/21)$,
$b - A\widehat{x}_W = (-1/21, 8/21, -4/21)$, $(A\widehat{x}_W)W^TW(b - A\widehat{x}_W) = 0$.

Problem Set 3.4

1. (a) $-4 = C - 2D$, $-3 = C - D$, $-1 = C + D$, $0 = C + 2D$. (b) Best line $-2 + t$ goes through all 4 points; $E^2 = 0$. (c) b is in the column space.

3. Projection on a_3: $(-2/3, 1/3, -2/3)$; the sum is b itself; notice that $a_1a_1^{\mathrm{T}}$, $a_2a_2^{\mathrm{T}}$, $a_3a_3^{\mathrm{T}}$ are projections onto three orthogonal directions. Their sum is projection onto the whole space and should be the identity.

5. $(I-2uu^{\mathrm{T}})^{\mathrm{T}}(I-2uu^{\mathrm{T}}) = I-4uu^{\mathrm{T}}+4uu^{\mathrm{T}}uu^{\mathrm{T}} = I$; $Q = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$.

7. $(x_1q_1 + \cdots + x_nq_n)^{\mathrm{T}}(x_1q_1 + \cdots + x_nq_n) = x_1^2 + \cdots + x_n^2 \Rightarrow \|b\|^2 = b^{\mathrm{T}}b = x_1^2 + \cdots + x_n^2$.

9. The combination closest to q_3 is $0q_1 + 0q_2$.

11. Q is upper triangular: column 1 has $q_{11} = \pm 1$; by orthogonality column 2 must be $(0, \pm 1, 0, \ldots)$; by orthogonality column 3 is $(0, 0, \pm 1, \ldots)$; and so on.

13. $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = QR.$

15. $q_1 = \begin{bmatrix} 1/3 \\ 2/3 \\ -2/3 \end{bmatrix}$, $q_2 = \begin{bmatrix} 2/3 \\ 1/3 \\ 2/3 \end{bmatrix}$, $q_3 = \begin{bmatrix} -2/3 \\ 2/3 \\ 1/3 \end{bmatrix}$ is in the left nullspace;

$\widehat{x} = \begin{bmatrix} q_1^{\mathrm{T}}b \\ q_2^{\mathrm{T}}b \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

17. $R\widehat{x} = Q^{\mathrm{T}}b$ gives $\begin{bmatrix} 3 & 3 \\ 0 & \sqrt{2} \end{bmatrix} \begin{bmatrix} \widehat{x} \end{bmatrix} = \begin{bmatrix} 5/3 \\ 0 \end{bmatrix}$ and $\widehat{x} = \begin{bmatrix} 5/9 \\ 0 \end{bmatrix}$.

19. $C^* - (q_2^{\mathrm{T}}C^*)\, q_2$ is $c - (q_1^{\mathrm{T}}c)\, q_1 - (q_2^{\mathrm{T}}c)\, q_2$ because $q_2^{\mathrm{T}}q_1 = 0$.

21. By orthogonality, the closest functions are $0 \sin 2x = 0$ and $0 + 0x = 0$.

23. $a_0 = 1/2$, $a_1 = 0$, $b_1 = 2/\pi$.

25. The closest line is $y = 1/3$ (horizontal since $(x, x^2) = 0$).

27. $(1/\sqrt{2}, -1/\sqrt{2}, 0, 0)$, $(1/\sqrt{6}, 1/\sqrt{6}, 2/\sqrt{6}, 0)$,
$(-1/2\sqrt{3}, -1/2\sqrt{3}, 1/2\sqrt{3}, -1/\sqrt{3})$.

29. $A = a = (1, -1, 0, 0)$; $B = b - p = \left(\frac{1}{2}, \frac{1}{2}, -1, 0\right)$; $C = c - p_A - p_B = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -1\right)$
Notice the pattern in those orthogonal vectors A, B, C. Next, $(1, 1, 1, 1)/4$.

31. (a) True. (b) True. $Qx = x_1q_1 + x_2q_2$. $\|Qx\|^2 = x_1^2 + x_2^2$ because $q_1^{\mathrm{T}}q_2 = 0$.

Problem Set 3.5

1. $F^2 = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 4 & 0 \\ 0 & 4 & 0 & 0 \end{bmatrix}$, $F^4 = \begin{bmatrix} 16 & 0 & 0 & 0 \\ 0 & 16 & 0 & 0 \\ 0 & 0 & 16 & 0 \\ 0 & 0 & 0 & 16 \end{bmatrix} = 4^2I.$

3. The submatrix is F_3.

5. $e^{ix} = -1$ for $x = (2k+1)\pi$, $e^{i\theta} = i$ for $\theta = 2k\pi + \pi/2$, k is integer.

7. $c = (1, 0, 1, 0)$.

9. (a) $y = F$ times $(1, 0, 0, 0)$ = column zero of $F = (1, 1, 1, 1)$.
(b) $c = (1, 1, 1, 1)/4$.

11. $c = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \rightarrow \begin{matrix} c_{\text{even}} = \\ c_{\text{odd}} = \end{matrix} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \rightarrow \begin{matrix} y' = \\ y'' = \end{matrix} \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \end{bmatrix} \rightarrow y = \begin{bmatrix} 2 \\ 0 \\ 2 \\ 0 \end{bmatrix}.$

13. $c_0 = (f_0+f_1+f_2+f_3)/4, c_1 = (f_0-if_1-f_2+if_3)/4, c_2 = (f_0-f_1+f_2-f_3)/4,$ $c_3 = (f_0+if_1-f_2-if_3)/4$; f odd means $f_0 = 0$, $f_2 = 0$, $f_3 = -f_1$. Then $c_0 = 0$, $c_2 = 0$, $c_3 = -c_1$, so c is also odd.

15. $F^{-1} = \begin{bmatrix} 1 & & & \\ & & 1 & \\ & 1 & & \\ & & & 1 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 & 1 & & \\ 1 & i^2 & & \\ & & 1 & 1 \\ & & 1 & i^2 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 & & 1 & \\ & 1 & & 1 \\ 1 & & -1 & \\ & -i & & i \end{bmatrix} = \frac{1}{4} F^{\mathrm{H}}.$

17. $D = \begin{bmatrix} 1 & & \\ & e^{2\pi i/6} & \\ & & e^{4\pi i/6} \end{bmatrix}$ and $F_3 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & e^{2\pi i/3} & e^{4\pi i/3} \\ 1 & e^{4\pi i/3} & e^{2\pi i/3} \end{bmatrix}.$

19. $\Lambda = \text{diag}(1, i, i^2, i^3)$; $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ and P^{T} lead to $\lambda^3 - 1 = 0$.

21. Eigenvalues $e_0 = 2-1-1 = 0$, $e_1 = 2-i-i^3 = 2$, $e_2 = 2-(-1)-(-1) = 4$, $e_3 = 2-i^3-i^9 = 2$. Check trace $0+2+4+2 = 8$.

23. The four components are $(c_0 + c_2) + (c_1 + c_3)$; then $(c_0 - c_2) + i(c_1 - c_3)$; then $(c_0 + c_2) - (c_1 + c_3)$; then $(c_0 - c_2) - i(c_1 - c_3)$. These steps are the **FFT**!

Problem Set 4.2

1. $\det(2A) = 8$ and $\det(-A) = (-1)^4 \det A = \frac{1}{2}$ and $\det(A^2) = \frac{1}{4}$ and $\det(A^{-1}) = 2$.

3. The row operations leave $\det A$ unchanged by Rule 5. Then multiplying a row by -1 (Rule 3) gives the row exchange rule: $\det B = -\det A$.

5. For the first matrix, two row exchanges will produce the identity matrix. The second matrix needs three row exchanges to reach I.

7. $\det A = 0$ (singular); $\det U = 16$; $\det U^{\mathrm{T}} = 16$; $\det U^{-1} = 1/16$; $\det M = 16$ (2 exchanges).

9. The new determinant is $(1 - m\ell)(ad - bc)$.

11. If $|\det Q|$ is not 1 then $\det Q^n = (\det Q)^n$ would blow up or approach zero. But Q^n remains an orthogonal matrix. So $\det Q$ must be 1 or -1.

13. (a) Rule 3 (factoring -1 from each row) gives $\det(K^{\mathrm{T}}) = (-1)^3 \det K$. Then $-\det K = \det K^{\mathrm{T}} = \det K$ gives $\det K = 0$.

(b) $\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{bmatrix}$ has det $= 1$.

15. Adding every column of A to the first column makes it a zero column, so $\det A = 0$. If every row of A adds to 1, then every row of $A - I$ adds to 0 and $\det(A - I) = 0$. But $\det A$ need not be 1: $A = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$ has $\det(A - I) = 0$, but $\det A = 0 \neq 1$.

17. $\det(A) = 10$, $\det(A^{-1}) = \frac{1}{10}$, $\det(A - \lambda I) = \lambda^2 - 7\lambda + 10 = 0$ for $\lambda = 5$ and $\lambda = 2$.

19. Taking determinants gives $(\det C)(\det D) = (-1)^n(\det D)(\det C)$. For n even the reasoning fails (because $(-1)^n = +1$) and the conclusion is wrong.

21. $\det(A^{-1}) = \det \begin{bmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{bmatrix} = \dfrac{ad-bc}{(ad-bc)^2} = \dfrac{1}{ad-bc}$.

23. Determinant $= 36$ and determinant $= 5$.

25. $\det(L) = 1$, $\det(U) = -6$, $\det(A) = -6$, $\det(U^{-1}L^{-1}) = -\frac{1}{6}$, and $\det(U^{-1}L^{-1}A) = 1$.

27. Row 3 − row 2 = row 2 − row 1 so A is singular.

29. A is rectangular so $\det(A^{\mathrm{T}}A) \neq (\det A^{\mathrm{T}})(\det A)$: these are not defined.

31. The Hilbert determinants are $1, 8 \times 10^{-2}, 4.6 \times 10^{-4}, 1.6 \times 10^{-7}, 3.7 \times 10^{-12}, 5.4 \times 10^{-18}, 4.8 \times 10^{-25}, 2.7 \times 10^{-33}, 9.7 \times 10^{-43}, 2.2 \times 10^{-53}$. Pivots are ratios of determinants, so the tenth pivot is near $10^{-53}/10^{-43} = 10^{-10}$: very small.

33. The largest determinants of 0–1 matrices for $n = 1, 2, \ldots,$ are 1, 1, 2, 3, 5, 9, 32, 56, 144, 320, on the web at *www.mathworld.wolfram.com/HadamardsMaximumDeterminantProblem.html* and also in the "On-Line Encyclopedia of Integer Sequences": *www.research.att.com*. With −1s and 1s, the largest 4 by 4 determinant (see Hadamard in index) is 16.

35. $\det(I + M) = 1 + a + b + c + d$. Subtract row 4 from rows 1, 2, and 3. Then subtract a(row 1) $+ b$(row 2) $+ c$(row 3) from row 4. This leaves a triangular matrix with 1, 1, 1, and $1 + a + b + c + d$ on its diagonal.

Problem Set 4.3

1. (a) $a_{12}a_{21}a_{34}a_{43} = 1$; *even*, so $\det A = 1$.
(b) $b_{13}b_{22}b_{31}b_{14} = 18$; *odd*, so $\det B = -18$.

3. (a) True (product rule). (b) False (all 1s).
(c) False (det[1 1 0; 0 1 1; 1 0 1] = 2).

5. The 1, 1 cofactor is F_{n-1}. The 1, 2 cofactor has a 1 in column 1, with cofactor F_{n-2}. Multiply by $(-1)^{1+2}$ and also -1 to find $F_n = F_{n-1} + F_{n-2}$. So the determinants are Fibonacci numbers, except F_n is the usual F_{n-1}.

7. Cofactor expansion: $\det = 4(3) - 4(1) + 4(-4) - 4(1) = -12$.

9. (a) $(n-1)n!$ (each term $n-1$). (b) $\left(1 + \frac{1}{2!} + \cdots + \frac{1}{(n-1)!}\right) n!$.
(c) $\frac{1}{3}(n^3 + 2n - 3)$.

11. $\begin{bmatrix} 0 & A \\ -B & I \end{bmatrix}\begin{bmatrix} I & 0 \\ B & I \end{bmatrix} = \begin{bmatrix} AB & A \\ 0 & I \end{bmatrix}$, $\det\begin{bmatrix} I & 0 \\ B & I \end{bmatrix} = 1 \Rightarrow \det\begin{bmatrix} 0 & A \\ -B & I \end{bmatrix} = \det\begin{bmatrix} AB & A \\ 0 & I \end{bmatrix} = \det(AB)$. Test $A = [1 \quad 2]$, $B = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\det\begin{bmatrix} 0 & A \\ -B & I \end{bmatrix} = 5 = \det(AB)$; $A = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $B = [1 \quad 2]$, $\det\begin{bmatrix} 0 & A \\ -B & I \end{bmatrix} = 0 = \det(AB)$. Singular: $\text{rank}(AB) \le \text{rank}(A) \le n < m$.

13. $\det A = 1 + 18 + 12 - 9 - 4 - 6 = 12$, so rows are independent; $\det B = 0$, so rows are dependent (row 1 + row 2 = row 3); $\det C = -1$, C has independent rows.

15. Each of the six terms in $\det A$ is zero; the rank is at most 2; column 2 has no pivot.

17. $a_{11}a_{23}a_{32}a_{44}$ has $-$, $a_{14}a_{23}a_{32}a_{41}$ has $+$, so $\det A = 0$;
$\det B = 2 \cdot 4 \cdot 4 \cdot 2 - 1 \cdot 4 \cdot 4 \cdot 1 = 48$.

19. (a) If $a_{11} = a_{22} = a_{33} = 0$ then four terms are sure zeros.
(b) Fifteen terms are zero.

21. Some term $a_{1\alpha}a_{2\beta}\cdots a_{n\omega}$ in the big formula is not zero! Move rows $1, 2, \ldots, n$ into rows $\alpha, \beta, \ldots, \omega$. Then these nonzero a's will be on the main diagonal.

23. $4!/2 = 12$ even permutations; $\det(I + P_{\text{even}}) = 16$ or 4 or 0 (16 comes from $I + I$).

25. $C = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}$ and $AC^{\mathrm{T}} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} = 4I$. Therefore $A^{-1} = \frac{1}{4}C^{\mathrm{T}}$.

27. $|B_n| = |A_n| - |A_{n-1}| = (n+1) - n = 1$.

29. We must choose 1s from columns 2 and 1, columns 4 and 3, and so on. Therefore n must be even to have $\det A_n \neq 0$. The number of exchanges is $\frac{1}{2}n$ so $C_n = (-1)^{n/2}$.

31. $S_1 = 3$, $S_2 = 8$, $S_3 = 21$. The rule looks like every second number in Fibonacci's sequence $\ldots, 3, 5, 8, 13, 21, 34, 55, \ldots$ so the guess is $S_4 = 55$. The five nonzero terms in the big formula for S_4 are (with 3s where Problem 39 has 2s) $81 + 1 - 9 - 9 - 9 = 55$.

33. Changing 3 to 2 in the corner reduces the determinant F_{2n+2} by 1 times the cofactor of that corner entry. This cofactor is the determinant of S_{n-1} (one size smaller), which is F_{2n}. Therefore changing 3 to 2 changes the determinant to $F_{2n+2} - F_{2n}$, which is F_{2n+1}.

35. (a) Every $\det L = 1$; $\det U_k = \det A_k = 2, 6, -6$ for $k = 1, 2, 3$.
(b) Pivots $5, \frac{6}{5}, \frac{7}{6}$.

37. The six terms are correct. Row 1 $-$ 2 row 2 + row 3 $= 0$, so the matrix is singular.

39. The five nonzero terms in $\det A = 5$ are

$$(2)(2)(2)(2) + (-1)(-1)(-1)(-1) - (-1)(-1)(2)(2) - (2)(2)(-1)(-1) - (2)(-1)(-1)(2).$$

41. With $a_{11} = 1$, the $-1, 2, -1$ matrix has det $= 1$ and inverse $(A^{-1})_{ij} = n + 1 - \max(i, j)$.

43. Subtracting 1 from the n, n entry subtracts its cofactor C_{nn} from the determinant. That cofactor is $C_{nn} = 1$ (smaller Pascal matrix). Subtracting 1 from 1 leaves 0.

Problem Set 4.4

1. $\det A = 20; C^{\mathrm{T}} = \begin{bmatrix} 20 & -10 & -12 \\ 0 & 5 & 0 \\ 0 & 0 & 4 \end{bmatrix}; AC^{\mathrm{T}} = 20I; A^{-1} = \frac{1}{20}\begin{bmatrix} 20 & -10 & -12 \\ 0 & 5 & 0 \\ 0 & 0 & 4 \end{bmatrix}$.

3. $(x, y) = (d/(ad - bc), -c/(ad - bc))$; $(x, y, z) = (3, -1, -2)$.

5. (a) The area of that parallelogram is $\det \begin{bmatrix} 2 & 2 \\ -1 & 3 \end{bmatrix}$, so the triangle ABC has area $\frac{1}{2}4 = 2$. (b) The triangle $A'B'C'$ has the same area; it is just moved to the origin.

7. The pivots of A are 2, 3, 6 from determinants 2, 6, 36; the pivots of B are 2, 3, 0.

9. (a) P^2 takes $(1, 2, 3, 4, 5)$ to $(3, 2, 5, 4, 1)$.
(b) P^{-1} takes $(1, 2, 3, 4, 5)$ to $(3, 4, 5, 2, 1)$.

11. The powers of P are all permutation matrices, so eventually one of those matrices must be repeated. If P^r is the same as P^s, then $P^{r-s} = I$.

13. (a) $\det A = 3$, $\det B_1 = -6$, $\det B_2 = 3$, so $x_1 = -6/3 = -2$ and $x_2 = 3/3 = 1$.
(b) $|A| = 4$, $|B_1| = 3$, $|B_2| = -2$, $|B_3| = 1$. So $x_1 = \frac{3}{4}$ and $x_2 = -\frac{1}{2}$ and $x_3 = \frac{1}{4}$.

15. (a) $x_1 = \frac{3}{0}$ and $x_2 = \frac{-2}{0}$: no solution (b) $x_1 = \frac{0}{0}$ and $x_2 = \frac{0}{0}$: *undetermined.*

17. If the first column in A is also the right-hand side b then $\det A = \det B_1$. Both B_2 and B_3 are singular, since a column is repeated. Therefore $x_1 = |B_1|/|A| = 1$ and $x_2 = x_3 = 0$.

19. If all cofactors $= 0$ (even in a single row or column), then $\det A = 0$ (no inverse). $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ has no zero cofactors but it is not invertible.

21. If $\det A = 1$ and we know the cofactors, then $C^{\mathrm{T}} = A^{-1}$ and also $\det A^{-1} = 1$. Since A is the inverse of A^{-1}, A must be the cofactor matrix for C.

23. Knowing C, Problem 22 gives $\det A = (\det C)^{\frac{1}{n-1}}$ with $n = 4$. So we can construct $A^{-1} = C^{\mathrm{T}}/\det A$ using the known cofactors. Invert to find A.

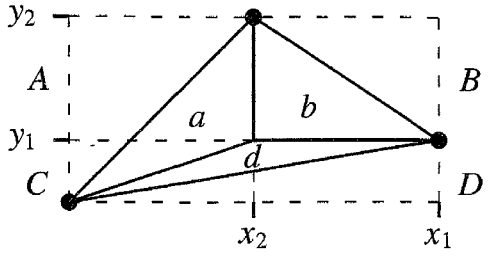
25. (a) Cofactors $C_{21} = C_{31} = C_{32} = 0$.
(b) $C_{12} = C_{21}, C_{31} = C_{13}, C_{32} = C_{23}$ make S^{-1} symmetric.

27. (a) Area $\begin{vmatrix} 3 & 2 \\ 1 & 4 \end{vmatrix} = 10$. (b) Triangle area $= 5$. (c) Triangle area $= 5$.

29. (a) Area $\frac{1}{2}\begin{vmatrix} 2 & 1 & 1 \\ 3 & 4 & 1 \\ 0 & 5 & 1 \end{vmatrix} = 5$. (b) $5 +$ new triangle area $\frac{1}{2}\begin{vmatrix} 2 & 1 & 1 \\ 0 & 5 & 1 \\ -1 & 0 & 1 \end{vmatrix} = 5 + 7 = 12$.

31. The edges of the hypercube have length $\sqrt{1+1+1+1} = 2$. The volume $\det H$ is $2^4 = 16$. ($H/2$ has orthonormal columns. Then $\det(H/2) = 1$ leads again to $\det H = 16$.)

33.



$x_1y_2 - x_2y_1$ = rectangles $A+B+D$ (not C).

Areas from same bases: rectangle $A = 2$(triangle a); rectangle $B = 2$(triangle b).

Areas from same heights: rectangle $D = 2$(triangle d).

So triangles $a + b + d = \frac{1}{2}(x_1y_2 - x_2y_1)$.

Check an example with $(a, b) = (3, 2)$, $(c, d) = (1, 4)$, and area $= 10$. The line from $(0, e)$ in step 3 has slope c/a and equation $y = e + cx/a$. Step 3 works because (b, d) is on that line! $d = e + cb/a$ is true, since $ad - bc =$ area ae at step 2.

35. The n-dimensional cube has 2^n corners, $n2^{n-1}$ edges, and $2n$ faces of dimension $n - 1$. The cube whose edges are the rows of $2I$ has volume 2^n.

37. $J = r$. The columns are orthogonal and their lengths are 1 and r.

39. $$\begin{vmatrix} \partial r/\partial x & \partial r/\partial y \\ \partial\theta/\partial x & \partial\theta/\partial y \end{vmatrix} = \begin{vmatrix} \cos\theta & \sin\theta \\ (-\sin\theta)/r & (\cos\theta)/r \end{vmatrix} = \frac{1}{r}.$$

41. $S = (2, 1, -1)$ gives a parallelogram, whose area is the length of a cross product: $\|PQ \times PS\| = \|(-2, -2, -1)\| = 3$. This comes also from a determinant! The other four corners could be $(0, 0, 0)$, $(0, 0, 2)$, $(1, 2, 2)$, $(1, 1, 0)$. The volume of the tilted box is $|\det| = 1$.

43. $$\det \begin{bmatrix} x & y & z \\ 3 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix} = 0 = 7x - 5y + z;$$ plane contains the two vectors.

45. VISA has the five reversals VI, VS, VA, IA, SA. And AVIS has the two reversals VI and VS. Since $5 - 2$ is odd, VISA and AVIS have opposite parity.

Problem Set 5.1

1. $\lambda = 2$ and $\lambda = 3$, trace $= 5$, determinant $= 6$.

3. $\lambda = -5$ and $\lambda = -4$; both λ's are reduced by 7, with unchanged eigenvectors.

5. $\lambda = 3, \lambda = 1, \lambda = 0$, with eigenvectors $(1, 0, 0)$, $(2, -1, 0)$, $(3, -2, 1)$; trace $= 4$, det $= 0$. $\lambda = 2, \lambda = 2, \lambda = -2$, with eigenvectors $(1, 1, 1)$, $(0, 1, 0)$, $(1, 0, -1)$; trace $= 2$, det $= -8$.

7. $Ax = \lambda x$ gives $(A - 7I)x = (\lambda - 7)x$; $Ax = \lambda x$ gives $x = \lambda A^{-1}x$, so $A^{-1}x = (1/\lambda)x$.

9. The coefficient of $(-\lambda)^{n-1}$ in $(\lambda_1 - \lambda)\cdots(\lambda_n - \lambda)$ is $\lambda_1 + \cdots + \lambda_n$. In $\det(A - \lambda I)$, a term that includes an off-diagonal a_{ij} excludes both $a_{ii} - \lambda$ and $a_{jj} - \lambda$. Such a term doesn't involve $(-\lambda)^{n-1}$. So the coefficient of $(-\lambda)^{n-1}$ in $\det(A - \lambda I)$ must come from the product down the main diagonal. That coefficient is $a_{11} + \cdots + a_{nn} = \lambda_1 + \cdots + \lambda_n$.

11. Transpose $A - \lambda I$: $\det(A - \lambda I) = \det(A - \lambda I)^{\mathrm{T}} = \det(A^{\mathrm{T}} - \lambda I)$.

13. The eigenvalues of A are 1, 2, 3, 7, 8, 9.

15. $\operatorname{rank}(A) = 1$, $\lambda = 0, \ldots, 0, n$ (trace n); $\operatorname{rank}(C) = 2$, $\lambda = 0, \ldots, n/2, -n/2$ (trace 0).

17. The third row contains 6, 5, 4.

19. A and A^2 and A^∞ all have the same eigenvectors. The eigenvalues are 1 and 0.5 for A, 1 and 0.25 for A^2, 1 and 0 for A^∞. Therefore A^2 is halfway between A and A^∞.

21. $\lambda_1 = 4$ and $\lambda_2 = -1$ (check trace and determinant) with $x_1 = (1, 2)$ and $x_2 = (2, -1)$. A^{-1} has the same eigenvectors as A, with eigenvalues $1/\lambda_1 = 1/4$ and $1/\lambda_2 = -1$.

23. (a) Multiply Ax to see λx which reveals λ. (b) Solve $(A - \lambda I)x = 0$ to find x.

25. (a) $Pu = (uu^{\mathrm{T}})u = u(u^{\mathrm{T}}u) = u$, so $\lambda = 1$. (b) $Pv = (uu^{\mathrm{T}})v = u(u^{\mathrm{T}}v) = 0$, so $\lambda = 0$. (c) $x_1 = (-1, 1, 0, 0)$, $x_2 = (-3, 0, 1, 0)$, $x_3 = (-5, 0, 0, 1)$ are orthogonal to u, so they are eigenvectors of P with $\lambda = 0$.

27. $\lambda^3 - 1 = 0$ gives $\lambda = 1$ and $\lambda = \frac{1}{2}(-1 \pm i\sqrt{3})$; the three eigenvalues are $1, 1, -1$.

29. (a) rank $= 2$. (b) $\det(B^{\mathrm{T}}B) = 0$. *Not* (c). (d) $(B + I)^{-1}$ has $(\lambda + 1)^{-1} = 1, \frac{1}{2}, \frac{1}{3}$.

31. $a = 0$, $b = 9$, $c = 0$ multiply $1, \lambda, \lambda^2$ in $\det(A - \lambda I) = 9\lambda - \lambda^3$: $A =$ *companion matrix.*

33. $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix}$. Always $A^2 =$ zero matrix if $\lambda = 0, 0$ (Cayley–Hamilton).

35. $Ax = c_1\lambda_1 x_1 + \cdots + c_n\lambda_n x_n$ equals $Bx = c_1\lambda_1 x_1 + \cdots + c_n\lambda_n x_n$ for all x. So $A = B$.

37. $\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} a+b \\ c+d \end{bmatrix} = (a+b)\begin{bmatrix} 1 \\ 1 \end{bmatrix}$; $\lambda_2 = d - b$ to produce trace $= a + d$.

39. We need $\lambda^3 = 1$ but not $\lambda = 1$ (to avoid I). With $\lambda_1 = e^{2\pi i/3}$ and $\lambda_2 = e^{-2\pi i/3}$, the determinant will be $\lambda_1\lambda_2 = 1$ and the trace is $\lambda_1 + \lambda_2 = \cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3} + \cos\frac{2\pi}{3} - i\sin\frac{2\pi}{3} = -1$. One matrix with this trace -1 and determinant 1 is $A = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}$.

Problem Set 5.2

1. $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}^{-1}$;

$\begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & -2 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 0 & -2 \end{bmatrix}^{-1}$.

3. $\lambda = 0, 0, 3$; the third column of S is a multiple of $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and the other columns are on the plane orthogonal to it.

5. A_1 and A_3 cannot be diagonalized. They have only one line of eigenvectors.

7. $A = \begin{bmatrix} 3 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 3 & 1 \\ 1 & -1 \end{bmatrix}^{-1}$ gives $A^{100} = \begin{bmatrix} 3 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 5^{100} & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 3 & 1 \\ 1 & -1 \end{bmatrix}^{-1} =$

$\frac{1}{4}\begin{bmatrix} 3\cdot 5^{100} + 1 & 3\cdot 5^{100} - 3 \\ 5^{100} - 1 & 5^{100} + 3 \end{bmatrix}$.

9. $\text{trace}(AB) = \text{trace}(BA) = aq+bs+cr+dt$. Then $\text{trace}(AB-BA) = 0$ (always). So $AB - BA = I$ is impossible for matrices, since I does not have trace zero.

11. (a) True; $\det A = 2 \neq 0$. (b) False; $\begin{bmatrix}1&1&1\\0&1&1\\0&0&2\end{bmatrix}$. (c) False; $\begin{bmatrix}1&0&0\\0&1&0\\0&0&2\end{bmatrix}$ is diagonal!

13. $A = \begin{bmatrix}1&1\\1&-1\end{bmatrix}\begin{bmatrix}9&0\\0&1\end{bmatrix}\begin{bmatrix}1&1\\1&-1\end{bmatrix}^{-1}$; $\begin{bmatrix}2&1\\1&2\end{bmatrix}$; four square roots.

15. $\begin{bmatrix}1&2\\0&3\end{bmatrix} = \begin{bmatrix}1&1\\0&1\end{bmatrix}\begin{bmatrix}1&0\\0&3\end{bmatrix}\begin{bmatrix}1&-1\\0&1\end{bmatrix}$; $\begin{bmatrix}1&1\\2&2\end{bmatrix} = \begin{bmatrix}1&1\\-1&2\end{bmatrix}\begin{bmatrix}0&0\\0&3\end{bmatrix}\begin{bmatrix}\frac{2}{3}&-\frac{1}{3}\\\frac{1}{3}&\frac{1}{3}\end{bmatrix}$.

17. $A = \begin{bmatrix}1&1\\0&1\end{bmatrix}\begin{bmatrix}2&0\\0&5\end{bmatrix}\begin{bmatrix}1&-1\\0&1\end{bmatrix} = \begin{bmatrix}2&3\\0&5\end{bmatrix}$.

19. (a) False: don't know λ's. (b) True. (c) True. (d) False: need eigenvectors of S!

21. The columns of S are multiples of (2, 1) and (0, 1) in either order. Same for A^{-1}.

23. A and B have $\lambda_1 = 1$ and $\lambda_2 = 1$. $A+B$ has $\lambda_1 = 1, \lambda_2 = 3$. Eigenvalues of $A+B$ *are not equal* to eigenvalues of A plus eigenvalues of B.

25. (a) True. (b) False. (c) False (A might have 2 or 3 independent eigenvectors).

27. $A = \begin{bmatrix}8&3\\-3&2\end{bmatrix}$ (or other), $A = \begin{bmatrix}9&4\\-4&1\end{bmatrix}$, $A = \begin{bmatrix}10&5\\-5&0\end{bmatrix}$; only eigenvectors are $(c, -c)$.

29. $S\Lambda^k S^{-1}$ approaches zero if and only if every $|\lambda| < 1$; $B^k \to 0$ from $\lambda = .9$ and $\lambda = .3$.

31. $\Lambda = \begin{bmatrix}.9&0\\0&.3\end{bmatrix}$, $S = \begin{bmatrix}3&-3\\1&1\end{bmatrix}$; $B^{10}\begin{bmatrix}3\\1\end{bmatrix} = (.9)^{10}\begin{bmatrix}3\\1\end{bmatrix}$, $B^{10}\begin{bmatrix}3\\-1\end{bmatrix} = (.3)^{10}\begin{bmatrix}3\\-1\end{bmatrix}$, $B^{10}\begin{bmatrix}6\\0\end{bmatrix}$ = sum of those two.

33. $B^k = \begin{bmatrix}1&1\\0&-1\end{bmatrix}\begin{bmatrix}3&0\\0&2\end{bmatrix}^k\begin{bmatrix}1&1\\0&-1\end{bmatrix} = \begin{bmatrix}3^k&3^k-2^k\\0&2^k\end{bmatrix}$.

35. $\text{trace}\,AB = (aq+bs)+(cr+dt) = (qa+rc)+(sb+td) = \text{trace}\,BA$. Proof for diagonalizable case: the trace of $S\Lambda S^{-1}$ is the trace of $(\Lambda S^{-1})S = \Lambda$, which is *the sum of the* λ*'s*.

37. The A's form a subspace, since cA and $A_1 + A_2$ have the same S. When $S = I$, the A's give the subspace of diagonal matrices. Dimension 4.

39. Two problems: The nullspace and column space can overlap, so x could be in both. There may not be r independent eigenvectors in the column space.

41. $A = \begin{bmatrix}1&1\\1&0\end{bmatrix}$ has $A^2 = \begin{bmatrix}2&1\\1&1\end{bmatrix}$, and $A^2 - A - I$ = zero matrix confirms Cayley–Hamilton.

43. By **5F**, B has the same eigenvectors $(1, 0)$ and $(0, 1)$ as A, so B is also diagonal. The equations $AB - BA = \begin{bmatrix} a & b \\ 2c & 2d \end{bmatrix} - \begin{bmatrix} a & 2b \\ c & 2d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ are $-b = 0$ and $c = 0$: rank 2.

45. A has $\lambda_1 = 1$ and $\lambda_2 = .4$ with $x_1 = (1, 2)$ and $x_2 = (1, -1)$. A^∞ has $\lambda_1 = 1$ and $\lambda_2 = 0$ (same eigenvectors). A^{100} has $\lambda_1 = 1$ and $\lambda_2 = (.4)^{100}$, which is near zero. So A^{100} is very near A^∞.

Problem Set 5.3

1. The Fibonacci numbers start even, odd, odd. Then *odd + odd = even*. The next two are odd (from odd + even and even + odd). Then repeat *odd + odd = even*.

3. $A^2 = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$, $A^3 = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$, $A^4 = \begin{bmatrix} 5 & 3 \\ 3 & 2 \end{bmatrix}$; $F_{20} = 6765$.

5. $A = S\Lambda S^{-1} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = \dfrac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{bmatrix}$ (notice S^{-1}).

$$S\Lambda^k S^{-1} = \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1^k & 0 \\ 0 & \lambda_2^k \end{bmatrix} \begin{bmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} - - - - - \\ (\lambda_1^k - \lambda_2^k)/(\lambda_1 - \lambda_2) \end{bmatrix}$$

7. Direct addition $L_k + L_{k+1}$ gives $L_0, \ldots, L_{10}$ as 2, 1, 3, 4, 7, 11, 18, 29, 47, 76, 123. My calculator gives $\lambda_1^{10} = (1.618\ldots)^{10} = 122.991\ldots$, which rounds off to $L_{10} = 123$.

9. The Markov transition matrix is $\begin{bmatrix} \frac{7}{12} & \frac{1}{6} & 0 \\ \frac{1}{6} & \frac{1}{2} & 0 \\ \frac{1}{4} & \frac{1}{3} & 1 \end{bmatrix}$. Fractions $\frac{7}{12}, \frac{1}{2}, 1$ don't move.

11. (a) $\lambda = 0, (1, 1, -2)$. (b) $\lambda = 1$ and -0.2. (c) limit $(3, 4, 4)$ = eigenvector for $\lambda = 1$.

13. (a) $0 \le a \le 1$, $0 \le b \le 1$.

(b) $$u_k = \begin{bmatrix} b/(1-a) & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1^k & 0 \\ 0 & (a-b)^k \end{bmatrix} \begin{bmatrix} b/(1-a) & 1 \\ 1 & -1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} \dfrac{2b}{b-a+1} - \dfrac{1-a-b}{b-a+1}(a-b)^k \\ \dfrac{2(1-a)}{b-a+1} - \dfrac{1-a-b}{b-a+1}(a-b)^k \end{bmatrix}.$$

(c) $u_k \to \begin{bmatrix} \dfrac{2b}{b-a+1} \\ \dfrac{2(1-a)}{b-a+1} \end{bmatrix}$ if $|a-b| < 1$; $a = 1/3$, $b = -1/3$, not Markov.

15. The components of Ax add to $x_1 + x_2 + x_3$ (each column adds to 1 and nobody is lost). The components of λx add to $\lambda(x_1 + x_2 + x_3)$. If $\lambda \neq 1$, $x_1 + x_2 + x_3$ must be zero.

17. $\begin{bmatrix} \alpha & \alpha \\ \alpha & \alpha \end{bmatrix}$ is unstable for $|\alpha| > 1/2$, and stable for $|\alpha| < 1/2$. Neutral for $\alpha = \pm 1/2$.

19. $A^2 = \begin{bmatrix} 0 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and $A^3 = 0$. So $(I - A)^{-1} = I + A + A^2 = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$.

21. If A is increased, then more goods are consumed in production and the expansion must be slower. Mathematically, $Ax \geq tx$ remains true if A is increased; $t_{\max}$ goes up.

23. $\begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$ and $A^k = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 5^k & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$.

25. $R = S\sqrt{\Lambda}S^{-1} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ has $R^2 = A$. $\sqrt{B}$ would have $\lambda = \sqrt{9}$ and $\lambda = \sqrt{-1}$, so its trace is not real. Note that $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ can have $\sqrt{-1} = i$ and $-i$, and real square root $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$.

27. $A = S\Lambda_1 S^{-1}$ and $B = S\Lambda_2 S^{-1}$. Diagonal matrices always give $\Lambda_1\Lambda_2 = \Lambda_2\Lambda_1$. Then $AB = BA$, from $S\Lambda_1 S^{-1}S\Lambda_2 S^{-1} = S\Lambda_1\Lambda_2 S^{-1} = S\Lambda_2\Lambda_1 S^{-1} = S\Lambda_2 S^{-1}S\Lambda_1 S^{-1} = BA$.

29. B has $\lambda = i$ and $-i$, so B^4 has $\lambda^4 = 1$ and 1; C has $\lambda = (1 \pm \sqrt{3}i)/2 = \exp(\pm\pi i/3)$, so $\lambda^3 = -1$ and -1. Then $C^3 = -I$ and $C^{1024} = -C$.

Problem Set 5.4

1. $\lambda_1 = -2$ and $\lambda_2 = 0$; $x_1 = (1, -1)$ and $x_2 = (1, 1)$;
$e^{At} = \frac{1}{2}\begin{bmatrix} e^{-2t} + 1 & -e^{-2t} + 1 \\ -e^{-2t} + 1 & e^{-2t} + 1 \end{bmatrix}$.

3. $u(t) = \begin{bmatrix} e^{2t} + 2 \\ -e^{2t} + 2 \end{bmatrix}$; as $t \to \infty$, $e^{2t} \to +\infty$.

5. (a) $e^{A(t+T)} = Se^{\Lambda(t+T)}S^{-1} = Se^{\Lambda t}e^{\Lambda T}S^{-1} = Se^{\Lambda t}S^{-1}Se^{\Lambda T}S^{-1} = e^{At}e^{AT}$.
(b) $e^A = I + A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, $e^B = I + B = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$, $A + B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ gives $e^{A+B} = \begin{bmatrix} \cos 1 & -\sin 1 \\ \sin 1 & \cos 1 \end{bmatrix}$ from Example 3 in the text, at $t = 1$. This matrix is different from $e^A e^B$.

7. $e^{At} = I + At = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$; $e^{\Lambda t}u(0) = \begin{bmatrix} 4t + 3 \\ 4 \end{bmatrix}$.

9. (a) $\lambda_1 = \frac{7+\sqrt{57}}{2}$, $\lambda_2 = \frac{7-\sqrt{57}}{2}$, $\operatorname{Re}\lambda_1 > 0$, unstable. (b) $\lambda_1 = \sqrt{7}$, $\lambda_2 = -\sqrt{7}$, $\operatorname{Re}\lambda_1 > 0$, unstable (c) $\lambda_1 = \frac{-1+\sqrt{13}}{2}$, $\lambda_2 = \frac{-1-\sqrt{13}}{2}$, $\operatorname{Re}\lambda_1 > 0$, unstable
(d) $\lambda_1 = 0$, $\lambda_2 = -2$, neutrally stable.

11. A_1 is unstable for $t < 1$, neutrally stable for $t \geq 1$. A_2 is unstable for $t < 4$, neutrally stable at $t = 4$, stable with real λ for $4 < t \leq 5$, and stable with complex λ for $t > 5$. A_3 is unstable for all $t > 0$, because the trace is $2t$.

13. (a) $u_1' = cu_2 - bu_3$, $u_2' = -cu_1 + au_3$, $u_3' = bu_1 - au_2$ gives $u_1'u_1 + u_2'u_2 + u_3'u_3 = 0$. (b) Because e^{At} is an orthogonal matrix, $\|u(t)\|^2 = \|e^{At}u(0)\|^2 = \|u(0)\|^2$ is

constant. (c) $\lambda = 0$ and $\pm(\sqrt{a^2+b^2+c^2})i$. Skew-symmetric matrices have pure imaginary λ's.

15. $u(t) = \frac{1}{2}\cos 2t \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \frac{1}{2}\cos\sqrt{6}t \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

17. $Ax = \lambda Fx + \lambda^2 x$ or $(A - \lambda F - \lambda^2 I)x = 0$.

19. Eigenvalues are real when $(\text{trace})^2 - 4\det \geq 0 \Rightarrow -4(-a^2 - b^2 + c^2) \geq 0 \Rightarrow a^2 + b^2 \geq c^2$.

21. $u_1 = e^{4t}\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $u_2 = e^t \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. If $u(0) = (5, -2)$, then $u(t) = 3e^{4t}\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2e^t \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

23. $\begin{bmatrix} y' \\ y'' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 4 & 5 \end{bmatrix}\begin{bmatrix} y \\ y' \end{bmatrix}$. Then $\lambda = \frac{1}{2}(5 \pm \sqrt{41})$.

25. $\lambda_1 = 0$ and $\lambda_2 = 2$. Now $v(t) = 20 + 10e^{2t} \to \infty$ as $t \to \infty$.

27. $A = \begin{bmatrix} 0 & 1 \\ -9 & 6 \end{bmatrix}$ has trace 6, det 9, $\lambda = 3$ and 3, with only one independent eigenvector $(1, 3)$. That gives $y = ce^{3t}$, $y' = 3e^{3t}$. Also te^{3t} solves $y'' = 6y' - 9y$.

29. $y(t) = \cos t$ starts at $y(0) = 1$ and $y'(0) = 0$. The vector equation has $u = (y, y') = (\cos t, -\sin t)$.

31. Substituting $u = e^{ct}v$ gives $ce^{ct}v = Ae^{ct}v - e^{ct}b$, or $(A - cI)v = b$, or $v = (A - cI)^{-1}b =$ particular solution. If c is an eigenvalue, then $A - cI$ is not invertible: this v fails.

33. $de^{At}/dt = A + A^2t + \frac{1}{2}A^3t^2 + \frac{1}{6}A^4t^3 + \cdots = A(I + At + \frac{1}{2}A^2t^2 + \frac{1}{6}A^3t^3 + \cdots) = Ae^{At}$.

35. The solution at time $t + T$ is also $e^{A(t+T)}u(0)$. Thus e^{At} times e^{AT} equals $e^{A(t+T)}$.

37. If $A^2 = A$ then $e^{At} = I + At + \frac{1}{2}At^2 + \frac{1}{6}At^3 + \cdots = I + (e^t - 1)A$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} e^t - 1 & e^t - 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} e^t & e^t - 1 \\ 0 & 1 \end{bmatrix}.$$

39. $A = \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 0 \end{bmatrix}\begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 0 & \frac{1}{2} \\ 1 & -\frac{1}{2} \end{bmatrix}$, then $e^{At} = \begin{bmatrix} e^t & \frac{1}{2}(e^{3t} - e^t) \\ 0 & e^{3t} \end{bmatrix} = I$ at $t = 0$.

41. (a) The inverse of e^{At} is e^{-At}. (b) If $Ax = \lambda x$ then $e^{At}x = e^{\lambda t}x$ and $e^{\lambda t} \neq 0$.

43. $\lambda = 2$ and 5 with eigenvectors $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Then $A = S\Lambda S^{-1} = \begin{bmatrix} -1 & 6 \\ -3 & 8 \end{bmatrix}$.

Problem Set 5.5

1. (b) sum $= 4 + 3i$; product $= 7 + i$. (c) $\overline{3 + 4i} = 3 - 4i$; $\overline{1 - i} = 1 + i$; $|3 + 4i| = 5$; $|1 - i| = \sqrt{2}$. Both numbers lie *outside* the unit circle.

3. $\overline{x} = 2 - i$, $x\overline{x} = 5$, $xy = -1 + 7i$, $1/x = 2/5 - (1/5)i$, $x/y = 1/2 - (1/2)i$; check that $|xy| = \sqrt{50} = |x||y|$ and $|1/x| = 1/\sqrt{5} = 1/|x|$.

5. (a) $x^2 = r^2e^{i2\theta}$, $x^{-1} = (1/r)e^{-i\theta}$, $\overline{x} = re^{-i\theta}$; $x^{-1} = \overline{x}$ gives $|x|^2 = 1$: on the unit circle.

7. $C = \begin{bmatrix} 1 & -i \\ -i & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & i & 0 \\ i & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & i & -i \\ -i & 1 & 0 \\ i & 0 & 1 \end{bmatrix}$, $C^{\mathrm{H}} = C$ because $(A^{\mathrm{H}}A)^{\mathrm{H}} = A^{\mathrm{H}}A$.

9. (a) $\det A^{\mathrm{T}} = \det A$ but $\det A^{\mathrm{H}} = \overline{\det A}$. (b) $A^{\mathrm{H}} = A$ gives $\overline{\det A} = \det A =$ real.

11. $P: \lambda_1 = 0, \lambda_2 = 1, x_1 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}, x_2 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$; $Q: \lambda_1 = 1, \lambda_2 = -1, x_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}, x_2 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$; $R: \lambda_1 = 5, \lambda_2 = -5, x_1 = \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}, x_2 = \begin{bmatrix} 1/\sqrt{5} \\ -2/\sqrt{5} \end{bmatrix}$.

13. (a) u, v, w are orthogonal to each other. (b) The nullspace is spanned by u; the left nullspace is the same as the nullspace; the row space is spanned by v and w; the column space is the same as the row space. (c) $x = v + \frac{1}{2}w$; not unique, we can add any multiple of u to x. (d) Need $b^{\mathrm{T}}u = 0$. (e) $S^{-1} = S^{\mathrm{T}}$; $S^{-1}AS =$ diag(0, 1, 2).

15. The dimension of S is $n(n+1)/2$, not n. Every symmetric matrix A is a combination of n projections, but the projections change as A changes. There is no basis of n fixed projection matrices, in the space **S** of symmetric matrices.

17. $(UV)^{\mathrm{H}}(UV) = V^{\mathrm{H}}U^{\mathrm{H}}UV = V^{\mathrm{H}}IV = I$. So UV is unitary.

19. The third column of U can be $(1, -2, i)/\sqrt{6}$, multiplied by any number $e^{i\theta}$.

21. A has $+1$ or -1 in each diagonal entry; eight possibilities.

23. Columns of Fourier matrix U are eigenvectors of P because $PU = \mathrm{diag}(1, w, w^2, w^3)U$ (and $w = i$).

25. n^2 steps for direct C times x; only $n \log n$ steps for F and F^{-1} by FFT (and n for Λ).

27. $A^{\mathrm{H}}A = \begin{bmatrix} 2 & 0 & 1+i \\ 0 & 2 & 1+i \\ 1-i & 1-i & 2 \end{bmatrix}$ and $AA^{\mathrm{H}} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$ are *Hermitian* matrices.

$(A^{\mathrm{H}}A)^{\mathrm{H}} = A^{\mathrm{H}}A^{\mathrm{HH}} = A^{\mathrm{H}}A$ again.

29. cA is still Hermitian for real c; $(iA)^{\mathrm{H}} = -iA^{\mathrm{H}} = -iA$ is skew-Hermitian.

31. $P^2 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, $P^3 = I$, $P^{100} = P^{99}P = P$; $\lambda =$ cube roots of $1 = 1$, $e^{2\pi i/3}, e^{4\pi i/3}$.

33. $C = \begin{bmatrix} 2 & 5 & 4 \\ 4 & 2 & 5 \\ 5 & 4 & 2 \end{bmatrix} = 2 + 5P + 4P^2$ has $\lambda(C) = \begin{Bmatrix} 2+5+4 \\ 2+5e^{2\pi i/3}+4e^{4\pi i/3} \\ 2+5e^{4\pi i/3}+4e^{8\pi i/3} \end{Bmatrix}$.

35. $A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & -1+i \\ 1+i & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1-i \\ -1-i & 1 \end{bmatrix}$.

$K = (iA^{\mathrm{T}}) = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & -1-i \\ 1-i & 1 \end{bmatrix} \begin{bmatrix} 2i & 0 \\ 0 & -i \end{bmatrix} \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ -1+i & 1 \end{bmatrix}$.

37. $V = \frac{1}{L}\begin{bmatrix} 1+\sqrt{3} & -1+i \\ 1+i & 1+\sqrt{3} \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\frac{1}{L}\begin{bmatrix} 1+\sqrt{3} & 1-i \\ -1-i & 1+\sqrt{3} \end{bmatrix}$ with $L^2 = 6+2\sqrt{3}$.

$V = V^{\mathrm{H}}$ gives real λ, unitary gives $|\lambda| = 1$, then trace zero gives $\lambda = 1, -1$.

39. Don't multiply e^{-ix} times e^{ix}; conjugate the first, then $\int_0^{2\pi} e^{2ix}\,dx = [e^{2ix}/2i]_0^{2\pi} = 0$.

41. $R + iS = (R+iS)^{\mathrm{H}} = R^{\mathrm{T}} - iS^{\mathrm{T}}$; R is symmetric but S is skew-symmetric.

43. [1] and [−1]; $\begin{bmatrix} a & b+ic \\ b-ic & -a \end{bmatrix}$ with $a^2+b^2+c^2 = 1$.

45. $(I-2uu^{\mathrm{H}})^{\mathrm{H}} = I-2uu^{\mathrm{H}}$; $(I-2uu^{\mathrm{H}})^2 = I-4uu^{\mathrm{H}}+4u(u^{\mathrm{H}}u)u^{\mathrm{H}} = I$; the matrix uu^{H} projects onto the line through u.

47. We are given $A+iB = (A+iB)^{\mathrm{H}} = A^{\mathrm{T}} - iB^{\mathrm{T}}$. Then $A = A^{\mathrm{T}}$ and $B = -B^{\mathrm{T}}$.

49. $A = \begin{bmatrix} 1-i & 1-i \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}\frac{1}{6}\begin{bmatrix} 2+2i & -2 \\ 1+i & 2 \end{bmatrix} = S\Lambda S^{-1}$. Real eigenvalues 1 and 4.

Problem Set 5.6

1. $C = N^{-1}BN = N^{-1}M^{-1}AMN = (MN)^{-1}A(MN)$; only $M^{-1}IM = I$ is similar to I.

3. If $\lambda_1, \ldots, \lambda_n$ are eigenvalues of A, then $\lambda_1+1, \ldots, \lambda_n+1$ are eigenvalues of $A+I$. So A and $A+I$ never have the same eigenvalues, and can't be similar.

5. If B is invertible, then $BA = B(AB)B^{-1}$ is similar to AB.

7. The (3, 1) entry of $M^{-1}AM$ is $g\cos\theta + h\sin\theta$, which is zero if $\tan\theta = -g/h$.

9. The coefficients are $c_1 = 1$, $c_2 = 2$, $d_1 = 1$, $d_2 = 1$; check $Mc = d$.

11. The reflection matrix with basis v_1 and v_2 is $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. The basis V_1 and V_2 (same reflection!) gives $B = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. If $M = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ then $A = MBM^{-1}$.

13. (a) $D = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$. (b) $D^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ = third derivative matrix. The third derivatives of 1, x, and x^2 are zero, so $D^3 = 0$. (c) $\lambda = 0$ (triple); only one independent eigenvector (1, 0, 0).

15. The eigenvalues are 1, 1, 1, −1. Eigenmatrices $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$.

17. (a) $TT^{\mathrm{H}} = U^{-1}AUU^{\mathrm{H}}A^{\mathrm{H}}(U^{-1})^{\mathrm{H}} = I$. (b) If T is triangular and unitary, then its diagonal entries (the eigenvalues) must have absolute value 1. Then all off-diagonal entries are zero because the columns are to be unit vectors.

19. The 1, 1 entries of $T^{\mathrm{H}}T = TT^{\mathrm{H}}$ give $|t_{11}|^2 = |t_{11}|^2 + |t_{12}|^2 + |t_{13}|^2$ so $t_{12} = t_{13} = 0$. Comparing the 2, 2 entries of $T^{\mathrm{H}}T = TT^{\mathrm{H}}$ gives $t_{23} = 0$. So T must be diagonal.

21. If $N = U\Lambda U^{-1}$, then $NN^{\mathrm{H}} = U\Lambda U^{-1}(U^{-1})^{\mathrm{H}}\Lambda^{\mathrm{H}}U^{\mathrm{H}}$ is equal to $U\Lambda\Lambda^{\mathrm{H}}U^{\mathrm{H}}$. This is the same as $U\Lambda^{\mathrm{H}}\Lambda U^{\mathrm{H}} = (U\Lambda U^{-1})^{\mathrm{H}}(U\Lambda U^{-1}) = N^{\mathrm{H}}N$. So N is normal.

23. The eigenvalues of $A(A-I)(A-2I)$ are 0, 0, 0.

25. Always $\begin{bmatrix} a^2+bc & ab+bd \\ ac+cd & bc+d^2 \end{bmatrix} - (a+d)\begin{bmatrix} a & b \\ c & d \end{bmatrix} + (ad-bc)\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$!

27. $M^{-1}J_3M = 0$, so the last two inequalities are easy. Trying for $MJ_1 = J_2M$ forces the first column of M to be zero, so M cannot be invertible. Cannot have $J_1 = M^{-1}J_2M$.

29. $A^{10} = 2^{10}\begin{bmatrix} 61 & 45 \\ -80 & -59 \end{bmatrix}$; $e^A = e^2\begin{bmatrix} 13 & 9 \\ -16 & -11 \end{bmatrix}$.

31. $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$, $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$ are similar; $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ by itself and $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ by itself.

33. (a) $(M^{-1}AM)(M^{-1}x) = M^{-1}(Ax) = M^{-1}0 = 0$. (b) The nullspaces of A and of $M^{-1}AM$ have the same *dimension*. Different vectors and different bases.

35. $J^2 = \begin{bmatrix} c^2 & 2c \\ 0 & c^2 \end{bmatrix}$, $J^3 = \begin{bmatrix} c^3 & 3c^2 \\ 0 & c^3 \end{bmatrix}$, $J^k = \begin{bmatrix} c^k & kc^{k-1} \\ 0 & c^k \end{bmatrix}$; $J^0 = I$, $J^{-1} = \begin{bmatrix} c^{-1} & -c^{-2} \\ 0 & c^{-1} \end{bmatrix}$.

37. $w(t) = \big(w(0) + tx(0) + \frac{1}{2}t^2y(0) + \frac{1}{6}t^3z(0)\big)e^{5t}$.

39. (a) Choose M_i = reverse diagonal matrix to get $M_i^{-1}J_iM_i = M_i^{\mathrm{T}}$ in each block
(b) M_0 has those blocks M_i on its diagonal to get $M_0^{-1}JM_0 = J^{\mathrm{T}}$.
(c) $A^{\mathrm{T}} = (M^{-1})^{\mathrm{T}}J^{\mathrm{T}}M^{\mathrm{T}}$ is $(M^{-1})^{\mathrm{T}}M_0^{-1}JM_0M^{\mathrm{T}} = (MM_0M^{\mathrm{T}})^{-1}A(MM_0M^{\mathrm{T}})$, and A^{T} is similar to A.

41. (a) True: One has $\lambda = 0$, the other doesn't. (b) False. Diagonalize a nonsymmetric matrix and Λ is symmetric. (c) False: $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ are similar.
(d) True: All eigenvalues of $A + I$ are increased by 1, thus different from the eigenvalues of A.

43. Diagonals 6 by 6 and 4 by 4; AB has all the same eigenvalues as BA plus $6-4$ zeros.

Problem Set 6.1

1. $ac - b^2 = 2 - 4 = -2 < 0$; $x^2 + 4xy + 2y^2 = (x+2y)^2 - 2y^2$ (difference of squares).

3. $\det(A - \lambda I) = \lambda^2 - (a+c)\lambda + ac - b^2 = 0$ gives $\lambda_1 = ((a+c) + \sqrt{(a-c)^2+b^2})/2$ and $\lambda_2 = ((a+c) - \sqrt{(a-c)^2+4b^2})/2)$; $\lambda_1 > 0$ is a sum of positive numbers; $\lambda_2 > 0$ because $(a+c)^2 > (a-c)^2 + 4b^2$ reduces to $ac > b^2$. Better way: product $\lambda_1\lambda_2 = ac - b^2$.

5. (a) Positive definite when $-3 < b < 3$.

(b) $\begin{bmatrix} 1 & b \\ b & 9 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ b & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 9-b^2 \end{bmatrix}\begin{bmatrix} 1 & b \\ 0 & 1 \end{bmatrix}$. (c) The minimum is $-\dfrac{1}{2(9-b^2)}$ when $\begin{bmatrix} 1 & b \\ b & 9 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, which is $\begin{bmatrix} x \\ y \end{bmatrix} = \dfrac{1}{9-b^2}\begin{bmatrix} -b \\ 1 \end{bmatrix}$. (d) No minimum, let $y \to \infty$, $x = -3y$, then $x - y$ approaches $-\infty$.

7. (a) $A_1 = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 1 \\ -1 & 1 & 1 \end{bmatrix}$ and $A_2 = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 2 & -2 \\ -1 & -2 & 11 \end{bmatrix}$.

(b) $f_1 = (x_1 - x_2 - x_3)^2 = 0$ when $x_1 - x_2 - x_3 = 0$.

(c) $f_2 = (x_1 - x_2 - x_3)^2 + (x_2 - 3x_3)^2 + x_3^2$; $L = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & -3 & 1 \end{bmatrix}$.

9. $A = \begin{bmatrix} 3 & 6 \\ 6 & 16 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$; the coefficients of the squares are the pivots in D, whereas the coefficients inside the squares are columns of L.

11. (a) Pivots are a and $c - |b|^2/a$ and $\det A = ac - |b|^2$. (b) Multiply $|x_2|^2$ by $(c - |b|^2/a)$. (c) Now $x^H Ax$ is a sum of squares. (d) det $= -1$ (indefinite) and det $= +1$ (positive definite).

13. $a > 1$ and $(a-1)(c-1) > b^2$. This means that $A - I$ is positive definite.

15. $f(x, y) = x^2 + 4xy + 9y^2 = (x + 2y)^2 + 5y^2$; $f(x, y) = x^2 + 6xy + 9y^2 = (x + 3y)^2$.

17. $x^T A^T Ax = (Ax)^T (Ax) =$ length squared $= 0$ only if $Ax = 0$. Since A has independent columns, this only happens when $x = 0$.

19. $A = \begin{bmatrix} 4 & -4 & 8 \\ -4 & 4 & -8 \\ 8 & -8 & 16 \end{bmatrix}$ has only one pivot $= 4$, rank $= 1$, eigenvalues 24, 0, 0, det $A = 0$.

21. $ax^2 + 2bxy + cy^2$ has a saddle point at $(0, 0)$ if $ac < b^2$. The matrix is *indefinite* ($\lambda < 0$ and $\lambda > 0$).

Problem Set 6.2

1. A is positive definite for $a > 2$. B is never positive definite: notice $\begin{bmatrix} 1 & 4 \\ 4 & 7 \end{bmatrix}$.

3. $\det A = -2b^3 - 3b^2 + 1$ is negative at (and near) $b = \frac{2}{3}$.

5. If $x^T Ax > 0$ and $x^T Bx > 0$ for any $x \neq 0$, then $x^T (A + B)x > 0$; condition (I).

7. Positive λ's because R is symmetric and $\sqrt{\Lambda} > 0$. $R = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$; $R = \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix}$.

9. $|x^T Ay|^2 = |x^T R^T Ry|^2 = |(Rx)^T Ry|^2 \leq$ (by the ordinary Schwarz inequality) $\|Rx\|^2 \|Ry\|^2 = (x^T R^T Rx)(y^T R^T Ry) = (x^T Ax)(y^T Ay)$.

11. $A = \begin{bmatrix} 3 & -\sqrt{2} \\ -\sqrt{2} & 2 \end{bmatrix}$ has $\lambda = 1$ and 4, axes $1 \begin{bmatrix} 1 \\ \sqrt{2} \end{bmatrix}$ and $\frac{1}{2} \begin{bmatrix} \sqrt{2} \\ -1 \end{bmatrix}$ along eigenvectors.

13. *Negative definite matrices*: (I) $x^T Ax < 0$ for all nonzero vectors x. (II) All the eigenvalues of A satisfy $\lambda_i < 0$. (III) $\det A_1 < 0$, $\det A_2 > 0$, $\det A_3 < 0$.

(IV) All the pivots (without row exchanges) satisfy $d_i < 0$. (V) There is a matrix R with independent columns such that $A = -R^{\mathrm{T}}R$.

15. False (Q must contain eigenvectors of A); True (same eigenvalues as A); True ($Q^{\mathrm{T}}AQ = Q^{-1}AQ$ is similar to A); True (eigenvalues of e^{-A} are $e^{-\lambda} > 0$).

17. Start from a_{jj} = (row j of R^{T})(column j of R) = length squared of column j of R. Then $\det A = (\det R)^2$ = (volume of the R parallelepiped)$^2 \le$ product of the lengths squared of all the columns of R. This product is $a_{11}a_{22}\cdots a_{nn}$.

19. $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ has pivots $2, \frac{3}{2}, \frac{4}{3}$; $A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$ is singular;

$$A\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

21. $x^{\mathrm{T}}Ax$ is not positive when $(x_1, x_2, x_3) = (0, 1, 0)$ because of the zero on the diagonal.

23. (a) Positive definite requires positive determinant (also: all $\lambda > 0$). (b) All projection matrices except I are singular. (c) The diagonal entries of D are its eigenvalues. (d) The negative definite matrix $-I$ has det $= +1$ when n is even.

25. $\lambda_1 = 1/a^2$ and $\lambda_2 = 1/b^2$, so $a = 1/\sqrt{\lambda_1}$ and $b = 1/\sqrt{\lambda_2}$. The ellipse $9x^2 + 16y^2 = 1$ has axes with half-lengths $a = \frac{1}{3}$ and $b = \frac{1}{4}$.

27. $A = \begin{bmatrix} 9 & 3 \\ 3 & 5 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$; $C = \begin{bmatrix} 2 & 0 \\ 4 & 3 \end{bmatrix}$ has $CC^{\mathrm{T}} = \begin{bmatrix} 4 & 8 \\ 8 & 25 \end{bmatrix}$.

29. $ax^2 + 2bxy + cy^2 = a\left(x + \frac{b}{a}y\right)^2 + \frac{ac-b^2}{a}y^2$; $2x^2 + 8xy + 10y^2 = 2(x+2y)^2 + 2y^2$.

31. $x^{\mathrm{T}}Ax = 2\left(x_1 - \frac{1}{2}x_2 - \frac{1}{2}x_3\right)^2 + \frac{3}{2}(x_2 - x_3)^2$; $x^{\mathrm{T}}Bx = (x_1 + x_2 + x_3)^2$. B has one pivot.

33. A and $C^{\mathrm{T}}AC$ have $\lambda_1 > 0$, $\lambda_2 = 0$. $C(t) = tQ + (1-t)QR$, $Q = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, $R = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$; C has one positive and one negative eigenvalue, but I has two positive eigenvalues.

35. The pivots of $A - \frac{1}{2}I$ are 2.5, 5.9, -0.81, so one eigenvalue of $A - \frac{1}{2}I$ is negative. Then A has an eigenvalue smaller than $\frac{1}{2}$.

37. rank$(C^{\mathrm{T}}AC) \le$ rank A, but also rank$(C^{\mathrm{T}}AC) \ge$ rank$((C^{\mathrm{T}})^{-1}C^{\mathrm{T}}ACC^{-1}) =$ rank A.

39. No. If C is not square, $C^{\mathrm{T}}AC$ is not the same size matrix as A.

41. $\det\begin{bmatrix} 6 - 4\lambda/18 & -3 - \lambda/18 \\ -3 - \lambda/18 & 6 - 4\lambda/18 \end{bmatrix} = 0$ gives $\lambda_1 = 54$, $\lambda_2 = \dfrac{54}{5}$.

Eigenvectors $\begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

43. *Groups*: orthogonal matrices; e^{tA} for all t; all matrices with det $= 1$. If A is positive definite, the group of all powers A^k contains only positive definite matrices.

Problem Set 6.3

1. $A^{\mathrm{T}}A = \begin{bmatrix} 5 & 20 \\ 20 & 80 \end{bmatrix}$ has only $\sigma_1^2 = 85$ with $v_1 = \begin{bmatrix} 1/\sqrt{17} \\ 4/\sqrt{17} \end{bmatrix}$, so $v_2 = \begin{bmatrix} 4/\sqrt{17} \\ -1/\sqrt{17} \end{bmatrix}$.

3. $A^{\mathrm{T}}A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ has eigenvalues $\sigma_1^2 = \dfrac{3+\sqrt{5}}{2}$ and $\sigma_2^2 = \dfrac{3-\sqrt{5}}{2}$.

Since $A = A^{\mathrm{T}}$, the eigenvectors of $A^{\mathrm{T}}A$ are the same as for A. Since $\lambda_2 = \frac{1}{2}(1-\sqrt{5})$ is *negative*, $\sigma_1 = \lambda_1$ but $\sigma_2 = -\lambda_2$. The unit eigenvectors are the same as in Section 6.2 for A, except for the effect of this minus sign (because we need $Av_2 = \sigma_2 u_2$):

$$u_1 = v_1 = \begin{bmatrix} \lambda_1/\sqrt{1+\lambda_1^2} \\ 1/\sqrt{1+\lambda_1^2} \end{bmatrix} \text{ and } u_2 = -v_2 = \begin{bmatrix} \lambda_2/\sqrt{1+\lambda_2^2} \\ 1/\sqrt{1+\lambda_2^2} \end{bmatrix}.$$

5. $AA^{\mathrm{T}} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ has $\sigma_1^2 = 3$ with $u_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$ and $\sigma_2^2 = 1$ with $u_2 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$.

$A^{\mathrm{T}}A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ has $\sigma_1^2 = 3$ with $v_1 = \begin{bmatrix} 1/\sqrt{6} \\ 2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}$, $\sigma_2^2 = 1$ with $v_2 = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{bmatrix}$,

and nullvector $v_3 = \begin{bmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$.

Then $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = [u_1 \quad u_2] \begin{bmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} [v_1 \quad v_2 \quad v_3]^{\mathrm{T}}$.

7. $A = 12uv^{\mathrm{T}}$ has one singular value $\sigma_1 = 12$.

9. Multiply $U\Sigma V^{\mathrm{T}}$ using columns (of U) times rows (of ΣV^{T}).

11. To make A singular, the smallest change sets its smallest singular value σ_2 to zero.

13. The singular values of $A + I$ are not $\sigma_j + 1$. They come from eigenvalues of $(A+I)^{\mathrm{T}}(A+I)$.

15. $A^+ = \begin{bmatrix} \frac{1}{4} \\ \frac{1}{4} \\ \frac{1}{4} \\ \frac{1}{4} \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $B^+ = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \end{bmatrix}$, $C^+ = \begin{bmatrix} \frac{1}{2} & 0 \\ \frac{1}{2} & 0 \end{bmatrix}$.

A^+ is the right-inverse of A; B^+ is the left-inverse of B.

17. $A^{\mathrm{T}}A = \begin{bmatrix} 10 & 6 \\ 6 & 10 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 16 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$, take square roots of 4 and 16 to obtain $S = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$ and $Q = AS^{-1} = \dfrac{1}{\sqrt{10}}\begin{bmatrix} 3 & 1 \\ -1 & 3 \end{bmatrix}$.

19. (a) With independent columns, the row space is all of $\mathbf{R}^n$; check $(A^{\mathrm{T}}A)A^+b = A^{\mathrm{T}}b$.
(b) $A^{\mathrm{T}}(AA^{\mathrm{T}})^{-1}b$ is in the row space because A^{T} times any vector is in that space; now $(A^{\mathrm{T}}A)A^+b = A^{\mathrm{T}}AA^{\mathrm{T}}(AA^{\mathrm{T}})^{-1}b = A^{\mathrm{T}}b$. Both cases give $A^{\mathrm{T}}Ax^+ = A^{\mathrm{T}}b$.

21. Take $A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$. Then $AB = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$. From C^+ in Problem 15 we have $A^+ = \begin{bmatrix} \frac{1}{2} & 0 \\ \frac{1}{2} & 0 \end{bmatrix}$, $B^+ = \begin{bmatrix} 0 & \frac{1}{2} \\ 0 & \frac{1}{2} \end{bmatrix} = (AB)^+$, and $(AB)^+ \neq B^+A^+$.

23. $A = Q_1\Sigma Q_2^{\mathrm{T}} \Rightarrow A^+ = Q_2\Sigma^+Q_1^{\mathrm{T}} \Rightarrow AA^+ = Q_1\Sigma\Sigma^+Q_1^{\mathrm{T}}$. Squaring gives $(AA^+)^2 = Q_1\Sigma\Sigma^+\Sigma\Sigma^+Q_1^{\mathrm{T}} = Q_1\Sigma\Sigma^+Q_1^{\mathrm{T}}$. So we have projections: $(AA^+)^2 = AA^+ = (AA^+)^{\mathrm{T}}$ and similarly for A^+A. AA^+ and A^+A project onto the column space and row space of A.

Problem Set 6.4

1. $P(x) = x_1^2 - x_1x_2 + x_2^2 - x_2x_3 + x_3^2 - 4x_1 - 4x_3$ has $\partial P/\partial x_1 = 2x_1 - x_2 - 4$, $\partial P/\partial x_2 = -x_1 + 2x_2 - x_3$, and $\partial P/\partial x_3 = -x_2 + 2x_3 - 4$.

3. $\partial P_1/\partial x = x + y = 0$ and $\partial P_1/\partial y = x + 2y - 3 = 0$ give $x = -3$ and $y = 3$. P_2 has no minimum (let $y \to \infty$). It is associated with the semidefinite matrix $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

5. Put $x = (1, \ldots, 1)$ in Rayleigh's quotient (the denominator becomes n). Since $R(x)$ is always between λ_1 and λ_n, we get $n\lambda_1 \leq x^{\mathrm{T}}Ax = \textit{sum of all } a_{ij} \leq n\lambda_n$.

7. Since $x^{\mathrm{T}}Bx > 0$ for all nonzero vectors x, $x^{\mathrm{T}}(A + B)x$ will be larger than $x^{\mathrm{T}}Ax$. So the Rayleigh quotient is larger for $A + B$ (in fact all n eigenvalues are increased).

9. Since $x^{\mathrm{T}}Bx > 0$, the Rayleigh quotient for $A + B$ is larger than the quotient for A.

11. The smallest eigenvalues in $Ax = \lambda x$ and $Ax = \lambda Mx$ are $\frac{1}{2}$ and $(3 - \sqrt{3})/4$.

13. (a) $\lambda_j = \min_{S_j}[\max_{x \text{ in } S_j} R(x)] > 0$ means that every S_j contains a vector x with $R(x) > 0$. (b) $y = C^{-1}x$ gives quotient $\overline{R}(y) = \dfrac{y^{\mathrm{T}}C^{\mathrm{T}}ACy}{y^{\mathrm{T}}y} = \dfrac{x^{\mathrm{T}}Ax}{x^{\mathrm{T}}x} = R(x) > 0$.

15. The extreme subspace S_2 is spanned by the eigenvectors x_1 and x_2.

17. If $Cx = C(A^{-1}b)$ equals d then $CA^{-1}b - d$ is zero in the correction term in equation (5).

Problem Set 6.5

1. $Ay = b$ is $4\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}\begin{bmatrix} 3/16 \\ 4/16 \\ 3/16 \end{bmatrix} = b = \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}$. The linear finite element $U = \frac{3}{16}V_1 + \frac{4}{16}V_2 + \frac{3}{16}V_3$ equals the exact $u = \frac{3}{16}, \frac{4}{16}, \frac{3}{16}$ at the nodes $x = \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$.

3. $A_{33} = 3$, $b_3 = \frac{1}{3}$. Then $A = 3\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$, $b = \dfrac{1}{3}\begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$, $Ay = b$ gives $y = \dfrac{1}{9}\begin{bmatrix} 5 \\ 8 \\ 9 \end{bmatrix}$.

5. Integrate by parts: $\int_0^1 -V_i''V_j\,dx = \int_0^1 V_i'V_j'\,dx - \left[V_i'V_j\right]_{x=0}^{x=1} = \int_0^1 V_i'V_j'\,dx =$ same A_{ij}.

7. $A = 4$, $M = \frac{1}{3}$. Their ratio 12 (Rayleigh quotient on the subspace of multiples of $V(x)$) is *larger* than the true eigenvalue $\lambda = \pi^2$.

9. The mass matrix M is $h/6$ times the 1, 4, 1 tridiagonal matrix.

Problem Set 7.2

1. If Q is orthogonal, its norm is $\|Q\| = \max \|Qx\|/\|x\| = 1$ because Q preserves length: $\|Qx\| = \|x\|$ for every x. Also Q^{-1} is orthogonal and has norm one, so $c(Q) = 1$.

3. $\|ABx\| \le \|A\|\|Bx\|$, by the definition of the norm of A, and then $\|Bx\| \le \|B\|\|x\|$. Dividing by $\|x\|$ and maximizing, $\|AB\| \le \|A\|\|B\|$. The same is true for the inverse, $\|B^{-1}A^{-1}\| \le \|B^{-1}\|\|A^{-1}\|$; $c(AB) \le c(A)c(B)$ by multiplying these inequalities.

5. In the definition $\|A\| = \max \|Ax\|/\|x\|$, choose x to be the particular eigenvector in question; $\|Ax\| = |\lambda|\|x\|$, so the ratio is $|\lambda|$ and *maximum* ratio is *at least* $|\lambda|$.

7. $A^{\mathrm{T}}A$ and AA^{T} have the same eigenvalues, since $A^{\mathrm{T}}Ax = \lambda x$ gives $AA^{\mathrm{T}}(Ax) = A(A^{\mathrm{T}}Ax) = \lambda(Ax)$. Equality of the largest eigenvalues means $\|A\| = \|A^{\mathrm{T}}\|$.

9. $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, $\lambda_{\max}(A+B) > \lambda_{\max}(A) + \lambda_{\max}(B)$ (since $1 > 0 + 0$), and $\lambda_{\max}(AB) > \lambda_{\max}(A)\lambda_{\max}(B)$. So $\lambda_{\max}(A)$ is not a norm.

11. (a) Yes, $c(A) = \|A\|\,\|A^{-1}\| = c(A^{-1})$, since $(A^{-1})^{-1}$ is A again. (b) $A^{-1}b = x$ leads to $\dfrac{\|\delta b\|}{\|b\|} \le \|A\|\,\|A^{-1}\|\dfrac{\|\delta x\|}{\|x\|}$. This is $\dfrac{\|\delta x\|}{\|x\|} \ge \dfrac{1}{c}\dfrac{\|\delta b\|}{\|b\|}$.

13. $\|A\| = 2$ and $c = 1$; $\|A\| = \sqrt{2}$ and c is infinite (singular !); $\|A\| = \sqrt{2}$ and $c = 1$.

15. If $\lambda_{\max} = \lambda_{\min} = 1$, then all $\lambda_i = 1$ and $A = SIS^{-1} = I$. The only matrices with $\|A\| = \|A^{-1}\| = 1$ are *orthogonal matrices*, because $A^{\mathrm{T}}A$ has to be I.

17. The residual $b - Ay = (10^{-7}, 0)$ is much smaller than $b - Az = (.0013, .0016)$. But z is much closer to the solution than y.

19. $x_1^2 + \cdots + x_n^2$ is not smaller than $\max(x_i^2) = (\|x\|_\infty)^2$ and not larger than $(|x_1| + \cdots + |x_n|)^2$, which is $(\|x\|_1)^2$. Certainly $x_1^2 + \cdots + x_n^2 \le n \max(x_i^2)$, so $\|x\| \le \sqrt{n}\|x\|_\infty$. Choose $y = (\operatorname{sign} x_1, \operatorname{sign} x_2, \ldots, \operatorname{sign} x_n)$ to get $x \cdot y = \|x\|_1$. By Schwarz, this is at most $\|x\|\|y\| = \sqrt{n}\|x\|$. Choose $x = (1, 1, \ldots, 1)$ for maximum ratios $\sqrt{n}$.

21. The exact inverse of the 3 by 3 Hilbert matrix is $A^{-1} = \begin{bmatrix} 9 & -36 & 30 \\ -36 & 192 & -180 \\ 30 & -180 & 180 \end{bmatrix}$.

23. The largest $\|x\| = \|A^{-1}b\|$ is $1/\lambda_{\min}$; the largest error is $10^{-16}/\lambda_{\min}$.

25. Exchange $\begin{bmatrix} 1 & 0 \\ 2 & 2 \end{bmatrix}$ to $\begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix} \to \begin{bmatrix} 2 & 2 \\ 0 & -1 \end{bmatrix} = U$ with $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and

$$L = \begin{bmatrix} 1 & 0 \\ .5 & 1 \end{bmatrix}. A \to \begin{bmatrix} 2 & 2 & 0 \\ 1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix} \to \begin{bmatrix} 2 & 2 & 0 \\ 0 & -1 & 1 \\ 0 & 2 & 0 \end{bmatrix} \to \begin{bmatrix} 2 & 2 & 0 \\ 0 & 2 & 0 \\ 0 & -1 & 1 \end{bmatrix} \to$$

$$\begin{bmatrix} 2 & 2 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = U. \text{Then } PA = LU \text{ with } P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \text{ and } L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ .5 & -.5 & 1 \end{bmatrix}.$$

Problem Set 7.3

1. $u_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, u_1 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, u_2 = \begin{bmatrix} 5 \\ -4 \end{bmatrix}, u_3 = \begin{bmatrix} 14 \\ -13 \end{bmatrix}; u_\infty = \dfrac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ normalized to unit vector.

3. $u_k/\lambda_1^k = c_1x_1 + c_2x_2(\lambda_2/\lambda_1)^k + \cdots + c_nx_n(\lambda_n/\lambda_1)^k \to c_1x_1$ if all ratios $|\lambda_i/\lambda_1| < 1$. The largest ratio controls, when k is large. $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ has $|\lambda_2| = |\lambda_1|$ and no convergence.

5. $Hx = x - (x - y)\dfrac{2(x - y)^{\mathrm{T}}x}{(x - y)^{\mathrm{T}}(x - y)} = x - (x - y) = y$. Then $H(Hx) = Hy$ is $x = Hy$.

7. $U = \begin{bmatrix} 1 & 0 \\ 0 & H \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -\frac{3}{5} & -\frac{4}{5} \\ 0 & -\frac{4}{5} & \frac{3}{5} \end{bmatrix} = U^{-1}$ and then $U^{-1}AU = \begin{bmatrix} 1 & -5 & 0 \\ -5 & \frac{9}{25} & \frac{12}{25} \\ 0 & \frac{12}{25} & \frac{16}{25} \end{bmatrix}$.

9. $\begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & 0 \end{bmatrix} = QR = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 1 & \cos\theta\sin\theta \\ 0 & -\sin^2\theta \end{bmatrix}$.

Then $RQ = \begin{bmatrix} c(1 + s^2) & -s^3 \\ -s^3 & -s^2c \end{bmatrix}$.

11. *Assume* that $(Q_0 \cdots Q_{k-1})(R_{k-1} \cdots R_0)$ is the QR factorization of A^k (certainly true if $k = 1$). By construction, $A_{k+1} = R_kQ_k$, so $R_k = A_{k+1}Q_k^{\mathrm{T}} = (Q_k^{\mathrm{T}} \cdots Q_0^{\mathrm{T}}AQ_0 \cdots Q_k)Q_k^{\mathrm{T}}$. Postmultiplying by $(R_{k-1} \cdots R_0)$, the assumption gives $R_k \cdots R_0 = Q_k^{\mathrm{T}} \cdots Q_0^{\mathrm{T}}A^{k+1}$. After moving the Q's to the left-hand side, this is the required result for A^{k+1}.

13. A has eigenvalues 4 and 2. Put one unit eigenvector in row 1 of P: it is either $\dfrac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ and $PAP^{-1} = \begin{bmatrix} 2 & -4 \\ 0 & 4 \end{bmatrix}$ or $\dfrac{1}{\sqrt{10}} \begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix}$ and $PAP^{-1} = \begin{bmatrix} 4 & -4 \\ 0 & 2 \end{bmatrix}$.

15. $P_{ij}A$ uses $4n$ multiplications (2 for each entry in rows i and j). By factoring out $\cos\theta$, the entries 1 and $\pm\tan\theta$ need only $2n$ multiplications, which leads to $\frac{2}{3}n^3$ for PR.

Problem Set 7.4

1. $D^{-1}(-L - U) = \begin{bmatrix} 0 & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 \end{bmatrix}$, eigenvalues $\mu = 0, \pm 1/\sqrt{2}$; $(D + L)^{-1}(-U) =$

$\begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 0 & \frac{1}{4} & \frac{1}{2} \\ 0 & \frac{1}{8} & \frac{1}{4} \end{bmatrix}$, eigenvalues $0, 0, 1/2$; $\omega_{\text{opt}} = 4 - 2\sqrt{2}$, reducing $\lambda_{\max}$ to $3 - 2\sqrt{2} \approx 0.2$.

3. $Ax_k = (2 - 2\cos k\pi h)x_k$; $Jx_k = \frac{1}{2}(\sin 2k\pi h, \sin 3k\pi h + \sin k\pi h, \ldots) = (\cos k\pi h)x_k$. For $h = \dfrac{1}{4}$, A has eigenvalues $2 - 2\cos\dfrac{\pi}{4} = 2 - \sqrt{2}$, $2 - \cos\dfrac{\pi}{2} = 2$, $2 - \cos\dfrac{3\pi}{4} = 2 + \sqrt{2}$.

5. $J = D^{-1}(L + U) = -\begin{bmatrix} 0 & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & \frac{1}{4} \\ \frac{2}{5} & \frac{2}{5} & 0 \end{bmatrix}$; the three circles have radius $r_1 = \dfrac{2}{3}$, $r_2 = \dfrac{1}{4}$, $r_3 = \dfrac{4}{5}$. Their centers are at zero, so all $|\lambda_i| \le 4/5 < 1$.

7. $-D^{-1}(L + U) = \begin{bmatrix} 0 & -b/a \\ -c/d & 0 \end{bmatrix}$ has $\mu = \pm\left(\dfrac{bc}{ad}\right)^{1/2}$; $-(D + L)^{-1}U = \begin{bmatrix} 0 & -b/a \\ 0 & bc/ad \end{bmatrix}$, $\lambda = 0, bc/ad$; $\lambda_{\max}$ does equal $\mu_{\max}^2$.

9. If $Ax = \lambda x$, then $(I - A)x = (1 - \lambda)x$. Real eigenvalues of $B = I - A$ have $|1 - \lambda| < 1$, provided that λ is between 0 and 2.

11. Always $\|AB\| \le \|A\|\|B\|$. Choose $A = B$ to find $\|B^2\| \le \|B\|^2$. Then choose $A = B^2$ to find $\|B^3\| \le \|B^2\|\|B\| \le \|B\|^3$. Continue (or use induction). Since $\|B\| \ge \max|\lambda(B)|$, it is no surprise that $\|B\| < 1$ gives convergence.

13. Jacobi has $S^{-1}T = \frac{1}{3}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ with $|\lambda|_{\max} = \frac{1}{3}$. Gauss–Seidel has $S^{-1}T = \begin{bmatrix} 0 & \frac{1}{3} \\ 0 & \frac{1}{9} \end{bmatrix}$ with $|\lambda|_{\max} = \frac{1}{9} = (|\lambda|_{\max}$ for Jacobi$)^2$.

15. Successive overrelaxation (SOR) in MATLAB.

17. The maximum row sums give all $|\lambda| \le .9$ and $|\lambda| \le 4$. The circles around diagonal entries give tighter bounds. First A: The circle $|\lambda - .2| \le .7$ contains the other circles $|\lambda - .3| \le .5$ and $|\lambda - .1| \le .6$ and all three eigenvalues. Second A: The circle $|\lambda - 2| \le 2$ contains the circle $|\lambda - 2| \le 1$ and all three eigenvalues $2 + \sqrt{2}$, 2, and $2 - \sqrt{2}$.

19. $r_1 = b - \alpha_1 Ab = b - (b^{\mathrm{T}}b/b^{\mathrm{T}}Ab)Ab$ is orthogonal to $r_0 = b$: *the residuals $r = b - Ax$ are orthogonal at each step.* To show that p_1 is orthogonal to $Ap_0 = Ab$, simplify p_1 to cP_1: $P_1 = \|Ab\|^2 b - (b^{\mathrm{T}}Ab)Ab$ and $c = b^{\mathrm{T}}b/(b^{\mathrm{T}}Ab)^2$. Certainly $(Ab)^{\mathrm{T}}P_1 = 0$, because $A^{\mathrm{T}} = A$. (That simplification put α_1 into $p_1 = b - \alpha_1 Ab + (b^{\mathrm{T}}b - 2\alpha_1 b^{\mathrm{T}}Ab + \alpha_1^2\|Ab\|^2)b/b^{\mathrm{T}}b$. For a good discussion see *Numerical Linear Algebra* by Trefethen and Bau.)

Problem Set 8.1

1. The corners are at (0, 6), (2, 2), (6, 0); see Figure 8.3.

3. The constraints give $3(2x + 5y) + 2(-3x + 8y) \le 9 - 10$, or $31y \le -1$. Can't have $y \ge 0$.

5. $x \ge 0$, $y \ge 0$, with added constraint that $x + y \le 0$ admits only the point (0, 0).

7. x (5% bonds) $= z$ (9% bonds) $= 20{,}000$ and y (6% bonds) $= 60{,}000$.

9. The cost to be minimized is $1000x + 2000y + 3000z + 1500u + 3000v + 3700w$. The amounts x, y, z to Chicago and u, v, w to New England satisfy $x + u = 1{,}000{,}000$; $y + v = 1{,}000{,}000$; $z + w = 1{,}000{,}000$; $x + y + z = 800{,}000$; $u + v + w = 2{,}200{,}000$.

Problem Set 8.2

1. At present $x_4 = 4$ and $x_5 = 2$ are in the basis, and the cost is zero. The entering variable should be x_3, to reduce the cost. The leaving variable should be x_5, since $2/1$ is less than $4/1$. With x_3 and x_4 in the basis, the constraints give $x_3 = 2$, $x_4 = 2$, and the cost is now $x_1 + x_2 - x_3 = -2$.

3. The "reduced costs" are $r = [1 \quad 1]$, so change is not good and the corner is optimal.

5. At P, $r = [-5 \quad 3]$; then at Q, $r = \left[\frac{5}{3} \quad -\frac{1}{3}\right]$; R is optimal because $r \geq 0$.

7. For a maximum problem the stopping test becomes $r \leq 0$. If this fails, and the ith component is the largest, then that column of N enters the basis; the rule **8C** for the vector leaving the basis is the same.

9. $BE = B[\cdots v \cdots] = [\cdots u \cdots]$, since $Bv = u$. So E is the correct matrix.

11. If $Ax = 0$, then $Px = x - A^{\mathrm{T}}(AA^{\mathrm{T}})^{-1}Ax = x$.

Problem Set 8.3

1. Maximize $4y_1 + 11y_2$, with $y_1 \geq 0$, $y_2 \geq 0$, $2y_1 + y_2 \leq 1$, $3y_2 \leq 1$; the primal has $x_1^* = 2$, $x_2^* = 3$, the dual has $y_1^* = \frac{1}{3}$, $y_2^* = \frac{1}{3}$, $cost = 5$.

3. The dual maximizes yb, with $y \geq c$. Therefore $x = b$ and $y = c$ are feasible, and give the same value cb for the cost in the primal and dual; by **8F** they must be optimal. If $b_1 < 0$, then the optimal x^* is changed to $(0, b_2, \ldots, b_n)$ and $y^* = (0, c_2, \ldots, c_n)$.

5. $b = [0 \quad 1]^{\mathrm{T}}$ and $c = [-1 \quad 0]$.

7. Since $cx = 3 = yb$, x and y are optimal by **8F**.

9. $x^* = [1 \quad 0]^{\mathrm{T}}$, $y^* = [1 \quad 0]$, with $y^*b = 1 = cx^*$. The second inequalities in both $Ax^* \geq b$ and $y^*A \leq c$ are strict, so the second components of y^* and x^* are zero.

11. (a) $x_1^* = 0$, $x_2^* = 1$, $x_3^* = 0$, $c^{\mathrm{T}}x = 3$. (b) It is the first quadrant with the tetrahedron in the corner cut off. (c) Maximize y_1, subject to $y_1 \geq 0$, $y_1 \leq 5$, $y_1 \leq 3$, $y_1 \leq 4$; $y_1^* = 3$.

13. Here $c = [1 \quad 1 \quad 1]$ with $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. No constraint $x \geq 0$ so the dual will have equality $yA = c$ (or $A^{\mathrm{T}}y = c^{\mathrm{T}}$). That gives $2y_1 = 1$ and $y_1 = 1$ and $y_2 = 2$ and *no feasible solution.* So the primal must have ∞ as maximum: $x_1 = -N$ and $x_2 = 2N$ and $x_3 = 0$ give Cost $= x_1 + x_2 + x_3 = N$ (arbitrarily large).

15. The columns of $\begin{bmatrix} 1 & 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 \end{bmatrix}$ or $\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix}$.

17. Take $y = [1 \quad -1]$; then $yA \geq 0$, $yb < 0$.

Problem Set 8.4

1. The maximal flow is 13, with the minimal cut separating node 6 from the other nodes.

3. Increasing the capacity of pipes from node 4 to node 6 or node 4 to node 5 will produce the largest increase in the maximal flow. The maximal flow increases from 8 to 9.

5. Assign capacities $= 1$ to all edges. The maximum number of disjoint paths from s to t then equals the maximum flow. The minimum number of edges whose removal disconnects s from t is the minimum cut. Then max flow $=$ min cut.

7. Rows 1, 4, and 5 violate Hall's condition; the 3 by 3 submatrix coming from rows 1, 4, 5, and columns 1, 2, 5 has $3 + 3 > 5$.

9. (a) The matrix has $2n$ 1s which cannot be covered by less than n lines because each line covers exactly two 1s. It takes n lines; there must be a complete matching.

(b) $\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$. The 1s can be covered with four lines; five marriages are not possible.

11. If each $m + 1$ marries the only acceptable man m, there is no one for #1 to marry (even though all are acceptable to #1).

13. Algorithm 1 gives 1–3, 3–2, 2–5, 2–4, 4–6, and algorithm 2 gives 2–5, 4–6, 2–4, 3–2, 1–3. These are equal-length shortest spanning trees.

15. (a) Rows 1, 3, 5 only have 1s in columns 2 and 4. (b) Columns 1, 3, 5 (in rows 2, 4). (c) Zero submatrix from rows 1, 3, 5 and columns 1, 3, 5. (d) Rows 2, 4 and columns 2, 4 cover all 1s.

Problem Set 8.5

1. $-10x_1 + 70(1 - x_1) = 10x_1 - 10(1 - x_1)$, or $x_1 = \frac{4}{5}, x_2 = \frac{1}{5}$; $-10y_1 + 10(1 - y_1) = 70y_1 - 10(1 - y_1)$, or $y_1 = \frac{1}{5}, y_2 = \frac{4}{5}$; average payoff $yAx = 6$.

3. If X chooses column j, Y will choose its smallest entry a_{ij} (in row i). X will not move, since this is the largest entry in that row. In Problem 2, $a_{12} = 2$ was an equilibrium of this kind. If we exchange the 2 and 4 below it, no entry has this property, and mixed strategies are required.

5. The best strategy for X combines the two lines to produce a horizontal line, guaranteeing this height of $7/3$. The combination is $\frac{2}{3}(3y + 2(1 - y)) + \frac{1}{3}(y + 3(1 - y)) = 7/3$, so X chooses the columns with frequencies $\frac{2}{3}, 0, \frac{1}{3}$.

7. For columns, we want $x_1a + (1 - x_1)b = x_1c + (1 - x_1)d = u$, so $x_1(a - b - c + d) = d - b$. For rows, $y_1a + (1 - y_1)c = y_1b + (1 - y_1)d = v$ exchanges b and c. Compare u with v:

$$u = x_1(a - b) + b = \frac{(a - b)(d - b)}{a - b - c + d} + b = \frac{ad - bc}{a - b - c + d} = \text{ same after } b \leftrightarrow c = v.$$

9. The inner maximum is the larger of y_1 and y_2; x concentrates on that one. Subject to $y_1 + y_2 = 1$, the minimum of the *larger* y is $\frac{1}{2}$. Notice $A = I$.

11. $Ax^* = \begin{bmatrix}\frac{1}{2} & \frac{1}{2}\end{bmatrix}^{\mathrm{T}}$ and $yAx^* = \frac{1}{2}y_1 + \frac{1}{2}y_2 = \frac{1}{2}$ for all strategies of Y; $y^*A = \begin{bmatrix}\frac{1}{2} & \frac{1}{2} & -1 & -1\end{bmatrix}$ and $y^*Ax = \frac{1}{2}x_1 + \frac{1}{2}x_2 - x_3 - x_4$, which cannot exceed $\frac{1}{2}$; in between is $y^*Ax^* = \frac{1}{2}$.

13. Value 0 (fair game). X chooses 2 or 3, y chooses odd or even: $x^* = y^* = \left(\frac{1}{2}, \frac{1}{2}\right)$.

Problem Set A

1. (a) Largest dim $(\mathbf{S} \cap \mathbf{T}) = 7$ when $\mathbf{S} \subset \mathbf{T}$. (b) Smallest dim $(\mathbf{S} \cap \mathbf{T}) = 2$.
(c) Smallest dim $(\mathbf{S} + \mathbf{T}) = 8$ when $\mathbf{S} \subset \mathbf{T}$.
(d) Largest dim $(\mathbf{S} + \mathbf{T}) = 13$ (all of $\mathbf{R}^{13}$).

3. $\mathbf{V} + \mathbf{W}$ and $\mathbf{V} \cap \mathbf{W}$ contain $\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ 0 & a_{32} & a_{33} & a_{34} \\ 0 & 0 & a_{43} & a_{44} \end{bmatrix}$ and $\begin{bmatrix} a_{11} & a_{12} & 0 & 0 \\ 0 & a_{22} & a_{23} & 0 \\ 0 & 0 & a_{23} & a_{34} \\ 0 & 0 & 0 & a_{44} \end{bmatrix}$.
dim $(\mathbf{V} + \mathbf{W}) = 13$ and dim $(\mathbf{V} \cap \mathbf{W}) = 7$; add to get $20 = \dim \mathbf{V} + \dim \mathbf{W}$.

5. The lines through $(1, 1, 1)$ and $(1, 1, 2)$ have $\mathbf{V} \cap \mathbf{W} = \{0\}$.

7. One basis for $\mathbf{V} + \mathbf{W}$ is v_1, v_2, w_1; dim $(\mathbf{V} \cap \mathbf{W}) = 1$ with basis $(0, 1, -1, 0)$.

9. The intersection of column spaces is the line through $y = (6, 3, 6)$:

$$y = \begin{bmatrix}1 & 5 \\ 3 & 0 \\ 2 & 4\end{bmatrix}\begin{bmatrix}1 \\ 1\end{bmatrix} = \begin{bmatrix}3 & 0 \\ 0 & 1 \\ 0 & 2\end{bmatrix}\begin{bmatrix}2 \\ 3\end{bmatrix} \text{ matches } [A \quad B]x = \begin{bmatrix}1 & 5 & 3 & 0 \\ 3 & 0 & 0 & 1 \\ 2 & 4 & 0 & 2\end{bmatrix}\begin{bmatrix}1 \\ 1 \\ -2 \\ -3\end{bmatrix} = 0.$$

The column spaces have dimension 2. Their sum and intersection give $3+1 = 2+2$.

11. $F_2 \otimes F_2 = \begin{bmatrix}F_2 & F_2 \\ F_2 & -F_2\end{bmatrix} = \begin{bmatrix}1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1\end{bmatrix}$.

13. $A_{3D} = (A_{1D} \otimes I \otimes I) + (I \otimes A_{1D} \otimes I) + (I \otimes I \otimes A_{1D})$.

Problem Set B

1. $J = \begin{bmatrix}2 & 0 \\ 0 & 0\end{bmatrix}$ (A is diagonalizable); $J = \begin{bmatrix}0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0\end{bmatrix}$ (eigenvectors $(1, 0, 0)$ and $(2, -1, 0)$).

3. $e^{Bt} = \begin{bmatrix}1 & t & 2t \\ 0 & 1 & 0 \\ 0 & 0 & 0\end{bmatrix} = I + Bt$ since $B^2 = 0$. Also $e^{Jt} = I + Jt$.

5. $J = \begin{bmatrix}1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 6\end{bmatrix}$ (distinct eigenvalues); $J = \begin{bmatrix}0 & 1 \\ 0 & 0\end{bmatrix}$ (B has $\lambda = 0, 0$ but rank 1).